

Derived Categories in Algebraic Geometry

Vihaan Dheer

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1 Background

1.1 Homological algebra

At least at the beginning, we will take the following very standard inputs from algebra:

1. The notion of an abelian category, which will usually be denoted by letters like $\mathcal{A}, \mathcal{B}$, etc. Moreover, if we make element-wise arguments then we may be implicitly using the fact that arguments can be extended via Mitchell embedding.
2. Elementary homological algebra facts like the five-lemma, the four lemmas, the snake lemma, and the fact that a short exact sequence of chain complexes gives rise to a long exact sequence in cohomology.
3. The spectral sequence associated to a filtration, which is described below.
4. The basic theory of cones, which is described below.

Let $\mathcal{A}$ be an abelian category. We denote by $\text{Ch}(\mathcal{A})$ the category of $\mathbb{Z}$ -graded chain complexes with values in $\mathcal{A}$, with cohomological indexing convention, meaning its objects are sequences $C^\bullet$ of objects $\{C^i : i \in \mathbb{Z}\}$ in $\mathcal{A}$ with differentials $d : C^i \rightarrow C^{i+1}$ satisfying $d^2 = 0$. Its cohomology is denoted $H^i(C^\bullet)$ as usual. The homotopy category $\text{K}(\mathcal{A})$ is the category with the same objects as $\mathcal{A}$, with morphisms as homotopy classes of maps.

1.1.1 Standard spectral sequences

Let $C^\bullet$ be an object of $\text{Ch}(\mathcal{A})$. A (*descending*) *filtration* of this complex is a sequence $\{F^p C^\bullet : p \in \mathbb{Z}\}$ of subcomplexes of $C^\bullet$ such that:

- (a) $F^{p+1} C^\bullet \subset F^p C^\bullet$ for all $p \in \mathbb{Z}$; and
- (b) $\bigcup_p F^p C^\bullet = C^\bullet$.

The filtration is *separated* if $\bigcap_p F^p C^\bullet = 0$. Given a filtration, we can form its *associated graded* $\text{gr}_F C^\bullet = \bigoplus_{p \in \mathbb{Z}} \text{gr}_F^p C^\bullet$, where $\text{gr}_F^p C^\bullet = F^p C^\bullet / F^{p+1} C^\bullet$. For each $p, q \in \mathbb{Z}$, write $E_0^{p,q} := (\text{gr}_F^p C^\bullet)^{p+q} = \text{gr}_F^p C^{p+q}$. Then the differential on $\text{gr}_F^p C$ goes between

$$d_0 : E_0^{p,q} = \text{gr}_F^p C^{p+q} \rightarrow \text{gr}_F^p C^{p+q+1} = E_0^{p,q+1},$$

i.e. it is a vertical differential. Now take cohomology: so,

$$E_1^{p,q} := H^{p+q}(\text{gr}_F^p C) = H^{p+q}(F^p C / F^{p+1} C).$$

Now there is a natural horizontal differential $d_1 : E_1^{p,q} \rightarrow E_1^{p+1,q}$ which works as in the following diagram:

$$\begin{array}{ccc}
 F^p C^{p+q+1} / F^{p+1} C^{p+q+1} & & F^{p+1} C^{p+q+2} / F^{p+2} C^{p+q+2} \\
 d_0 \uparrow & & d_0 \uparrow \\
 F^p C^{p+q} / F^{p+1} C^{p+q} & & F^{p+1} C^{p+q+1} / F^{p+2} C^{p+q+1} \\
 d_0 \uparrow & & d_0 \uparrow \\
 F^p C^{p+q-1} / F^{p+1} C^{p+q-1} & & F^{p+1} C^{p+q} / F^{p+2} C^{p+q}
 \end{array}$$

This is part of the E_0 page, and to get to E_1 we take cohomology at each step. Suppose we have cocycle in $E_1^{p,q}$ represented by an element $[x]$ with

$x \in F^p C^{p+q}$. Since $d[x] = 0$, $dx \in F^{p+1} C^{p+q+1}$, which we claim represents a well-defined element of $E_1^{p+1,q}$. Here's what we have to check:

- i. $[dx] \in F^{p+1} C^{p+q+1} / F^{p+2} C^{p+q+1}$ is a cocycle: $d[dx] = [d^2 x] = 0$.
- ii. If $x \in F^{p+1} C^{p+q}$, then dx is by definition in the image of the differential $E_0^{p+1,q-1} \rightarrow E_0^{p+1,q}$, so we obtain a well-defined map $Z_1^{p,q} \rightarrow E_1^{p+1,q}$.
- iii. Finally, if $x = dy$ for $y \in F^p C^{p+q-1}$, then $dx = 0$, so this descends to a well-defined map $d_1: E_1^{p,q} \rightarrow E_1^{p+1,q}$.

Lemma 1.1.1 *Let $C^\bullet$ be an object of $\text{Ch } \mathcal{A}$ and $F^* C^\bullet$ be a descending filtration. There is a canonical spectral sequence $(E_r^{\bullet,\bullet}, d_r)$ with initial pages*

$$E_0^{p,q} = \text{gr}_F^p C^{p+q} \quad E_1^{p,q} = H^{p+q}(\text{gr}_F^p C^\bullet),$$

and with d_0 and d_1 as described above. In general, the differential has signature

$$d_r: E_r^{p,q} \rightarrow E_r^{p+r, q-r+1}.$$

This is called the spectral sequence associated to the filtration F of $C^\bullet$.

Now we state the standard conditions for convergence. First of all, the filtration $F^* C^\bullet$, because it respects the differential, induces a descending filtration $F^* H^\bullet(C)$ on the cohomology of C , which is induced from the chain map $F^p C^\bullet \rightarrow C^\bullet$ as follows:

$$F^p H^n(C) := \text{im} (H^n(F^p C) \rightarrow H^n(C)).$$

We say that the filtration is *finite in cohomological degree* if for each $n \in \mathbb{Z}$, there are integers $a_n \leq b_n$ such that $F^p C^n = C^n$ for $p \leq a_n$, and $F^p C^n = 0$ for $p \geq b_n$. In other words, for every n , the set

$$\{p \in \mathbb{Z} : \text{gr}_F^p C^n \neq 0\}$$

is finite; equivalently, along each diagonal $\{(p, q) : p + q = n\}$ of $\mathbb{Z}^2$, $E_0^{p,q}$ is nonzero for only finitely many terms. So, this includes cases like when E_0 is contained in the first or third quadrant, or e.g. if it is contained in an infinitely long horizontal strip but bounded above and below.

Now in general, at a point (p, q) , the differentials leaving this point for all r are of the form $(p, q) \rightarrow (p + r, q - r + 1)$, which always adds to $p + q + 1$. In particular, they all lie on a fixed diagonal, and likewise for those differentials entering (p, q) . Thus under this condition, the point (p, q) will stop changing once $r \gg 0$ (depending on p and q). So in this case, the E_∞ page is well-defined without having to take any limits.

Lemma 1.1.2 *Let $C^\bullet$ be an object of $\text{Ch } \mathcal{A}$, let F be a descending filtration of $C^\bullet$, and let (E_r, d_r) be the associated spectral sequence. If F is finite in cohomological degree, then there is a canonical isomorphism*

$$E_\infty^{p,q} \cong F^p H^{p+q}(C) / F^{p+1} H^{p+q}(C) =: \text{gr}_F^p H^{p+q}(C).$$

In other words, the spectral sequence converges

$$E_r^{p,q} \implies H^{p+q}(C^\bullet)$$

with the associated filtration F of the cohomology.

Recall how this convergence is used: to get information about the n th cohomology of C , we look along the diagonal $E_\infty^{p,q}$ for $p+q=n$, which will necessarily be contained in the finite horizontal strip $a_n \leq p \leq b_n$. If we take the direct sum along this diagonal, we get the graded object $\mathrm{gr}_F H^n(C)$ in p .

There are a few immediate deductions: if there's at most one object $E_\infty^{p,q}$ in the n -diagonal which is nonzero, then $H^n(C) \cong E_\infty^{p,q}$, because the filtration must be of the form

$$H^n = F^{a_n} H^n \supset \dots \supset F^p H^n \supset \dots \supset F^{b_n} H^n = 0,$$

where all terms to the right (and excluding) $F^p H^n$ are zero, and those to the left are $E_\infty^{p,q}$. In general, we have to solve some extension problems to conclude: namely, for each p we have an exact sequence

$$0 \rightarrow F^{p+1} H^n \rightarrow F^p H^n \rightarrow E_\infty^{p,n-p} \rightarrow 0,$$

and starting with $F^{b_n} H^n = 0$, so $F^{b_n-1} H^n = E_\infty^{b_n-1, n+1-b_n}$, we can try to solve downwards for $F^p H^n$ until we've gotten to $F^{a_n} H^n = H^n$. In practice, it is worth mentioning that when we are working with finite-dimensional vector spaces, we have $\dim \mathrm{gr}_F H^n(C) = \dim H^n(C)$, so it is especially easy to conclude dimensions from the spectral sequence.

In these notes, we won't care too much about this spectral sequence in this level of generality; instead we will use much more the *spectral sequence associated to a bicomplex or double complex*. A double complex is a bigraded collection of objects $\{C^{i,j} : i, j \in \mathbb{Z}\}$ of an abelian category $\mathcal{A}$, equipped with a vertical differential d_v and a horizontal differential d_h satisfying the following properties:

- (a) For each (i, j) , the vertical differentials

$$d_v : C^{i,j} \rightarrow C^{i,j+1} \quad j \in \mathbb{Z}$$

makes $(C^{i,\bullet}, d_v)$ a chain complex for all i .

- (b) Likewise, the horizontal differentials $d_h : C^{i,j} \rightarrow C^{i+1,j}$ for all i, j make $(C^{\bullet,j}, d_h)$ a chain complex for all j .

- (c) The differentials satisfy $d_v d_h + d_h d_v = 0$ for all i, j .

If $C^{\bullet,\bullet}$ is a bicomplex with values in $\mathcal{A}$, then there is an associated chain complex called the *totalization*, given by

$$(\mathrm{Tot} C)^n := \bigoplus_{p+q=n} C^{p,q},$$

where the differential $(\text{Tot } C)^n \rightarrow (\text{Tot } C)^{n+1}$ is defined with (p, q) th component $C^{p,q} \rightarrow \bigoplus_{r+s=n+1} C^{r,s}$ given by

$$C^{p,q} \xrightarrow{d_v + d_h} C^{p+1,q} \oplus C^{p,q+1} \hookrightarrow (\text{Tot } C)^{n+1}.$$

Then $(d_v + d_h)^2 = d_v^2 + d_v d_h + d_h d_v + d_h^2 = 0$ by definition.

We will have legitimate interest complexes which are totalizations of bicomplexes.

Example 1.1.3 Suppose $\mathcal{A}$ has a symmetric monoidal structure compatible with the abelian structure, and let A, B be chain complexes. There is a natural bicomplex given on objects by $C^{p,q} := A^p \otimes B^q$, where the horizontal differential is

$$d_h = d_A \otimes B^q: A^p \otimes B^q \rightarrow A^{p+1} \otimes B^q,$$

and the vertical differential is

$$d_v = (-1)^p A^p \otimes d_B: A^p \otimes B^q \rightarrow A^p \otimes B^{q+1}.$$

Without the sign change on d_v , the two differentials would have commuted, but as defined they anticommute. Thus $C^{\bullet,\bullet}$ forms a bicomplex, and we define the *tensor product* $A \otimes B := \text{Tot } C$. Explicitly,

$$(A \otimes B)^n = \bigoplus_{p+q=n} A^p \otimes B^q$$

with differential $d = d_A \otimes 1 + (-1)^p 1 \otimes d_B$. It would be useful to know the relation between the cohomology of the tensor product complex and the cohomologies of the individual complexes.

Now, let C be a bicomplex. There are two natural filtrations of the associated totalization $\text{Tot } C$, induced by two filtrations on the bicomplex as follows. Arrange $C^{p,q}$ in the plane $\mathbb{Z}^2$: the horizontal filtration $F_h C$ is defined by setting $F_h^p C$ to be the same bicomplex except we set everything to the left of the line $\{(p, q) : q \in \mathbb{Z}\}$ to zero. Likewise, $F_v^q C$ is defined by setting everything below $\{(p, q) : p \in \mathbb{Z}\}$ to zero. This induces filtrations on the totalizations:

$$F_h^p \text{Tot } C := \text{Tot } F_h^p C \quad F_v^q \text{Tot } C := \text{Tot } F_v^q C,$$

which explicitly on n th cohomological components are

$$F_h^r \text{Tot } C^n = \bigoplus_{p+q=n, p \geq r} C^{p,q} \quad F_v^s \text{Tot } C^n = \bigoplus_{p+q=n, q \geq s} C^{p,q}.$$

These filtrations, by virtue of [Lemma 1.1.1](#), are going to induce two spectral sequences associated to the totalization. Our goal is to compute the E_0, E_1 , and E_2 pages of these two sequences. From the above description, clearly:

$$\begin{aligned} F_h^{p+1}(\text{Tot } C)^n &= C^{p,n-p} \oplus F_h^p(\text{Tot } C)^n \\ F_h^{q+1}(\text{Tot } C)^n &= C^{n-q,q} \oplus F_v^q(\text{Tot } C)^n \end{aligned}$$

So, if the associated spectral sequences are $E_{h,r}$ and $E_{v,r}$, then we will have $E_{h,0}^{p,q} = \text{gr}_{F_h}(\text{Tot } C)^{p+q} = C^{p,q}$ and likewise $E_{v,0}^{q,p} = C^{p,q}$. Thus, it is clear that the differentials on the first page are just the vertical and horizontal differentials respectively:

$$E_{h,0}^{p,q} = C^{p,q} \xrightarrow{d_v} E_{h,0}^{p,q+1} = C^{p,q+1} \quad E_{v,0}^{q,p} = C^{p,q} \xrightarrow{d_h} E_{v,0}^{q,p+1} = C^{p+1,q}$$

This is because we took the totalization differential $d_v + d_h$ but quotiented the target of the d_h differential (and likewise for the E_v case). Thus we take cohomology vertical (resp. horizontal) cohomology to get to E_1 :

$$E_{h,1}^{p,q} = H^q(C^{p,\bullet}, d_v) =: H_v^q(C^{p,\bullet}) \quad E_{v,1}^{q,p} = H^p(C^{\bullet,q}, d_h) = H_h^p(C^{\bullet,q}).$$

Now return to the definition of the E_1 -page differential: we just used the cocycle condition and took the total differential of the objects before taking the filtration quotients. This corresponds to simply viewing the horizontal (resp. vertical) differentials as chain maps and looking at the induced map on cohomology. Thus our E_1 differentials are

$$d_h: H_v^q(C^{p,\bullet}) \rightarrow H_v^q(C^{p+1,\bullet}) \quad d_v: H_h^p(C^{\bullet,q}) \rightarrow H_h^p(C^{\bullet,q+1}).$$

In particular, we have shown the following:

Theorem 1.1.4 *Let $C^{\bullet,\bullet}$ be a bicomplex with values in $\mathcal{A}$. There are two spectral sequences $(E_{h,r}, d_r)$ and $(E_{v,r}, d_r)$ as described above, which in particular have E_2 -pages given by*

$$E_{h,2}^{p,q} = H_h^p(H_v^q(C)) \quad E_{v,2}^{q,p} = H_v^q(H_h^p(C)).$$

If for each $n \in \mathbb{Z}$, $\{(p, q) : p + q = n, C^{p,q} \neq 0\}$ is finite, then both spectral sequences converge to $H^{p+q}(\text{Tot } C)$, where in the first case the associated filtration is induced by F_h , and in the second it is induced by F_v .

Remark 1.1.5 This data comes with some natural ‘edge maps’: for simplicity, suppose we are in the circumstance where $C^{p,q}$ is a first-quadrant bicomplex. Let us just consider the first spectral sequence. There are two edge maps

$$H^n(\text{Tot } C) \rightarrow H_h^0(H_v^n(C)) \quad H_h^n H_v^0(C) \rightarrow H^n(\text{Tot } C)$$

which either project a total cocycle to the $(0, n)$ component, or include an $(n, 0)$ cocycle in the total cohomology. These natural maps factor through E_∞ and the filtration on cohomology. So, if some of these edge components survive till E_∞ , then we already know how they are subquotients of the total cohomology.

Example 1.1.6 Suppose we’re in the situation of [Example 1.1.3](#). An object X of $\mathcal{A}$ is *flat* if $X \otimes (-)$ is an exact functor. So, if $B^\bullet$ is a complex which has no cohomology (i.e. it is an exact sequence), then $X \otimes B^\bullet$ also has no cohomology. Now suppose $A^\bullet$ is a complex of flat objects; does the tensor product $A \otimes B$ also have no cohomology?

Let's answer this under the additional assumption that $A^\bullet$ and $B^\bullet$ are bounded above complexes. Form the bicomplex $C^{i,j} = A^i \otimes B^j$ as in [Example 1.1.3](#), and observe that our hypothesis ensures that bicomplex is cutoff at some point along the x -axis and the y -axis, so this looks like a shifted third quadrant spectral sequence. Thus by [Thm. 1.1.4](#), we have a spectral sequence

$$E_2^{p,q} = H_h^p(H_v^q(C)) \implies H^{p+q}(A \otimes B).$$

Note that $H_v^q(C)^p = H^q(A^p \otimes B^\bullet, d_B) = 0$ because each A^p is flat, so $E_2^{p,q} = 0$, and thus $H^n(A \otimes B)$ is zero for all $n \in \mathbb{Z}$. Note that this simple fact would be somewhat difficult to prove just using the definitions, but has a trivial proof via spectral sequences.

Remark 1.1.7 Let us note that [Lemma 1.1.2](#) and [Thm. 1.1.4](#) actually hold in a slightly more general setting: namely, e.g. for the former, we don't need the filtration to finite in cohomological degree, only bounded towards the right. In other words, we should have the condition that for fixed n , $F^p C^n = 0$ for $p \gg 0$. In our bicomplex situation, this means that we assume $C^{p,n-p} = 0$ for $p \gg 0$ for the first sequence, or $C^{n-q,q} = 0$ for $q \gg 0$ in the second one. However, in this case, to get the convergence result, we need to assume that countable direct sums are exact in the abelian category. In the examples we're interested in this will always hold, because filtered colimits will be exact.

The subtlety is that at a given point $(p, q) \in \mathbb{Z}^2$, the sequence may not stabilize anymore. However, the differentials leaving $E_r^{p,q}$ are all going further down and to the right along the $p + q + 1$ diagonal, so eventually they are zero. Thus, we do have a well-defined cocycle space $Z_\infty^{p,q}$. To take the image of the incoming differential, for large enough r we use the image $B_r^{p,q}$ in $Z_\infty^{p,q}$, assuming $Z_r^{p,q} = Z_\infty^{p,q}$. Then $B_\infty^{p,q} = \bigcup_r B_r^{p,q}$, and $E_{p,q}^\infty = Z_\infty^{p,q} / B_\infty^{p,q}$.

In practice, we usually don't need to worry about this because we hope that the whole sequence degenerates quickly. For example, using this we can now extend [Example 1.1.6](#) to the case where $A^\bullet$ is still a bounded above complex of flat objects, but $B^\bullet$ is an arbitrary acyclic complex.

Remark 1.1.8 Again let $C^{\bullet,\bullet}$ be a bicomplex. Sometimes we care instead about the *product totalization*, denoted $\text{Tot}^\Pi(C)$,

$$\text{Tot}^\Pi(C)^n := \prod_{p+q=n} C^{p,q}$$

which is defined if $\mathcal{A}$ has countable products. The differential component mapping to $C^{p,q}$ for $p + q = n + 1$, is given by

$$(dx)_{p,q} = d_h(x_{p-1,q}) + d_v(x_{p,q-1}),$$

which we can still write as $d_h + d_v$. Then again $d^2 = 0$ and this forms a complex. We still want a spectral sequence in this case, so we can use the following trick:

observe that $D^{p,q} := C^{-p,-q}$ is a bicomplex in the opposite category $\mathcal{A}^{\text{op}}$, and

$$(\text{Tot}_{\mathcal{A}^{\text{op}}} D^{\bullet,\bullet})^n = \bigoplus_{p+q=n}^{\mathcal{A}^{\text{op}}} C^{-p,-q} = \prod_{-p-q=-n} C^{-p,-q},$$

so $(\text{Tot}_{\mathcal{A}^{\text{op}}} D^{\bullet,\bullet})^n = \text{Tot}_{\mathcal{A}}^{\Pi}(C^{\bullet,\bullet})^{-n}$, and the differentials coincide with what we just wrote down. So, we have two spectral sequences with reversed indices as before, and let us focus on the first one. The boundedness assumption from the previous remark was that along all diagonals, we must eventually get zero if we go far enough to the right. Thus the condition flips, and we obtain the statement

$$E_2^{p,q} = H_h^p(H_v^q(C)) \implies H^{p+q}(\text{Tot } C)$$

assuming that for each n , $C^{p,n-p} = 0$ for $p \ll 0$. If we do the second spectral sequence, then the condition should be that $C^{n-q,q} = 0$ for $q \ll 0$. We will see an example of this in use when defining the Hom complex. Since we used the opposite category, in this case we have to assume that countable products are exact. This will *not* always be true for the categories we care about, so we should be careful.

1.1.2 Cones and homotopy

Let's review some standard facts about cones. Again, fix $\mathcal{A}$ an abelian category, and we generally work in either the category of cochain complexes $\text{Ch}(\mathcal{A})$, or its homotopy category $K(\mathcal{A})$.

Let A be a chain complex. For $i \in \mathbb{Z}$, we define $A[i]$ to be $(A[i])^n = A^{i+n}$, with $d_{A[i]} = (-1)^i d_A$. Note that e.g. $A[1]$ is the shift of A to the *left*, not the right. This defines a functor $\text{Ch}(\mathcal{A}) \rightarrow \text{Ch}(\text{src } \mathcal{A})$. It is clear that $H^n(A[i]) = H^{i+n}(A)$, and in particular $H^0(A[i]) = H^n(A)$.

Let $f: A \rightarrow B$ be a chain map. Then $\text{Cone}(f)$ is the chain complex with

$$\text{Cone}(f)^i := A^{i+1} \oplus B^i \quad d_{\text{Cone}(f)} = \begin{pmatrix} -d_A & 0 \\ f & d_B \end{pmatrix} : A^{i+1} \oplus B^i \rightarrow A^{i+2} \oplus B^{i+1}.$$

There is a canonical exact sequence

$$0 \rightarrow B \rightarrow \text{Cone}(f) \rightarrow A[1] \rightarrow 0 \quad (1)$$

where the maps on the object level are the obvious ones: $B^i \rightarrow A^{i+1} \oplus B^i$ and $A^{i+1} \oplus B^i \rightarrow A^{i+1}$, so this sequence is 'termwise-split', but not split globally because of the extra f component in the differential of the cone. Considering the corresponding long exact sequence, we obtain

$$\cdots \rightarrow H^i(B) \rightarrow H^i(\text{Cone}(f)) \rightarrow H^{i+1}(A) \xrightarrow{\delta} H^{i+1}(B) \rightarrow \cdots,$$

where δ is the connecting map; it is easy to check in this case that $\delta = f_*$ is the induced map, and thus we see that f is a quasi-isomorphism if and only if $\text{Cone}(f)$ is acyclic.

Many properties of maps are easily translated into properties of cones. The following properties are important to define the derived category as a well-behaved localization of the homotopy category of chain complexes. They will also be used later to prove that the latter is a triangulated category.

Lemma 1.1.9 *Let $f: A \rightarrow B$ be a chain map. Then there is a natural homotopy equivalence*

$$\text{Cone}(B \xrightarrow{\tau} \text{Cone}(f)) \simeq A[1],$$

where in fact $A[1]$ is a deformation retract of $\text{Cone}(f)$.

Proof (Sketch) Let us describe what the deformation retract is:

$$A^{i+1} \xrightarrow{(-f \quad 1 \quad 0)^T} B^{i+1} \oplus (A^{i+1} \oplus B^i) \xrightarrow{(0 \quad 1 \quad 0)} A^{i+1}.$$

Then the composite written above is the identity on $A[1]$, and the other composite is a 3×3 matrix which is homotopy equivalent to the identity. $\square$

As in the lemma, let $\tau: B \rightarrow \text{Cone}(f)$ be the canonical map. Then we have a commutative diagram:

$$\begin{array}{ccccccc} A & \longrightarrow & B & \longrightarrow & \text{Cone}(f) & \longrightarrow & A[1] \xrightarrow{-f} B[1] \\ & & \parallel & & \parallel & & \downarrow_{he} \\ & & B & \longrightarrow & \text{Cone}(f) & \longrightarrow & \text{Cone}(\tau) \longrightarrow B[1] \end{array}$$

This diagram will tell us that we can shift distinguished triangles in the homotopy category. More precisely, we will want to study triangles sequences of the form $A \rightarrow B \rightarrow \text{Cone}(f) \rightarrow A[1]$, and if we ‘rotate’ such a triangle we get

$$B \rightarrow \text{Cone}(f) \rightarrow A[1] \xrightarrow{-f} B[1].$$

The diagram above and the lemma shows that this rotated triangle is isomorphic to a triangle of the first form.

Lemma 1.1.10 *Let $f: A \rightarrow B$ be a chain map, and consider the natural map $\eta: \text{Cone}(f)[-1] \rightarrow A$. Then $\text{Cone}(\eta)$ is canonically homotopy equivalent to $B[1]$.*

Proof Follows from [Lemma 1.1.9](#) and the fact homotopy invariance of cones under homotopy equivalences, which is standard. $\square$

Next, we will prove a lemma which is explained by the next section:

Lemma 1.1.11 *Suppose we have morphisms $X \xrightarrow{f} Y \xleftarrow{g} Z$, where f is a quasi-isomorphism. Then we can extend this data to a commutative diagram*

$$\begin{array}{ccc} T & \xrightarrow{f'} & Z \\ g' \downarrow & & g \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

in $K(\mathcal{A})$ where f' is a quasi-isomorphism.

Proof (Sketch) Let $\tau: Y \rightarrow \text{Cone}(f)$ be the canonical map. By our previous results, we have a diagram:

$$\begin{array}{ccccccc} \text{Cone}(\tau g)[-1] & \longrightarrow & Z & \xrightarrow{\tau g} & \text{Cone}(f) & \longrightarrow & \text{Cone}(\tau g) \\ \downarrow \bar{g}'[-1] & & \downarrow g & & \parallel & & \downarrow \bar{g} \\ \text{Cone}(\tau)[-1] & \longrightarrow & Y & \xrightarrow{\tau} & \text{Cone}(f) & \longrightarrow & \text{Cone}(\tau) \\ \simeq \downarrow he & & \parallel & & \parallel & & \simeq \downarrow he \\ X & \xrightarrow{f} & Y & \xrightarrow{\tau} & \text{Cone}(f) & \longrightarrow & X[1] \end{array}$$

Importantly, the two squares which involve a term that says $he, \simeq$ only commute up to homotopy, because in fact the truly commutative square is the one where we replace them by their homotopy inverses. Also, $\text{Cone}(\tau g) \rightarrow \text{Cone}(\tau)$ is the natural map, induced by commutativity of the middle upper square, and this map makes the upper left square commute because we could have added a downward arrow $Z[1] \rightarrow Y[1]$ on the right side of the diagram.

To conclude, we want to check that $\text{Cone}(\tau g)[-1] \rightarrow Z$ is a qis. Since f is a qis, $\text{Cone}(f)$ is acyclic. Then [Lemma 1.1.10](#) tells us that the cone of $\text{Cone}(\tau g)[-1] \rightarrow Z$ is homotopy equivalent to $\text{Cone}(f)$, hence acyclic. $\square$

Lemma 1.1.12 *Let $f, g: A \rightarrow B$ be chain maps. Then there exists a quasi-isomorphism q for which $qf \sim qg$ if and only if there exists a quasi-isomorphism q' for which $fq' \sim gq'$.*

Proof We may assume $g = 0$. The point is that both conditions are equivalent to f factoring through an acyclic complex. $\square$

Lemma 1.1.13 *Let f, g, h be composable morphisms for which fg and gh are quasi-isomorphisms. Then g is a quasi-isomorphism.*

Proof To check this we have to check it after applying H^n for all n , after which fg and gh become isomorphisms. But this property is true for isomorphisms in a category. $\square$

1.1.3 Localizing classes

In this section we define localizations of categories in the case of the existence of a localizing class. Let $\mathcal{C}$ be a category and S a subclass of a morphisms.

By analogy with rings, S is *multiplicative* if it includes all identity morphisms, and is closed under composition.

Recall the definition of the localization $\gamma: \mathcal{C} \rightarrow \mathcal{C}[S^{-1}]$, which is a category $\mathcal{C}[S^{-1}]$ equipped with a functor γ such that for every category $\mathcal{D}$, restriction along γ induces an equivalence of categories

$$\gamma^*: \text{Fun}(\mathcal{C}[S^{-1}], \mathcal{D}) \xrightarrow{\sim} \text{Fun}^S(\mathcal{C}, \mathcal{D}),$$

where Fun^S denotes the full subcategory of functors which send elements of S to isomorphisms. Recall moreover that if we allow locally large categories, then we can always construct this category with the same objects as $\mathcal{C}$, so that γ is an identity-on-objects functor, and γ^* above is an isomorphism of categories, not just an equivalence.

The following setup is one way to make localizations behave well (another is to go directly to triangulated categories, and a third is to use model categories).

Definition 1.1.14 Let $\mathcal{C}$ be a category and $S \subset \text{Mor}(\mathcal{C})$ a multiplicative subclass of morphisms. S is a *localizing class* if the following additional conditions hold:

- (a) For every $s \in S$ and map $f: X \rightarrow Y$, we can extend this data to a commutative diagram

$$\begin{array}{ccc} T & \dashrightarrow & Z \\ \downarrow s' & & \downarrow s \\ X & \xrightarrow{f} & Y \end{array} \qquad \begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow s & & \downarrow s' \\ Z & \dashrightarrow & T \end{array}$$

so that in both cases, $s' \in S$ as well.

- (b) Given two parallel maps $f, g: X \rightarrow Y$, there exists $s \in S$ for which $sf = sg$ iff there exists $s' \in S$ for which $fs' = gs'$.

The point of this definition is that we have a very explicit description of the localization. Let S be multiplicative subclass of morphisms; a *roof diagram* from A to B is a pair of arrows of the form $A \xleftarrow{s} X \xrightarrow{f} B$ for which $s \in S$. We temporarily denote this set $\text{Roof}_S(A, B)$, then there is natural equivalence relation on this set, where

$$(A \xleftarrow{s} X \rightarrow B) \sim (A \xleftarrow{s'} X' \rightarrow B)$$

iff there exists a commutative diagram of the form

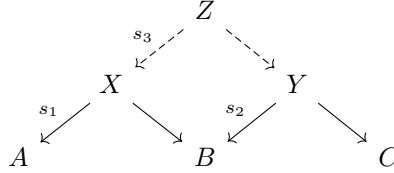
$$\begin{array}{ccccc} A & \xleftarrow{s} & X & \xleftarrow{t} & Y \\ & \swarrow s' & & & \parallel \\ B & \xleftarrow{t'} & X' & \xleftarrow{t'} & Y \end{array}$$

for which $t, t' \in S$.

Now suppose in addition that S is localizing. We are going to construct a category $\mathcal{C}[S^{-1}]$ with objects equal to those objects of $\mathcal{C}$, and morphisms

$$\mathrm{Hom}_{\mathcal{C}[S^{-1}]}(A, B) := \mathrm{Roof}_S(A, B) / \sim$$

being equivalence classes of roof diagrams with respect to S . The identity maps $A \rightarrow A$ will be represented by the roof diagrams $A \xleftarrow{1} A \xrightarrow{1} A$. The composition of two roof diagrams is defined by the following diagram:

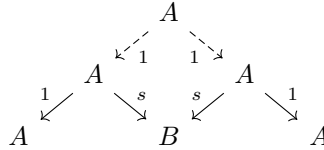


In particular, we use the first axiom of a localizing class to choose an element $s_3 \in S$ and $Z \rightarrow Y$ as in the diagram, and then our composite is the outer perimeter of the triangle.

Lemma 1.1.15 *Let S be a localizing class of morphisms in $\mathcal{C}$. The composition law defined above is well-defined on equivalence classes on roofs and independent of any choices. With this composition, $\mathcal{C}[S^{-1}]$ is a well-defined category, possibly locally large.*

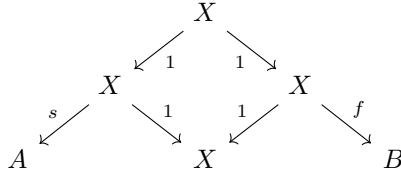
Remark 1.1.16 In general, note that if $\mathcal{C}$ was small then the localization is small. Thus we will ignore all set-theoretic technicalities by subtly referring to Grothendieck universes.

Given this construction, there is a natural map $\gamma: \mathcal{C} \rightarrow \mathcal{C}[S^{-1}]$, which is the identity on the objects, and on morphisms we send $A \rightarrow B$ to the roof diagram $A \xleftarrow{1} A \rightarrow B$. Under this map, note that everything in S becomes invertible. If $s \in S$, the corresponding map $A \xleftarrow{1} A \xrightarrow{s} B$, and we claim the inverse is $B \xleftarrow{s} A \xrightarrow{1} A$. Indeed, we can take as our composite roof:



The same thing works in the other direction. More generally, if we are given any roof of the form $A \xleftarrow{s} X \xrightarrow{f} B$, we can always write it as the following

composite:



In particular, this is equal to $\gamma(f) \circ \gamma(s)^{-1}$ as we just showed.

Lemma 1.1.17 *Keep the notation above. Then:*

- (a) *Every morphism of $\mathcal{C}[S^{-1}]$ is of the form $\gamma(f) \circ \gamma(s)^{-1}$ for some morphism f and $s \in S$.*
- (b) *γ exhibits $\mathcal{C}[S^{-1}]$ as the categorical localization of $\mathcal{C}$ with respect to the class S .*

Proof We just proved (a). We also shows that γ sends elements of S to isomorphisms. To verify the universal property, let $F: \mathcal{C} \rightarrow \mathcal{D}$ be a functor which sends S to isomorphisms. An extension along γ is clearly unique by (a), and moreover, defining $F(\gamma(f) \circ \gamma(s)^{-1}) = F(f) \circ F(s)^{-1}$ is (we claim) well-defined. Checking this comes down to using the equivalence relation on roof diagrams, and this gives the result. $\square$

Now let us study the basic properties of the roofs more closely. Once again, we will fix γ to be our localization functor. First, if a map $g = \gamma(f)\gamma(s)^{-1}$ is represented by the roof consisting of s and f , then by the first axiom of localizing classes we can write $s'f = f's$ for some $s' \in S$, which implies that

$$g = \gamma(f)\gamma(s)^{-1} = \gamma(s')^{-1}\gamma(f').$$

So, any element can also be represented as a fraction with the denominator coming first as well.

Second, suppose f, g are parallel arrows in $\mathcal{C}$, we want to know when $\gamma(f) = \gamma(g)$ holds. According to the roof diagrams, this happens iff there is a commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{s'} & A \\ s \downarrow & & \downarrow 1 \\ A & \xrightarrow{1} & A \end{array} \quad \begin{array}{c} \searrow g \\ \nearrow f \end{array} \quad \begin{array}{c} \\ B \end{array}$$

as so with $s, s' \in S$. Thus $s = s'$, and so this happens iff $gs = fs$ for some element s , which happens iff $s'g = s'f$ for some element $s' \in S$, by Axiom 2 of a localizing class.

Third, given morphisms f, g in $\mathcal{C}[S^{-1}]$, we can always write them as a roof with the same denominator, i.e. we can write them in the form

$$f = \gamma(a)\gamma(s)^{-1} \quad g = \gamma(b)\gamma(s)^{-1}$$

for a single s (and similarly the other way too). To see this, first write $f = \gamma(a)\gamma(s_1)^{-1}$ and $g = \gamma(b)\gamma(s_2)^{-1}$, and then by axiom 1 we may write $s_1s_3 = s_2r$ for some s_3, r . Let $s := s_1s_3$. Then

$$f = \gamma(a)\gamma(s_3)\gamma(s_3)^{-1}\gamma(s_1)^{-1} = \gamma(as_3)\gamma(s)^{-1},$$

and likewise $g = \gamma(br)\gamma(s)^{-1}$. More generally, we can do this simultaneously for n elements: in the induction step, we'll have

$$f_i = \gamma(a_i)\gamma(s)^{-1} \quad f_n = \gamma(b)\gamma(t)^{-1}$$

for $i \leq n-1$, then we do the same trick as in the $n=2$ case to obtain the desired representation.

Lemma 1.1.18 *Keep the same notation as above. The following are equivalent:*

i. *S satisfies the 2-out-of-6 property, namely if f, g, h are composable in $\mathcal{C}$, then*

$$fg, gh \in S \implies g \in S.$$

ii. *If g is in $\mathcal{C}$ and $\gamma(g)$ is an isomorphism, then $f \in S$.*

If any of these conditions hold, we say S is saturated. In this case, the isomorphisms in $\mathcal{C}[S^{-1}]$ are exactly the elements of the form $\gamma(s)\gamma(s')^{-1}$.

Proof If (i) holds, then it also satisfies 2-out-of-3, because we can take f or h to be the identity. If g has an inverse of the form $\gamma(h)\gamma(s)^{-1}$, then $\gamma(fh) = \gamma(s)$, so $fhs' = ss' \in S$, hence by 2-of-3 we have $gh \in S$. Likewise, it has a left inverse of the form $\gamma(s_2)^{-1}\gamma(f)$, so $fg \in S$ as well. Thus $g \in S$.

If (ii) holds, then $\gamma(f)\gamma(g)$ and $\gamma(g)\gamma(h)$ are isomorphisms, so $\gamma(g)$ is both a split epi and split mono, thus an isomorphism. Hence $g \in S$.

Lastly, if e.g. the second holds, then consider a morphism $f = \gamma(g)\gamma(s)^{-1}$ which is an isomorphism. Thus $\gamma(g)$ is an isomorphism, so $g \in S$. $\square$

Lemma 1.1.19 *Let $\mathcal{C}$ be an additive category and S a localizing class. Then $\mathcal{C}[S^{-1}]$ is an additive category, and the localization functor $\gamma: \mathcal{C} \rightarrow \mathcal{C}[S^{-1}]$ is an additive functor. Addition of morphisms is uniquely specified by the rule*

$$\gamma(f)\gamma(s)^{-1} + \gamma(g)\gamma(s)^{-1} = \gamma(f+g)\gamma(s)^{-1}$$

for parallel morphisms f and g in $\mathcal{C}$.

Proof We already know that the specification is unique, because any parallel arrows in $\mathcal{C}[S^{-1}]$ can be written in the form on the LHS. The only thing remaining to check, in light of the above remarks, is that this is well-defined. This follows (e.g. one step at a time) from the equivalence relations on roofs. $\square$

We will very quickly state some final results about localizing subcategories.

Remark 1.1.20 Via opposites (and the symmetry in the definition), it is clear that everything said here is equivalent if we use ‘right roofs’ instead of left roofs. So, e.g., morphisms in $\mathcal{C}[S^{-1}]$ can be represented as equivalence classes of diagrams

$$A \xrightarrow{f} X \xleftarrow{s} B$$

where $s \in S$, where we define things similarly as in the previous case.

Proposition 1.1.21 *Let S be a localizing class in $\mathcal{C}$ and suppose $\mathcal{B} \subset \mathcal{A}$ is a full subcategory. Suppose $S \cap \mathcal{B}$ is a localizing class in $\mathcal{B}$, and that one of the following conditions hold:*

- (a) *For any $s: X' \rightarrow X$ with $s \in S$ and X in $\mathcal{B}$, there exists $f: X'' \rightarrow X'$ with X'' in $\mathcal{B}$ such that $sf \in S$; or*
- (b) *The same as (a) but with arrows reversed.*

Then, writing $S_{\mathcal{B}} := S \cap \mathcal{B}$, the canonical map $\mathcal{B}[S_{\mathcal{B}}^{-1}] \rightarrow \mathcal{C}[S^{-1}]$ is a fully faithful embedding.

1.2 The Derived Category

Let $\mathcal{A}$ be an abelian category, and as usual we denote by $K(\mathcal{A})$ the homotopy category of chain complexes of $\mathcal{A}$. We will from now also write

$$K^+(\mathcal{A}) \quad K^-(\mathcal{A}) \quad K^b(\mathcal{A})$$

for the full (additive) subcategory of bounded below, bounded above, and bounded complexes. Note that for chain maps inside one of these categories, taking the cone lands us back inside the same category. Therefore, [Lemma 1.1.11](#) and [Lemma 1.1.12](#) tell us that the class of quasi-isomorphisms in $K(\mathcal{A})$ forms a localizing class of morphisms, and from the above remark it is also clear that the same is true for the three categories written above.

Definition 1.2.1 The additive categories $D(\mathcal{A})$, $D^+(\mathcal{A})$, $D^-(\mathcal{A})$, and $D^b(\mathcal{A})$ are all defined as the localizations of the respective homotopy categories with respect to the quasi-isomorphisms. Any of these is called the *derived category* of $\mathcal{A}$. We always consider these categories with their natural functor from $K(\mathcal{A})$ or $\text{Ch}(\mathcal{A})$, or a subcategory of one of these.

Using [Prop. 1.1.21](#), we identify the latter three categories as full subcategories of $D(\mathcal{A})$. [Lemma 1.1.13](#) with [Lemma 1.1.18](#) shows that the quasi-isomorphisms form a saturated class in all four of the categories. Thus a map is a quasi-isomorphism iff it becomes an isomorphism in the derived category.

For the sake of time, we will very briefly state some important results regarding the homotopy categories and derived categories (to be filled in later).

1.2.1 The triangulated structure

In any of the categories $D^*(\mathcal{A})$ or $K^*(\mathcal{A})$, a *triangle* will be a sequence of maps

$$A \rightarrow B \rightarrow C \rightarrow A[1].$$

A morphism of triangles is a commutative diagram

$$\begin{array}{ccccccc} A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & A[1] \\ \downarrow a & & \downarrow b & & \downarrow c & & \downarrow a[1] \\ A' & \longrightarrow & B' & \longrightarrow & C' & \longrightarrow & A'[1] \end{array}$$

A *standard* triangle is one of the form

$$A \xrightarrow{f} B \rightarrow \text{Cone}(f) \rightarrow A[1],$$

and a *distinguished* (or exact) triangle is one isomorphic to a standard triangle.

Lemma 1.2.2 *A short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ of complexes induces a distinguished triangle in $D^*(\mathcal{A})$. More precisely, there is a commutative diagram*

$$\begin{array}{ccccc} A & \xrightarrow{i} & B & \longrightarrow & \text{Cone}(i) \\ \parallel & & \parallel & & \downarrow \\ A & \xrightarrow{i} & B & \longrightarrow & C \end{array}$$

where $\text{Cone}(i) \rightarrow C$ is a quasi-isomorphism.

Proof The map $\text{Cone}(i)^n = A^{n+1} \oplus B^n \rightarrow C^n$ is given by projecting to the B factor and applying $B \rightarrow C$. This is surjective, and the kernel is termwise of the form $A^{n+1} \oplus A^n$, so we get an exact sequence

$$0 \rightarrow \text{Cone}(1_A) \rightarrow \text{Cone}(i) \rightarrow C \rightarrow 0.$$

Since $\text{Cone}(1_A)$ is acyclic the map is a quasi-isomorphism. $\square$

Lemma 1.2.3 *Let $A \xrightarrow{f} B \rightarrow \text{Cone}(f) \rightarrow A[1]$ be a standard triangle in $K^*(\mathcal{A})$. Then the rotations*

$$B \rightarrow \text{Cone}(f) \rightarrow A[1] \xrightarrow{-f} B[1]$$

and

$$\text{Cone}(f)[-1] \rightarrow A \rightarrow B \rightarrow \text{Cone}(f)$$

are both distinguished in $K^*(\mathcal{A})$. In particular, the same is true in $D^*(\mathcal{A})$.

Proof We have already discussed this via a deformation retract. $\square$

Lemma 1.2.4 *Every distinguished triangle in $D^*(\mathcal{A})$ comes from an exact sequence in the sense of [Lemma 1.2.2](#).*

Proof Every distinguished triangle is isomorphic to a standard triangle, but it's also isomorphic to the rotation of some standard triangle. Indeed, if $A \rightarrow B \rightarrow C \rightarrow A[1]$ is distinguished, then by the above lemma so is $C[-1] \rightarrow A \rightarrow B \rightarrow C$, and thus this is isomorphic to a standard triangle.

In particular, they are all isomorphic to a triangle of the form $B \rightarrow \text{Cone}(f) \rightarrow A[1] \xrightarrow{-f} B[1]$, and we know that this comes from the short exact sequence

$$0 \rightarrow B \rightarrow \text{Cone}(f) \rightarrow A[1] \rightarrow 0$$

in $\text{Ch}(\mathcal{A})$. □

As a special case of this, we see that every distinguished triangle in $D^*(\mathcal{A})$ induces a long exact sequence in cohomology. We will now partially axiomatize the above results and revisit this more carefully.

Let $\mathcal{C}$ be a category equipped with an autoequivalence $T: \mathcal{C} \rightarrow \mathcal{C}$. For simplicity, we will assume T is actually an automorphism. Just as above, we can define triangles in $\mathcal{C}$ to be sequences of the form $A \rightarrow B \rightarrow C \rightarrow T(A)$, and likewise morphisms.

Definition 1.2.5 A *triangulated category* is an additive category $\mathcal{C}$ equipped with an additive automorphism T and a class of triangles called *distinguished* or *exact triangles* satisfying the following properties:

- (TR1) (a) For A in $\mathcal{C}$, $A \xrightarrow{1_A} A \rightarrow 0 \rightarrow T(A)$ is distinguished.
- (b) Any triangle isomorphic to a distinguished triangle is distinguished.
- (c) Any morphism $f: A \rightarrow B$ can be completed to a distinguished triangle $A \rightarrow B \rightarrow C \rightarrow T[1]$.
- (TR2) A triangle $A \xrightarrow{u} B \xrightarrow{v} C \xrightarrow{w} T(A)$ is distinguished iff its *rotation*

$$B \xrightarrow{v} C \xrightarrow{w} T(A) \xrightarrow{-T(u)} T(B)$$

is distinguished.

- (TR3) Given two distinguished triangles as in the diagram below

$$\begin{array}{ccccccc} A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & T(A) \\ \downarrow f & & \downarrow g & & \downarrow h & & \downarrow T(f) \\ A' & \longrightarrow & B' & \longrightarrow & C' & \longrightarrow & T(A') \end{array}$$

with f and g given, we can find an h making the diagram commute.

- (TR4) The octahedral axiom, which we will skip.

One often calls a pre-triangulated category a category satisfying all axioms except (TR4), and in fact we will mostly just be dealing with pre-triangulated categories when we discuss basic properties.

Proposition 1.2.6 *Let $\mathcal{C}$ be a triangulated category.*

- (a) *The composite of any two morphisms in a distinguished triangle is zero.*
 (b) *For any distinguished triangle $A \rightarrow B \rightarrow C \rightarrow T(A)$, we have long exact sequences*

$$\cdots \rightarrow \mathcal{C}(X, T^i(A)) \rightarrow \mathcal{C}(X, T^i(B)) \rightarrow \mathcal{C}(X, T^i(C)) \rightarrow \mathcal{C}(X, T^{i+1}(A)) \rightarrow$$

and likewise in the contravariant direction for any object X in $\mathcal{C}$.

- (c) *Given a diagram as in (TR3), if f and g are isomorphisms, then h is an isomorphism.*
 (d) *The distinguished triangle of (TR1) is unique up to a non-canonical isomorphism.*
 (e) *If in a morphism of triangles, any two of the first three maps are isomorphisms, the third is an isomorphism.*

Proof (a) By rotation, consider $A \rightarrow B \rightarrow C \rightarrow T(A)$ and we show that $A \rightarrow B \rightarrow C$ is zero. The axioms imply that we must be able to complete the diagram below,

$$\begin{array}{ccccccc} A & \xrightarrow{1} & A & \longrightarrow & 0 & \longrightarrow & T(A) \\ \downarrow 1 & & \downarrow f & & \downarrow & & \downarrow T(f) \\ A & \xrightarrow{f} & B & \longrightarrow & C & \longrightarrow & T(A) \end{array}$$

but then $A \rightarrow B \rightarrow C$ factors through 0.

- (b) By rotation, we just have to check that if $A \rightarrow B \rightarrow C \rightarrow T(A)$ is distinguished, then

$$\mathrm{Hom}(X, A) \rightarrow \mathrm{Hom}(X, B) \rightarrow \mathrm{Hom}(X, C)$$

is exact in the middle. By (a) the composite is zero. If $a: X \rightarrow B$ satisfies $X \rightarrow B \rightarrow C = 0$, then consider the diagram of rotated triangles

$$\begin{array}{ccccccc} X & \longrightarrow & 0 & \longrightarrow & T(X) & \longrightarrow & T(X) \\ a \downarrow & & \downarrow & & \downarrow h & & \downarrow \\ B & \longrightarrow & C & \longrightarrow & T(A) & \longrightarrow & T(B) \end{array}$$

where our assumption means that the first square commutes. Then we win by using $\pm h[-1]$.

- (c) By functoriality of T and the LES of (a), we obtain a diagram of LES's and the 5-lemma shows that for every X , $\mathrm{Hom}(X, h)$ is an isomorphism. Thus h is an isomorphism.

- (d) It's unique because if we had two such C or C' , we can use (TR3) to get a map between them with maps on A and B being identities, and thus we're done by (c).

(e) Consider a morphism of triangles:

$$\begin{array}{ccccccc} A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & TA \\ \downarrow f & & \downarrow g & & \downarrow h & & \downarrow Tf \\ A' & \longrightarrow & B' & \longrightarrow & C' & \longrightarrow & TA' \end{array}$$

We have just explained the case when f and g are isomorphisms. If f and h are isos or g and h are isos, then rotate once to the right or once to the left respectively to get into that case. $\square$

Let $\mathcal{C}$ be triangulated and $\mathcal{A}$ be abelian. An additive functor $F: \mathcal{C} \rightarrow \mathcal{A}$ is *cohomological* if for any distinguished triangle $A \rightarrow B \rightarrow C \rightarrow T(A)$ the induced sequence

$$F(A) \rightarrow F(B) \rightarrow F(C)$$

is exact in $\mathcal{A}$. Given a distinguished triangle, we can always extend it to a sequence

$$\cdots \rightarrow T^i(A) \rightarrow T^i(B) \rightarrow T^i(C) \rightarrow T^{i+1}(A) \rightarrow T^{i+1}(B) \rightarrow \cdots,$$

where any subset of four terms is a distinguished triangle. Thus, a cohomological functor induces a long exact sequence in $\mathcal{A}$. For example, we have shown that $\text{Hom}(X, -)$ into abelian groups is cohomological for any X , and so is $\text{Hom}(-, X)$ if we view it as a functor $\mathcal{C} \rightarrow \mathbf{Ab}^{\text{op}}$.

Here are two related notions:

Definition 1.2.7 Let $\mathcal{C}, \mathcal{C}'$ be triangulated categories, and $\mathcal{A}$ be an abelian category.

- (a) An *exact* or *triangulated* functor $F: \mathcal{C} \rightarrow \mathcal{C}'$ is an additive functor F together with a natural isomorphism

$$\mu: F \circ T \xrightarrow{\sim} T' \circ F$$

such that whenever $A \rightarrow B \rightarrow C \rightarrow T(A)$ is distinguished, the induced triangle

$$F(A) \rightarrow F(B) \rightarrow F(C) \rightarrow (F(T(A)) \xrightarrow{\mu} T'(F(A)))$$

is distinguished.

- (b) A morphism of triangulated functors is a natural transformation which commutes with the natural isomorphisms μ above.
- (c) A δ -functor $G: \mathcal{A} \rightarrow \mathcal{C}$ is an additive functor which assigns to each short exact sequence

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

a map $\delta: G(C) \rightarrow G(A)[1]$ satisfying the following properties:

- i. $GA \rightarrow GB \rightarrow GC \xrightarrow{\delta} GA[1]$ is a distinguished triangle.
- ii. δ is natural with respect to morphisms of exact sequences.

Lemma 1.2.8 *Let $F: \mathcal{C} \rightarrow \mathcal{C}'$ be a triangulated functor which is also an equivalence of categories. Then F is a triangulated equivalence.*

Proof After unwinding the definitions, and even making the simplifying assumption that F commutes with the translation functor, the main thing to prove is that if

$$FA \xrightarrow{Ff} FB \rightarrow FC \rightarrow T'(FA) = F(T(A))$$

is distinguished in $\mathcal{C}'$, then $A \rightarrow B \rightarrow C \rightarrow TA$ was already distinguished. Indeed, to see this choose a distinguished triangle $A \xrightarrow{f} B \rightarrow D \rightarrow TA$, and apply F to it; we are going to get a diagram:

$$\begin{array}{ccccccc} FA & \xrightarrow{Ff} & FB & \longrightarrow & FC & \longrightarrow & F(TA) \\ \parallel & & \parallel & & \downarrow & & \parallel \\ FA & \xrightarrow{Ff} & FB & \longrightarrow & FD & \longrightarrow & T'F(A) \end{array}$$

Since both rows are distinguished we obtain a map $FC \rightarrow FD$, which by the previous proposition is an isomorphism. By full faithfulness, this is of the form $Fi: FC \rightarrow FD$ for an isomorphism $i: C \xrightarrow{\sim} D$, so $A \rightarrow B \rightarrow C \rightarrow TA$ was isomorphic to a distinguished triangle already. $\square$

The definition of a δ -functor is made so that for every cohomological functor $\mathcal{C} \rightarrow \mathcal{D}$, the composite $\mathcal{A} \rightarrow \mathcal{C} \rightarrow \mathcal{D}$ is a δ -functor of abelian categories. We will discuss some general properties of such functors later.

Our main example of triangulated categories are the homotopy categories $K^*(\mathcal{A})$ and their localizations $D^*(\mathcal{A})$. Once we know that they are triangulated, it will be obvious that the localization map $K^*(\mathcal{A}) \rightarrow D^*(\mathcal{A})$ is a triangulated functor.

It will also be obvious that $H^0: D^*(\mathcal{A}) \rightarrow \mathcal{A}$ is a cohomological functor: we may show the property on the middle portion of the sequence $A \rightarrow B \rightarrow \text{Cone}(f) \rightarrow A[1]$, and then we are done because we know that $H^0(B) \rightarrow H^0(\text{Cone}(f)) \rightarrow H^0(A[1])$ is part of a long exact sequence in cohomology. This gives another proof of the fact that any distinguished triangle gives a long exact sequence in cohomology.

Third, it will be clear that $\text{Ch}^*(\mathcal{A}) \rightarrow D^*(\mathcal{A})$ is a δ -functor. Indeed, we have proven in [Lemma 1.2.2](#) that for each SES $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$, we have a corresponding exact triangle

$$A \xrightarrow{i} B \rightarrow \text{Cone}(i) \rightarrow A[1]$$

induced because of the quasi-isomorphism $q_i: \text{Cone}(i) \rightarrow C$. Therefore, $A \rightarrow B \rightarrow C \rightarrow A[1]$ is an exact triangle in the derived category, and the map δ is the natural map corresponding to $\text{Cone}(i) \rightarrow A[1]$. This is called *the universal δ -functor* (not to be confused with universal δ -functors between abelian categories).

Theorem 1.2.9 *Let $\mathcal{A}$ be an abelian category. The homotopy categories $K^*(\mathcal{A})$ are all triangulated categories with the shift functor $[1]$ and the distinguished triangle as defined.*

Proof (TR1) The cone $C(1_A)$ is homotopy equivalent to 0, so the standard triangle $A \rightarrow A \rightarrow C(1_A) \rightarrow A[1]$ is isomorphic to the triangle $A \rightarrow A \rightarrow 0 \rightarrow A[1]$. (b) is obvious, and we have (c) by taking $C = \text{Cone}(A \rightarrow B)$.

(TR2) We already proved that both rotations of cones are cones up to homotopy equivalence.

(TR3) By replacing with a homotopy equivalent diagram, we may assume that $C = \text{Cone}(A \rightarrow B)$, etc. Then $h = f[1] \oplus g$ and we are done.

(TR4) As usual, we skip the octahedral axiom. $\square$

Let us now state a general means for constructing localizations in the presence of a triangulated structure on a category. We will say a localizing class on a triangulated category is *compatible with the triangulation* if $s \in S$ iff $T(s) \in S$ for all s , and if in TR3, when both $f, g \in S$, and then we can take $h \in S$ as well.

Proposition 1.2.10 *Let $\mathcal{C}$ be a triangulated category, and S localizing class of morphisms which is compatible with the triangulation. The localization $\mathcal{C}[S^{-1}]$ admits a natural triangulated structure making*

$$\mathcal{C} \rightarrow \mathcal{C}[S^{-1}]$$

a triangulated functor. A triangle in $\mathcal{C}[S^{-1}]$ is distinguished if and only if it is isomorphic to the image of a distinguished triangle in $\mathcal{C}$.

Proof Everything is easy to check using roofs with the exception of (TR4), which we skip as usual. $\square$

Corollary 1.2.11 *Let $\mathcal{A}$ be an abelian category. The derived categories $D^*(\mathcal{A})$ are triangulated categories with the shift functor $[1]$ and exact triangles as defined previously. Moreover, the localization functors*

$$K^*(\mathcal{A}) \rightarrow D^*(\mathcal{A})$$

are all triangulated functors. The functors

$$\text{Ch}^*(\mathcal{A}) \rightarrow D^*(\mathcal{A})$$

are δ -functors, and a triangle in $D^(\mathcal{A})$ is exact iff it is isomorphic to the image of an exact sequence in $\text{Ch}^*(\mathcal{A})$.*

Proof We just have to check the hypotheses of the previous theorem for quasi-isomorphisms. Clearly, a map is a qis iff its shifts are qis's, and for the (TR3) property, we have said that we can take $f[1] \oplus g$ to be the third map. If f and g are qis, then this map is also a qis. We have already explained the second statement. $\square$

Any general lemmas about the basic structure of the derived category will be added here.

Lemma 1.2.12 *Let $\mathcal{A}$ be an abelian category and $f: X \rightarrow Y$ a morphism in $D^*(\mathcal{A})$. Let $Q: K^*(\mathcal{A}) \rightarrow D^*(\mathcal{A})$ denote the localization functor. The following are equivalent:*

- (a) *f is an isomorphism.*
- (b) *$f = Q(s) \circ Q(t)^{-1}$ for s and t quasi-isomorphisms.*
- (c) *$f = Q(s)^{-1} \circ Q(t)$ for s and t quasi-isomorphisms.*
- (d) *$H^i(f)$ is an isomorphism in $\mathcal{A}$ for all $i \in \mathbb{Z}$.*

Proof We already know that (a) = (b) because of isomorphisms in a saturated localization, [Lemma 1.1.18](#). If (c) holds then clearly it is an isomorphism, and conversely if (b) holds, we can choose a diagram

$$\begin{array}{ccc} A & \xrightarrow{s} & Y \\ t \downarrow & & \downarrow q \\ X & \xrightarrow{-r-} & B \end{array}$$

by Axiom 1 of a localizing class, with q a quasi-isomorphism. Then by 2-out-of-3 r is a quasi-isomorphism, and we get (c). Finally, if (d) holds, then write f in the form $f = Q(b) \circ Q(s)^{-1}$, so that $f \circ Q(s) = Q(b)$. Applying H^i , this implies b is a quasi-isomorphism, so we're in the case of (b). $\square$

1.2.2 Derived functors

We are now going to define derived functors on the derived categories of an abelian category $\mathcal{A}$. We will at first not do this using injective/projective resolutions, and instead define adapted classes directly. In the next section, we will explain what assuming the existence of enough injectives (or projectives) buys us.

Let $\mathcal{A}$ be an abelian category, and let $F: \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor. What we say now will depend on a specific parity, so by convention our chosen ‘parity’ is left exact functor, right derived functor, etc., and the other case is delegated to parentheses.

Definition 1.2.13 Let $F: \mathcal{A} \rightarrow \mathcal{B}$ be a left (resp. right) exact functor of abelian categories. A subclass $\mathcal{R} \subset \text{Ob } \mathcal{A}$ of objects is *adapted to F* if the following properties hold:

- (a) $\mathcal{R}$ is closed under finite direct sums (meaning it includes 0).
- (b) For each acyclic complex $C^\bullet$ in $\text{Ch}^+(\mathcal{R})$ (resp. $\text{Ch}^-(\mathcal{R})$), the complex $F(C^\bullet)$ in $\text{Ch}^+(\mathcal{B})$ (resp. $\text{Ch}^-(\mathcal{B})$) is acyclic.

- (c) $\mathcal{C}$ has *enough objects in $\mathcal{R}$* , or $\mathcal{R}$ is large enough, meaning every object of $\mathcal{C}$ admits a monomorphism into (resp. an epimorphism from) an object of $\mathcal{R}$.

If $\mathcal{R}$ only satisfies conditions (a) and (c), then it is called *preadapted*, independent of the functor F .

The main theorem which will allow us to define derived functors is the following:

Theorem 1.2.14 *Let $\mathcal{R}$ be a preadapted class of objects of $\mathcal{A}$. Then $S_{\mathcal{R}}$, the class of quasi-isomorphisms between complexes with terms in $\mathcal{R}$, is a localizing class, and the canonical functor*

$$K^+(\mathcal{R})[S_{\mathcal{R}}^{-1}] \rightarrow D^+(\mathcal{A})$$

is an equivalence of categories. Moreover, $K^+(\mathcal{R})$ is a triangulated subcategory of $K^+(\mathcal{A})$, $S_{\mathcal{R}}$ is compatible with the triangulation, and the equivalence above is an equivalence of triangulated categories.

Proof We do the case that everything has a monomorphism into $\mathcal{R}$. $S_{\mathcal{R}}$ is a localizing class, because to prove that qis's were a localizing class in the homotopy category we constructed it out of shifts and cones of some maps given. If those maps lie in $K^+(\mathcal{R})$, then since $\mathcal{R}$ is closed under finite direct sums the cones (and shifts) are also in the same category. Thus $S_{\mathcal{R}}$ is a localizing class in $K^+(\mathcal{R})$.

Claim For any complex A in $K^+(\mathcal{A})$ there exists a qis $A \rightarrow R$ where R is in $K^+(\mathcal{R})$.

Once we have shown the claim, to check that this canonical functor is fully faithful, we apply [Prop. 1.1.21\(b\)](#), so we need to show that if $s: X \rightarrow X'$ is a qis with X in $\mathcal{R}$, then we can find $f: X' \rightarrow X''$ such that fs is a qis in $\mathcal{R}$. Indeed, just take a f a quasi-isomorphism in the claim where R is in $\mathcal{R}$.

Next, to check that this is essentially surjective, the claim says that every object in $D^+(\mathcal{A})$ is isomorphic to an object in $K^+(\mathcal{R})$. This means it's an equivalence. For the statements about the triangulated structure, note that we have said $K^+(\mathcal{R})$ is closed under shifts, cones (and contains 0 by definition). This is all we needed to establish the triangulated structure on the homotopy category. To prove that $S_{\mathcal{R}}$ is compatible, we already said its closed under shifts, and the other condition is one on maps, not on objects.

It is from this clear that the natural map is an exact functor of triangulated categories, and hence it is a triangulated equivalence. Thus we have reduced to proving the claim. This will be deduced from the following lemma. $\square$

Lemma 1.2.15 *Let $\mathcal{A}$ be an abelian category, and $\mathcal{R}$ a class of objects of $\mathcal{A}$ such that every object of $\mathcal{A}$ admits a monomorphism into an object of $\mathcal{R}$. Then for each $A^\bullet$ in $\text{Ch}^+(\mathcal{A})$, there exists $R^\bullet$ in $\text{Ch}^+(\mathcal{R})$ and a quasi-isomorphism*

$A^\bullet \rightarrow R^\bullet$, which we may take to be a monomorphism. Moreover, if $A^i = 0$ for $i < k$, then we may assume that $R^i = 0$ for $i < k$.

Proof Assume $A^i = 0$ for $i < 0$ choose a mono $f^0: A^0 \rightarrow R^0$, and take a pushout as in the following diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & 0 & \longrightarrow & A^0 & \xrightarrow{d_A^1} & A^1 = A^1 \\ & & \downarrow & & \downarrow f^0 & \downarrow \lrcorner & \downarrow f^1 \\ 0 & \longrightarrow & 0 & \longrightarrow & R^0 & \dashrightarrow & P \hookrightarrow R^1 \end{array}$$

The bottom composite $R^0 \rightarrow R^1$ is what we define as d_R^0 . Since pushouts of monics are monic in an abelian category, $A^1 \rightarrow P$ is monic, and thus so are f^0 and f^1 . Now if we've constructed the i th step of R already, we do a similar thing:

$$\begin{array}{ccccccc} A^{i-1} & \xrightarrow{d_A} & A^i & \xrightarrow{q} & \text{cok } d_A^{i-1} & \xrightarrow{\overline{d_A^i}} & A^{i+1} = A^{i+1} \\ \downarrow f^{i-1} & & \downarrow f^i & & \downarrow \overline{f^i} & & \downarrow a \\ R^{i-1} & \xrightarrow{d_R} & R^i & \xrightarrow{p} & \text{cok}(d_R^{i-1}) & \dashrightarrow & P \hookrightarrow R^{i+1} \end{array}$$

Namely, construct the same pushout, except first pass to the cokernels. This ensures $d^2 = 0$ because we factor through the cokernel of d^{i-1} . Moreover, we have the relation $\overline{d_A^i}q = d_A^i$. Then define $d_R^i = cbp$. This construction defines a map $f: A^\bullet \rightarrow R^\bullet$. Observe also that if f^i is monic, then a is monic, so $f^{i+1} = ca$ is monic.

We have to show that f is a quasi-isomorphism and that all the f^i are monos. To do this, in the diagram above we are going to show that $H^i(f)$ is an epimorphism, and then that the map $\text{cok } d_A^i \rightarrow \text{cok } d_R^i$ induced by f^{i+1} is monic, which will imply that its restriction to the subobjects $H^i(A) \rightarrow H^i(R)$ is monic as well. As explained in the construction this will keep each f^i monic.

For the former, let $[x] \in H^i(R)$ so $cbp(x) = 0$, so $bp(x) = 0$, meaning that the image of $(p(x), 0)$ in P is 0. Thus we can find $\overline{y}$ such that $(p(x), 0) = (\overline{f^i}, \overline{d_A^i})\overline{y}$, where we may assume $\overline{y} = q(y)$. Thus $(p(x), 0) = (pf^i(y), d_A^i(y))$, so $[y] \in H^i(A)$, and $x - f^i(y) \in \ker p = \text{im } d_R^{i-1}$, i.e. $[x] = [f^i(y)]$.

To show that the induced map $\overline{f^{i+1}}$ on cokernels is monic, we want to show that if $x \in A^{i+1}$ maps to the image of d_R^i , then it is in the image of d_A^i . So suppose $ca(x) = cbp(x)$, i.e. $a(x) = bp(x)$, then the element $(p(y), -x)$ maps to zero in P , so is of the form $(\overline{f^i}, -\overline{d_A^i})\overline{z}$, where we may assume $\overline{z} = q(z)$. Thus $-d_A^i(z) = -x$, as desired. $\square$

Here's another statement about preadapted classes which is sometimes useful.

Lemma 1.2.16 *Let $\mathcal{R}$ be a (right) preadapted class in $\mathcal{A}$. Let $X \in \mathbf{K}(\mathcal{A})$ satisfy the property that*

$$\text{Hom}_{\mathbf{K}(\mathcal{A})}(X, R) = 0$$

for every acyclic R in $K^+(\mathcal{R})$. Then the natural map

$$\mathrm{Hom}_{K(\mathcal{A})}(X, B) \rightarrow \mathrm{Hom}_{D(\mathcal{A})}(X, B)$$

is an isomorphism for all $B \in K^+(\mathcal{R})$

Proof Let $q: B \rightarrow B'$ be a quasi-isomorphism in $K^+(\mathcal{R})$. Then $B \rightarrow B' \rightarrow \mathrm{Cone}(q) \rightarrow B[1]$ is an exact triangle in $K(\mathcal{A})$, so applying the cohomological functor $\mathrm{Hom}_{K(\mathcal{A})}(X, -)$ any noting that $\mathrm{Cone}(q)$ is acyclic, we deduce that

$$q_*: \mathrm{Hom}_{K(\mathcal{A})}(X, B) \rightarrow \mathrm{Hom}_{K(\mathcal{A})}(X, B')$$

is an isomorphism. Now consider the natural map in the lemma statement: represent a map $X \rightarrow B$ by a right roof $X \rightarrow A \leftarrow B$ with $B \rightarrow A$ a quasi-isomorphism. By consider $A \rightarrow \tau_{\geq m} A$ for $m \gg 0$, we may assume A is bounded below. Choosing a quasi-isomorphism $A \rightarrow B'$ with B' in $K^+(\mathcal{R})$, we may assume we have a roof

$$X \rightarrow B' \xleftarrow{q} B$$

for q a quasi-isomorphism in $K^+(\mathcal{R})$. By the above remark, there's a unique map $X \rightarrow B$ in $K(\mathcal{A})$ making the diagram commute.

Conversely, if a map $X \rightarrow B$ in $K(\mathcal{A})$ is sent to zero in $D(\mathcal{A})$, we can repeat the argument above to get a qis $q: B \rightarrow B'$ in $K^+(\mathcal{R})$ such that $X \rightarrow B \rightarrow B' = 0$. By the above remark, $X \rightarrow B$ is zero in the homotopy category. $\square$

Now we can come to the definition and construction of derived functors in the presence of an adapted class. As usual, we state the left exact case, where the other one follows just as well.

Definition 1.2.17 Let $F: \mathcal{A} \rightarrow \mathcal{B}$ be a left-exact functor of abelian categories, and denote the standard localization functors by

$$Q_{\mathcal{A}}: K^+(\mathcal{A}) \rightarrow D^+(\mathcal{A}) \quad Q_{\mathcal{B}}: K^+(\mathcal{B}) \rightarrow D^+(\mathcal{B}).$$

A (total) right derived functor of F is a triangulated left Kan extension of $Q_{\mathcal{B}} \circ K^+(F)$ along $Q_{\mathcal{A}}$. More precisely, it is a triangulated functor

$$RF: D^+(\mathcal{A}) \rightarrow D^+(\mathcal{B})$$

equipped with a natural transformation $\varepsilon: Q_{\mathcal{B}} \circ K^+(F) \rightarrow RF \circ Q_{\mathcal{A}}$ with the universal property that for every triangulated functor $G: D^+(\mathcal{A}) \rightarrow D^+(\mathcal{B})$ and map $\pi: Q_{\mathcal{B}} \circ K^+(F) \rightarrow G \circ Q_{\mathcal{A}}$, there is a unique map $\eta: RF \rightarrow G$ such that

$$\begin{array}{ccc} Q_{\mathcal{B}} \circ K^+(F) & & \\ \varepsilon \downarrow & \searrow \pi & \\ RF \circ Q_{\mathcal{A}} & \xrightarrow{\eta \circ Q_{\mathcal{A}}} & G \circ Q_{\mathcal{A}} \end{array}$$

commutes.

In particular, the derived functor, if it exists, is unique up to canonical isomorphism commuting with the natural transformation.

Example 1.2.18 The universal property is made exactly so that natural transformations induce natural maps of our derived functors. More specifically, suppose $F, G: \mathcal{A} \rightarrow \mathcal{B}$ are functors for which (RF, ε_F) and (RG, ε_G) are derived functors. Suppose we have a map $u: F \rightarrow G$. Then this induces a natural map $u: K^+(F) \rightarrow K^+(G)$, so by the universal property there is a unique natural map $\eta: RF \rightarrow RG$ such that

$$\begin{array}{ccc} Q_{\mathcal{B}} \circ K^+(F) & \xrightarrow{u} & Q_{\mathcal{B}} \circ K^+(G) \\ \downarrow \varepsilon_F & & \downarrow \varepsilon_G \\ RF \circ Q_{\mathcal{A}} & \xrightarrow{\eta} & RG \circ Q_{\mathcal{A}} \end{array}$$

commutes. More concretely, for every $A^\bullet$ in $K^+(\mathcal{A})$, the diagram

$$\begin{array}{ccc} F(A^\bullet) & \xrightarrow{u} & G(A^\bullet) \\ \downarrow & & \downarrow \\ RF(A^\bullet) & \xrightarrow{\eta} & RG(A^\bullet) \end{array}$$

commutes. Usually, this is the best way to define maps out of/into derived functors, and in particular out of/into their cohomology.

We now show the main theorem of this section, that the derived functor exists (and how to construct it) in the presence of an adapted class.

Theorem 1.2.19 *Let $F: \mathcal{A} \rightarrow \mathcal{B}$ be a left exact functor of abelian categories, and $\mathcal{R}$ an F -adapted class of objects of $\mathcal{A}$. Then the derived functor RF exists with the following construction: For any choice Φ of inverse to the canonical equivalence of [Thm. 1.2.14](#), define RF by the diagram:*

$$\begin{array}{ccccc} K^+(\mathcal{R}) & \longrightarrow & K^+(\mathcal{R})[S_{\mathcal{R}}^{-1}] & \xleftarrow{\Phi} & D^+(\mathcal{A}) \\ & \searrow K^+(F) & \downarrow & & \downarrow RF \\ & & K^+(\mathcal{B}) & \xrightarrow{Q_{\mathcal{B}}} & D^+(\mathcal{B}) \end{array}$$

Then there is a canonical construction of ε making (RF, ε) the right derived functor of F , which satisfies the property that for each $R^\bullet$ in $K^+(\mathcal{R})$, the component

$$\varepsilon_R: F(R^\bullet) \xrightarrow{\sim} RF(R^\bullet)$$

is an isomorphism in $D^+(\mathcal{B})$.

To prove this theorem, let us make a few observations. Firstly, it is clear from the definition that for any additive functor F , we have $F(\text{Cone}(f)) = \text{Cone}(Ff)$ (where we are implicitly using F for the functor on $\text{Ch}^*(\mathcal{A})$ and $K^*(\mathcal{A})$ as well). Therefore, since F preserves acyclics between objects of $\mathcal{R}$, it preserves quasi-isomorphisms between objects of $\mathcal{R}$. Hence it extends along the localization $K^+(\mathcal{R}) \rightarrow K^+(\mathcal{R})[S_{\mathcal{R}}^{-1}]$, justifying the left triangle in the diagram

above. Note that this extension is triangulated because F preserves cones, so if $A \rightarrow B \rightarrow \text{Cone}(f) \rightarrow A[1]$ is distinguished then

$$FA \rightarrow FB \rightarrow F(\text{Cone } f) = \text{Cone}(Ff) \rightarrow FA[1]$$

is distinguished. Then [Thm. 1.2.14](#) shows that $\Psi: D^+(\mathcal{A}) \rightarrow K^+(\mathcal{R})[S_{\mathcal{R}}^{-1}]$, so we can choose a homotopy inverse Φ . With this choice, the diagram above indeed makes sense and we get a well-defined functor RF between the derived categories. By the above, RF is triangulated because Ψ is a triangulated equivalence (so Φ is one too).

Now we construct the natural map ε . By definition, there is a natural isomorphism

$$\beta: \text{Id}_{D^+(\mathcal{A})} \xrightarrow{\sim} \Psi \circ \Phi,$$

so for any A in $K^+(\mathcal{A})$, viewed as an object of $D^+(\mathcal{A})$, we have an isomorphism $\beta_A: A \xrightarrow{\sim} \Psi(\Phi(A))$, where $\Phi(A) = Y$ is some object of $K^+(\mathcal{R})$. On objects, $\Psi(Y) = Y$. Thus we may write $\beta_A = Q_{\mathcal{A}}(s)^{-1}Q_{\mathcal{A}}(t)$ for some quasi-isomorphisms s, t , in a commutative diagram

$$\begin{array}{ccc} A & & \\ Q_{\mathcal{A}}(t) \downarrow & \searrow \beta_A & \\ X & \xrightarrow{Q_{\mathcal{A}}(s)^{-1}} & Y \end{array}$$

By taking a quasi-isomorphism $X \rightarrow R$ for some R in $K^+(\mathcal{R})$ and composing, we may assume that X is in $K^+(\mathcal{R})$. Now define ε_A to be

$$F(A) \xrightarrow{Q_{\mathcal{B}}(F(t))} F(X) \xrightarrow{Q_{\mathcal{B}}(s)^{-1}} F(Y) =: RF(A),$$

viewed inside $D^+(\mathcal{B})$, where the second map is defined because s was a qis between objects of $K^+(\mathcal{R})$, so $F(s)$ is a qis. First of all, note immediately that if A was in $K^+(\mathcal{R})$, the left map would also be a quasi-isomorphism so $Q_{\mathcal{B}}F(A) \xrightarrow{\sim} RFQ_{\mathcal{A}}(A)$ would be an isomorphism in $D^+(\mathcal{B})$. Second, we claim that this map is independent of the choice of middle object X in $K^+(\mathcal{R})$.

This is clear, because for any two X_1 and X_2 middle objects over $\mathcal{R}$ we can complete this to a commutative diagram of the form:

$$\begin{array}{ccccc} & & X_1 & & \\ & \nearrow t_1 & \downarrow & \nwarrow s_1 & \\ A & \xrightarrow{t} & X & \xleftarrow{s} & Y \\ & \searrow t_2 & \uparrow & \swarrow s_2 & \\ & & X_2 & & \end{array}$$

Here X may not be over $\mathcal{R}$, but once we apply F , s_1 and s_2 become invertible, so commutativity of the diagram gives the desired statement. It is not difficult from this to check naturality of ε . Finally, checking the universal property of the left Kan extension is also a standard roof argument, where at each step we

can assume our right-roofs have middle term in $\mathcal{R}$. This completes the sketch of [Thm. 1.2.19](#).

Example 1.2.20 Suppose we are in the situation of [Thm. 1.2.19](#), and A is a complex for which $A^i = 0$ for $i < k$. Then $H^i(RF(A)) = 0$ for $i < k$. Indeed, by [Lemma 1.2.15](#), we can choose our resolution $R^\bullet$ to have no terms below degree k , and then the cohomology of $F(R^\bullet)$ will have the same property.

Example 1.2.21 Suppose A and B are complexes. Since $\text{Ch}^*(\mathcal{A}) \rightarrow D^*(\mathcal{A})$ is additive, $A \oplus B$ can refer unambiguously to the termwise direct sum complex. Since H^i is an additive functor (e.g. on complexes or the derived category), we have $H^i(A) \oplus H^i(B) = H^i(A \oplus B)$. Since the derived functors are (in particular) additive, $RF(A \oplus B) = RF(A) \oplus RF(B)$.

In the rest of this section, we discuss what happens when we want to compute a composition of two derived functors. This is the first place where the derived category makes something significantly easier than classical derived functors.

Theorem 1.2.22 *Consider two left exact functors between abelian categories*

$$\mathcal{A} \xrightarrow{F} \mathcal{B} \xrightarrow{G} \mathcal{C}.$$

Let $\mathcal{R}_\mathcal{A}$ be a class adapted to F , and $\mathcal{R}_\mathcal{B}$ a class adapted to G . If $F(\mathcal{R}_\mathcal{A}) \subset \mathcal{R}_\mathcal{B}$, then the three derived functors RF, RG , and $R(G \circ F)$ all exist on the (bounded below) derived categories, and the natural map

$$R(G \circ F) \xrightarrow{\sim} RG \circ RF$$

is a natural isomorphism.

Proof RF and RG exist by [Thm. 1.2.19](#). Note that $\mathcal{R}_\mathcal{A}$ is adapted to $G \circ F$, since if $A \in K^+(\mathcal{R}_\mathcal{A})$ is acyclic, then $F(A)$ is acyclic and in $K^+(\mathcal{R}_\mathcal{B})$, so $GF(A)$ is acyclic. Thus $R(G \circ F)$ exists; note $RG \circ RF$ is exact and the natural maps $\varepsilon_G(B): G(B) \rightarrow RG(B)$ and $\varepsilon_F(A): F(A) \rightarrow RF(A)$ in the derived category give a composite

$$\pi(A) = GF(A) \xrightarrow{\varepsilon_G(FA)} RG(FA) \xrightarrow{RG(\varepsilon_F(A))} RG(RF(A)).$$

Hence there is a map $R(G \circ F) \rightarrow RG \circ RF$ such that

$$\begin{array}{ccc} (G \circ F)A & \longrightarrow & RG(FA) \\ \downarrow & & \downarrow \\ R(G \circ F)A & \dashrightarrow & RG(RF(A)) \end{array}$$

commutes in $D^+(\mathcal{C})$ for each $A^\bullet$ in $K^+(\mathcal{A})$. If we take A to be in $K^+(\mathcal{R}_\mathcal{A})$, then the left map is an isomorphism, and $FA \rightarrow RF(A)$ is an isomorphism so the right one is too, and FA is in $K^+(\mathcal{R}_\mathcal{B})$ so the top map is an isomorphism too. Now we are done since every object is isomorphic to one of this form. $\square$

So, in practice, we now have a good idea of how to compute derived functors in these situations: pick a quasi-isomorphism $A^\bullet \rightarrow R^\bullet$ where $R^\bullet$ has terms in our adapted class, and then $RF(A) \xrightarrow{\sim} RF(R) = F(R)$. Finally, let us discuss the trivial case of the above.

Lemma 1.2.23 *Suppose $F: \mathcal{A} \rightarrow \mathcal{B}$ is an exact functor of abelian categories. The class of all objects of $\mathcal{A}$ is adapted to F , and the right/left derived functors of F both exist and coincide with the restrictions of the natural induced functor*

$$F: D(\mathcal{A}) \rightarrow D(\mathcal{B}).$$

Proof This natural functor exists because F sends acyclic complexes to acyclic complexes, and hence preserves quasi-isomorphisms. Thus we clearly get a triangulated functor as above on the full derived category. Now if we restrict e.g. to $D^+(\mathcal{A})$, it is clear that ε is an isomorphism of F and RF because it is an isomorphism on all objects, since indeed the class of all objects is adapted by definition. $\square$

1.2.3 Basic objects of the derived category

Here are some useful things to work with the derived category $D^*(\mathcal{A})$ of an abelian category $\mathcal{A}$. First, we have some standard filtrations associated to any complex $A^\bullet$: The *stupid filtrations* are $\sigma_{\geq n}A$ and $\sigma_{\leq n}A$, which are given by truncating the complex.

$$(\sigma_{\geq n}A)^i = \begin{cases} 0 & i < n \\ A^i & i \geq n \end{cases} \quad (\sigma_{\leq n}A)^i = \begin{cases} 0 & i < n \\ A^i & i \leq n \end{cases},$$

and the differentials are the obvious ones. There are some obvious properties, namely there is an inclusion $\sigma_{\geq n}C \hookrightarrow C$, and

$$H^i(\sigma_{\geq n}A) = \begin{cases} 0 & i < n \\ \ker d^n & i = n \\ H^i(A) & i > n \end{cases}.$$

For the upper filtration we likewise have a quotient map $C \rightarrow \sigma_{\leq n}C$ and a similar statement as above. Finally, the main property is that we have an exact sequence

$$0 \rightarrow \sigma_{\geq n+1}C \rightarrow \sigma_{\geq n}C \rightarrow C^n[-n] \rightarrow 0$$

of complexes.

The *smart* or *canonical filtration* is $\tau_{\geq n}C$, defined as follows:

$$\cdots \rightarrow 0 \rightarrow 0 \rightarrow \operatorname{cok}(d^{n-1}) \xrightarrow{d^n} C^{n+1} \xrightarrow{d^{n+1}} C^{n+2} \rightarrow \cdots$$

The differential is well-defined because of $d^2 = 0$ in the original complex, and now the n th cohomology is

$$\ker d^n / \operatorname{cok}(d^{n-1}) = \ker d^n / \operatorname{im}(d^{n-1}) = H^n(C),$$

with the natural map $C \rightarrow \tau_{\geq n}C$ identifying cohomology $\geq n$. There is a natural map $\tau_{\geq n}C \rightarrow \tau_{\geq n+1}C$ because the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \operatorname{cok} d^{n-1} & \xrightarrow{\bar{d}^n} & C^{n+1} & \longrightarrow & C^{n+2} \\ & & \downarrow & & \downarrow & & \parallel \\ 0 & \longrightarrow & \operatorname{cok} d^n & \xrightarrow{\bar{d}^{n+1}} & C^{n+2} & & \end{array}$$

commutes. This map is surjective, and the kernel is clearly

$$0[n-1] \rightarrow \operatorname{cok} d^{n-1}[n] \rightarrow \operatorname{im} d^n[n+1] \rightarrow 0.$$

The nontrivial differential is surjective, so the H^{n+1} of this complex is zero, and $H^n \cong H^n(C)$. Thus the kernel is quasi-isomorphic to $H^n(C)[-n]$, and hence there is a distinguished triangle

$$H^n(C)[-n] \rightarrow \tau_{\geq n}C \rightarrow \tau_{\geq n+1}C \rightarrow H^n(C)[-n+1]$$

in the derived category. Likewise, the upper filtration $\tau_{\leq n}C$ is given by

$$\dots \rightarrow C^{n-2} \rightarrow C^{n-1} \rightarrow \ker d^n \rightarrow 0 \rightarrow 0 \rightarrow \dots,$$

so we have a natural inclusion $\tau_{\leq n}C \rightarrow C$ preserving cohomology in degrees $\leq n$, and a similar distinguished triangle

$$\tau_{\leq n-1}C \rightarrow \tau_{\leq n}C \rightarrow H^n(C)[-n] \rightarrow \tau_{\leq n-1}C[1]$$

in $D^*(\mathcal{A})$.

Example 1.2.24 If $RF: D^+(\mathcal{A}) \rightarrow D^+(\mathcal{B})$ is a right derived functor in the presence of an adapted class, [Example 1.2.20](#) showed that if A has no terms below degree k , then $RF(A)$ has no cohomology below degree k . Now we can weaken the assumption to A having no *cohomology* in degree less than k . Indeed in this case, $A \rightarrow \tau_{\geq k}A$ is a quasi-isomorphism, and [Example 1.2.20](#) applies to this latter complex.

We next would like to discuss complexes concentrated at 0 (called 0-complexes), and complexes whose cohomology is concentrated in degree 0, called H^0 -complexes. Obviously, $\mathcal{A} \rightarrow \operatorname{Ch}^*(\mathcal{A})$ sending $A \mapsto A[0]$ is a fully faithful embedding. Composing this with $\operatorname{Ch}^*(\mathcal{A}) \rightarrow \operatorname{K}^*(\mathcal{A})$ remains a fully faithful embedding, because a homotopy between 0-complexes must be 0.

On the other hand, suppose $A^\bullet$ is an H^0 -complex. Then $A \rightarrow \tau_{\geq 0}C$ is a quasi-isomorphism, and then $\tau_{\leq 0}\tau_{\geq 0}C \rightarrow \tau_{\geq 0}C$ is a quasi-isomorphism, so A is isomorphic in the derived category to a complex concentrated in degree zero, namely $H^0(A)[0]$.

Lemma 1.2.25 *The natural map $\mathcal{A} \rightarrow D^*(\mathcal{A})$ defined by $A \mapsto A[0]$ is a fully faithful embedding, giving an equivalence to the full subcategory of H^0 -complexes of $D^*(\mathcal{A})$.*

Proof Consider the cohomology functor $H^0: D^*(\mathcal{A}) \rightarrow \mathcal{A}$, and restrict to the H^0 -complexes. The composite $\mathcal{A} \rightarrow D^*(\mathcal{A}) \rightarrow \mathcal{A}$ is the identity, so our functor is at least faithful. It is also essentially surjective because of the remark made above.

Now let $\alpha: A[0] \rightarrow B[0]$ be a map in the derived category, and write it as a roof $A[0] \xleftarrow{s} X \xrightarrow{u} B[0]$ in the homotopy category. Then X is an H^0 -complex, so by precomposing our roof with the natural inclusion $\tau_{\leq 0}X \rightarrow X$ (which is a quasi-isomorphism here), we may assume X has no positive degree terms.

The quasi-isomorphism $X \rightarrow \tau_{\geq 0}X = H^0(X)[0]$ sends X^i to 0 for $i \neq 0$, and $X^0 \rightarrow \text{cok } d^{-1} = H^0(X)$. On the other hand, since $B[0]$ is concentrated in 0, the map $X \rightarrow B[0]$ satisfies $u^0 d^{-1} = 0$, namely u^0 factors through the cokernel $H^0(X)$. Thus u factors through $X \rightarrow H^0(X)[0]$ and likewise for s , so we have an equivalent roof

$$A[0] \xleftarrow{s'} H^0(X)[0] \xrightarrow{u'} B[0].$$

Since the left map is a quasi-isomorphism, it must be an isomorphism of complexes, so we have found the representative $u' \circ s'^{-1}$ as a chain map for $A[0] \rightarrow B[0]$. This shows full faithfulness. $\square$

Here is an application. If A and B are objects of $\mathcal{A}$, we define

$$\text{Ext}_{\mathcal{A}}^i(A, B) := \text{Hom}_{D^*(\mathcal{A})}(A[0], B[i]) \quad i \in \mathbb{Z}$$

(as an abelian group), where this is independent of $*$ because the respective embeddings are fully faithful. Since the shift functors $[k]$ are additive equivalences of categories, it is also isomorphic to look at $\text{Hom}_{D^*(\mathcal{A})}(A[k], B[i+k])$ instead of 0.

Lemma 1.2.26 *Let $\mathcal{A}$ be an abelian category and A, B be objects of $\mathcal{A}$. Then*

- (a) $\text{Ext}_{\mathcal{A}}^i(A, B) = 0$ for all $i < 0$.
- (b) $\text{Ext}_{\mathcal{A}}^0(A, B) = \text{Hom}_{\mathcal{A}}(A, B) \equiv \mathcal{A}(A, B)$.
- (c) *Given a short exact sequence $0 \rightarrow B' \rightarrow B \rightarrow B'' \rightarrow 0$ in $\mathcal{A}$, there is a long exact sequence*

$$0 \rightarrow \mathcal{A}(A, B') \rightarrow \mathcal{A}(A, B) \rightarrow \mathcal{A}(A, B'') \rightarrow \text{Ext}_{\mathcal{A}}^1(A, B') \rightarrow \dots$$

of abelian groups.

- (d) *Given a short exact sequence $0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0$ in $\mathcal{A}$, there is a long exact sequence*

$$\dots \leftarrow \text{Ext}_{\mathcal{A}}^1(A'', B) \leftarrow \mathcal{A}(A', B) \leftarrow \mathcal{A}(A, B) \leftarrow \mathcal{A}(A'', B) \leftarrow 0$$

of abelian groups.

Proof We already proved (b) above, and (c) and (d) follow from (a) and the fact that Homs in either variable are a cohomological functor to abelian groups

in any triangulated category. For (a), represent a map $A[0] \rightarrow B[-i]$ by a roof

$$A[0] \xleftarrow{s} X \xrightarrow{f} B[-i]$$

with s a quasi-isomorphism. Since $\tau_{\leq 0}X \rightarrow X$ is a quasi-isomorphism, we may assume X has no positive terms. But $B[-i]$ is concentrated in degree $i > 0$, hence $f = 0$, so $A[0] \rightarrow B[-i]$ is zero in $D^*(\mathcal{A})$. $\square$

From the lemma, it is obvious that if $\text{Ext}_{\mathcal{A}}^1(P, B) = 0$ for every B in $\mathcal{A}$, then $\text{Hom}_{\mathcal{A}}(P, -)$ is exact, so P is projective. Likewise, if $\text{Ext}_{\mathcal{A}}^1(A, I) = 0$ for all A , then I is injective. We will in greater depth explore the converse in the next section. These are especially nice functors on the derived category, because they exist even independently of injective or projective resolutions.

Here are some elementary observations we can make in the case of Ext . The proofs will also illustrate general techniques for working with roofs. A main idea will be the following remark: Suppose we have an exact sequence

$$0 \rightarrow P^{-1} \rightarrow P^0 \rightarrow A \rightarrow 0$$

and a map $P^{-1} \rightarrow B$. Then we can extend this to a diagram of exact sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & P^{-1} & \longrightarrow & P^0 & \longrightarrow & A \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \parallel \\ 0 & \longrightarrow & B & \dashrightarrow & B' & \longrightarrow & A \longrightarrow 0 \end{array}$$

by simply taking B' to be the pushout (since the pushout of monos in an abelian category is a mono), where the bottom row is still exact by the pasting lemma (paste the square with 0, so $B'/B = P^0/P^{-1} = A$). Thus, if we have an exact sequence $0 \rightarrow P^{-i} \rightarrow \cdots \rightarrow P^0 \rightarrow A \rightarrow 0$ and a map $P^{-i} \rightarrow B$, we can do the same thing: First split this up into short exact sequences

$$0 \rightarrow P^{-i} \rightarrow P^{-i+1} \rightarrow C^{-i+1} \rightarrow 0 \quad 0 \rightarrow C^k \rightarrow P^{k+1} \rightarrow C^{k+1} \rightarrow 0$$

ending with $0 \rightarrow C^{-1} \rightarrow P^0 \rightarrow A \rightarrow 0$. Then fill in each sequence, compose the squares together, and recombine the exact sequence to obtain a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & P^{-i} & \longrightarrow & \cdots & \longrightarrow & P^0 \longrightarrow A \longrightarrow 0 \\ & & \downarrow & & & & \downarrow & \parallel \\ 0 & \longrightarrow & B & \longrightarrow & \cdots & \longrightarrow & B' \longrightarrow A \longrightarrow 0 \end{array}$$

with exact rows. Now the result:

Lemma 1.2.27 *Let A, B be objects of an abelian category $\mathcal{A}$. Morphisms in the derived category from $A[0]$ to $B[i]$, i.e. elements of $\text{Ext}_{\mathcal{A}}^i(A, B)$, are equivalence classes of roofs of the form*

$$A[0] \xleftarrow{q} P \rightarrow B[i]$$

where P is concentrated in the interval $[-i, 0]$ and q is a quasi-isomorphism, with $P^{-i} = A$ and the map $P^{-i} \rightarrow A$ is the identity.

Proof If a map is represented by a roof $A[0] \leftarrow Q \rightarrow B[i]$, then we may immediately precompose the roof with the map $\tau_{\geq 0}Q \rightarrow Q$ (which is a qis since Q is an H^0 complex), so we may assume Q is concentrated in $(-\infty, 0]$. Now we want to factor this roof through the quasi-isomorphism $Q \rightarrow \tau_{\geq -i}Q$; the map $Q \rightarrow A[0]$ obviously factors through $\tau_{\geq -i}Q$ because A is zero in degree < 0 .

For $B[i]$, we are considering a diagram:

$$\begin{array}{ccccc} Q^{-i-1} & \xrightarrow{d^{-i-1}} & Q^{-i} & \xrightarrow{d^{-i}} & Q^{-i+1} \\ \downarrow & & \downarrow f & & \downarrow \\ 0 & \longrightarrow & B & \longrightarrow & 0 \end{array}$$

The fact that $fd^{-i-1} = 0$ means exactly that f factors through the cokernel of this differential, which is how we defined $\tau_{\geq -i}Q$. This gives an equivalent roof where the middle term is concentrated in $[-i, 0]$.

Equivalently, we may view this as an exact sequence $0 \rightarrow P^{-i} \rightarrow \dots \rightarrow P^0 \rightarrow A$ equipped with a map $P^{-i} \rightarrow B$, and now by the remark above we can choose another complex Q concentrated in $[-i, 0]$ with $Q \rightarrow A[0]$ a quasi-isomorphism, $Q^{-i} = B$, and a quasi-isomorphism $P \rightarrow Q$ so that the diagram

$$\begin{array}{ccc} & P & \\ \swarrow & \downarrow & \searrow \\ A[0] & \leftarrow Q & \longrightarrow B[i] \end{array}$$

commutes. This shows what we wanted. $\square$

This lemma shows that elements of $\text{Ext}_{\mathcal{A}}^i(A, B)$ are roof equivalence classes of exact sequences,

$$0 \rightarrow B = P^{-i} \rightarrow P^{-i+1} \rightarrow \dots \rightarrow P^0 \rightarrow A \rightarrow 0.$$

Let us examine what the equivalence relation is: Two roofs $A[0] \leftarrow P \rightarrow B[-i]$ and $A[0] \leftarrow Q \rightarrow B[-i]$ of this form are equivalent iff there are quasi-isomorphisms $q: R \rightarrow P$ and $r: R \rightarrow Q$ making the diagrams

$$\begin{array}{ccc} R^0 & \xrightarrow{r^0} & Q^0 \\ q^0 \downarrow & & \downarrow \\ P^0 & \longrightarrow & A \end{array} \quad \begin{array}{ccc} R^{-i} & \xrightarrow{r^{-i}} & Q^{-i} \\ q^{-i} \downarrow & & \parallel \\ P^{-i} & \longrightarrow & B \end{array}$$

commute. By the standard trick we may assume R is concentrated in negative degrees, which as before implies $R^0 \rightarrow A$ is an epimorphism. This looks like a big commutative diagram below, the middle resolution row R is exact, as are the other two rows. Moreover, the two maps $R^{-i} \rightarrow B$ on either side are the same.

$$\begin{array}{ccccccccccc}
0 & \longrightarrow & B & \longrightarrow & Q^{-i+1} & \longrightarrow & \cdots & \longrightarrow & Q^0 & \longrightarrow & A \longrightarrow 0 \\
\uparrow & & \uparrow & & \uparrow & & & & \uparrow & & \parallel \\
\cdots & \longrightarrow & R^{-i-1} & \longrightarrow & R^{-i} & \longrightarrow & R^{-i+1} & \longrightarrow & \cdots & \longrightarrow & R^0 \longrightarrow A \longrightarrow 0 \\
\downarrow & & \downarrow & & \downarrow & & & & \downarrow & & \parallel \\
0 & \longrightarrow & B & \longrightarrow & P^{-i+1} & \longrightarrow & \cdots & \longrightarrow & P^0 & \longrightarrow & A \longrightarrow 0
\end{array}$$

Given such a diagram, commutativity means that we may replace R^{-i} with $\text{cok } R^{-i-1} \rightarrow R^{-i}$ and the lower R terms with zero without changing the diagram. Thus we can also assume $R^k = 0$ for $k < -i$. Finally, we can use the same trick from earlier: pushout the map $R^{-i} \rightarrow B$ along the sequence $0 \rightarrow R^{-i} \rightarrow R^{-i+1} \rightarrow C \rightarrow 0$. This will give a diagram like this:

$$\begin{array}{ccccccc}
0 & \longrightarrow & R^{-i} & \longrightarrow & R^{-i+1} & \longrightarrow & C \longrightarrow 0 \\
& & \downarrow & & \downarrow & & \parallel \\
0 & \longrightarrow & B & \dashrightarrow & X & \longrightarrow & C \longrightarrow 0
\end{array}$$

Now replace $R^{-i} \rightarrow R^{-i+1}$ with $B \rightarrow X$, which leaves the middle row exact and the maps still factor through as desired.

Corollary 1.2.28 *Let $\mathcal{A}$ be an abelian category with objects A, B . Then elements of $\text{Ext}_{\mathcal{A}}^i(A, B)$ correspond canonically to equivalence classes of acyclic complexes $P \in \text{Ch}^{[-i, 1]}(\mathcal{A})$ such that $P^{-i} = B$ and $P^1 = A$, as follows:*

- (a) P, Q are equivalent iff there are morphisms $P \leftarrow R \rightarrow Q$ with R also in $\text{Ext}_{\mathcal{A}}^i(A, B)$.
- (b) Given such an acyclic complex P , and considering the associated quasi-isomorphism $\sigma_{\leq 0} P \rightarrow A$, its Ext class is the one corresponding under

$$\text{Ext}^0(A, A) \xleftarrow{\cong} \text{Hom}_{D(\mathcal{A})}(A, \sigma_{\leq 0} P) \rightarrow \text{Ext}^i(A, B),$$

to the identity map, where the second map is $\text{Hom}(A, -)$ applied to the projection $\sigma_{\leq 0} P \rightarrow B[i]$.

Proof We already proved (a). Note that the arrows in (b) are obtained by applying $\text{Hom}_{D(\mathcal{A})}(A, -)$ to $A[0] \leftarrow \sigma_{\leq 0} P \rightarrow B[i]$, which is exactly the roof that corresponds to the sequence P . Applying Hom and taking the element corresponding to 1 recovers the map associated to this roof. $\square$

Notice that this puts a natural abelian group structure on the equivalence classes of exact sequences from B to A . To construct the addition in the general case is a bit annoying, but we can at least write down the zero element by copying the proofs above. Namely, we should start with $A[0] \leftarrow A[0] \xrightarrow{0} B[i]$, and then pushout repeatedly.

So, if $i > 1$, then the zero sequence looks like

$$0 \rightarrow B \xrightarrow{1} B \xrightarrow{0} 0 \rightarrow \cdots \rightarrow 0 \rightarrow A \xrightarrow{1} A \rightarrow 0.$$

However, if $i = 1$ then we get

$$0 \rightarrow B \rightarrow B \oplus A \rightarrow A \rightarrow 0,$$

the canonical split exact sequence. Moreover, in the Ext^1 case, note that two short exact sequences induce the same element of Ext^1 iff they both connect to another complex R as in:

$$\begin{array}{ccccccc} 0 & \longrightarrow & B & \longrightarrow & P & \longrightarrow & A \longrightarrow 0 \\ & & \parallel & & \downarrow & & \parallel \\ 0 & \longrightarrow & B & \longrightarrow & R & \longrightarrow & A \longrightarrow 0 \end{array}$$

But by the five lemma, the existence of such a diagram already gives an isomorphism $P \xrightarrow{\sim} R$, and hence an isomorphism of exact sequences. Thus Ext^1 classifies short exact sequences up to isomorphism, with the zero element being the split sequence. A nice thing in the Ext^1 case which follows immediately from (b) of the above corollary is the fact that the extension class of an exact sequence $0 \rightarrow B \rightarrow C \rightarrow A \rightarrow 0$ is the image of 1_C under the map

$$\text{Hom}_{\mathcal{A}}(A, A) \xrightarrow{\delta} \text{Ext}_{\mathcal{A}}^1(A, B)$$

in the long exact sequence. In this case, we can describe the addition more clearly. Recall that in an additive category, to add to morphisms $f, g: X \rightarrow Y$, we use the following recipe:

$$X \xrightarrow{\Delta} X \oplus X \xrightarrow{f \oplus g} Y \oplus Y \xrightarrow{\nabla} Y.$$

For addition of extension of A by B , we do a similar thing. First take the sum of the two sequences as in the bottom of the diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & B & \dashrightarrow & Q & \longrightarrow & A \longrightarrow 0 \\ & & \uparrow \nabla & & \uparrow & & \parallel \\ 0 & \longrightarrow & B \oplus B & \longrightarrow & P & \dashrightarrow & A \longrightarrow 0 \\ & & \parallel & & \downarrow & & \downarrow \Delta \\ 0 & \longrightarrow & B \oplus B & \longrightarrow & E \oplus F & \longrightarrow & A \oplus A \longrightarrow 0 \end{array}$$

Then pushout upwards along the addition map ∇ . We claim the resulting sequence at the top is the sum of E and F . Indeed, this data induces a commutative diagram:

$$\begin{array}{ccc} \text{Hom}(A, A) & \longrightarrow & \text{Ext}^1(A, B) \\ \parallel & & \uparrow \\ \text{Hom}(A, A) & \longrightarrow & \text{Ext}^1(A, B \oplus B) \\ \downarrow & & \parallel \\ \text{Hom}(A, A \oplus A) & \longrightarrow & \text{Ext}^1(A, B \oplus B) \end{array}$$

Since Hom and Ext^1 preserve sums in the second variable, the image of $(1, 0)$ and $(0, 1)$ in the bottom right Ext^1 respectively are $[E]$ and $[F]$, and the left-lower vertical map sends $1 \mapsto (1, 1)$. Therefore the upper horizontal map sends $1 \mapsto [E] + [F]$ as desired.

Example 1.2.29 (This is misplaced.) Let $\mathcal{R}$ be a right preadapted class in $\mathcal{A}$. Let us prove two easy related statements.

1. If every object of $\mathcal{A}$ has a bounded resolution in $\mathcal{R}$, then every complex in $\text{Ch}^b(\mathcal{A})$ has a bounded right resolution in $\mathcal{R}$. If every object of $\mathcal{A}$ has $\mathcal{R}$ -resolution of length d , then every complex in $\text{Ch}^{[a,b]}(\mathcal{A})$ has a right resolution in $\text{Ch}^{[a,b+d]}(\mathcal{R})$.

Let $A^\bullet \in K^+(\mathcal{A})$ be bounded, so that $A^i = 0$ for $i > N$, and take a resolution $A \rightarrow X$ where $X \in K^+(\mathcal{R})$. The map $A^N \rightarrow X^N$ composed with d_X^N factors through $A^{N+1} = 0$, so it factors as $A^N \rightarrow \ker d_X^N \hookrightarrow X^N$. In particular, $A \rightarrow X$ factors as $A \rightarrow \tau_{\leq N} X \rightarrow X$.

Since X can only have cohomology in degrees $\leq N$, the second map is a quasi-isomorphism, so the first one is too. $\tau_{\leq N} X$ has all but its last term $\ker d_X^N$ in $\mathcal{R}$. Take an exact sequence

$$0 \rightarrow \ker d_X^N \rightarrow Y^0 \rightarrow \dots \rightarrow Y^d \rightarrow 0$$

with $Y^i \in \mathcal{R}$, and then form the new complex

$$Y := (\dots \rightarrow X^{N-2} \rightarrow X^{N-1} \rightarrow Y^0 \rightarrow Y^1 \rightarrow \dots)$$

where the middle differential is $X^{N-1} \rightarrow \ker d_X^N \rightarrow Y^0$. Then Y has top degree $N + d$, and the obvious map $\tau_{\leq N} X \rightarrow Y$ is a quasi-isomorphism because Y is exact in $[N + 1, d + N]$ and has N th cohomology $\ker d_X^N / \text{im } d_X^{N-1}$, as desired.

Here's a less direct statement on the level of cohomology. Now we'll assume that $\mathcal{R}$ is a right adapted class for a left exact functor $F: \mathcal{A} \rightarrow \mathcal{B}$. If A is an object of $\mathcal{A}$, its RF -dimension is the largest i for which $R^i F(A) \neq 0$ (or ∞). If $A^\bullet$ is in $\text{Ch}^+(\mathcal{A})$, then the RF -amplitude of A is the interval $[a, b]$ in which $RF(A)$ has nonzero cohomology.

2. If every object of $\mathcal{A}$ has finite RF -dimension, then every complex in $\text{Ch}^b(\mathcal{A})$ has bounded RF -amplitude. If every object of $\mathcal{A}$ has RF -dimension $\leq d$, then every complex in $\text{Ch}^{[a,b]}(\mathcal{A})$ has RF -amplitude inside $[a, b + d]$.

To see this, let A be a bounded complex, and consider the triangle

$$\tau_{\leq n-1} A \rightarrow \tau_{\leq n} A \rightarrow H^n(A)[-n] \rightarrow \tau_{n-1} A[1]$$

in the derived category of $\mathcal{A}$. Applying the triangulated functor RF to this, we deduce that if $\tau_{\leq n-1} A$ had cohomology concentrated in $[a, b_0 + d]$, and $H^n(A)[0]$ has no cohomology after degree d , then $\tau_{\leq n} A$ is concentrated in $[a, b_0 + 1 + d]$. Thus we conclude by induction.

1.3 Homological algebra revisited

1.3.1 Basic notions

We now have quite a big toolkit to do homological algebra, and yet we have gotten no real results yet. Now we come to this. Let $\mathcal{A}$ be an abelian category, and suppose we have a short exact sequence

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

of chain complexes. If $F: \mathcal{A} \rightarrow \mathcal{B}$ is a left exact functor with an adapted class $\mathcal{R}$, then RF exists and is triangulated on the derived categories. In particular if A, B, C are bounded below, then $A \rightarrow B \rightarrow C \rightarrow A[1]$ is an exact triangle, so $RF(A) \rightarrow RF(B) \rightarrow RF(C) \rightarrow RF(A)[1]$ is an exact triangle. In particular, there is a long exact sequence in cohomology:

$$\cdots \rightarrow H^i(RF(A)) \rightarrow H^i(RF(B)) \rightarrow H^i(RF(C)) \rightarrow H^{i+1}(RF(A)) \rightarrow \cdots$$

Likewise, if F was left exact, and A, B, C were bounded above, then the left derived functor $LF: D^-(\mathcal{A}) \rightarrow D^-(\mathcal{B})$ exists and we have an exact triangle

$$LF(A) \rightarrow LF(B) \rightarrow LF(C) \rightarrow LF(A)[1]$$

in $D^-(\mathcal{B})$, producing a similar long exact sequence in cohomology. In practice, we usually care most about objects of $\mathcal{A}$ themselves, and not chain complexes. In the situation above, we define the *classical right derived functors* by

$$R^i F := \mathcal{A} \hookrightarrow D^+(\mathcal{A}) \xrightarrow{RF} D^+(\mathcal{B}) \xrightarrow{H^i} \mathcal{B}.$$

So, if $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ is an exact sequence of objects in $\mathcal{A}$, then we get a long exact sequence

$$0 \rightarrow F(A) \rightarrow F(B) \rightarrow F(C) \rightarrow R^1 F(A) \rightarrow R^1 F(B) \rightarrow \cdots$$

of right-derived functors, natural in the exact sequence. The first few terms of this are justified by the following lemma.

Lemma 1.3.1 *In the situation above, $R^i F = 0$ for $i < 0$, and there is a natural isomorphism $F \xrightarrow{\sim} R^0 F$.*

Proof For A in $\mathcal{A}$, then e.g. from [Lemma 1.2.15](#) we have a quasi-isomorphism $A[0] \rightarrow X^\bullet$ with $X^i = 0$ for $i < 0$ and X^i in $\mathcal{R}$ for all i . We have the commutative diagram on the left below,

$$\begin{array}{ccc} F(A)[0] & \longrightarrow & RF(A[0]) \\ \downarrow & & \downarrow \cong \\ F(X) & \xrightarrow{\cong} & RF(X) \end{array} \qquad \begin{array}{ccc} F(A) & \longrightarrow & R^0 F(A) \\ \downarrow & & \cong \downarrow \\ H^0(F(X)) & \xrightarrow{\cong} & H^0(RF(X)) \end{array}$$

and on the right is what we get by applying H^0 . First, if we apply H^i for $i < 0$, then $H^i(F(X)) = 0$ because $F(X^i) = 0$ for $i < 0$. The natural map $F \rightarrow R^0 F$

is the top of the right-hand square. Since F is left exact, the map

$$F(A) \rightarrow H^0(F(X)) = \ker(F(d_X^0)) = F(\ker(d_X^0))$$

is given by applying F to the isomorphism $A \rightarrow \ker(d_X^0)$ (since $A \rightarrow X$ is a quasi-isomorphism), so is an isomorphism. $\square$

Let $F: \mathcal{A} \rightarrow \mathcal{B}$ be a left exact functor, and suppose its right derived functor $RF: D^+(\mathcal{A}) \rightarrow D^+(\mathcal{B})$ exists. We say an object A of $\mathcal{A}$ is *F-acyclic* if $R^i F(A) = 0$ for all $i \neq 0$. *F-acyclics* behave nicely in long exact sequences:

Lemma 1.3.2 *Let $F: \mathcal{A} \rightarrow \mathcal{B}$ be a left exact functor and $\mathcal{R}$ an F -adapted class so that RF exists.*

(a) *If $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ is an exact sequence of objects of $\mathcal{A}$ and A is F -acyclic, then B is F -acyclic iff C is.*

(b) *Given a (bounded below) long exact sequence*

$$0 \rightarrow A^0 \rightarrow A^1 \rightarrow A^2 \rightarrow \dots$$

of F -acyclic objects, the corresponding sequence

$$0 \rightarrow F(A^0) \rightarrow F(A^1) \rightarrow F(A^2) \rightarrow \dots$$

is exact as well.

The corresponding statements for right exact functors holds as well.

Proof (a) The LES gives an exact sequence $0 \rightarrow R^i F(B) \rightarrow R^i F(C) \rightarrow 0$ for $i > 0$.

(b) Break up the sequence into short exact sequences:

$$0 \rightarrow A^0 \rightarrow A^1 \rightarrow C^1 \rightarrow 0, \quad 0 \rightarrow C^i \rightarrow A^{i+1} \rightarrow C^{i+1} \rightarrow 0 \quad i \geq 1.$$

Since $R^1 F(A^0) = 0$, F preserves the first exact sequence. Moreover, since A^0, A^1 are F -acyclic, (a) shows that C^1 is F -acyclic. But inductively, if C^i is F -acyclic, then C^{i+1} is too by the second SES and (a). Thus applying F to all of the exact sequences above we get exact sequences, and then we can put them back together. $\square$

Next we prove a powerful result about F -acyclics. The first proof is by trickery, so we provide another direct proof afterwards. We'll give a third proof in a special case once we construct Cartan-Eilenberg resolutions.

Theorem 1.3.3 *Let $F: \mathcal{A} \rightarrow \mathcal{B}$ be a left exact functor and $\mathcal{R}$ an F -adapted class, so that its right derived functor $RF: D^+(\mathcal{A}) \rightarrow D^+(\mathcal{B})$ exists. For each complex $X^\bullet$ of F -acyclic object, the natural map*

$$F(X) \rightarrow RF(X) \quad \text{in } D^+(\mathcal{B})$$

is an isomorphism. In fact, if $\overline{\mathcal{R}}$ the class of all F -acyclic objects, then $\overline{\mathcal{R}}$ is the unique largest class of objects adapted to F , and it is closed under taking summands, quotients (of F -acyclics by F -acyclics), and extensions.

Proof 1 The second statement, particularly that $\overline{\mathcal{R}}$ is an adapted class, implies the first immediately by uniqueness of the derived functor. $\overline{\mathcal{R}}$ contains 0 and is closed under biproducts. Part (b) of the lemma above was exactly the statement that F takes acyclic complexes in $K^+(\overline{\mathcal{R}})$ to acyclic complexes.

Now, if $\mathcal{R}'$ is any F -adapted class, then for C in $\mathcal{R}'$, $C[0]$ is in $K^+(\mathcal{R}')$, so $F(C)[0] \xrightarrow{\sim} RF(C[0])$ in the derived category, hence C is F -acyclic. Thus $\mathcal{R}' \subset \overline{\mathcal{R}}$. In particular, $\mathcal{R} \subset \overline{\mathcal{R}}$, so every object admits a mono to an object in $\overline{\mathcal{R}}$. Lastly, the statement about quotients and extensions follows from the previous lemma, and the statement about summands follows from the fact that if $A \oplus B$ is acyclic, then $0 = R^i F(A \oplus B) = R^i F(A) \oplus R^i F(B)$ for $i > 0$. $\square$

In most cases, we will have either injective or projective resolutions of objects. Here is the main set of lemmas telling us that they behave well. To make this especially simple, we first define the Hom complex.

Given $A^\bullet$ and $B^\bullet$ in $\text{Ch}(\mathcal{A})$, we define the complex $\underline{\text{Hom}}_{\mathcal{A}}(A, B)$ by

$$\underline{\text{Hom}}_{\mathcal{A}}^i(A, B) := \prod_{p \in \mathbb{Z}} \text{Hom}_{\mathcal{A}}(A^p, B^{p+i}),$$

(a complex of abelian groups), where the differential is given by

$$df = d_B f - (-1)^i f \circ f_A,$$

or more precisely $(df)^p = d_B^{p+i} \circ f^p - (-1)^i f^{p+1} \circ d_A^p$. An element of $\underline{\text{Hom}}^0(A, B)$ is a sequence of maps $f^p: A^p \rightarrow B^p$ for all p ; if $df = 0$, then $df = fd$, i.e. f is a chain map. An element of $\underline{\text{Hom}}^{-1}(A, B)$ is a sequence of degree -1 maps $h^p: A^p \rightarrow B^{p-1}$, and if a chain map f is of the form dh , then $f = df + fd$, i.e. f is nullhomotopic. In other words,

$$\ker d^0 = \text{Hom}_{\text{Ch}(\mathcal{A})}(A, B) \quad \text{im } d^{-1} = \{f : A \rightarrow B : f \sim 0\},$$

so $H^0(\underline{\text{Hom}}(A, B)) = \text{Hom}_{K(\mathcal{A})}(A, B)$. One checks easily that

$$\underline{\text{Hom}}(A, B)[n] = \underline{\text{Hom}}(A, B[n]) = \underline{\text{Hom}}(A[-n], B).$$

Now we construct a bicomplex $C^{\bullet, \bullet}$ whose product totalization is the Hom complex. Define $C^{p, q} := \text{Hom}_{\mathcal{A}}(A^{-p}, B^q)$, and let

$$d_v(\phi) = d_B^q \circ \phi \quad d_h(\phi) = (-1)^{p+q+1} \phi \circ d_A^{-p-1}.$$

Then in each direction we have $d^2 = 0$, and the vertical and horizontal differentials anticommute. It is then clear from the definition that

$$\underline{\text{Hom}}_{\mathcal{A}}(A, B) = \text{Tot}^{\Pi}(C).$$

Therefore we can apply our spectral sequence associated to a product totalization. Recall that in this case (e.g. for the first sequence) $C^{p, n-p} = 0$ for $p \ll 0$.

This translates to asking for $\mathrm{Hom}(A^p, B^{n+p}) = 0$ for $p \gg 0$, e.g. if A or B is bounded above. For the second sequence, we get the condition $\mathrm{Hom}(A^{q-n}, B^q)$ for $q \ll 0$, which is satisfied e.g. if A or B is bounded below. Here, we are using the fact that arbitrary products are exact in the category modules. Thus:

Lemma 1.3.4 *Suppose A or B is a bounded above complex in $\mathrm{Ch}(\mathcal{A})$. Then there is a spectral sequence*

$$E_2^{p,q} = H_h^q(H_v^p(\mathrm{Hom}(A^*, B^*))) \implies H^{p+q}(\underline{\mathrm{Hom}}_{\mathcal{A}}(A, B)).$$

If A or B is bounded below, then we have a spectral sequence

$$E_2^{p,q} = H_v^q(H_h^p(\mathrm{Hom}(A^*, B^*))) \implies H^{p+q}(\underline{\mathrm{Hom}}_{\mathcal{A}}(A, B)).$$

In both cases, the filtration on the resulting object is described by the totalization above and [Thm. 1.1.4](#).

Recall that an object I (resp. P) of $\mathcal{A}$ is called *injective* (resp. *projective*) if $\mathrm{Hom}_{\mathcal{A}}(-, I)$ (resp. $\mathrm{Hom}_{\mathcal{A}}(P, -)$) is an exact functor.

Corollary 1.3.5 *Let $I^\bullet$ be a bounded below complex of injective objects of $\mathcal{A}$.*

(a) *If $A^\bullet$ is an acyclic complex, then $\underline{\mathrm{Hom}}_{\mathcal{A}}(A, I)$ is acyclic.*

(b) *If $q: B \rightarrow C$ is a quasi-isomorphism, then*

$$\mathrm{Hom}_{K(\mathcal{A})}(C, I) \rightarrow \mathrm{Hom}_{K(\mathcal{A})}(B, I)$$

is an isomorphism of abelian groups.

(c) *Any quasi-isomorphism $q: I \rightarrow A$ has a one-sided homotopy inverse, i.e. a quasi-isomorphism $r: A \rightarrow I$ such that $rq \sim 1_I$. In particular, a quasi-isomorphism between bounded-below complexes of injective objects is a homotopy equivalence.*

(d) *The natural map*

$$\mathrm{Hom}_{K(\mathcal{A})}(A, I) \rightarrow \mathrm{Hom}_{D(\mathcal{A})}(A, I)$$

is an isomorphism of abelian groups for any A in $D(\mathcal{A})$.

The corresponding statements hold for a bounded-above complex of projectives.

Proof If $A^\bullet$ is acyclic, then since I^n is injective for each n , the sequence

$$\cdots \leftarrow \mathrm{Hom}(A^{i-1}, I^n) \leftarrow \mathrm{Hom}(A^i, I^n) \leftarrow \mathrm{Hom}(A^{i+1}, I^n) \leftarrow \cdots$$

is exact for all $n \in \mathbb{Z}$. Thus $H_h^q(\mathrm{Hom}(A^*, I^n)) = 0$ for all n , so by the spectral sequence of [Lemma 1.3.4](#), $\underline{\mathrm{Hom}}_{\mathcal{A}}(A, I)$ is acyclic. Since cones commute with functors, this means that for any quasi-isomorphism $B \rightarrow C$, the map

$$\underline{\mathrm{Hom}}(C, I) \rightarrow \underline{\mathrm{Hom}}(B, I)$$

is a quasi-isomorphism. Taking H^0 gives the statement of (b). For (c), we have an isomorphism of abelian groups

$$- \circ q: \operatorname{Hom}_{K(\mathcal{A})}(A, I) \xrightarrow{\sim} \operatorname{Hom}_{K(\mathcal{A})}(I, I),$$

so there is a map $r: A \rightarrow I$ such that $rq \sim 1_I$. Since q and 1_I are quasi-isomorphisms, r is as well. So, q is a split monomorphism in $K(\mathcal{A})$ and r is a split epimorphism. In the case that $A = I'$ is also a bounded below complex of injectives, r satisfies the same hypotheses and hence is a split monomorphism itself. Thus r is a homotopy equivalence, so q is too.

For the last result, any map $A \rightarrow I$ is represented by a right roof

$$A \rightarrow X \xleftarrow{q} I$$

for some quasi-isomorphism q . By (c), there is a one-sided homotopy inverse $r: X \rightarrow I$, so we get an equivalent roof (commutative in $K(\mathcal{A})$):

$$\begin{array}{ccccc} A & \longrightarrow & X & \xleftarrow{q} & I \\ & \searrow \text{dashed} & \downarrow r & \swarrow 1_I & \\ & & I & & \end{array}$$

Then the map $A \rightarrow I$ on the bottom right is equal to the one we started with, and induced from $K(\mathcal{A})$. Similarly, if $a: A \rightarrow I$ becomes zero in $D(\mathcal{A})$, then there's a quasi-isomorphism $q: I \rightarrow X$ such that the composite $A \xrightarrow{a} I \xrightarrow{q} X$ is 0. Then $rq \sim 1_I$ shows that $a \sim 0$. $\square$

The above result will be central to the rest of these notes. Let us start to deduce some consequences. If A is an object of $\mathcal{A}$, then an *injective resolution* is a quasi-isomorphism $A[0] \rightarrow I$ where I is a complex of injectives with $I^i = 0$ for $i < 0$. Likewise a projective resolution is one of the form $P \rightarrow A[0]$ with $P^i = 0$ for $i > 0$.

Proposition 1.3.6 *Let $\mathcal{A}$ be an abelian category. Let $A, B \in \operatorname{Ob} \mathcal{A}$.*

- (a) *An object I is injective iff $\operatorname{Ext}_{\mathcal{A}}^1(-, I) = 0$ iff $\operatorname{Ext}_{\mathcal{A}}^i(-, I) = 0$ for all $i > 0$.*
- (b) *An object P is projective iff $\operatorname{Ext}_{\mathcal{A}}^1(P, -) = 0$ iff $\operatorname{Ext}_{\mathcal{A}}^i(P, -) = 0$ for all $i > 0$.*
- (c) *If $B[0] \rightarrow I^\bullet$ is an injective resolution, then there is a natural isomorphism*

$$\operatorname{Ext}_{\mathcal{A}}^i(A, B) \xrightarrow{\sim} H^i(\operatorname{Hom}_{\mathcal{A}}(A, I^\bullet))$$

in A . Likewise, if $P^\bullet \rightarrow A$ is a projective resolution, then there is a natural isomorphism

$$H^i(\operatorname{Hom}_{\mathcal{A}}(P^\bullet, B)) \xrightarrow{\sim} \operatorname{Ext}_{\mathcal{A}}^i(A, B)$$

in B .

Proof (a)/(b) and (c)/(d) are dual pairs, so we prove the injective versions. If I is injective, then $I[i]$ is a bounded below complex of injectives, so

$$\mathrm{Hom}_{D^*(\mathcal{A})}(A[0], I[i]) = \mathrm{Hom}_{K(\mathcal{A})}(A[0], I[i]) = 0,$$

since for $i > 0$ every chain map $A[0] \rightarrow I[i]$ is zero.

(c) If $B \rightarrow I^\bullet$ is a qis, then we have isomorphisms

$$\begin{aligned} \mathrm{Hom}_{D(\mathcal{A})}(A[0], B[i]) &\xrightarrow{\sim} \mathrm{Hom}_{D(\mathcal{A})}(A[0], I[i]) \\ &\cong \mathrm{Hom}_{K(\mathcal{A})}(A[0], I[i]) \\ &\cong H^i(\underline{\mathrm{Hom}}_{\mathcal{A}}(A[0], I)), \end{aligned}$$

but since $A[0]$ is a 0-complex, we have $\underline{\mathrm{Hom}}^i(A[0], I) = \mathrm{Hom}(A, I^i)$ and the differentials coincide as well. $\square$

Example 1.3.7 It is not too difficult to show that elements of $\mathrm{Ext}^1(C, A)$ are in natural correspondence with extensions of C by A . Regardless, if $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ is such an extension, then we define the *extension class* to be the image of 1_C in $\mathrm{Ext}_{\mathcal{A}}^1(C, A)$ under the induced map:

$$\cdots \rightarrow \mathrm{Hom}_{\mathcal{A}}(C, B) \rightarrow \mathrm{Hom}_{\mathcal{A}}(C, C) \xrightarrow{\delta} \mathrm{Ext}_{\mathcal{A}}^1(C, A) \rightarrow \cdots$$

Note that the extension splits iff the extension class is zero, because the latter condition is (by exactness) equivalent to the existence of a section of $B \rightarrow C$. So, e.g., I is injective iff every SES with I as the first term splits.

Example 1.3.8 Consider the exact sequence $0 \rightarrow \mathbb{Z} \xrightarrow{-2} \mathbb{Z} \rightarrow \mathbb{Z}/2 \rightarrow 0$ over $\mathbb{Z}$. This is a projective resolution of $\mathbb{Z}/2$, so to compute $\mathrm{Ext}^*(\mathbb{Z}/2, \mathbb{Z})$ we apply $\mathrm{Hom}(-, \mathbb{Z})$ to $0 \rightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \rightarrow 0$ to get the same thing back, and the cohomology of this is

$$\mathrm{Ext}^0(\mathbb{Z}/2, \mathbb{Z}) = 0 \quad \mathrm{Ext}^1(\mathbb{Z}/2, \mathbb{Z}) = \mathbb{Z}/2 \quad \mathrm{Ext}^i(\mathbb{Z}/2, \mathbb{Z}) = 0, i > 0.$$

Since the sequence above was not split, its extension class was the unique nonzero element of $\mathrm{Ext}^1(\mathbb{Z}/2, \mathbb{Z})$.

Continuing in our development of the theory, we consider the role of injectives/projectives for computing derived functors.

Proposition 1.3.9 *Let $F: \mathcal{A} \rightarrow \mathcal{B}$ be a left exact functor between abelian categories. Let $\mathcal{R}$ be an F -adapted class, so that $R\mathcal{F}$ exists.*

(a) *Injective objects of $\mathcal{A}$ are F -acyclic.*

(b) *If $I^\bullet$ is a bounded-below complex of injective objects, then the natural map $F(I) \rightarrow R\mathcal{F}(I)$ in $D^+(\mathcal{B})$ is an isomorphism. In particular, if $A^\bullet \rightarrow I^\bullet$ is an injective resolution, then canonically $R\mathcal{F}(A) \cong F(I)$.*

In general, the class $\mathcal{I}_{\mathcal{A}}$ of injective objects is adapted to any left exact functor F if (and only if) $\mathcal{A}$ has enough injectives. The corresponding statements hold for projectives.

Proof (b) immediately follows from (a). For (c), consider an exact sequence

$$0 \rightarrow I^0 \rightarrow I^1 \rightarrow \dots$$

(bounded below) of injectives, and break it up into short exact sequences

$$0 \rightarrow I^0 \rightarrow I^1 \rightarrow C^1 \rightarrow 0 \quad 0 \rightarrow C^i \rightarrow I^{i+1} \rightarrow C^{i+1} \rightarrow 0.$$

Applying $\text{Ext}^i(A, -)$ to the first sequence shows that $\text{Ext}^i(-, C^1)$, i.e. C^1 is injective. Doing it to the second sequence shows that C^{i+1} is injective if C^i . Hence by induction, all of the sequences split, so they're all preserved by F , and we can recombine. This if $\mathcal{A}$ has enough injectives $\mathcal{I}_{\mathcal{A}}$ is F -adapted for any F .

For (a), embed an injective $0 \rightarrow I \rightarrow X \rightarrow Y \rightarrow 0$ for X in $\mathcal{R}$. Then the sequence splits, so I is a direct summand of X , which is F -acyclic, so I is F -acyclic. $\square$

Remark 1.3.10 Let $\mathcal{A}$ have enough injectives. Then $\mathcal{I}_{\mathcal{A}}$ is a preadapted class, so in the composite

$$K^+(\mathcal{I}_{\mathcal{A}}) \rightarrow K^+(\mathcal{I}_{\mathcal{A}})[S_{\mathcal{I}_{\mathcal{A}}}^{-1}] \rightarrow D^+(\mathcal{A})$$

the second map is an equivalence. But the first map is also an equivalence, because all quasi-isomorphisms between bounded above complexes of injectives are homotopy equivalences. Because of this, much of the work of the previous (and this) section would be a lot easier if we made this assumption, but we went to all this trouble because in algebraic geometry one doesn't have enough projectives.

Example 1.3.11 Let $\mathcal{A}$ be a *semisimple* abelian category, which means that every exact sequence splits, so equivalently every object is projective, or equivalently every object is injective. Thus $K^+(\mathcal{A}) \xrightarrow{\sim} D^+(\mathcal{A})$ is an equivalence, and likewise $K^-(\mathcal{A}) = D^-(\mathcal{A})$. For a complex $C^\bullet$, let $H^\bullet(C)$ denote the complex:

$$\dots \rightarrow H^{i-1}(C) \xrightarrow{0} H^i(C) \xrightarrow{0} H^{i+1}(C) \rightarrow \dots$$

By choosing splittings $H^i(C) \oplus \text{im } d^{i-1} = \ker d^i$, including into the kernel gives a map $H^\bullet(C) \rightarrow C^\bullet$ which is a quasi-isomorphism. Thus every object in the derived category is isomorphic to a complex with zero differentials. In the bounded above or bounded below case, we know this must be a homotopy equivalence. By definition, for such a complex a nullhomotopic map is necessarily zero, so this gives an equivalence $K^+(\mathcal{A}) = D^+(\mathcal{A})$ with $\mathbb{Z}$ -graded bounded above complexes, and likewise in the bounded below case.

If we look more closely we can get this for the full derived category. The map $H^\bullet(C) \rightarrow C$ constructed above is a termwise injection, with some acyclic

kernel A as in

$$0 \rightarrow H^\bullet(C) \rightarrow C \rightarrow A \rightarrow 0.$$

If we additionally split $0 \rightarrow \ker d^i \rightarrow C^i \rightarrow \operatorname{im} d^i \rightarrow 0$, we get isomorphisms

$$C^i = H^i(C) \oplus \operatorname{im} d^{i-1} \oplus \operatorname{im} d^i \equiv H^i(C) \oplus A,$$

where A is the exact sequence $\operatorname{im} d^{i-1} \oplus \operatorname{im} d^i \rightarrow \operatorname{im} d^i \oplus \operatorname{im} d^{i+1} \rightarrow \dots$ that is the identity on the common factor and zero otherwise. Thus A is precisely the cone of the complex

$$\dots \rightarrow \operatorname{im} d^{i-1} \xrightarrow{0} \operatorname{im} d^i \xrightarrow{0} \operatorname{im} d^{i+1} \rightarrow \dots$$

Since the sequence is termwise split, there is an exact triangle $H(C) \rightarrow C \rightarrow A \rightarrow H(C)[1]$ already in $\mathbf{K}(\mathcal{A})$ with $A \cong 0$, so $H(C) \rightarrow C$ is a homotopy equivalence. (The termwise split detail will be explained later.)

The point of this example is that we will often end up with an object of $D(k)$, i.e. the derived category of k -vector spaces, and this tells us that two such objects are isomorphic iff they have isomorphic cohomology.

1.3.2 Classical derived functors

Let us briefly revisit the classical theory of derived functors using our new language. In the last section, we defined Ext without any reference to derived functors, so let us explain the connection now.

In the previous section, [Cor. 1.3.5\(a\)](#) showed that if $I^\bullet \in \mathbf{K}^+(\mathcal{I}_{\mathcal{A}})$, then there is a well-defined functor

$$\underline{\operatorname{Hom}}_{\mathcal{A}}(-, I): D(\mathcal{A})^{\operatorname{op}} \rightarrow D(\mathbb{Z}),$$

where $D(\mathbb{Z}) = D(\mathbf{Ab})$. Likewise, if P was a bounded above complex of projectives, then $\underline{\operatorname{Hom}}(P, -)$ was well-defined on $D(\mathcal{A})$. If $I \rightarrow I'$ is a chain map, then we get a corresponding natural transformation $\underline{\operatorname{Hom}}_{\mathcal{A}}(-, I) \rightarrow \underline{\operatorname{Hom}}_{\mathcal{A}}(-, I')$ on complexes, and thus as functors on the derived category above. Such a transformation is independent of the homotopy class of $I \rightarrow I'$, because the hom complex is already well-defined on the homotopy categories. So this construction gives a functor

$$\underline{\operatorname{Hom}}_{\mathcal{A}}: D(\mathcal{A})^{\operatorname{op}} \times \mathbf{K}^+(\mathcal{I}_{\mathcal{A}}) \rightarrow D(\mathbb{Z})$$

Thus, if $\mathcal{A}$ has enough injectives, then by choosing an inverse $D^+(\mathcal{A}) \rightarrow \mathbf{K}^+(\mathcal{I}_{\mathcal{A}})$ we get a functor

$$\operatorname{RHom}_{\mathcal{A}}: D(\mathcal{A})^{\operatorname{op}} \times D^+(\mathcal{A}) \rightarrow D(\mathbb{Z}).$$

The choice of an inverse gives, for each $B \in \mathbf{K}^+(\mathcal{A})$, a natural map

$$\underline{\operatorname{Hom}}_{\mathcal{A}}(-, B) \rightarrow \operatorname{RHom}_{\mathcal{A}}(-, B)$$

of functors, and hence a natural map $\underline{\operatorname{Hom}}_{\mathcal{A}}(A, B) \rightarrow \operatorname{RHom}_{\mathcal{A}}(A, B)$ in $D(\mathbb{Z})$ for all $A \in D(\mathcal{A})^{\operatorname{op}}$ and $B \in D^+(\mathcal{A})$. By definition, it is an isomorphism if B is a bounded-below complex of injectives.

Likewise, if there are enough projectives in $\mathcal{A}$ then we get another well-defined functor

$$\mathrm{RHom}_{\mathcal{A}} : D^-(\mathcal{A})^{\mathrm{op}} \times D(\mathcal{A}) \rightarrow D(\mathbb{Z})$$

with the same name. Note that we haven't really done anything complicated here: to compute $\mathrm{RHom}(A, B)$, say if the right entry is bounded above, take an injective resolution $B \rightarrow I$, then $\mathrm{RHom}(A, B) = \underline{\mathrm{Hom}}(A, I)$, the Hom-complex. The 'overlapping' case works as follows: we will stick to the case where we are assuming enough injectives, however.

$$\begin{array}{ccccc} \mathrm{RHom}(A, B) & \xrightarrow{\cong} & \mathrm{RHom}(A, I) & = & \underline{\mathrm{Hom}}(A, I) \\ \cong \downarrow & & \downarrow \cong & & \cong \downarrow \\ \mathrm{RHom}(P, B) & \xrightarrow{\cong} & \mathrm{RHom}(P, I) & = & \underline{\mathrm{Hom}}(P, I) \\ \uparrow \cong & & \parallel & & \\ \underline{\mathrm{Hom}}(P, B) & \xrightarrow{\cong} & \underline{\mathrm{Hom}}(P, I) & & \end{array}$$

In the diagram, since we're taking the injective definition, our definition for $\mathrm{RHom}(A, B)$ is to first take an injective resolution $B \rightarrow I$ and compute the Hom complex. Now if $P \rightarrow A$ is any quasi-isomorphism, then the manipulations on the right side are valid. If $P \rightarrow A$ is a bounded above projective resolution (e.g. even if A is unbounded in both directions), then the bottom steps are valid. This shows that as long as we replace one of the entries, we can compute using whatever resolution we want.

Sustain the enough injectives assumption. As we have already seen, taking cohomology of RHom computes maps in the derived category:

$$\begin{aligned} H^i(\mathrm{RHom}(A, B)) &= \mathrm{Hom}_{K(\mathcal{A})}(A, I[i]) \\ &= \mathrm{Hom}_{D(\mathcal{A})}(A, I[i]) \\ &= \mathrm{Hom}_{D(\mathcal{A})}(A, B[i]) \end{aligned}$$

(for possibly unbounded A , and bounded below B). Likewise, if we have enough projectives, the same thing works via the first variable for bounded above A and possibly unbounded B .

Proposition 1.3.12 *Suppose $\mathcal{A}$ has enough injectives. For any object A of $\mathcal{A}$, the functor*

$$\mathrm{RHom}(A, -) : D^+(\mathcal{A}) \rightarrow D^+(\mathbb{Z})$$

is canonically the right derived functor of $\mathrm{Hom}(A, -) : \mathcal{A} \rightarrow \mathbf{Ab}$. Its classical right derived functors agree with Ext :

$$H^i \circ \mathrm{RHom}(A, -) \cong R^i \mathrm{Hom}(A, -) \cong \mathrm{Ext}_{\mathcal{A}}^i(A, -).$$

Likewise, a similar statement holds if $\mathcal{A}$ has enough projectives.

Proof Since $\mathcal{A}$ has enough injectives, the supposed right derived functor is well-defined. The natural map $\mathrm{Hom}(A, -) \rightarrow \mathrm{RHom}(A, -)$ is an isomorphism on bounded above complexes of injectives, and we claim the same is true for $\mathrm{RHom}(A, -) \rightarrow \mathrm{Hom}(A, -)$. Indeed, putting a complex I of injectives

we obtain $\underline{\text{Hom}}(A[0], I) = \text{Hom}(A, I^*)$ as desired. The second statement is a consequence of the fact that we showed Ext^i could be computed by an injective resolution in the second variable. $\square$

The tensor product is a bit more intuitive. The fundamental lemma we need is something we proved a long time ago:

Lemma 1.3.13 *Let $\mathcal{A}$ be endowed with a compatible monoidal structure, and let P be a bounded above complex of flat objects. Assume that taking countable direct sums in $\mathcal{A}$ is exact. Then $P \otimes A \cong A \otimes P$ is acyclic for any acyclic complex A .*

Proof This was [Example 1.1.6](#). $\square$

Since the tensor product complex commutes with cones (check this), the functor $P \otimes (-)$ preserves quasi-isomorphisms if P is a bounded above complex of flats. In particular, for each such P we get functors

$$P \otimes (-), (-) \otimes P: D(\mathcal{A}) \rightarrow D(\mathcal{A})$$

on the full derived category. Now assume $\mathcal{R}$ denotes the class of flat objects, and assume $\mathcal{A}$ has enough flat objects so that $\mathcal{R}$ is a pre-adapted class. Then we have a triangulated equivalence

$$K^-(\mathcal{R})[S_{\mathcal{R}}^{-1}] \rightarrow D^-(\mathcal{A}).$$

The functor $K^-(\mathcal{R}) \times D(\mathcal{A}) \rightarrow D(\mathcal{A})$ (and the flipped one) satisfies the property that for P a bounded above acyclic complex of flats, and $B \in D^-(\mathcal{A})$, $P \otimes B$ is acyclic. Indeed, take a quasi-isomorphism $Q \rightarrow B$ with $Q \in K^-(\mathcal{R})$, then $P \otimes Q \rightarrow P \otimes B$ is a quasi-isomorphism, but P is acyclic so $P \otimes Q$ is acyclic. It follows that we get a functor:

$$(-) \otimes^{\mathbf{L}} (-): D^-(\mathcal{A}) \times D^-(\mathcal{A}) \rightarrow D^-(\mathcal{A}).$$

Let us further assume that tensoring with a fixed object is right exact (e.g. our category is closed monoidal). Then we have:

Lemma 1.3.14 *Keep the notation and assumptions as above. For X, Y, Z in $D^-(\mathcal{A})$:*

- (a) *We have natural isomorphisms $X \otimes^{\mathbf{L}} Y \cong Y \otimes^{\mathbf{L}} X$.*
- (b) *We have natural isomorphisms $(X \otimes^{\mathbf{L}} Y) \otimes^{\mathbf{L}} Z \cong X \otimes^{\mathbf{L}} (Y \otimes^{\mathbf{L}} Z)$.*
- (c) *A is an object of $\mathcal{A}$, then the functor $A \otimes^{\mathbf{L}} (-): D^-(\mathcal{A}) \rightarrow D^-(\mathcal{A})$ is the left derived functor of $A \otimes (-): \mathcal{A} \rightarrow \mathcal{A}$, a functor for which $\mathcal{R}$ is an adapted class. The same holds the other way around.*

Proof (a) $P^\bullet \rightarrow X^\bullet$ is a flat resolution, then

$$X \otimes^{\mathbf{L}} Y \cong P \otimes Y \cong Y \otimes P \cong Y \otimes^{\mathbf{L}} X.$$

For (b), let $P \rightarrow X$ and $Q \rightarrow Z$ be flat resolutions, then by functoriality

$$(X \otimes^{\mathbf{L}} Y) \otimes^{\mathbf{L}} Z \cong (P \otimes^{\mathbf{L}} Y) \otimes^{\mathbf{L}} Z \cong (P \otimes^{\mathbf{L}} Y) \otimes^{\mathbf{L}} Q.$$

The latter expression is equal to $(P \otimes Y) \otimes Q$ which is $P \otimes (Y \otimes Q)$, and then we are done by the same argument backwards.

For (c), note that indeed $\mathcal{R}$ is an adapted class: a special case of what we have said above is that tensoring with A sends bounded above acyclic complexes of flats to acyclics. $\square$

In light of this, we denote the higher left derived functors by

$$\mathrm{Tor}_{\mathcal{A}}^i(A, -) = L^i(A \otimes -) \quad \mathrm{Tor}_{\mathcal{A}}^i(-, B) = L^i(- \otimes B),$$

both of which coincide with $H^i(A \otimes^{\mathbf{L}} B)$.

Lemma 1.3.15 *In the situation above, for all A, B in $\mathcal{A}$:*

(a) $\mathrm{Tor}_{\mathcal{A}}^i(A, B) = 0$ for $i > 0$, and $\mathrm{Tor}_{\mathcal{A}}^0(A, B) = A \otimes B$

(b) $\mathrm{Tor}_{\mathcal{A}}^i(A, B) \cong \mathrm{Tor}_{\mathcal{A}}^i(B, A)$ for all A, B .

(c) An object A is flat iff $\mathrm{Tor}_{\mathcal{A}}^i(A, -) = \mathrm{Tor}_{\mathcal{A}}^i(-, A) = 0$ for $i < 0$, (iff $\mathrm{Tor}_{\mathcal{A}}^{-1}(A, -) = 0$). Consequently, any projective object is flat.

Proof (a) is because Tor's are the classical left derived functors of a functor, and (b) is a direct consequence of Lemma 1.3.14(a). For (c), if Tor^{-1} is zero then tensoring with A is exact by the LES. Conversely, an object of the adapted class for F is always F -acyclic. $\square$

Example 1.3.16 Suppose $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ is an exact sequence in $\mathcal{A}$ and C is flat. For any X in $\mathcal{A}$,

$$0 = \mathrm{Tor}^{i-1}(X, C) \rightarrow \mathrm{Tor}^i(X, A) \rightarrow \mathrm{Tor}^i(X, B) \rightarrow \mathrm{Tor}^i(X, C) = 0$$

is exact, so the natural map $\mathrm{Tor}^i(-, A) \rightarrow \mathrm{Tor}^i(-, B)$ is an isomorphism for all $i < 0$. Therefore B is flat iff C is flat.

Moreover, for any such X , the first part of this sequence is

$$0 = \mathrm{Tor}^{-1}(X, C) \rightarrow X \otimes A \rightarrow X \otimes B \rightarrow X \otimes C \rightarrow 0,$$

so tensoring any object with the sequence is exact if C is flat. Note that we are using the symmetry of Tor in a crucial way here.

Remark 1.3.17 If we didn't yet know that projectives were flat, but we had enough projectives, then we note that projectives are automatically an adapted class for $A \otimes (-)$ for all A in $\mathcal{A}$, so we could still define the derived functors $\mathrm{Tor}^*(A, -)$ and say $\mathrm{Tor}^*(-, A)'$. But, it's *not* true that projectives would have to be flat, and thus it's not true that these two functors would coincide.

Example 1.3.18 Let's compute $\mathbb{Z}/m \otimes^{\mathbf{L}} \mathbb{Z}/n$ in the derived category $D^-(\mathbb{Z})$. We can pick one of them to resolve, but for fun let's resolve both simultaneously and take the tensor product complex:

$$\begin{aligned} \mathbb{Z}/m \otimes^{\mathbf{L}} \mathbb{Z}/n &= (0 \rightarrow \mathbb{Z} \xrightarrow{m} \mathbb{Z} \rightarrow 0) \otimes (0 \rightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \rightarrow 0) \\ &= (0 \rightarrow \mathbb{Z} \xrightarrow{(m \quad -n)^T} \mathbb{Z}^2 \xrightarrow{(n \quad m)} \mathbb{Z} \rightarrow 0), \end{aligned}$$

where the right most $\mathbb{Z}$ has degree 0. If $m, n \neq 0$ and $d = \gcd(m, n)$, then the cohomology of this complex is

$$\mathrm{Tor}^0(\mathbb{Z}/m, \mathbb{Z}/n) = \mathrm{Tor}^{-1}(\mathbb{Z}/m, \mathbb{Z}/n) = \mathbb{Z}/d \quad \mathrm{Tor}^{-i}(\mathbb{Z}/m, \mathbb{Z}/n) = 0, i > 2.$$

Example 1.3.19 Let k be a field, consider the ring $A = k[\epsilon]/(\epsilon^2)$, and view $k = A/(\epsilon)$ as an A -module. Consider the sequence

$$\cdots \rightarrow A \xrightarrow{\epsilon} A \xrightarrow{\epsilon} A \rightarrow A/(\epsilon) = k \rightarrow 0.$$

The kernel of the last map is (ϵ) , which is the image of the previous map. But $\epsilon: A \rightarrow A$ has a kernel: $\epsilon(a + b\epsilon) = a\epsilon$, so its kernel is (ϵ) . Thus this repeats and we get an infinite free resolution as above. Therefore,

$$\begin{aligned} k \otimes^{\mathbf{L}} k &= k \otimes [\cdots \rightarrow A \xrightarrow{\epsilon} A \xrightarrow{\epsilon} A \rightarrow 0] \\ &= [\cdots \rightarrow k \xrightarrow{0} k \xrightarrow{0} k \rightarrow 0], \end{aligned}$$

because ϵ acts by zero on k . Thus $k \otimes^{\mathbf{L}} k = \bigoplus_{i \geq 0} k[i]$ has cohomology in every degree ≤ 0 . As we will discuss, we have just computed the derived self-intersection of a point inside a double point. This also gives an example of a module with infinite Tor-dimension, and hence infinite projective dimension.

Now let us study what we can say given two adjoint functors $F: \mathcal{A} \rightarrow \mathcal{B} : G$ with $F \dashv G$. Given complexes $A^\bullet$ and $B^\bullet$, we have a natural isomorphism

$$\mathrm{Hom}_{\mathcal{B}}(F(A^i), B^{n+i}) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{A}}(A^i, G(B^{n+i})),$$

and naturality in both components means that this commutes with the differential. It follows that we get a (natural) isomorphism

$$\underline{\mathrm{Hom}}_{\mathcal{B}}^\bullet(F(A), B) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{\mathcal{A}}^\bullet(A, G(B))$$

of Hom complexes. If we take zero-cycles or H^0 , we obtain

$$\mathrm{Hom}_{\mathrm{Ch}(\mathcal{B})}(F(A), B) \cong \mathrm{Hom}_{\mathrm{Ch} \mathcal{A}}(A, G(B))$$

and

$$\mathrm{Hom}_{\mathrm{K}(\mathcal{B})}(F(A), B) \cong \mathrm{Hom}_{\mathrm{K}(\mathcal{A})}(A, G(B))$$

respectively, so we expect these adjunctions to pass to the derived category. If we had been working with unbounded derived functors, this would be a lot easier. In our setting, we will not be able to get this in full generality, because our derived functors aren't even defined on the same categories. However, if we assume e.g. that $\mathcal{B}$ has enough injectives, then we can say something substantive.

Theorem 1.3.20 *Let $F: \mathcal{A} \rightleftarrows \mathcal{B}: G$ be an adjunction between abelian categories $\mathcal{A}$ and $\mathcal{B}$. Assume F has a (left) adapted class $\mathcal{R}_F$ and $\mathcal{B}$ has enough injectives, so that the derived functors*

$$LF: D^-(\mathcal{A}) \rightarrow D^-(\mathcal{B}) \quad RG: D^+(\mathcal{B}) \rightarrow D^+(\mathcal{A})$$

both exist. Then for $A \in D^-(\mathcal{A})$ and $B \in D^+(\mathcal{B})$ there is a natural isomorphism

$$\mathrm{Hom}_{D(\mathcal{B})}(LF(A), B) \cong \mathrm{Hom}_{D(\mathcal{A})}(A, RG(B)).$$

If $\mathcal{A}$ has either enough injectives or enough projectives, then there is a natural isomorphism

$$\mathrm{RHom}_{\mathcal{B}}(LF(A), B) \cong \mathrm{RHom}_{\mathcal{A}}(A, RG(B))$$

in $D(\mathbb{Z})$ for $A \in D^-(\mathcal{A})$ and $B \in D^+(\mathcal{B})$ as well.

Proof Let $X \rightarrow A$ be a qis with $X \in K^-(\mathcal{R}_F)$ and $B \rightarrow I$ a qis with I a bounded below complex of injectives. Then

$$\begin{aligned} \mathrm{Hom}_{D(\mathcal{B})}(LF(A), B) &= \mathrm{Hom}_{D(\mathcal{B})}(F(X), I) \\ &= \mathrm{Hom}_{K(\mathcal{B})}(F(X), I) \\ &= \mathrm{Hom}_{K(\mathcal{A})}(X, G(I)). \end{aligned}$$

Note that if C is an acyclic complex in $K^-(\mathcal{R}_F)$, then

$$\mathrm{Hom}_{K(\mathcal{A})}(C, G(I)) = \mathrm{Hom}_{K(\mathcal{B})}(F(C), I) = 0$$

because $F(C)$ is acyclic and I is K -injective. Hence by [Lemma 1.2.16](#), we have

$$\mathrm{Hom}_{K(\mathcal{A})}(X, G(I)) = \mathrm{Hom}_{D(\mathcal{A})}(X, G(I)) = \mathrm{Hom}_{D(\mathcal{A})}(A, RG(B)),$$

so this proves the first statement. For the second one, note that we need enough injectives or projectives in $\mathcal{A}$ simply to define RHom . Since LF preserves bounded above-ness and RG preserves bounded below-ness, the two sides of the desired isomorphism are defined.

Almost all of the steps we did before still work:

$$\begin{aligned} \mathrm{RHom}_{\mathcal{B}}(LF(A), B) &= \mathrm{RHom}_{\mathcal{B}}(F(X), I) \\ &= \underline{\mathrm{Hom}}_{\mathcal{B}}(F(X), I) \\ &= \underline{\mathrm{Hom}}_{\mathcal{A}}(X, G(I)), \end{aligned}$$

and we have the canonical map

$$\underline{\mathrm{Hom}}_{\mathcal{A}}(X, G(I)) \rightarrow \mathrm{RHom}_{\mathcal{A}}(X, G(I)) = \mathrm{RHom}_{\mathcal{A}}(A, RG(B)).$$

Combining these gives a natural map

$$\mathrm{RHom}_{\mathcal{B}}(LF(A), B) \rightarrow \mathrm{RHom}_{\mathcal{A}}(A, RG(B))$$

which we claim is an isomorphism. We can check this after applying H^i for all $i \in \mathbb{Z}$, which gives the map

$$\mathrm{Hom}_{D(\mathcal{B})}(LF(A), B[i]) \rightarrow \mathrm{Hom}_{D(\mathcal{A})}(A, RG(B)[i]).$$

Since $RG(B)[i] = RG(B[i])$, it is easy to see that this recovers the isomorphism proved in the first part. $\square$

To conclude this section, we are going to state an important result which holds in the case of enough injectives, which is the existence of a *Cartan-Eilenberg* resolution. From this we will deduce a very important spectral sequence which relates the classical higher derived functors to the cohomology of the derived functor applied to a complex. A special case of this will be the Grothendieck spectral sequence. To existence theorem, whose proof is purely classical, will be discussed in the next section.

Definition 1.3.21 Let $\mathcal{A}$ be an abelian category and $C^\bullet$ a bounded below complex. A *Cartan-Eilenberg resolution* of C is a bicomplex $I^{\bullet,\bullet}$ equipped with a morphism $\epsilon: C^\bullet \rightarrow I^{\bullet,0}$ of complexes satisfying the following properties:

- (a) $I^{p,q} = 0$ for $q < 0$, and there is k such that $I^{p,q} = 0$ for $p < k$ and all q .
- (b) The natural maps $C^p[0] \rightarrow I^{p,\bullet}$ are injective resolutions for all $p \in \mathbb{Z}$.
- (c) The natural maps

$$\ker d_C^p[0] \rightarrow \ker d_h^{p,\bullet} \quad \text{im } d_C^{p-1}[0] \rightarrow \text{im } d_h^{p-1,\bullet}$$

are injective resolutions for all $p \in \mathbb{Z}$.

- (d) The maps $H^p(C) \rightarrow H_h^p(I^{\bullet,\bullet})$ are injective resolutions for all $p \in \mathbb{Z}$.

A dual definition is made in the projective/bounded above case.

The point is to have an object-wise injective resolution which remains an injective resolution after taking cohomology. The picture is as follows:

$$\begin{array}{ccccccc} & & \vdots & & \vdots & & \vdots \\ & & \uparrow & & \uparrow & & \uparrow \\ \dots & \longrightarrow & I^{p-1,1} & \longrightarrow & I^{p,1} & \longrightarrow & I^{p+1,1} \longrightarrow \dots \\ & & \uparrow & & \uparrow & & \uparrow \\ \dots & \longrightarrow & I^{p-1,0} & \longrightarrow & I^{p,0} & \longrightarrow & I^{p+1,0} \longrightarrow \dots \\ & & \vdots & & \vdots & & \vdots \\ \dots & \longrightarrow & C^{p-1} & \longrightarrow & C^p & \longrightarrow & C^{p+1} \longrightarrow \dots \end{array}$$

So, each vertical sequence provides an injective resolution of C^p . Once we take horizontal cohomology everywhere, so that there are only vertical differentials remaining, we will obtain an injective resolution of $H(C^p)$. Note also that (a) in the definition implies that the bicomplex is a shifted first-quadrant complex.

Theorem 1.3.22 *Let $\mathcal{A}$ be an abelian category with enough injectives. Then every bounded below complex admits a Cartan-Eilenberg resolution. Likewise, if $\mathcal{A}$ has enough projectives, then the same holds for bounded above complexes. Moreover, it can be chosen functorially.*

The idea of the proof is simple, namely split up the complex into pairs of short exact sequences, choose compatible injective resolutions of each term, and then glue this data together. Since we will need some standard lemmas, we push this to the next section.

If we're given a Cartan-Eilenberg resolution $C \rightarrow I^{\bullet,0}$, the totalization behaves nicely as well. First, since it's a shifted first-quadrant complex, the totalization $\text{Tot } I$ involves only finite direct sums and therefore always exists. Moreover, we have a canonical map

$$C^n \xrightarrow{\epsilon} I^{n,0} \hookrightarrow \bigoplus_{p+q=n} I^{p,q}$$

which commutes with the differentials, hence a chain map $\tilde{\epsilon}: C \rightarrow \text{Tot } I$. Note that each term in $\text{Tot } I$ is an injective object. Now take vertical cohomology, so we can apply the first spectral sequence associated to a bicomplex. Since the resolutions are exact at degrees 1 and beyond, only the first row remains, and the canonical maps $C^p \rightarrow E_1^{p,0}$ are isomorphisms. Thus $C^\bullet \cong E_1^{\bullet,0}$, and the sequence degenerates at E_2 :

$$E_2^{p,0} = H^p(C) \quad E_2^{p,q} = 0 \quad q > 0.$$

The edge maps induce $\tilde{\epsilon}: H^p(C) \xrightarrow{\sim} H^p(\text{Tot } I)$ an isomorphism, hence the natural map $\tilde{\epsilon}$ is an injective resolution of C .

Theorem 1.3.23 *Let $F: \mathcal{A} \rightarrow \mathcal{B}$ be a left exact functor of abelian categories with an adapted class $\mathcal{R}$, so that RF exists. Suppose $\epsilon: C^\bullet \rightarrow I^{\bullet,0}$ is a Cartan-Eilenberg resolution of a bounded below complex C . The two spectral sequences associated to the bicomplex $F(I^{\bullet,\bullet})$ are bounded, and satisfy*

$$\begin{aligned} E_{h,1}^{p,q} = R^q F(C^p) &\implies H^{p+q}(RF(C)) \\ E_{v,2}^{p,q} = R^p F(H^q(C)) &\implies H^{p+q}(RF(C)) \end{aligned}$$

each converging to degree-wise finite filtrations.

Proof It is clear from the definition that $F(\text{Tot } I) = \text{Tot}(F(I))$ since F is additive. We have just shown that $C \rightarrow \text{Tot } I$ is an injective resolution, and both sequences converge to the cohomology of $\text{Tot } F(I) = F(\text{Tot } I) = RF(C)$, so both RHS's above are correct. For the first sequence, if we apply F and simply take vertical cohomology, this is just taking the cohomology of F applied to an injective resolution of C^p , which is exactly $R^q F(C^p)$.

Now let's take horizontal cohomology first. Our E_0 page has the portion $F(I^{p-1,q}) \rightarrow F(I^{p,q}) \rightarrow F(I^{p+1,q})$, and the underlying I differentials come with an exact sequence

$$0 \rightarrow \text{im } d_h^{p-1,q} \rightarrow \ker d_h^{p,q} \rightarrow H_h^p(I^{\bullet,q}) \rightarrow 0,$$

where our assumption is that all three of these terms are injective. In particular the sequence splits, so applying F keeps it exact. Therefore, the E_1 page looks

like:

$$\begin{array}{ccccc}
 & \vdots & & \vdots & & \vdots \\
 & \uparrow & & \uparrow & & \uparrow \\
 \cdots & \longrightarrow & F(J^{p-1,q+1}) & & F(J^{p,q+1}) & & F(J^{p+1,q+1}) \longrightarrow \cdots \\
 & \uparrow & & \uparrow & & \uparrow \\
 \cdots & \longrightarrow & F(J^{p-1,q}) & & F(J^{p,q}) & & F(J^{p+1,q}) \longrightarrow \cdots
 \end{array}$$

Here, $J^{p,q} = H_h^p(I^{\bullet,q})$, so as q varies these are the terms in a vertical injective resolution for $H^p(C)$. Hence, when we take vertical cohomology to get to E_2 , the (p, q) th spot will become $R^q F(H^p(C))$. The indices of this second spectral sequence get flipped, so this gives the statement in the theorem. $\square$

We will often directly use the first spectral sequence in the theorem, even when we are interested in eventually computing classical derived functors themselves. The reason is that we can try to find a resolution $P^\bullet \rightarrow A$ of an object A , and then the theorem gives us a spectral sequence

$$E_1^{p,q} = R^q F(P^p) \implies H^{p+q}(RF(P)) = R^{p+q} F(A).$$

This is especially nice because we even know the (horizontal) differentials on the first page, namely they are $R^q F$ applied to the differentials in P . This technique is useful because we usually can't write down injective resolutions, but we can write down e.g. locally free resolutions.

We can also use the first sequence when we want hypercohomology, which we'll do when discussing Hodge-to-de-Rham. The second sequence gives us the Grothendieck spectral sequence.

Corollary 1.3.24 *Let $F: \mathcal{A} \rightarrow \mathcal{B}$ and $G: \mathcal{B} \rightarrow \mathcal{C}$ be left exact functors of abelian categories such that $\mathcal{R}_F$ is an adapted class for F , and $\mathcal{B}$ has enough injectives. Assume that for each $X \in \mathcal{R}_F$, $F(X)$ is G -acyclic. Then there is a bounded spectral sequence*

$$E_2^{p,q} = R^p G(H^q(RF(A))) \implies H^{p+q}(R(G \circ F)(A))$$

for every A in $\text{Ch}^+(\mathcal{A})$ (converging to a finite filtration). In particular, if A is an object of $\mathcal{A}$, then

$$E_2^{p,q} = R^p G(R^q F(A)) \implies R^{p+q}(GF)(A)$$

as a special case.

Proof Since $\mathcal{B}$ has enough injectives, RG exists and we may define it with the maximal adapted class $\mathcal{R}_G$ of all G -acyclic objects in $\mathcal{B}$. Then $F(\mathcal{R}_F) \subset \mathcal{R}_G$, so by [Thm. 1.2.22](#) we can identify $RG \circ RF = R(G \circ F)$. Let $A \in \text{Ch}^+(\mathcal{A})$. Then $RF(A)$ in $D^+(\mathcal{B})$ has a Cartan-Eilenberg resolution by [Thm. 1.3.22](#) since $\mathcal{B}$ has enough injectives, so applying the second sequence of [Thm. 1.3.23](#) to $C = RF(A)$, are done. $\square$

Example 1.3.25 Let us demonstrate an example showing that, in spite of the Grothendieck spectral sequence, knowing $R^q F(A)$ does not determine $R^n(GF)(A)$. We will instead do it using left derived functors. First some elementary observations: if $R \rightarrow S$ is a ring map, then the functor

$$F = (-) \otimes_R S: \text{Mod}(R) \rightarrow \text{Mod}(S)$$

preserves free modules, which are an adapted class for tensor products on both sides. Using this and the fact that restriction of scalars is exact, it follows that $LF(A)|_R = A \otimes_R^{\mathbf{L}} S$ for any $A \in D^-(R)$.

Now let $S = k[\epsilon]/(\epsilon^2)$ which is a quotient of $R = S[t]$ by t . Moreover, k is a quotient of S by (ϵ) . Consider the functors

$$F = (-) \otimes_R S: \text{Mod}(R) \rightarrow \text{Mod}(S) \quad G = (-) \otimes_S k: \text{Mod}(S) \rightarrow \text{Mod}(k).$$

Lastly, consider the R -modules $M = R/(\epsilon - t)$ and $N = R/(t, \epsilon) = k$. The consideration about free modules above shows that the composition satisfies the hypotheses for composing derived functors, and the Grothendieck spectral sequence.

Now let's compute: t is a nonzerodivisor on R , so $S = [R \xrightarrow{t} R]$ in $D^-(R)$. Thus the underlying R -modules of $LF(M)$ and $LF(N)$ are

$$M \otimes_R^{\mathbf{L}} S = [M \xrightarrow{t} M] \cong [S \xrightarrow{\epsilon} S]$$

because $k[\epsilon, t]/(\epsilon^2, t - \epsilon) = k[\epsilon]/(\epsilon^2)$ as S -modules. On the other hand,

$$N \otimes_R^{\mathbf{L}} S = [k \xrightarrow{0} k] = k \oplus k[1].$$

So, $L^0 F(M) = L^{-1} F(M) = k$ and lower degrees are zero, and the same is true for N . However, we showed previously that $LG(k)$ has cohomology in every degree, so the same holds for $LG(LF(N))$. On the other hand, our presentation for $M \otimes_R^{\mathbf{L}} S$ is already a free resolution, so applying LG gives $k \xrightarrow{0} k$, which has cohomology in only two degrees.

Let us see what is happening in the Grothendieck spectral sequence, which by construction has the same E_2 -page in both cases:

$$\begin{array}{ccccc|c} \cdots & -3 & -2 & -1 & 0 & \\ \cdots & k & k & k & k & 0 \\ \cdots & k & k & k & k & -1 \end{array}$$

The differentials go 2 spaces to the right and 1 down, and since this is a spectral sequence of k -vector spaces the nonzero ones are each either 0 or isomorphisms. No matter what the sequence will degenerate on E_3 . Now we can see that in the case of M , all of the nonzero differentials have to be isomorphisms so that we end up with only k at $(-1, 0)$ and $(0, 0)$. For N , all of the differentials are 0, so this is already E_∞ and we get the expected two-dimensional cohomology in all strictly negative degrees.

1.4 Appendix: Classical homological algebra

This appendix will have a few short sections discussing some basic results from the classic theory (in contrast to the previous section, which recast the classical theory in our new language). In the last section, we will discuss any results from commutative algebra which are helpful in establishing the cohomology theory which we would like to setup in the next chapter.

1.4.1 More on resolutions

In the previous section we stated [Thm. 1.3.22](#) on the existence of Cartan-Eilenberg resolutions. In this section, we will complete the proof of this and introduce some useful homological techniques along the way.

A short exact sequence $0 \rightarrow A \xrightarrow{i} B \xrightarrow{p} C \rightarrow 0$ of complexes is *termwise split* if $0 \rightarrow A^i \rightarrow B^i \rightarrow C^i \rightarrow 0$ is a split exact sequence for each $i \in \mathbb{Z}$. These behave especially nicely because the connecting map in the long exact sequence is already exists in the chain complex category.

To see this, choose a splitting $B^i = A^i \oplus C^i$ and let $s^i: C^i \rightarrow B^i$ and $\pi^i: B^i \rightarrow A^i$ be the splitting maps. Then we claim that the family of maps

$$\delta^n := \pi^{n+1} \circ d_B^n \circ s^n: C \rightarrow A^{n+1}$$

define a morphism $\delta: C \rightarrow A[1]$ of complexes. To see this, first compute the matrix of the differential $A^i \oplus C^i \rightarrow A^{i+1} \oplus C^{i+1}$ of B : on the A^i component, we have $d_B^i = id_A$ i.e. the canonical injection. On the C^i component the map is $d_B s$, which has components

$$p(d_B s) = d_C p s = d_C \quad \pi d_B s = \delta.$$

This shows that

$$d_B = \begin{pmatrix} d_A & \delta \\ 0 & d_B \end{pmatrix}.$$

Doing matrix multiplication, $d^2 = 0$ implies that

$$0 = d_A \delta + \delta d_C = -d_{A[1]} \delta + \delta d_C,$$

so indeed $\delta: C \rightarrow A[1]$ is a chain map. From the explicit differential, we immediately conclude that $B = \text{Cone}(-\delta)[-1]$ compatible with A , so there is an exact triangle

$$C \xrightarrow{-\delta} A[1] \rightarrow \text{Cone}(-\delta) \cong B[1] \rightarrow C[1]$$

in $K(\mathcal{A})$. Shifting and rotating this we obtain $A \rightarrow B \rightarrow C \xrightarrow{\delta} A[1]$, thus a termwise split sequence gives an exact triangle in $K(\mathcal{A})$. Conversely, we know that all triangles are of this form up to isomorphism, because $0 \rightarrow B \rightarrow \text{Cone } f \rightarrow A[1] \rightarrow 0$ is a termwise split exact sequence.

In particular, if we pass to the derived category our triangle has the same form, so applying the cohomological functor H^0 gives the usual long exact sequence with the connecting map being induced by δ . Of course, this is non-unique and depends exactly on the choice of splitting.

Lemma 1.4.1 *Let $\mathcal{A}$ be an abelian category.*

- (a) *If $q: B \rightarrow C$ is a quasi-isomorphism which is a termwise epimorphism, then $\ker d_B^i \rightarrow \ker d_C^i$ is an epimorphism for all i .*
- (b) *If $i: A \rightarrow B$ is a quasi-isomorphism which is a termwise monomorphism, and $I \in \text{Ch}^+(\mathcal{I}_\mathcal{A})$, then every map $A \rightarrow I$ extends to B .*

Proof (a) Consider the commutative diagrams with exact rows:

$$\begin{array}{ccccccc}
 B^i & \longrightarrow & \text{im } d_B^i & \longrightarrow & 0 & & \text{im } d_B^{i-1} \longrightarrow \ker d_B^i \longrightarrow H^i(B) \longrightarrow 0 \\
 q \downarrow & & q \downarrow & & & & \downarrow \quad \downarrow \quad \downarrow \cong \quad \downarrow \\
 C^i & \longrightarrow & \text{im } d_C^i & \longrightarrow & 0 & & \text{im } d_C^{i-1} \longrightarrow \ker d_C^i \longrightarrow H^i(C) \longrightarrow 0
 \end{array}$$

The left-hand diagram shows that the map on coboundaries is an epi for all i , and then the diagram on the right proves the claim by the epimorphism 4-lemma.

(b) Then $\underline{\text{Hom}}_\mathcal{A}(B, I) \rightarrow \underline{\text{Hom}}_\mathcal{A}(A, I)$ is a quasi-isomorphism, and it is a termwise epi because $\text{Hom}(C^i, I^{k+i}) \rightarrow \text{Hom}(B^i, I^{k+i})$ is surjective. Thus this induces a surjection on 0-cocycles by (a), i.e.

$$\text{Hom}_{\text{Ch}(\mathcal{A})}(B, I) \rightarrow \text{Hom}_{\text{Ch}(\mathcal{A})}(A, I)$$

is surjective, so every map lifts as desired. $\square$

We are going to apply this to the following powerful resolution extension lemma, which is the key step in finding Cartan-Eilenberg resolutions.

Lemma 1.4.2 *Let $\mathcal{A}$ be an abelian category, with $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ a short exact sequence in $\text{Ch}^+(\mathcal{A})$. Let*

$$A \rightarrow I_1 \quad C \hookrightarrow I_3$$

be injective resolutions with $I_1, I_3 \in \text{Ch}^+(\mathcal{I}_\mathcal{A})$, and assume the latter is a monomorphism. Then there exists $I_2 \in \text{Ch}^+(\mathcal{I}_\mathcal{A})$ and a commutative diagram

$$\begin{array}{ccccccc}
 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & I_1 & \dashrightarrow & I_2 & \dashrightarrow & I_3 \longrightarrow 0
 \end{array}$$

of complexes in $\text{Ch}^+(\mathcal{A})$ with exact rows, such that $B \rightarrow I_2$ is an injective resolution. If I_1, I_3 are concentrated in $[a, b]$, then we may assume I_2 is as well.

If the given sequence was split, then we could just take the direct sum of the injective resolutions. We can't reduce to this case, but we can reduce to the case that it is termwise split, where we can take the direct sum termwise, and then modify the differential.

Proof (of Lemma 1.4.2) First take the pushout in the category of complexes (which is done termwise), so that we get a commutative diagram of exact sequences of complexes:

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \parallel \\ 0 & \longrightarrow & I_1 & \dashrightarrow & P & \longrightarrow & C \longrightarrow 0 \end{array}$$

By triangulated 2-out-of-3 in $D^+(\mathcal{A})$, the map $B \rightarrow P$ is a quasi-isomorphism. Since I_1 is termwise injective, the bottom row is termwise split, so by choosing a splitting $P^i = I_1^i \oplus C^i$, we can take the map $\delta: C \rightarrow I_1[1]$ as above. This identifies $P = \text{Cone}(-\delta)[-1]$. We construct a resolution for the bottom row termwise as follows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & I_1^i & \longrightarrow & I_1^i \oplus C^i & \longrightarrow & C^i \longrightarrow 0 \\ & & \parallel & & \downarrow & & \downarrow \\ 0 & \longrightarrow & I_1^i & \longrightarrow & I_1^i \oplus I_3^i & \longrightarrow & I_3^i \longrightarrow 0 \end{array}$$

All we have to do is equip $I_2^i := I_1^i \oplus I_3^i$ with a differential so that $P \rightarrow I_2$ is a chain map: Once this is done, triangulated 2/3 will say that this is a diagram of quasi-isomorphisms, and then by composing it with the initial diagram we are done.

By the lemma we just proved, the map $\delta: C \rightarrow I_1[1]$ extends along the quasi-isomorphic inclusion $q: C \rightarrow I_3$ to a map $\delta_2: I_3 \rightarrow I_1[1]$. In other words, $(1_{I_1[1]}, q)$ is a morphism of arrows $\delta \rightarrow \delta_2$, so this induces a map $\text{Cone}(-\delta)[-1] \rightarrow \text{Cone}(-\delta_2)[-1]$ which componentwise is exactly the middle map of the diagram above. $\square$

At last, we finish the proof that Cartan-Eilenberg resolutions exist in an abelian category $\mathcal{A}$ with enough injectives.

Proof (of Thm. 1.3.22) We use B and Z notation for coboundaries and cocycles. Assume for simplicity that our complex C has $C^i = 0$ for $i < 0$, and we inductively construct compatible resolutions for $i \geq 0$. For simplicity, in this proof we use the word ‘resolution’ to mean injective resolution of an object of $\mathcal{A}$ which extends only to the right (so is termwise injective).

For the i th step, we’ll use the exact sequences

$$0 \rightarrow Z^{i-1} \rightarrow C^{i-1} \rightarrow B^i \rightarrow 0 \quad 0 \rightarrow B^i \rightarrow Z^i \rightarrow H^i(C) \rightarrow 0,$$

and assume that we are given a resolution $Z^{i-1}[0] \rightarrow J_Z^{i-1, \bullet}$ from the previous step. Choose a resolution $B^i[0] \rightarrow J_B^{i, \bullet}$, and using our lemma extend this to a diagram of extensions like so:

$$\begin{array}{ccccccc} 0 & \longrightarrow & J_Z^{i-1, \bullet} & \dashrightarrow & I^{i-1, \bullet} & \dashrightarrow & J_B^{i, \bullet} \longrightarrow 0 \\ & & \uparrow & & \uparrow & & \uparrow \\ 0 & \longrightarrow & Z^{i-1} & \longrightarrow & C^{i-1} & \longrightarrow & B^i \longrightarrow 0 \end{array}$$

Of course, if $i = 0$ then we take the left two vertical maps and the resolution to be zero. To construct the next J_Z resolution, choose a resolution $H^i(C)[0] \rightarrow J_H^{i,\bullet}$ and extend over the second sequence:

$$\begin{array}{ccccccc} 0 & \longrightarrow & J_B^{i,\bullet} & \dashrightarrow & J_Z^{i,\bullet} & \dashrightarrow & J_H^{i,\bullet} \longrightarrow 0 \\ & & \uparrow & & \uparrow & & \uparrow \\ 0 & \longrightarrow & B^i & \longrightarrow & Z^i & \longrightarrow & H^i(C) \longrightarrow 0 \end{array}$$

Now we are done: pasting the successive diagrams of exact sequences of the first kind together gives a commutative diagram with vertical maps $\epsilon: C^i[0] \rightarrow I^{i,\bullet}$ injective resolutions, so that when we take coboundaries, cocycles, and then cohomology, we get the injective resolutions J_B^i, J_Z^i, J_H^i . This gives us a bicomplex up to our sign convention, so we should flip the sign of the vertical differential in the odd columns. $\square$

1.4.2 Cohomological δ -functors

Here we quickly discuss the classical topic of δ -functors, and then relate it to what we defined as a δ -functor from an abelian category to a triangulated category.

Definition 1.4.3 A (cohomological) δ -functor $T: \mathcal{A} \rightarrow \mathcal{B}$ between abelian categories is a sequence $(T^i)_{i \geq 0}$ of additive functors $T^i: \mathcal{A} \rightarrow \mathcal{B}$ equipped with maps

$$\delta = \delta_{A \rightarrow B \rightarrow C}: T^i(C) \rightarrow T^{i+1}(A) \quad i \geq 0$$

for every short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ in $\mathcal{A}$, such that:

- (a) For every short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ in $\mathcal{A}$, the sequence

$$0 \rightarrow T^0(A) \rightarrow T^0(B) \rightarrow T^0(C) \xrightarrow{\delta} T^1(A) \rightarrow \dots$$

is exact in $\mathcal{B}$.

- (b) Given a morphism

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & A' & \longrightarrow & B' & \longrightarrow & C' \longrightarrow 0 \end{array}$$

of short exact sequences in $\mathcal{A}$, the corresponding diagram

$$\begin{array}{ccc} T^i(C) & \xrightarrow{\delta} & T^{i+1}(A) \\ \downarrow & & \downarrow \\ T^i(C') & \xrightarrow{\delta} & T^{i+1}(A') \end{array}$$

in $\mathcal{B}$ commutes for all $i \geq 0$.

Note that in particular if T is a δ -functor, then T^0 has to be left exact. The primary example of δ -functors come from the triangulated δ -functors in the

sense of Def. 1.2.7. Namely, if $\mathcal{A} \rightarrow \mathcal{C}$ is a δ -functor from an abelian category to a triangulated category $\mathcal{C}$, and $H: \mathcal{C} \rightarrow \mathcal{B}$ is a cohomological functor, then

$$T^i := \mathcal{A} \rightarrow \mathcal{C} \xrightarrow{H \circ [i]} \mathcal{B}$$

is a cohomological δ -functor, essentially by definition. So, for example:

- (a) Since $\mathrm{Ch}^*(\mathcal{A}) \rightarrow D^*(\mathcal{A})$ are δ -functors, whenever we have a cohomological functor $D^*(\mathcal{A}) \rightarrow \mathcal{B}$ we get a δ -functor $\mathrm{Ch}^*(\mathcal{A}) \rightarrow \mathcal{B}$.
- (b) Since the inclusion $\mathcal{A} \rightarrow D^*(\mathcal{A})$ is a δ -functor, any cohomological functor on the derived category gives a δ -functor $\mathcal{A} \rightarrow \mathcal{B}$. For example, using the functor $\mathrm{Hom}_{D^*(\mathcal{A})}(A[0], -)$ produces $\{\mathrm{Ext}_{\mathcal{A}}^i(A, -)\}$ from $\mathcal{A}$ to $\mathbf{Ab}$.
- (c) In particular, if $F: \mathcal{A} \rightarrow \mathcal{B}$ is a left exact functor of abelian categories admitting an adapted class, so that $RF: D^+(\mathcal{A}) \rightarrow D^+(\mathcal{B})$ exists, then the classical derived functors

$$(R^i F)_{i \geq 0} \quad R^i F: \mathcal{A} \rightarrow \mathcal{B}$$

form a δ -functor, because they are the following composite

$$\mathcal{A} \rightarrow D^+(\mathcal{A}) \xrightarrow{RF} D^+(\mathcal{B}) \xrightarrow{H^0 \circ [i]} \mathcal{B},$$

where the composition of the first two functors is a triangulated δ -functor since the first one is, and RF is triangulated.

Definition 1.4.4 Let T and S be cohomological δ -functors from $\mathcal{A}$ to $\mathcal{B}$.

- (a) A morphism $f: T \rightarrow S$ is a sequence of natural transformations $f^i: T^i \rightarrow S^i$ such that the diagram

$$\begin{array}{ccc} T^i(C) & \xrightarrow{\delta_T} & T^{i+1}(A) \\ f^i \downarrow & & \downarrow f^i \\ S^i(C) & \xrightarrow{\delta_S} & S^{i+1}(A) \end{array}$$

commutes for every exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ in $\mathcal{A}$.

- (b) The δ -functor T is called *universal* if for every δ -functor S and morphism $f^0: T^0 \rightarrow S^0$, there is a unique morphism $f: T \rightarrow S$ extending f^0 .

Clearly, a universal δ -functor is uniquely determined by its first component, meaning that if T, S are universal, and $T^0 \cong S^0$, then there's a unique isomorphism $T \cong S$ of δ -functors extending the given one. In the special case of the δ -functor associated to classical derived functors, we have *some* kind of universal property but it is a universality among exact functors of triangulated categories.

Sometimes, it is enough to check things on the level of classical derived functors to get information about the derived category, and to construct maps between such functors we don't want to have to start out with the full data

of a triangulated functor. For example, if A, B are objects of $\mathcal{A}$ and there is a map $A \rightarrow B$ such that $R^i F(A) \rightarrow R^i F(B)$ is an isomorphism, then we already know that $RF(A[0]) \rightarrow RF(B[0])$ is an isomorphism (because it is one on cohomology). We need one final definition before proving the main result.

Definition 1.4.5 An additive functor $F: \mathcal{A} \rightarrow \mathcal{B}$ of abelian categories is *effaceable* if for each A in $\mathcal{A}$ there exists a monomorphism $i: A \rightarrow X$ for some X in $\mathcal{A}$ such that $F(i) = 0$.

The following is the main theorem on δ -functors.

Theorem 1.4.6 *Let T be a δ -functor from $\mathcal{A}$ to $\mathcal{B}$, and suppose that T^i is effaceable for $i > 0$. Then T is a universal δ -functor.*

Proof Let S be another δ -functor and assume we have a natural transformation $f^0: T^0 \rightarrow S^0$. Suppose inductively that we have constructed a natural map $f^n: T^n \rightarrow S^n$. To define f^{n+1} on an object A , let $A \rightarrow X$ be an *effacement* (of degree $n+1$), i.e. a mono as in the definition of effaceable. We have a diagram of exact sequences

$$\begin{array}{ccccccc} T^n X & \longrightarrow & T^n C & \longrightarrow & T^{n+1} A & \xrightarrow{0} & T^{n+1} X \\ f^n \downarrow & & f^n \downarrow & & \downarrow & & \\ S^n X & \longrightarrow & S^n C & \longrightarrow & S^{n+1} A & \longrightarrow & S^{n+1} X \end{array}$$

(without the dotted line). The 0 means that $T^{n+1} A$ is the cokernel of the map $T^n X \rightarrow T^n C$, and any dotted map filling in the square would have to factor through the cokernel of $S^n X \rightarrow S^n C$, so it is uniquely determined. We are going to define this as f^{n+1} and demonstrate the required properties, but note that this morphism is already unique for any extension of f^n to a morphism of δ -functors.

1. f^{n+1} is independent of the chosen effacement. Suppose we are given two (degree $n+1$) effacements $A \rightarrow X$ and $A \rightarrow X'$. Form the pushout P , then we still have a mono $A \rightarrow P$, and let $D = P/A$ so that we have a morphism

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \longrightarrow & X & \longrightarrow & C \longrightarrow 0 \\ & & \parallel & & \downarrow & & \downarrow \\ 0 & \longrightarrow & A & \longrightarrow & P & \longrightarrow & D \longrightarrow 0 \end{array}$$

of exact sequences. If we consider the associated long exact sequences, we get a commutative diagram:

$$\begin{array}{ccccccc} T^n X & \longrightarrow & T^n C & \longrightarrow & T^{n+1} A & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ ? & & ? & & ? & & ? \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ S^n P & \longrightarrow & S^n D & \longrightarrow & S^{n+1} A & \longrightarrow & S^{n+1} P \end{array}$$

There are two natural choices for the middle row: either we apply T to the bottom row, or we apply S to the top row. The first choice, since

T is a δ -functor, shows that $A \rightarrow P$ is also an effacement. That f^n is a natural transformation is exactly the statement that the first two vertical composites $T^n X \rightarrow S^n P$ and $T^n C \rightarrow S^n D$ are the same no matter which middle row we take. It follows from the same argument above that any map $T^{n+1} A \rightarrow S^{n+1} A$ filling in the diagram are equal.

For the two choices of middle rows, the possible composites are

$$T^{n+1} A \xrightarrow{1} T^{n+1} A \rightarrow S^{n+1} A \quad T^{n+1} A \rightarrow S^{n+1} A \xrightarrow{1} S^{n+1} A,$$

where in the first case the second map is induced by the effacement $A \rightarrow P$, and in the second the first map is induced by the effacement $A \rightarrow X$. Therefore these two effacements induce the same morphism f^{n+1} . Since the same argument applies to $0 \rightarrow A \rightarrow X' \rightarrow X'/A \rightarrow 0$ instead of the current top row, this shows that $A \rightarrow X$ and $A \rightarrow X'$ induce the same map f^{n+1} .

2. f^{n+1} commutes with δ . Start with a sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$, and choose an effacement $B \rightarrow X$. The composite $A \rightarrow X$ is again a monomorphism, so writing $D = X/A$ we have a morphism of exact sequences:

$$\begin{array}{ccccccc} 0 & \rightarrow & A & \rightarrow & B & \rightarrow & C \rightarrow 0 \\ & & \parallel & & \downarrow & & \downarrow \\ 0 & \rightarrow & A & \rightarrow & X & \rightarrow & D \rightarrow 0 \end{array}$$

One portion of the long exact sequence is the diagram

$$\begin{array}{ccc} T^{n+1} A & \longrightarrow & T^{n+1} B \\ \parallel & & \downarrow 0 \\ T^{n+1} A & \longrightarrow & T^{n+1} X \end{array}$$

which shows that $A \rightarrow X$ is also an effacement for X . We have another commutative diagram as follows:

$$\begin{array}{ccccc} T^n C & \longrightarrow & T^n D & \longrightarrow & T^{n+1} A \\ \downarrow & & \downarrow & & \downarrow \\ S^n C & \longrightarrow & S^n D & \longrightarrow & S^{n+1} A \end{array}$$

The right-hand square is the definition of f^{n+1} using $A \rightarrow X$, and the left-hand square is naturality of f^n . However, since we have a morphism of exact sequences, the composites of both rows are exactly the δ maps for T and S , which is what we wanted.

3. f^{n+1} is natural. This is similar to Step 1: Given an effacement $A \rightarrow X$ and any map $A \rightarrow B$, pushout along the map to obtain a morphism of exact sequences:

$$\begin{array}{ccccccc} 0 & \rightarrow & A & \rightarrow & X & \rightarrow & C \rightarrow 0 \\ & & \downarrow & & \downarrow & & \parallel \\ 0 & \rightarrow & B & \rightarrow & P & \rightarrow & C \rightarrow 0 \end{array}$$

The long exact sequence gives us:

$$\begin{array}{ccccccc}
 T^n X & \longrightarrow & T^n C & \longrightarrow & T^{n+1} A & \longrightarrow & 0 \\
 \downarrow & & \downarrow & & \downarrow & & \\
 ? & & ? & & ? & & \\
 \downarrow & & \downarrow & & \downarrow & & \\
 S^n P & \longrightarrow & S^n C & \longrightarrow & S^{n+1} B & &
 \end{array}$$

As before, we have two ways to fill this in, either applying T to the bottom row or S to the top row. In either case, naturality of f^n shows that the two maps $T^n X \rightarrow S^n P$ and $T^n C \rightarrow S^n C$ have to coincide, so any two extensions filling in $T^{n+1} A \rightarrow S^{n+1} B$ are the same. But these two possibilities are exactly:

$$T^{n+1} A \rightarrow T^{n+1} B \xrightarrow{f^{n+1}} S^{n+1} B \quad T^{n+1} A \xrightarrow{f^{n+1}} S^{n+1} B \rightarrow S^{n+1} B$$

Filling in the diagram in the second case is by definition of f^{n+1} , but in the first case we are applying part (2) to deduce that the bottom right square commutes. It follows that the above maps are equal, which is what we wanted.

We have constructed f^{n+1} given f^n , which completes the induction step, and we have already explained why it is uniquely determined by f^n . In particular, all the f^i are uniquely determined by f^0 , so this shows T is universal. $\square$

Corollary 1.4.7 *Let $\mathcal{A}, \mathcal{B}$ be abelian categories, $F: \mathcal{A} \rightarrow \mathcal{B}$ a left exact functor which admits an adapted class. The classical higher derived functors $(R^i F)_{i \geq 0}: \mathcal{A} \rightarrow \mathcal{B}$ form a universal δ -functor.*

In particular, if $T = (T^i)_{i \geq 0}$ is a universal δ -functor from $\mathcal{A}$ to $\mathcal{B}$, and the functor T^0 admits an adapted class (e.g. $\mathcal{A}$ has enough injectives), then the canonical map $T \rightarrow (R^i T^0)_{i \geq 0}$ is an isomorphism.

Proof The second statement is simply uniqueness of universal δ -functors. To check universality, we need to show that the higher derived functors are effaceable, which is clear because members of an adapted class are F -acyclic. $\square$

Remark 1.4.8 Note from the proof here that derived functors with an adapted class satisfied a much stronger condition than effaceability: we found a mono to a single object which made all higher functors vanish, whereas we only needed it to send the map to zero, and the effacement was allowed to depend on the degree. Thus, this gives a general criterion to check whether a δ -functor is given by a derived functor.

Remark 1.4.9 We note in passing that, as is immediate from the definition, we can always precompose or postcompose a δ -functor with an exact functor and get a δ -functor.

1.4.3 Modules over a ring

Let R be a ring. Recall that we have made the following definitions so far:

- (a) I is injective if $\text{Hom}_R(-, I)$ is exact.
- (b) P is projective if $\text{Hom}_R(P, -)$ is exact.
- (c) F is flat if $F \otimes_R (-)$ is exact.

$\text{Mod}(R)$ denotes the abelian category of R -modules. We have shown that I is injective iff $\text{Ext}_R^i(-, I) = 0$ for $i > 0$, and P is projective iff $\text{Ext}_R^i(P, -) = 0$ for $i > 0$. Observe that if P is a free module, i.e. $P \cong R^{\oplus I}$ for some index set I , then P is projective because

$$\text{Hom}_R(P, M) \cong \prod_I \text{Hom}_R(R, M) \cong \prod_I M,$$

and it is clear that this functor is exact: if $M \rightarrow N \rightarrow 0$ is a surjection, then we can lift a sequence $(y_i)_{i \in I}$ termwise to a sequence in M . Moreover, every module M admits a surjection from a free module, namely $R^{(M)} \rightarrow M \rightarrow 0$, so $\text{Mod}(R)$ has enough projectives.

Note that $R^{\oplus I} \otimes M \cong (R \otimes M)^{\oplus I} \cong M^{\oplus I}$, and this functor is clearly exact, because if $M \subset N$ and a sequence $(x_i)_{i \in I}$ is zero in M , then it is zero in N too. Thus $\text{Mod}(R)$ has enough flats by virtue of having enough frees, so Tor is well-defined and satisfies the standard properties. In particular, projective modules are flat. In this section we will use the notation $\text{Tor}_i^R(M, N) := \text{Tor}_{\text{Mod}(R)}^{-i}(M, N)$. We have also shown that F is flat iff $\text{Tor}_i^R(F, -) = 0$ for $i > 0$.

In such a setting we have also demonstrated the following equivalent conditions for injectivity, projectivity, and flatness; I is injective iff every short exact sequence of the form $0 \rightarrow I \rightarrow M \rightarrow M'' \rightarrow 0$, P is projective iff every short exact sequence of the form $0 \rightarrow M' \rightarrow M \rightarrow P \rightarrow 0$ splits, and F is flat iff for every short exact sequence $0 \rightarrow M' \rightarrow M \rightarrow F \rightarrow 0$, the sequence

$$0 \rightarrow M' \otimes N \rightarrow M \otimes N \rightarrow F \otimes N \rightarrow 0$$

is exact. From this we deduce that a module P is projective iff it is a direct summand of a free module, where the second direction follows from $\text{Ext}^1(P \oplus P', M) = \text{Ext}^1(P, M) \oplus \text{Ext}^1(P', M)$ and the fact that free modules are projective. The first goal of this section is the following theorem:

Theorem 1.4.10 *Let R be a ring. Then $\text{Mod}(R)$ has enough injectives. Moreover, injective embeddings can be chosen functorially.*

We will deduce this by first studying abelian groups. To do this, we will use a few times the right adjoint to restriction of scalars: recall that if $R \rightarrow S$ is a ring map, then we have a canonical functor

$$\text{Hom}_R(S, -): \text{Mod}(R) \rightarrow \text{Mod}(S),$$

where for an R -module M we view $\text{Hom}_R(S, M)$ as an S -module by pre-multiplication in the first factor. The adjunction gives isomorphisms

$$\text{Hom}_S(N, \text{Hom}_R(S, M)) = \text{Hom}_R(N, M)$$

of restricted R -modules, or of S -modules by doing the same thing as above on the RHS. Here are some basic properties:

Lemma 1.4.11 *Let R be a ring.*

- (a) *A product of injective modules is injective.*
- (b) *If $R \rightarrow S$ is a flat map, and E is injective over S , then $E|_R$ is injective over R .*
- (c) *Let $R \rightarrow S$ is an epimorphism of rings and E an S -module. If $E|_R$ is injective, then E is injective.*
- (d) *If $R \rightarrow S$ is a ring map and E is injective over R , then $\text{Hom}_R(S, E)$ is injective over S .*

Proof (a) Note that $\text{Hom}(-, \prod_i E_i) = \prod_i \text{Hom}(-, E_i)$, and we already explained why taking products was exact.

(b) We have $\text{Hom}_R(M, E|_R) \cong \text{Hom}_S(M \otimes_R S, E)$, and both $M \mapsto M \otimes_R S$ is exact and $\text{Hom}_S(-, E)$ is exact.

(c) The forgetful map $\text{Hom}_S(M, E) \rightarrow \text{Hom}_R(M, E|_R)$ is of course injective, but it is surjective as well by assumption.

(d) Note that $\text{Hom}_S(-, \text{Hom}_R(S, M)) = \text{Hom}_R(-, M)$ as functors of S -modules, and restriction of scalars on the right is just $\text{Hom}_R((-)|_R, M)$. $\square$

A less trivial characterization is the ideal-wise criterion for injectivity:

Lemma 1.4.12 *Let E be an R -module. The following are equivalent:*

- (a) *E is injective.*
- (b) *For any ideal $I \subset R$, every map $I \rightarrow E$ extends to R .*
- (c) *$\text{Ext}_R^1(R/I, E) = 0$ for every ideal $I \subset R$.*

Proof (a) $\Rightarrow$ (b) $\text{Hom}(-, E)$ is supposed to preserve $0 \rightarrow I \rightarrow R$.

(b) $\Leftrightarrow$ (c) We have an exact sequence

$$\text{Hom}(R, E) \rightarrow \text{Hom}(I, E) \rightarrow \text{Ext}_R^1(R/I, E),$$

so surjectivity of the first map is controlled by $\text{Ext}_R^1 = 0$.

Let us add our own extra statement:

(b') *For a strict inclusion $M \subsetneq N$ and map $\phi: M \rightarrow E$, there exists $M' \subset N$ with $M \subsetneq M'$ and $\phi': M' \rightarrow E$ extending ϕ .*

(b') $\Rightarrow$ (a) Let $M \subset N$ and $\phi: M \rightarrow E$. Consider the set of pairs $\{(M', \phi')\}$ with $M \subset M' \subset N$ and $\phi'|_{M'} = \phi$, which is nonempty. Order this by inclusion on the first factor and agreement of restriction on the second. Clearly if we have an ascending chain of submodules then we can extend our map to the union. We are done by Zorn's lemma, because the hypothesis guarantees that a maximal element must be N itself.

(b) $\Rightarrow$ (b') Take $x \in N \setminus M$ and consider the ideal $I = \{r \in R : rx \in M\}$. The map $I \rightarrow E$ given by $r \mapsto \phi(rx)$ can be extended to $\psi: R \rightarrow E$, and it is easy to check that the extension $M + \langle x \rangle \rightarrow E$ sending $a + rx \mapsto \phi(a) + \psi(r)$ works, so we are done. $\square$

Recall that if R is a domain, then an R -module M is *divisible* if for each nonzero $r \in R$ and $x \in M$ there exists $y \in M$ such that $ry = x$.

Corollary 1.4.13 *Let R be a domain. Then an injective R -module is divisible. If R is a PID, then any divisible R -module is injective.*

Proof For nonzero $r \in R$, we have an exact sequence $0 \rightarrow R \xrightarrow{r} R \rightarrow R/(r) \rightarrow 0$, and thus an exact sequence:

$$\begin{array}{ccccccc} \mathrm{Hom}(R, E) & \longrightarrow & \mathrm{Hom}(R, E) & \longrightarrow & \mathrm{Ext}^1(R/(r), E) & \longrightarrow & \mathrm{Ext}^1(R, E) \\ \parallel & & & & & & \parallel \\ E & \xrightarrow{r \cdot (-)} & E & & & & 0 \end{array}$$

The 0 on the right is because R is projective. Divisibility is the statement that for all r , the left hand map is surjective, i.e. $\mathrm{Ext}^1(R/(r), E) = 0$. If E is injective then this holds; conversely if R is PID, then all ideals are either zero or of this form. But for the zero ideal, $\mathrm{Ext}^1(R, E)$ is already 0, so we conclude by [Lemma 1.4.12](#). $\square$

Example 1.4.14 It is clear from the definition that a quotient of a divisible module is divisible. If R is a PID, then $\mathrm{Frac}(R)$ is injective, because it is clearly divisible; thus the same is true of any R -quotient of it. In the case $R = \mathbb{Z}$, this provides the two most important injective abelian groups, $\mathbb{Q}$ and $\mathbb{Q}/\mathbb{Z}$. This also gives us an example of an injective resolution $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$. Note that this is also a flat resolution of $\mathbb{Q}/\mathbb{Z}$.

Example 1.4.15 Here are some weird properties of the groups mentioned above. Consider the sequence $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$. It is a trivial exercise that $\mathrm{Hom}(\mathbb{Q}, \mathbb{Z}) = 0$ and $\mathrm{Hom}(\mathbb{Q}, \mathbb{Q}) \rightarrow \mathrm{Hom}(\mathbb{Z}, \mathbb{Q})$ is an isomorphism, so we deduce $\mathrm{Hom}(\mathbb{Q}/\mathbb{Z}, \mathbb{Q}) = 0$. More difficult is the group $G = \mathrm{Hom}(\mathbb{Q}, \mathbb{Q}/\mathbb{Z})$: One description is via the sequence

$$0 \rightarrow \mathbb{Q} \rightarrow G \rightarrow \mathrm{Ext}_{\mathbb{Z}}^1(\mathbb{Q}, \mathbb{Z}) \rightarrow 0,$$

which splits because $\mathbb{Q}$ is injective. It is a theorem that $\mathrm{Ext}_{\mathbb{Z}}^1(\mathbb{Q}, \mathbb{Z}) \cong \mathbb{R}$, then $G = \mathbb{Q} \oplus \mathbb{R} \cong \mathbb{R}$. Alternatively, one can use the simple fact that $\mathbb{Q}/\mathbb{Z} =$

$\varinjlim_n \mathbb{Z}/n\mathbb{Z}$ as the colimit of its n -torsion parts, and then

$$\mathrm{Hom}(\mathbb{Q}/\mathbb{Z}, \mathbb{Q}/\mathbb{Z}) = \varprojlim_n \mathrm{Hom}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Q}/\mathbb{Z}) = \varprojlim_n \mathbb{Z}/n\mathbb{Z} = \widehat{\mathbb{Z}}.$$

We can combine this with the exact sequence

$$0 \rightarrow \mathrm{Hom}(\mathbb{Q}/\mathbb{Z}, \mathbb{Q}/\mathbb{Z}) = \widehat{\mathbb{Z}} \rightarrow \mathrm{Hom}(\mathbb{Q}, \mathbb{Q}/\mathbb{Z}) \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0,$$

which exhibits $\mathbb{R}$ as an extension of $\mathbb{Q}/\mathbb{Z}$ by $\widehat{\mathbb{Z}}$.

Let R be a ring, and M be an R -module. Via the unique map $\mathbb{Z} \rightarrow R$, we define the R -module $M^\vee := \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{Q}/\mathbb{Z})$, which is naturally a functor $\mathrm{Mod}(R)^{\mathrm{op}} \rightarrow \mathrm{Mod}(R)$. This is exact because $\mathbb{Q}/\mathbb{Z}$ is injective over $\mathbb{Z}$. Note that $R^\vee$ is injective over R by a previous lemma.

The double dual $(M^\vee)^\vee = \mathrm{Hom}_{\mathbb{Z}}(M^\vee, \mathbb{Q}/\mathbb{Z})$ has for each $x \in M$ a distinguished element $\mathrm{ev}(x): M^\vee \rightarrow \mathbb{Q}/\mathbb{Z}$ given by $\phi \mapsto \phi(x)$. This defines a homomorphism $\mathrm{ev}: M \rightarrow (M^\vee)^\vee$, which is in fact a monomorphism.

Claim: $\mathrm{ev}: M \rightarrow (M^\vee)^\vee$ is monic.

Indeed, if $x \neq 0$, then let $G \subset M$ be the abelian subgroup generated by x . Then either $G \cong \mathbb{Z}/n\mathbb{Z}$ or $G \cong \mathbb{Z}$. In the former case, a nonzero map $G \rightarrow \mathbb{Q}/\mathbb{Z}$ is given by choosing an n -torsion element, such as $1/n$. In the latter case, any nonzero element gives such a nonzero map. By injectivity, this extends to a nonzero $\mathbb{Z}$ -map $M \rightarrow \mathbb{Q}/\mathbb{Z}$, and therefore $\mathrm{ev}(x)$ is nonzero.

Using this, we can immediately deduce the existence of enough injectives in $\mathrm{Mod}(R)$. If we cared, we could note easily that the construction is functorial.

Proof (of Thm. 1.4.10) Take a surjection $\bigoplus_I R \rightarrow M^\vee \rightarrow 0$, and apply the exact functor $(-)^\vee$ to obtain an injection

$$(M^\vee)^\vee \rightarrow \mathrm{Hom}_{\mathbb{Z}}\left(\bigoplus_I R, \mathbb{Q}/\mathbb{Z}\right) = \prod_I \mathrm{Hom}_{\mathbb{Z}}(R, \mathbb{Q}/\mathbb{Z}) = \prod_I R^\vee.$$

We just proved that M injects into the LHS, and on the other hand the RHS is a product of injective modules $R^\vee$, hence injective. Lastly, it is clear from the constructions that this is functorial. $\square$

Now all of our derived functor theory from the previous section applies. One difference is that we now view RHom as a functor on the derived category of the ring itself, rather than to abelian groups. (...finish this discussion...).

Some finiteness conditions

Let us briefly define some conditions and prove basic properties that we will return to later. A module M is *pseudo-coherent* if it has a resolution by finite projective modules. It is *finitely presented* if there is an exact sequence $F \rightarrow F' \rightarrow M \rightarrow 0$ where F, F' are finite free, i.e. there is a sequence $0 \rightarrow K \rightarrow F \rightarrow M \rightarrow 0$ where F is finite free and K is finitely generated.

We are going to use frequently and without mention that the forgetful functor $\text{Mod}(R) \rightarrow \mathbf{Set}$ creates filtered colimits. Recall that an object M is compact if $\text{Hom}_R(M, -)$ preserves filtered colimits as a functor to $\mathbf{Set}$, which by the above remark is equivalent to $\text{Hom}_R(M, -)$ preserving filtered colimits. It is clear that the class of compact objects is closed under finite colimits because filtered colimits commute with finite limits (in $\mathbf{Set}$ or $\text{Mod}(R)$).

Example 1.4.16 All finitely presented R -modules are compact. Indeed, observe that $\text{Hom}_R(R, -) \cong \text{Id}$ so R is compact, hence finite free modules are compact, and a finitely presented module is a cokernel of a map between finite free modules.

Lemma 1.4.17 *Every R -module is a filtered colimit of its finitely generated submodules. Each R -module is a filtered colimit of finitely presented modules.*

Proof The first claim is elementary. For the second, let M be an R -module. Consider the collection of pairs (S, E) where $S \subset M$ is a finite subset of M , and - letting $K_S := \ker(R^{\oplus S} \rightarrow M)$ - E is a finite subset of K_S .

Let $(S, E) \leq (S', E')$ if $S \subset S'$, and the image of $E \subset K_S$ under the natural map $R^{\oplus S} \rightarrow R^{\oplus S'}$ is contained in E' . For each (S, E) , let $C_{S,E}$ be the cokernel of the natural map $R^{\oplus E} \rightarrow R^{\oplus S}$, so $C_{S,E}$ is finitely presented. If $(S, E) \leq (S', E')$, then by construction we get a commutative diagram:

$$\begin{array}{ccccccc} R^{\oplus E} & \longrightarrow & R^{\oplus S} & \longrightarrow & C_{S,E} & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \\ R^{\oplus E'} & \longrightarrow & R^{\oplus S'} & \longrightarrow & C_{S',E'} & \longrightarrow & 0 \end{array}$$

The maps $R^{\oplus S} \rightarrow M$ send the relations K_S to zero, hence factor through maps $C_{S,E} \rightarrow M$. We claim $\varinjlim_{(S,E)} C_{S,E} \rightarrow M$ is an isomorphism. Taking the filtered colimit of the above diagram of exact sequences, we get the exact sequence

$$R^{\oplus K} \rightarrow R^{\oplus M} \rightarrow \varinjlim_{(S,E)} C_{S,E} \rightarrow 0$$

where K is the kernel of the map $R^{\oplus M} \rightarrow M$, so we're done. $\square$

Lemma 1.4.18 *Let M be a finite R -module. Then there is a finite filtration*

$$M = M_0 \subset \cdots \subset M_n = M$$

such that $M_{i+1}/M_i \cong R/I_i$ for some ideals $I_i \subset R$.

Proof We induct on the number of generators of M , the case of a cyclic module being clear. If $M = \langle x_1, \dots, x_m \rangle$, then take a filtration by induction for $M/\langle x_1 \rangle$, which is of the form

$$0 = N_0/\langle x_1 \rangle \subset N_1/\langle x_1 \rangle \subset \cdots \subset M/\langle x_1 \rangle$$

with cyclic quotients, then we win by taking

$$0 \subset \langle x_1 \rangle \subset N_1 \subset \cdots \subset N_n \subset M$$

with the same quotients. $\square$

We say that M is FP_n if there is an exact sequence

$$P^{-n} \rightarrow \cdots P^0 \rightarrow M \rightarrow 0$$

with each P^i finite projective. For example, if R is Noetherian (or more generally coherent), then any finitely generated module M is FP_∞ , because if we have a free resolution $R^{n_k} \rightarrow \cdots \rightarrow R^{n_0} \rightarrow M$, the left hand kernel remains finitely generated so we can extend it.

Proposition 1.4.19 *Let R be a ring.*

- (a) $\text{Ext}^j(M, -)$ preserves arbitrary products for all j .
- (b) $\text{Ext}^j(-, N)$ preserves arbitrary coproducts for all j .
- (c) If M is FP_{n+1} , then the functors

$$\text{Ext}^0(M, -), \text{Ext}^1(M, -), \dots, \text{Ext}^n(M, -)$$

preserve filtered colimits.

- (d) If M is FP_{n+1} and Q is flat, then we have

$$\text{Ext}^i(M, N \otimes Q) = \text{Ext}^i(M, N) \otimes Q$$

for $i = 0, \dots, n$. If Q is finite projective and M is arbitrary, then

$$\text{Ext}^i(M \otimes Q^\vee, N) = \text{Ext}^i(M, N \otimes Q) = \text{Ext}^i(M, N) \otimes Q$$

for every $i \geq 0$, where $Q^\vee := \text{Hom}_R(Q, R)$.

- (e) If $R \rightarrow S$ is a flat map and M is FP_{n+1} , then

$$\text{Ext}_R^i(M, N) \otimes_R S = \text{Ext}_S^i(M \otimes_R S, N \otimes_R S)$$

for $i = 0, \dots, n$ as S -modules.

Proof (a) and (b): For the former, let $P^\bullet \rightarrow M$ be a projective resolution, so that $\text{Ext}^*(M, N)$ is the cohomology of

$$0 \rightarrow \text{Hom}(P^0, N) \rightarrow \text{Hom}(P^{-1}, N) \rightarrow \cdots$$

If $N = \prod_\alpha N_\alpha$, then we can move this out of the Hom and use the fact that products are exact (so commute with cohomology). Likewise, if $N \rightarrow I^\bullet$ is an injective resolution, then $\text{Ext}^*(M, N)$ is cohomology of

$$0 \rightarrow \text{Hom}(M, I^0) \rightarrow \text{Hom}(M, I^1) \rightarrow \cdots,$$

and taking $M = \bigoplus_\alpha M_\alpha$ we conclude in the way as above.

(c) We may choose a projective resolution $P^\bullet \rightarrow M$ so that P^i is finite for $-i \leq n+1$. Let N_α be a filtered diagram of modules, then to compute Ext we take cohomology of:

$$0 \rightarrow \text{Hom}(P^0, N_\alpha) \rightarrow \text{Hom}(P^{-1}, N_\alpha) \rightarrow \dots$$

Since P^{-i} is finite projective for $i \leq n$, it is finitely presented hence compact, so $\text{Hom}(P^{-i}, -)$ preserve filtered colimits. Since filtered colimits are exact (hence commute with cohomology), for the terms $0, \dots, n$ we can take cohomology to obtain the result.

(d) For the first part, just as in (c) (because $(-) \otimes Q$ is exact) we reduce immediately to showing that

$$\text{Hom}(M, N \otimes Q) = \text{Hom}(M, N) \otimes Q$$

if Q is flat and M is finitely presented. There is an obvious natural map from the RHS to the LHS, and let $R^q \rightarrow R^r \rightarrow M \rightarrow 0$ be a presentation, and now for any P we have an exact sequence

$$0 \rightarrow \text{Hom}(M, P) \rightarrow P^r \rightarrow P^q.$$

On one hand, we could set $P = N$ and tensor by Q , so the right two terms because $(Q \otimes N)^r \rightarrow (Q \otimes N)^q$. On the other hand, we could set $P = N \otimes Q$ and the right two terms become the same thing.

For the second part, again because Q is still flat, we reduce to showing that

$$\text{Hom}(M \otimes Q^\vee, N) = \text{Hom}(M, N \otimes Q) = \text{Hom}(M, N) \otimes Q$$

with no hypotheses on M . The point is to reduce to the free case: For the first equality, note that

$$\text{Hom}(M \otimes Q^\vee, N) = \text{Hom}(M, \text{Hom}(Q^\vee, N)),$$

and if $Q \oplus Q' = R^n$ is free, then $R^n = (R^n)^\vee = Q^\vee \oplus (Q')^\vee$ implies $Q^\vee$ is still finite projective. Thus by replacing Q with its dual we may show that $Q^\vee \otimes N = \text{Hom}(Q, N)$. There's an evaluation map α_Q in the forward direction which is natural in Q ; choosing Q' such that $Q \oplus Q'$ is free we see $\alpha_{Q \oplus Q'} = \alpha_Q \oplus \alpha_{Q'}$ is an isomorphism, so α_Q is an isomorphism.

For the second equality, there's a canonical map

$$\beta_Q: \text{Hom}(M, N) \otimes Q \rightarrow \text{Hom}(M, N \otimes Q)$$

which is natural in Q . If $Q \cong R^n$ were finite free, then this map would be the isomorphism

$$\text{Hom}(M, N) \otimes R^n \cong \text{Hom}(M, N^n) \cong \text{Hom}(M, N^n).$$

So for Q finite projective there exists Q' such that $\beta_Q \oplus \beta_{Q'}$ is an isomorphism, hence β_Q is an isomorphism.

(e) Observe that if $R \rightarrow S$ is a map, then we have

$$\mathrm{Ext}_R^i(M, N|_R) \cong \mathrm{Ext}_S^i(M \otimes_R S, N)$$

by reducing to the case of Homs and applying the extension-restriction adjunction. Thus we conclude by applying this to (d). $\square$

As an example of how to use this, we'll prove the basic structure theorem on finite projective modules.

Theorem 1.4.20 *Let M be an R -module.*

- (a) *If M is (finite) projective, then $M[S^{-1}]$ is (finite) projective over $R[S^{-1}]$ for any S .*
- (b) *Every finite projective module over a local ring is finite free.*
- (c) *If M is finitely presented, then M is finite projective if and only if $M_{\mathfrak{p}}$ is finite free.*
- (d) *M is finite projective if and only if there exist $f_1, \dots, f_n$ generating the unit ideal of R such that M_{f_i} is finite free.*

Proof (a) If $M \oplus P = R^{\oplus I}$ is (finite) free, then $M[S^{-1}] \oplus P[S^{-1}] = (R[S^{-1}])^{\oplus I}$.

(b) Let $(R, \mathfrak{m})$ be local, M finite projective, and choose $\{x_1, \dots, x_n\}$ a minimal generating set, which defines an exact sequence $0 \rightarrow K \rightarrow R^n \rightarrow M \rightarrow 0$. Thus $R^n = K \oplus M$; if $(r_1, \dots, r_n) \in K$, then

$$r_1 x_1 + \dots + r_n x_n = 0$$

by definition, so none of the r_i are units by minimality. In particular $K \subset \mathfrak{m}^n$, which means $K \subset \mathfrak{m}K \oplus \mathfrak{m}M$, so $K/\mathfrak{m}K = 0$. Since K is finitely generated, it is zero by Nakayama's lemma.

(c) One direction is (a) and (b), and the other is the fact that as functors

$$\mathrm{Hom}_R(M, -)_{\mathfrak{p}} = \mathrm{Hom}_{R_{\mathfrak{p}}}(M_{\mathfrak{p}}, (-)_{\mathfrak{p}}),$$

so if $M_{\mathfrak{p}}$ is finite projective over $A_{\mathfrak{p}}$ for all $\mathfrak{p}$, then the functor above is exact for all $\mathfrak{p}$, hence $\mathrm{Hom}_R(M, -)$ is exact.

(d) In this case $\mathrm{Spec} A = \bigcup_i D(f_i)$, so for all $\mathfrak{p}$ let M_{f_i} be free over A_{f_i} for $\mathfrak{p} \in D(f_i)$, then $M_{\mathfrak{p}} = (M_{f_i})_{\mathfrak{p}}$ is free over $A_{\mathfrak{p}}$. Conversely, suppose M is finite projective, so $M_{\mathfrak{p}}$ is finite free for each $\mathfrak{p}$. Consider a basis $A_{\mathfrak{p}}^n \rightarrow M_{\mathfrak{p}}$, which by clearing denominators we may assume is the localization of a map $A^n \rightarrow M$. The cokernel C is a finite module so its support is closed, and thus there is some f such that $A_f^n \rightarrow M_f \rightarrow 0$ is exact. Its kernel is another finite projective module, and since its support is closed there is some g such that $A_{fg}^n \rightarrow M_{fg}$ is an isomorphism. $\square$

Basic results on flatness Now we'll discuss some basic properties of Tor , which will give some nice flatness criterion. As in the last statement, we really want to be making many of these statements in the derived category of modules.

Lemma 1.4.21 *Let R be a ring. For any M the functors*

$$\text{Tor}_i^R(M, -) \quad \text{Tor}_i^R(-, M) \quad i \geq 0$$

preserve filtered colimits.

Proof Let $(N_i)_{i \in I}$ be a filtered diagram. Let $P^\bullet \rightarrow M[0]$ be a flat resolution, then

$$\text{Tor}_n(M, \varinjlim_i N_i) = H^{-n}(P^\bullet \otimes_R (\varinjlim_i N_i)) = \varinjlim_i H^{-n}(P^\bullet \otimes_R N_i)$$

because cohomology commutes with filtered colimits. $\square$

As an example, we will prove the following result. I like this result because the statement is simple yet it demonstrates nice properties of classical derived functors.

Proposition 1.4.22 *Let M an R -module. The following are equivalent:*

- (a) *M is flat.*
- (b) *For all finitely generated ideals $I \subset R$, the map $I \otimes_R M \xrightarrow{\sim} IM$ is an isomorphism.*
- (c) *$\text{Tor}_1^R(R/I, M) = 0$ for all finitely generated ideals $I \subset R$.*

Proof For an ideal I , consider the sequence $0 \rightarrow I \rightarrow R \rightarrow R/I \rightarrow 0$. Tensor this with M to obtain

$$0 = \text{Tor}_1^R(R, M) \rightarrow \text{Tor}_1^R(R/I, M) \rightarrow I \otimes M \rightarrow M \rightarrow M/IM \rightarrow 0.$$

The image of $I \otimes M \rightarrow M$ is IM , so (b) is equivalent to saying $I \otimes M \rightarrow M$ is injective for finitely generated I . Thus (b) is equivalent to (c) by the sequence, and clearly (a) implies (c). Assuming (c), let I be any ideal, then $I = \varinjlim_J J$ is a filtered colimit of its finitely generated submodules, so by exactness $R/I = \varinjlim_J R/J$, and hence $\text{Tor}_1^R(R/I, M) = 0$ for all I by the lemma.

Now, for any finite module N , take a filtration

$$0 = N_0 \subset \cdots \subset N_r = N$$

with cyclic quotients by [Lemma 1.4.18](#). For each $i \geq 0$, there is an exact sequence

$$\text{Tor}_1(N_i, M) \rightarrow \text{Tor}_1(N_{i+1}, M) \rightarrow \text{Tor}_1(R/I, M)$$

for some ideal I . The third term is zero, and inductively assuming that $\text{Tor}_1(N_i, M) = 0$ the middle term is zero. Thus $\text{Tor}_1(N, M) = 0$ for every finite

module N . Since any module is the filtered colimit of its finite submodules, $\mathrm{Tor}_1(N, M) = 0$ for all N , and thus M is flat. $\square$

Corollary 1.4.23 *Let R be a Bézout domain (e.g. a PID). An R -module M is flat iff it is torsion-free.*

Proof A module M is torsion free iff the map $M \xrightarrow{r} M$ is injective for all nonzero $r \in R$, which is $(-) \otimes_R M$ applied to the exact sequence $0 \rightarrow R \xrightarrow{r} R \rightarrow R/(r) \rightarrow 0$. Thus

$$0 \rightarrow \mathrm{Tor}_1(R/(r)) \rightarrow M \xrightarrow{r} M$$

is exact, and all finitely generated ideals of R are principal, so we're done by the ideal-wise criterion for flatness. $\square$

Here's an easy observation. If F is flat, then

$$M \otimes^{\mathbf{L}} (N \otimes F) = M \otimes^{\mathbf{L}} N \otimes^{\mathbf{L}} F = (M \otimes F) \otimes^{\mathbf{L}} N = (M \otimes^{\mathbf{L}} N) \otimes F.$$

Since $(-) \otimes F$ is exact, it commutes with cohomology, so

$$\mathrm{Tor}^i(M \otimes F, N) = \mathrm{Tor}^i(M, N \otimes F) = \mathrm{Tor}^i(M, N) \otimes F$$

for all $i \geq 0$. If $R \rightarrow S$ is a flat map so $(-) \otimes_R S$ is exact, then

$$(M \otimes_R^{\mathbf{L}} N) \otimes_R S = (M \otimes_R S) \otimes_R^{\mathbf{L}} (N \otimes_R S),$$

and therefore

$$\mathrm{Tor}_i^R(M, N) \otimes_R S = \mathrm{Tor}_i^S(M \otimes_R S, N \otimes_R S)$$

as S -modules. In particular,

$$\mathrm{Tor}_i^R(M, N)_{\mathfrak{p}} = \mathrm{Tor}_i^{R_{\mathfrak{p}}}(M_{\mathfrak{p}}, N_{\mathfrak{p}}) \quad \mathrm{Tor}_i^R(M, N)_f = \mathrm{Tor}_i^{R_f}(M_f, N_f)$$

for all $f \in R$ and $\mathfrak{p} \subset R$ prime. We have the following immediate corollary:

Proposition 1.4.24 *Let M be an R -module. The following are equivalent:*

- (a) M is flat.
- (b) $M_{\mathfrak{p}}$ is flat over $R_{\mathfrak{p}}$ for all primes $\mathfrak{p}$ (or $\mathfrak{m}$ maximal).
- (c) M_f is flat over A_f for all $f \in A$.
- (d) M_{f_i} is flat over A_{f_i} for $(f_1, \dots, f_r) = (1)$ a Zariski cover.

Proof Immediate from the above, because these properties all correspond to criterion for a module to be zero. $\square$

A map $M \rightarrow N$ is *universally injective* if $M \otimes P \rightarrow N \otimes P$ is injective for all P . More commonly we say M is a *pure* submodule of N , and e.g. if $0 \rightarrow M \rightarrow N \rightarrow P \rightarrow 0$ is exact, we say it's *pure-exact* if $M \subset N$ is pure.

Lemma 1.4.25 *Let $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ be an exact sequence of R -modules. The following are equivalent:*

- (a) $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ is pure-exact.
 (b) For every finitely presented module F ,

$$0 \rightarrow \operatorname{Hom}_R(F, M') \rightarrow \operatorname{Hom}(F, M) \rightarrow \operatorname{Hom}(F, M'') \rightarrow 0$$

is exact.

- (c) There is a filtered family of split exact sequences

$$0 \rightarrow M'_i \rightarrow M_i \rightarrow M''_i \rightarrow 0$$

whose filtered colimit is $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$.

Proof Note that (c) easily implies (a), because tensor products commute with filtered colimits and tensoring with a split sequence remains split, and (b) because F is compact so $\operatorname{Hom}_R(F, -)$ preserves the filtered colimit (and split sequences).

(a) $\Rightarrow$ (b) Let $S^\bullet$ be the exact sequence viewed as a bounded acyclic complex, so that purity is equivalent to $Q \otimes S^\bullet$ being acyclic for all Q . If F is finitely presented, then let $Q_1 \rightarrow Q_2 \rightarrow F \rightarrow 0$ be a (finite free) presentation of F , so that $\operatorname{Hom}_R(Q_i, -) \cong Q_i^\vee \otimes_R (-)$, then choose a cokernel C as

$$0 \rightarrow F^\vee \rightarrow Q_2^\vee \rightarrow Q_1^\vee \rightarrow C \rightarrow 0.$$

Now apply $\operatorname{Hom}_R(-, S^\bullet)$ to the original presentation to obtain

$$0 \rightarrow \operatorname{Hom}_R(F, S^\bullet) \rightarrow Q_2^\vee \otimes_R S^\bullet \rightarrow Q_1^\vee \otimes_R S^\bullet \rightarrow C \otimes_R S^\bullet \rightarrow 0.$$

Exactness is checked termwise, and on the right four terms is by right-exactness of tensor product and the cokernel sequence above. By assumption, the right three complexes are acyclic, so the first one is too.

(b) $\Rightarrow$ (c) Write $M'' = \varinjlim_i M''_i$ as a filtered colimit of finitely presented modules. We can pullback the exact sequence along the natural maps $M''_i \rightarrow M''$ to obtain a commutative diagram of exact sequences as follows:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & M' & \longrightarrow & M_i & \dashrightarrow & M''_i & \longrightarrow & 0 \\ & & \parallel & & \downarrow & \lrcorner & \downarrow & & \\ 0 & \longrightarrow & M' & \longrightarrow & M & \longrightarrow & M'' & \longrightarrow & 0 \end{array}$$

By functoriality of pullback, the top row is a filtered diagram of exact sequences. Note that if we apply $\varinjlim_i$, the 5-lemma shows that $\varinjlim_i M_i \xrightarrow{\sim} M$ and we obtain the original diagram. To see that each of the top sequences

splits, apply $\text{Hom}(M_i'', -)$:

$$\begin{array}{ccccc} \text{Hom}(M_i'', M_i) & \longrightarrow & \text{Hom}(M_i'', M_i'') & \longrightarrow & \text{Ext}^1(M_i'', M_i) \\ \downarrow & & \downarrow & & \parallel \\ \text{Hom}(M_i'', M) & \twoheadrightarrow & \text{Hom}(M_i'', M'') & \xrightarrow{0} & \text{Ext}^1(M_i'', M_i) \end{array}$$

Our supposition is that the bottom left map is surjective, so that the bottom right map is zero. Thus the top right map is zero, which means that the Ext^1 class of the top sequences are 0. $\square$

This gives us a nice way to characterize various properties of purity, but most importantly gives a new flatness result.

Corollary 1.4.26 *Let F and P be R -modules.*

- (a) *If F is finitely presented, then every pure exact sequence $0 \rightarrow M' \rightarrow M \rightarrow F \rightarrow 0$ splits.*
- (b) *P is flat iff every exact sequence $0 \rightarrow M' \rightarrow M \rightarrow P \rightarrow 0$ is pure exact.*
- (c) *A finitely presented module is flat iff it is (finite) projective.*
- (d) *If P is flat, then a short exact sequence $0 \rightarrow M' \rightarrow P \rightarrow M'' \rightarrow 0$ is pure exact iff M'' is flat. So, pure submodules of flat modules are flat.*

Proof (a) If F is finitely presented and the sequence is pure, then the map $\text{Hom}(F, F) \rightarrow \text{Ext}^1(F, P')$ is zero, so the extension is trivial.

(b) We already know one direction. Pure exactness is trivially equivalent to $\text{Tor}_1(M, N) \rightarrow \text{Tor}_1(P, N) \rightarrow 0$ being surjective for all N . In particular, if we choose $M \rightarrow P$ to be a surjection from a free module, then $\text{Tor}_1(P, N) = 0$ for all N so N is flat.

(c) If P is finitely presented flat, take a presentation $0 \rightarrow K \rightarrow F \rightarrow P \rightarrow 0$ with F free. Then this is pure, hence split, so P is a summand of a free module.

(d) One direction is known, and the other is by the condition in (b). $\square$

Example 1.4.27 Let $R = C^\infty(\mathbb{R})$ be the ring of smooth functions on $\mathbb{R}$, and

$$\mathfrak{m} := \{f \in R : f(0) = 0\} \quad I := \{f \in R : \exists \epsilon > 0, f(x) = 0 \forall |x| < \epsilon\}.$$

Consider the finite module $M = R/I$. We claim M is R -flat but not projective. Note that the R -map $R \rightarrow R_{\mathfrak{m}}$ has kernel those f which are annihilated by an element of $R - \mathfrak{m}$, i.e. f such that $gf \equiv 0$ for $g(0) \neq 0$. If f is zero in a nbhd of 0, then we can clearly arrange this, and conversely if this holds, then f is zero on the nbhd $g^{-1}(\mathbb{R} - 0)$. Thus we get an injective map $M \hookrightarrow R_{\mathfrak{m}}$, and if f/g is an element of the codomain, with $g(0) \neq 0$, then we can find a function h which agrees with f/g on a nbhd of 0, hence gets sent to f/g under localization.

M is flat because it is a localization. If M were projective, then I would be a summand of R , so $R \rightarrow I \rightarrow R$ would give an idempotent e which generates I . If $e^2 = e$, then $e(e - 1) = 0$ as functions, so $e = 0$ or 1 . This is a contradiction, because $I \neq 0$ and $I \neq R$.

2 Algebraic geometry

2.1 Sheaf cohomology

In this section, we are going to discuss the basic theory of sheaf cohomology using the language of the previous section. The category we will focus on is $\text{Mod}(\mathcal{O}_X)$, the category of sheaves of $\mathcal{O}_X$ -modules, where $(X, \mathcal{O}_X)$ is a ringed space. In passing we will also care about $\text{PMod}(\mathcal{O}_X)$, which is the category of presheaves of $\mathcal{O}_X$ -modules. We will consider the derived categories $D^*(\mathcal{O}_X) := D^*(\text{Mod}(\mathcal{O}_X))$, and the derived functors of standard functors associated to maps of spaces, global sections, and tensor product. We will also construct Čech cohomology as a means for computing sheaf cohomology.

2.1.1 General setup

Let us recall the basic setup of the theories of ringed spaces. Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be ringed spaces. If $f: X \rightarrow Y$ is a morphism of ringed spaces, then we have the pushforward map

$$f_*: \text{Mod}(\mathcal{O}_X) \rightarrow \text{Mod}(\mathcal{O}_Y)$$

defined the same way on presheaves. We also have the inverse image map

$$f^{-1}: \text{Mod}(\mathcal{O}_Y) \rightarrow \text{Mod}(f^{-1}\mathcal{O}_Y),$$

which is an exact functor, along with the pullback map

$$f^*: \text{Mod}(\mathcal{O}_Y) \rightarrow \text{Mod}(\mathcal{O}_X)$$

obtained by base change $f^{-1}\mathcal{F} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$ along $f^\flat: f^{-1}\mathcal{O}_Y \rightarrow \mathcal{O}_X$. We have an adjunction $f^* \dashv f_*$, so f^* is right exact and f_* is left exact. If $U \subset X$ is an open subset, then we have the sections functor

$$\Gamma(U, -): \text{Mod}(\mathcal{O}_X) \rightarrow \text{Mod}(\mathcal{O}_X(U)),$$

which is left exact. Slightly more generally, we usually consider the case where $\mathcal{O}_X$ is a sheaf of algebras over some base ring R , i.e. $\mathcal{O}_X(X)$ is an R -algebra, in which case we consider the composition of the functor above with $\text{Mod}(\mathcal{O}_X(U)) \rightarrow \text{Mod}(R)$ by restricting scalars. The same functor as given on presheaves is by definition exact.

The categories are also naturally symmetric monoidal, with tensor products $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$ given by sheafifying the sectionwise tensor products. Likewise we have an internal hom given sectionwise by

$$\text{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})(U) := \text{Hom}_{\mathcal{O}_X|_U}(\mathcal{F}|_U, \mathcal{G}|_U).$$

This satisfies a tensor-hom adjunction, an internal tensor hom adjunction, and so on.

Consider the case of an inclusion $j: U \hookrightarrow X$ of an open subset, where U is considered as the ringed space $(U, \mathcal{O}_X|_U)$. Then we have some more explicit

formulas

$$(j_*\mathcal{F})(V) = \mathcal{F}(U \cap V) \quad j^{-1}\mathcal{F} = \mathcal{F}|_U \quad j^*\mathcal{F} = \mathcal{F}|_U,$$

where the second equality is to be interpreted alongside the fact that the natural map $j^{-1}\mathcal{O}_X \rightarrow \mathcal{O}_U$ is an isomorphism. Thus in this case, j^* is an exact functor, and this is exhibited by an explicit *extension by zero* functor which is left adjoint to $j^* = j^{-1}$,

$$j_! : \text{Mod}(\mathcal{O}_U) \rightarrow \text{Mod}(\mathcal{O}_X),$$

defined on $\mathcal{F}$ as the sheafification of the presheaf $j_!^p \mathcal{F}$ by:

$$(j_!^p \mathcal{F})(V) = \begin{cases} \mathcal{F}(V) & V \subset U \\ 0 & V \not\subset U \end{cases}$$

In particular $j_! \mathcal{F}$ has stalks either $\mathcal{F}_x$ if $x \in U$, or 0 otherwise. This is witnessed by the counit map $j_! \mathcal{F}|_U \rightarrow \mathcal{F}$ which is an isomorphism in U and zero elsewhere. Note in particular that $j_!$ is an exact functor by the explicit description of its stalks.

We also mention that if $\mathbb{Z}$ denotes the constant sheaf at $\mathbb{Z}$, viewed as a sheaf of rings on X , then $\text{Mod}(X, \mathcal{O}_{\mathbb{Z}})$ is just the category $\text{Ab}(X)$ of sheaves of abelian groups on X . In this special case, we use f^{-1} and f^* interchangeably, and all of our functors go between $\text{Ab}(X) \rightleftarrows \text{Ab}(Y)$.

We recall in this case that more can be said for closed embeddings $i: Z \rightarrow X$ where the unit map $i^{-1}i_*\mathcal{F} \rightarrow \mathcal{F}$ is a stalkwise isomorphism inside Z , and $i_*\mathcal{F}|_{X-Z} = 0$. Thus in this case, i_* is an exact functor. The inverse image and pushforward functors yield an equivalence of categories between abelian sheaves on Z and abelian sheaves on X supported in Z .

To define the derived functors of all of the objects above, we want to have enough injectives and projectives. Sheaf categories unfortunately don't have enough projectives, but since all of the non-left exact functors above were defined with tensor products, it will suffice to consider flat resolutions. On the other hand, they do have enough injectives.

Theorem 2.1.1 *The categories $\text{Mod}(\mathcal{O}_X)$ and $\text{PMod}(\mathcal{O}_X)$ have enough injectives. In both cases, an injective embedding can be chosen functorially. Moreover, the inclusion functor $\text{Mod}(\mathcal{O}_X) \rightarrow \text{PMod}(\mathcal{O}_X)$ preserves injectives.*

To prove this (and other things), we will make a simple observation: If $F: \mathcal{A} \rightleftarrows \mathcal{B} : G$ is an adjunction between abelian categories (so $F \dashv G$), and F is an exact functor, then G preserves injectives. Indeed, if I is injective in $\mathcal{A}$, then

$$\text{Hom}_{\mathcal{B}}(-, G(I)) \cong \text{Hom}_{\mathcal{A}}(F(-), I)$$

is a composition of two exact functors. As an example, since sheafification is exact, we deduce immediately that the inclusion $\text{Mod}(\mathcal{O}_X) \hookrightarrow \text{PMod}(\mathcal{O}_X)$

preserves injectives. A second example is that if $f: X \rightarrow Y$ is a flat morphism of ringed spaces, meaning that f^* is exact, then f_* preserves injectives. Another useful fact is that an arbitrary product (which exists) of injectives remains injective. Indeed,

$$\mathrm{Hom}(-, \prod_{\alpha} I_{\alpha}) = \prod_{\alpha} \mathrm{Hom}(-, I_{\alpha})$$

as abelian groups, and arbitrary products of abelian groups are exact.

Proof (of Thm. 2.1.1) The presheaf and sheaf case are similar in structure but the former is sectionwise and the latter is stalkwise. Let's start with presheaves: For each $U \subset X$, the sections over U functor $\Gamma_U: \mathrm{PMod}(\mathcal{O}_X) \rightarrow \mathrm{PMod}(\mathcal{O}_X(U))$ is exact on presheaves. Define a functor

$$Q_U: \mathrm{Mod}(\mathcal{O}_X(U)) \rightarrow \mathrm{PMod}(\mathcal{O}_X)$$

by sending

$$Q_U(M)(V) = \begin{cases} M & U \subset V \\ 0 & U \not\subset V \end{cases}$$

(note the difference in direction from extension by zero). It is easy to check that maps $\mathcal{F} \rightarrow Q_U(M)$ are in correspondence with maps $\mathcal{F}(U) \rightarrow M$, and hence that $\Gamma_U \dashv Q_U$. To conclude, fix a presheaf $\mathcal{F}$ and an inclusion $\mathcal{F}(U) \hookrightarrow I_U$ into an injective $\mathcal{O}_X(U)$ -module. Then $Q_U(I_U)$ is injective by the aforementioned adjunction property, and it is equipped with a canonical map $\mathcal{F} \rightarrow Q_U(\mathcal{F}(U)) \rightarrow Q_U(I_U)$. This induces a map

$$\mathcal{F} \rightarrow \prod_{U \subset X} Q_U(I_U)$$

where the RHS is injective, so it suffices to show that this is a monomorphism. That is clear because over each open set, one of the factors in the product is the inclusion $\mathcal{F}(U) \subset I_U$.

For the sheaf case, the stalk functor $(-)_x: \mathrm{Mod}(\mathcal{O}_X) \rightarrow \mathrm{Mod}(\mathcal{O}_{X,x})$ is exact. If $i_x: \{x\} \rightarrow X$ is the inclusion, and we view $\{x\}$ as a ringed space with structure sheaf $\mathcal{O}_{X,x}$, then the stalk functor is $i_x^{-1} \cong i_x^*$, so its right adjoint is $i_{x,*}$, which is the skyscraper sheaf construction. Thus $i_{x,*}$ preserves injectives; fix $\mathcal{F}$ a sheaf and for each x choose a monomorphism $\mathcal{F}_x \rightarrow I_x$ where I_x is an injective $\mathcal{O}_{X,x}$ -module.

We have a natural map $\mathcal{F} \rightarrow i_{x,*}\mathcal{F}_x \rightarrow i_{x,*}I_x$, which in particular induces a map into the product

$$\mathcal{F} \rightarrow \prod_{x \in X} i_{x,*}I_x,$$

where the RHS is injective. We can check that this is a mono sectionwise: over U , the map is

$$\mathcal{F}(U) \rightarrow \prod_{x \in U} \mathcal{F}_x \rightarrow \prod_{x \in U} I_x,$$

where the first map is monic for sheaves, and the second one is a product of monomorphisms of modules.

Note that in each case, as long as we chose the injective embedding of $\mathcal{F}(U)$ or $\mathcal{F}_x$ functorially, as we may do for modules over a ring (Thm. 1.4.10), it is clear that the resulting injective embedding of $\mathcal{F}$ is functorial. $\square$

While we're at it, let us also discuss the existence of enough flat objects, whose proof has a very similar structure to the one above. Recall that a sheaf $\mathcal{F}$ is flat if $\mathcal{F} \otimes_{\mathcal{O}_X} (-)$ is exact. Note that because tensor products commute with taking stalks, $\mathcal{F}$ is flat iff $\mathcal{F}_x$ is flat for all x . From this definition and the tensor hom adjunction, it is clear that a direct sum of flats is flat.

Lemma 2.1.2 *The category $\text{Mod}(\mathcal{O}_X)$ has enough flat objects. The same is true for $\text{PMod}(\mathcal{O}_X)$. In both cases, a surjection from a flat object can be chosen functorially.*

Proof Start with $\text{Mod}(\mathcal{O}_X)$: for each $j_U: U \hookrightarrow X$, the extension by zero $j_{U,!}\mathcal{O}_U$ is flat, because its stalks are either $\mathcal{O}_{X,x}$ or 0. By adjunction,

$$\text{Hom}_{\mathcal{O}_X}(j_!\mathcal{O}_U, \mathcal{F}) \cong \text{Hom}_{\mathcal{O}_U}(\mathcal{O}_U, \mathcal{F}|_U) = \mathcal{F}(U),$$

so each section $s \in U$ gives a map $f_{U,s}: j_{U,!}\mathcal{O}_U \rightarrow \mathcal{F}$. On stalks, this is either the zero map, or if $x \in U$ then it is the map $\mathcal{O}_{X,x} \rightarrow \mathcal{F}_x$ sending $1 \mapsto s_x$. Consider the induced map

$$f: \bigoplus_{(U,s)} j_{U,!}\mathcal{O}_U \rightarrow \mathcal{F}$$

from the coproduct, where the LHS is flat. To show surjectivity, check at a point $x \in X$: each $s_x \in \mathcal{F}_x$, coming from $s \in \mathcal{F}(U)$, is in the image of the corresponding map $f_{U,s}$ on the stalk at $x \in U$, so we are done.

In the case of presheaves of $\mathcal{O}_X$ -modules, repeat the same proof as above, but replace $j_{U,!}$ with $j_{U,!}^p$, and note that for presheaves, flatness is checked sectionwise. But the map we constructed above from the coproduct is clearly surjective sectionwise in the presheaf case. Lastly, it is clear that this construction is functorial in either case. $\square$

We will study the derived functors that the preceding two results give us in more detail in the next section. In this section, we will focus on derived global sections: if $(X, \mathcal{O}_X)$ is a ringed space, and Γ_U is the sections-over- U , functor we denote the corresponding right derived functor

$$R\Gamma(U, -): D^+(\mathcal{O}_X) \rightarrow D^+(\mathcal{O}_X(U)).$$

If $A^\bullet$ is a bounded above complex of $\mathcal{O}_X$ -modules, we usually write

$$\mathbb{H}^i(U, A^\bullet) := H^i(R\Gamma(U, A^\bullet))$$

and call this the *hypercohomology* of A over U . In the special case $A^\bullet = \mathcal{F}[0]$ is just an object of $\text{Mod}(\mathcal{O}_X)$, we write

$$H^i(U, \mathcal{F}) := \mathbb{H}^i(R\Gamma(U, \mathcal{F}[0]))$$

for the classical higher derived functors and call this the *sheaf cohomology* of $\mathcal{F}$ over U .

First of all, let us see that our notation is not ambiguous. Let $j: U \hookrightarrow X$ be an inclusion, so that j^* is the restriction functor $(-)|_U$ on $\mathcal{O}_X$ -modules. Since j^* is right adjoint to the exact functor $j_!$, it preserves injectives. Suppose $V \subset U \subset X$. Then we have a commutative diagram

$$\begin{array}{ccc} \mathrm{Mod}(\mathcal{O}_X) & \xrightarrow{j^*} & \mathrm{Mod}(\mathcal{O}_X|_U) \\ & \searrow \Gamma_V & \downarrow \Gamma_V \\ & & \mathrm{Mod}(\mathcal{O}_X(V)) \end{array}$$

of functors. Since j^* is exact, j^* descends to the derived category, and since j^* preserves injectives, we deduce that the diagram

$$\begin{array}{ccc} D^+(\mathcal{O}_X) & \xrightarrow{j^*} & D^+(\mathcal{O}_X|_U) \\ & \searrow R\Gamma_V & \downarrow R\Gamma_V \\ & & D^+(\mathcal{O}_X(V)) \end{array}$$

commutes. In particular, $R\Gamma(V, -)$ depends only on $(V, \mathcal{O}_V)$ and not the ambient space; likewise for $\mathbb{H}^*(V, -)$. Since $H^0(U, -)$ coincides with the global sections functor on sheaves, it is customary to at this stage write H^0 for sections.

Next, viewing Γ_U and Γ_V as functors on $\mathrm{Mod}(\mathcal{O}_X)$ with $V \subset U$, restriction provides a natural transformation $\Gamma_U \rightarrow \Gamma_V$, and hence a natural map $R\Gamma(U, -) \rightarrow R\Gamma(V, -)$ which is restriction on a bounded below complex of Γ -acyclics. Thus we obtain maps $\mathbb{H}^n(U, \mathcal{F}) \rightarrow \mathbb{H}^n(V, \mathcal{F})$ natural in $\mathcal{F}^\bullet$, which by the above description form a presheaf:

$$\underline{\mathbb{H}}^n(X, \mathcal{F}^\bullet) := (U \mapsto \mathbb{H}^n(U, \mathcal{F}^\bullet)).$$

These are called the *hypercohomology presheaves*. Here is the basic observation: let $i: \mathrm{Mod}(\mathcal{O}_X) \hookrightarrow \mathrm{PMod}(\mathcal{O}_X)$ be the natural inclusion, which is left exact because it is right adjoint to sheafification. We have already discussed the fact that i preserves injectives.

Thus, if $\mathcal{F}^\bullet \rightarrow \mathcal{I}^\bullet$ is an injective resolution, $Ri(\mathcal{F}^\bullet) = \mathcal{I}^\bullet$ as a complex of presheaves, and the n th cohomology of this is a presheaf satisfying

$$U \mapsto \ker(\mathcal{I}^n(U) \rightarrow \mathcal{I}^{n+1}(U)) / \mathrm{im}(\mathcal{I}^{n-1}(U) \rightarrow \mathcal{I}^n(U)),$$

but the RHS is $H^n(\Gamma(U, \mathcal{I}^\bullet)) = \mathbb{H}^n(U, \mathcal{F}^\bullet)$, which shows that we have a canonical identification

$$H^n(Ri(\mathcal{F}^\bullet)) = \underline{\mathbb{H}}^n(X, \mathcal{F}^\bullet),$$

and in particular $R^n i(\mathcal{F}) = \underline{H}^n(X, \mathcal{F})$. In the rest of this section, we will consider the theory of flasque sheaves. We will consider applications of flasque sheaves in the section on derived functors of sheaves.

Let $\mathcal{F}$ be an object of $\text{Mod}(\mathcal{O}_X)$. $\mathcal{F}$ is called *flasque* if for each $V \subset U$, the restriction map $\mathcal{F}(U) \rightarrow \mathcal{F}(V)$ is surjective. Equivalently, all restriction maps $\mathcal{F}(X) \rightarrow \mathcal{F}(V)$ are surjective, because we can lift to $\mathcal{F}(X)$ first and then restrict down to $\mathcal{F}(U)$.

Flasqueness is a very non-geometric condition on a sheaf, since we usually expect to obtain more ‘functions’ when we restrict to a smaller open. We care about flasque sheaves mainly because they are Γ -acyclic, so we can compute e.g. hypercohomology on flasque resolutions. The following is the main structure theorem on flasque sheaves:

Lemma 2.1.3 *Let X be a ringed space.*

1. *A finite direct sum of flasque sheaves is flasque.*
2. *If $\mathcal{F}$ is flasque, then $\mathcal{F}|_U$ is flasque for every $U \subset X$.*
3. *If $f: X \rightarrow Y$, is a morphism of ringed spaces and $\mathcal{F}$ is flasque, then $f_*\mathcal{F}$ is flasque.*

Let $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ be an exact sequence of $\mathcal{O}_X$ -modules.

4. *If $\mathcal{F}'$ is flasque, then*

$$0 \rightarrow \mathcal{F}'(U) \rightarrow \mathcal{F}(U) \rightarrow \mathcal{F}''(U) \rightarrow 0$$

is exact for every $U \subset X$.

5. *If $\mathcal{F}'$ and $\mathcal{F}$ are flasque, then $\mathcal{F}''$ is flasque.*
6. *In the situation of (5), if $f: X \rightarrow Y$ is a morphism of ringed spaces, then $0 \rightarrow f_*\mathcal{F}' \rightarrow f_*\mathcal{F} \rightarrow f_*\mathcal{F}'' \rightarrow 0$ is an exact sequence of $\mathcal{O}_Y$ -modules.*

Proof (1) A direct sum of surjections is a surjection. (2) Flasqueness of $\mathcal{F}|_U$ is included in the definition of flasqueness of $\mathcal{F}$. (3) The restriction maps $H^0(U, f_*\mathcal{F}) \rightarrow H^0(V, f_*\mathcal{F})$ are just the restriction maps

$$H^0(f^{-1}(U), \mathcal{F}) \rightarrow H^0(f^{-1}(V), \mathcal{F})$$

of $\mathcal{F}$, which are surjective. (5) assuming (4): We have a commutative diagram involving the restriction maps

$$\begin{array}{ccc} \mathcal{F}(U) & \longrightarrow & \mathcal{F}''(U) \\ \downarrow & & \downarrow \\ \mathcal{F}(V) & \longrightarrow & \mathcal{F}''(V) \end{array}$$

where the horizontal maps are surjective by (4), and the left-hand vertical map is surjective. Thus the right-hand one is too. (6) This is a sequence of flasque sheaves by (3), hence it is exact over every open set by (5), thus exact as $\mathcal{O}_Y$ -modules.

(4) We may take $U = X$; let $s \in \mathcal{F}''(X)$ be a section and consider the collection $\{(U, t)\}$ of opens $U \subset X$ and sections $t \in \mathcal{F}(U)$ such that $t \mapsto s|_U$. Order this

collection by setting $(U, t) \leq (U', t')$ if $U \subset U'$ and $t'|_U = t$. The fact that $\mathcal{F} \rightarrow \mathcal{F}''$ is surjective means that the opens in this collection cover X . Also, given a totally ordered subset $\{(U_\alpha, t_\alpha)\}$ of this collection, the sections t_α glue to a section t over $\bigcup_\alpha U_\alpha$, by definition of the ordering.

Thus there is a maximal element (U, t) by Zorn's lemma, so assume for contradiction $U \neq X$. Then we can find an element (U', t') such that $U' \subsetneq U$, so $t - t' \mapsto s - s = 0$ over $U \cap U'$, which means there exists $r \in \mathcal{F}'(U \cap U')$ such that $r \mapsto t - t'$. By flasqueness, extend r to a section over U' , and consider its image $i(r) \in \mathcal{F}(U)$. Then t and $t' + i(r)$ have the same restriction to $U \cap U'$, so they glue to a section of $\mathcal{F}$ over $U \cup U'$. The image of this section in $\mathcal{F}''(U \cap U')$ is s , because the composite $\mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}''$ is zero, which contradicts the maximality. $\square$

Corollary 2.1.4 *Let X be a ringed space. An injective $\mathcal{O}_X$ -module is flasque. Moreover, the flasque $\mathcal{O}_X$ -modules form an adapted class for Γ , and f_* for any morphism $f: X \rightarrow Y$. In particular, any flasque $\mathcal{O}_X$ -module is Γ - and f_* -acyclic.*

Proof Let $U \subset X$ and $j: U \hookrightarrow X$ be the inclusion, then $j_!\mathcal{O}_U \rightarrow \mathcal{O}_X$ is a monomorphism, so

$$\mathrm{Hom}_{\mathcal{O}_X}(j_!\mathcal{O}_U, \mathcal{I}) = \mathrm{Hom}_{\mathcal{O}_U}(\mathcal{O}_U, \mathcal{I}) = \mathcal{I}(U) \rightarrow \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{I}) = \mathcal{I}(X)$$

is surjective for any injective sheaf $\mathcal{I}$, i.e. $\mathcal{I}$ is flasque.

To show the second claim, note that the flasque sheaves are a (right) preadapted class by [Lemma 2.1.3\(1\)](#) and what we just proved. Thus, let $\mathcal{F}^\bullet$ be a bounded below acyclic complex of flasque $\mathcal{O}_X$ -modules; we have to show Γ and f_* preserve acyclicity.

Assume $\mathcal{F}^i = 0$ for $i < n$, and split into exact sequences

$$0 \rightarrow \mathcal{G}^i \rightarrow \mathcal{F}^{i+1} \rightarrow \mathcal{G}^{i+1} \rightarrow 0$$

where $\mathcal{G}^n = \mathcal{F}^n$ and the lower $\mathcal{G}^i$ are zero. Since $\mathcal{G}^n$ is flasque, by induction and [Lemma 2.1.3\(5\)](#), all $\mathcal{G}^i$ are flasque. Thus Γ and f_* preserve each sequence above, by [Lemma 2.1.3\(4\)](#) and (6), and hence they both preserve acyclicity of the original complex. $\square$

Here's an example of how to use this, which we will see more of in two sections. Suppose $i: Z \rightarrow X$ is a morphism of ringed spaces which is a 'closed topological embedding', meaning the underlying map of spaces is a closed embedding. Then we know that i_* is an exact functor, because $(i_*\mathcal{F})_x$ is either 0, or $\mathcal{F}_z$ if $x = i(z)$. Thus this descends to a functor on the derived categories $i_*: D^*(\mathcal{O}_Z) \rightarrow D^*(\mathcal{O}_X)$.

Moreover, this comes with a natural ring homomorphism $\mathcal{O}_X(X) \rightarrow \mathcal{O}_Z(Z)$, along which restriction of scalars is exact. Thus this gives a map $D^*(\mathcal{O}_Z(Z)) \rightarrow$

$D^*(\mathcal{O}_X(X))$ on the derived categories, which is simply termwise restriction of scalars. We can use this to formulate a basic ‘change of scalars’ result:

Lemma 2.1.5 *Let Z and X be ringed spaces and $i: Z \rightarrow X$ be a closed topological embedding. Then the diagram*

$$\begin{array}{ccc} D^+(\mathcal{O}_Z) & \xrightarrow{i_*} & D^+(\mathcal{O}_X) \\ R\Gamma \downarrow & & \downarrow R\Gamma \\ D^+(\mathcal{O}_Z(Z)) & \longrightarrow & D^+(\mathcal{O}_X(X)) \end{array}$$

commutes (up to natural isomorphism). In particular, there is a canonical identification

$$\mathbb{H}^n(Z, \mathcal{F}^\bullet) = \mathbb{H}^n(X, i_*\mathcal{F}^\bullet)$$

of hypercohomology for all $n \geq 0$.

Proof We claim both directions of the square are the derived functors of the composite $\text{Mod}(\mathcal{O}_Z) \rightarrow \text{Mod}(\mathcal{O}_Z(Z)) \rightarrow \text{Mod}(\mathcal{O}_X(X))$. One way is clear because the second functor is exact, so all modules are acyclic for it. For the other way, note that the composite is equal to

$$\text{Mod}(\mathcal{O}_Z) \xrightarrow{i_*} \text{Mod}(\mathcal{O}_X) \rightarrow \mathcal{O}_X(X)$$

But flasque sheaves form an adapted class for both i_* and $R\Gamma$, and i_* preserves flasque sheaves, so we are done by the results on composition of derived functors. $\square$

Remark 2.1.6 Note that if $f: X \rightarrow Y$ is any map of ringed spaces, then the exact same proof can be repeated with f instead of i , except pushing forward may not be exact, so we have to replace the top functor with

$$Rf_*: D^+(\mathcal{O}_X) \rightarrow D^+(\mathcal{O}_Y)$$

and we’ll get a similar commutative diagram. In this case,

$$\mathbb{H}^n(X, \mathcal{F}^\bullet) = \mathbb{H}^n(Y, Rf_*\mathcal{F}^\bullet)$$

for all $n \geq 0$. We will study the properties of derived pushforward in the next two sections.

Example 2.1.7 Suppose we have a morphism $\mathcal{O}_1 \rightarrow \mathcal{O}_2$ of sheaves of rings on X . Then this gives a morphism of ringed spaces $(X, \mathcal{O}_1) \rightarrow (X, \mathcal{O}_2)$, which on underlying spaces is the identity map. Thus the lemma applies, and shows that the hypercohomology of a complex of $\mathcal{O}_2$ -modules viewed as an $\mathcal{O}_1(X)$ -module, is the same as its hypercohomology by first viewing it as a complex of $\mathcal{O}_1$ -modules. If we take $\mathbb{Z} \rightarrow \mathcal{O}_X$, then this shows that the underlying abelian group of cohomology as $\mathcal{O}_X$ -modules agrees with abelian sheaf cohomology.

2.1.2 Čech cohomology

Recall that in a category with enough injectives, we defined

$$\mathrm{RHom}_{\mathcal{A}} : D(\mathcal{A})^{\mathrm{op}} \times D^+(\mathcal{A}) \rightarrow D(\mathbb{Z})$$

via the equivalence $K^+(\mathcal{I}_{\mathcal{A}}) \xrightarrow{\sim} D^+(\mathcal{A})$, and showed that it can be computed via an injective resolution in the second factor, or a projective resolution in the first factor. If we consider $\mathcal{A} = \mathrm{Mod}(\mathcal{O}_X)$ or $\mathrm{PMod}(\mathcal{O}_X)$, the hom complex functor $\underline{\mathrm{Hom}}_{\mathcal{O}_X}(-, \mathcal{I})$ factors through $\mathrm{Mod}(\mathcal{O}_X(X))$ because each Hom group is an $\mathcal{O}_X(X)$ -module, so we can consider RHom as a functor

$$\mathrm{RHom}_{\mathcal{O}_X} : D(\mathcal{O}_X)^{\mathrm{op}} \times D^+(\mathcal{O}_X) \rightarrow D(\mathcal{O}_X(X)),$$

and likewise for the derived categories of $\mathrm{PMod}(\mathcal{O}_X)$. This is because any relevant exactness/acyclic-ness is checked on the underlying abelian group. Via the exact restriction of scalars functor $D(\mathcal{O}_X(X)) \rightarrow D(\mathbb{Z})$, the diagram

$$\begin{array}{ccc} D(\mathcal{O}_X)^{\mathrm{op}} \times D^+(\mathcal{O}_X) & \longrightarrow & D(\mathcal{O}_X(X)) \\ & \searrow & \downarrow \\ & & D(\mathbb{Z}) \end{array}$$

commutes (and again, likewise for $\mathrm{PMod}(\mathcal{O}_X)$). This is because, if we fix $\mathcal{I}$ a bounded below complex of injectives, the functor is just the one induced by $\underline{\mathrm{Hom}}(-, \mathcal{I})$ on the localizations. Note that in particular, if $A, B \in D^+(\mathcal{O}_X)$, then

$$H^0(\mathrm{RHom}_{\mathcal{O}_X}(A, B)) = \mathrm{Hom}_{D^+(\mathcal{O}_X)}(A, B)$$

naturally has the structure of an $\mathcal{O}_X(X)$ -module.

Almost all of the properties of RHom carry over even when we consider it as an $\mathcal{O}_X(X)$ -module. For example, for a fixed sheaf $\mathcal{F}$, $\mathrm{RHom}(\mathcal{F}, -)$ viewed as a functor $D^+(\mathcal{O}_X) \rightarrow D^+(\mathcal{O}_X(X))$ is still the right derived functor of the left exact functor

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}, -) : \mathrm{Mod}(\mathcal{O}_X) \rightarrow \mathrm{Mod}(\mathcal{O}_X(X)).$$

We prove this the same way as before: the right derived functor on the derived category $\mathrm{RHom}_{\mathcal{O}_X}(\mathcal{F}, -)$ exists, and on equivalent category $K^+(\mathcal{I}_{\mathrm{Mod}(\mathcal{O}_X)})$ it is isomorphic to $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}, -)$, but the same is true for $\underline{\mathrm{Hom}}(\mathcal{F}[0], -) = \mathrm{Hom}(\mathcal{F}, -)$. Likewise, the same holds for $\mathrm{PMod}(\mathcal{O}_X)$. Combining the above remarks shows that hypercohomology of a complex $\mathcal{F}^\bullet$ is just maps $\mathcal{O}_X \rightarrow \mathcal{F}^\bullet$ in the derived category. Also, by this discussion we view $\mathrm{Ext}^i(\mathcal{F}, \mathcal{G})$ as an $\mathcal{O}_X(X)$ -module.

Now, let $\mathcal{U} = \{U_i\}$ be an open cover of the ringed space X . For indices $i_0, \dots, i_p$, we use the notation

$$U_{i_0 \dots i_p} := U_{i_0} \cap \dots \cap U_{i_p}.$$

If $\mathcal{F}$ is an object of $\text{PMod}(\mathcal{O}_X)$, we define

$$\check{C}^p(\mathcal{U}, \mathcal{F}) := \prod_{i_0, \dots, i_p} \mathcal{F}(U_{i_0 \dots i_p}) \quad p \geq 0.$$

We make this into a complex in $\text{Ch}^+(\mathcal{O}_X(X))$ by defining the differential as

$$d(s)_{(i_0, \dots, i_{p+1})} = \sum_{k=0}^{p+1} (-1)^k s_{i_0 \dots \widehat{i_k} \dots i_{p+1}},$$

where on the RHS we implicitly consider the restriction of the section s to the open $U_{i_0 \dots i_{p+1}}$. Here is $d^2 = 0$:

$$\begin{aligned} d^2(s)_{i_0 \dots i_{p+2}} &= \sum_{l=0}^{p+2} (-1)^l d(s)_{i_0 \dots \widehat{i_l} \dots i_{p+2}} \\ &= \sum_{l=0}^{p+2} \left(\sum_{k < l} (-1)^{k+l} s_{i_0 \dots \widehat{i_k} \dots \widehat{i_l} \dots i_{p+2}} + \sum_{k \geq l} (-1)^{k+l} s_{i_0 \dots \widehat{i_l} \dots \widehat{i_{k+1}} \dots i_{p+2}} \right) \\ &= \sum_{k < l} (-1)^{k+l} s_{i_0 \dots \widehat{i_k} \dots \widehat{i_l} \dots i_{p+2}} + \sum_{k > l} (-1)^{k-1+l} s_{i_0 \dots \widehat{i_l} \dots \widehat{i_k} \dots i_{p+2}} \\ &= \sum_{k < l} (-1)^{k+l} s_{i_0 \dots \widehat{i_k} \dots \widehat{i_l} \dots i_{p+2}} - \sum_{k < l} (-1)^{k+l} s_{i_0 \dots \widehat{i_k} \dots \widehat{i_l} \dots i_{p+2}} \\ &= 0. \end{aligned}$$

For the third equality, we re-indexed k to $k+1$, and for the fourth we flipped k and l . Thus, $\check{C}^\bullet(\mathcal{U}, \mathcal{F})$ is a complex, called the *Čech complex of $\mathcal{F}$ with respect to $\mathcal{U}$* . We write

$$\check{H}^p(\mathcal{U}, \mathcal{F}) := H^p(\check{C}^\bullet(\mathcal{U}, \mathcal{F}))$$

for its cohomology, called the *Čech cohomology of $\mathcal{F}$ with respect to $\mathcal{U}$* . Čech cohomology behaves well with respect to products. Indeed, since taking sections of presheaves (or sheaves) commutes with arbitrary products, and products commute with products, we have

$$\check{C}^\bullet(\mathcal{U}, \prod_{\alpha} \mathcal{F}_{\alpha}) = \prod_{\alpha} \check{C}^\bullet(\mathcal{U}, \mathcal{F}_{\alpha}).$$

Since arbitrary products are exact in the category of modules, we conclude $\check{H}^n(\mathcal{U}, \prod_{\alpha} \mathcal{F}_{\alpha}) = \prod_{\alpha} \check{H}^n(\mathcal{U}, \mathcal{F}_{\alpha})$ for all n .

Our next goal is to relate this cohomology to the sheaf cohomology defined above, which will be through a spectral sequence. Čech cohomology is not itself a derived functor on $\text{Mod}(\mathcal{O}_X)$, but it is one on $\text{PMod}(\mathcal{O}_X)$. The result is the following: Let $j_i: U_i \rightarrow X$ be the inclusion, and consider the presheaf extension-by-zero functors $j_{i,!}^p$. We write $\mathcal{O}_{\mathcal{U}}^{\text{pre}}$ for the image presheaf of the canonical map

$$\bigoplus_i j_{i,!}^p \mathcal{O}_{U_i} \rightarrow \mathcal{O}_X$$

induced by the counits $j_{i,!}^p \mathcal{O}_{U_i} \rightarrow \mathcal{O}_X$, i.e. the sectionwise image. Then:

Proposition 2.1.8 *The diagram of functors*

$$\begin{array}{ccc}
 \mathrm{PMod}(\mathcal{O}_X) & \xrightarrow{\check{C}(\mathcal{U}, -)} & \mathrm{Ch}^+(\mathrm{Mod}(\mathcal{O}_X(X))) \\
 \downarrow & & \downarrow \\
 D^+(\mathrm{PMod}(\mathcal{O}_X)) & \xrightarrow{\mathrm{RHom}_{\mathcal{O}_X}^{\mathrm{pre}}(\mathcal{O}_{\mathcal{U}}, -)} & D^+(\mathcal{O}_X(X))
 \end{array}$$

commutes up to canonical natural isomorphism. Moreover, there is a canonical identification $\mathrm{Hom}_{\mathcal{O}_X}^{\mathrm{pre}}(\mathcal{O}_{\mathcal{U}}, -) = \check{H}^0(\mathcal{U}, -)$. Therefore

$$R\check{H}^0(\mathcal{U}, -) = \mathrm{Ext}_{\mathcal{O}_X}^{\mathrm{pre}, n}(\mathcal{O}_{\mathcal{U}}, -) = \check{H}^n(\mathcal{U}, -)$$

for all $n \geq 0$.

Let us demonstrate this in two simple steps. First, since we are working in the presheaf category, $\mathcal{O}_X$ is projective, because it is projective sectionwise. So, $\mathcal{O}_U$ is projective in $\mathrm{PMod}(\mathcal{O}_U)$, and since $j_!$ is a right adjoint to restriction, which is exact, $j_!^p \mathcal{O}_U$ is projective.

Now, let $j_{i_0 \dots i_p}$ be the inclusion of $U_{i_0 \dots i_p}$. Write

$$K^{-n}(\mathcal{U}) := \bigoplus_{i_0, \dots, i_n} j_{i_0 \dots i_n}^p j_{i_0 \dots i_n, !} \mathcal{O}_{U_{i_0 \dots i_n}},$$

so each $K^{-n}(\mathcal{U})$ is projective. Moreover,

$$\begin{aligned}
 \mathrm{Hom}_{\mathcal{O}_X}^{\mathrm{pre}}(K^{-n}(\mathcal{U}), \mathcal{F}) &= \prod_{i_0, \dots, i_n} \mathrm{Hom}_{\mathcal{O}_X}^{\mathrm{pre}}(j_{i_0 \dots i_n}^p j_{i_0 \dots i_n, !} \mathcal{O}_{U_{i_0 \dots i_n}}, \mathcal{F}) \\
 &= \prod_{i_0, \dots, i_n} \mathrm{Hom}^{\mathrm{pre}}(\mathcal{O}_{U_{i_0 \dots i_n}}, \mathcal{F}|_{U_{i_0 \dots i_n}}) \\
 &= \prod_{i_0, \dots, i_n} \mathcal{F}(U_{i_0 \dots i_n}) = \check{C}^p(\mathcal{U}, \mathcal{F})
 \end{aligned}$$

for any presheaf $\mathcal{F}$, by adjunction. Since the Čech differential $\check{C}^n(\mathcal{U}, -) \rightarrow \check{C}^{n+1}(\mathcal{U}, -)$ is natural, it corresponds by Yoneda to a map $K^{-n-1}(\mathcal{U}) \rightarrow K^{-n}(\mathcal{U})$, which also satisfies $d^2 = 0$ by Yoneda. We modify this differential by multiplying the $-p$ th one by $(-1)^p$, so that

$$\underline{\mathrm{Hom}}(K^\bullet(\mathcal{U}), \mathcal{F}) = \check{C}^\bullet(\mathcal{U}, \mathcal{F})$$

holds as complexes (according to our convention for Hom complexes). Since $K^\bullet$ is a bounded above complex of projectives, we may replace the left-hand side with $\mathrm{RHom}(K(\mathcal{U}), \mathcal{F})$ after passing to the derived category. It remains to show that $K(\mathcal{U})$ is isomorphic in the derived category to $\mathcal{O}_{\mathcal{U}}^{\mathrm{pre}}$.

This is the second step: form the augmented complex

$$\dots \rightarrow K^{-2}(\mathcal{U}) \rightarrow K^{-1}(\mathcal{U}) \rightarrow K^0(\mathcal{U}) = \bigoplus_i j_{i, !}^p \mathcal{O}_{U_i} \rightarrow \mathcal{O}_{\mathcal{U}} \rightarrow 0$$

We claim this is an exact sequence of presheaves; if this holds, then by applying $\mathrm{Hom}(-, \mathcal{F})$ and using one-sided exactness, we deduce that $\mathrm{Hom}(\mathcal{O}_{\mathcal{U}}, \mathcal{F}) \cong \check{H}^0(\mathcal{U}, \mathcal{F})$ as both sides are kernels of the same map. To prove the claim, we may show that evaluating the sequence at each open set W produces a contractible complex of $\mathcal{O}_X(W)$ -modules.

Proof (of Prop. 2.1.8) This proof is essentially copied from Lemma 01EM in the Stacks Project. As above, fix an open set W , and let I be the collection of indices $\{i\}$ such that $W \subset U_i$, so that by definition of the functors j_i^p , the augmented complex above becomes:

$$\cdots \rightarrow \bigoplus_{i_0, \dots, i_{p+1} \in I} \mathcal{O}_X(W) \rightarrow \bigoplus_{i_0, \dots, i_p \in I} \mathcal{O}_X(W) \rightarrow \cdots$$

The differential here is

$$d(s)_{i_0 \dots i_p} = \sum_{j=0}^{p+1} \sum_{i \in I} (-1)^j s_{i_0 \dots i_{j-1} i i_j \dots i_p}.$$

Now, choose an element $\ell \in I$ which we will use to define the contracting homotopy. Set:

$$h(s)_{i_0 \dots i_{p+1}} = \begin{cases} s_{i_1 \dots i_{p+1}} & i_0 = \ell \\ 0 & i_0 \neq \ell \end{cases}$$

Then it is an elementary check that

$$dh + hd(s)_{i_0 \dots i_p} = s_{i_0 \dots i_p}$$

by splitting the sum into components where $i_0 = \ell$ and $i_0 \neq \ell$, where the two sums cancel as in the proof that $d^2 = 0$. $\square$

Here are some immediate deductions from this result. In fact, these can be deduced in a more direct way by explicitly constructing homotopies, which we will do e.g. in the case of the alternating Čech complex.

Example 2.1.9 Let $\mathcal{U}$ be an open cover of X , and suppose that $X \in \mathcal{U}$ is itself one of the elements of the cover. Then for each presheaf $\mathcal{F}$, the Čech complex $\check{C}^\bullet(\mathcal{U}, \mathcal{F})$ just has a zeroth cohomology group $\check{H}^0(\mathcal{U}, \mathcal{F}) = \mathcal{F}(X)$ with higher cohomology all zero. Indeed, in this case the presheaf $\mathcal{O}_{\mathcal{U}}^{\mathrm{pre}}$ is (by definition) all of $\mathcal{O}_X$, and $\mathrm{Hom}^{\mathrm{pre}}(\mathcal{O}_X, -)$ is the global sections functor which is exact on presheaves. This will be the key observation in the proof that quasicoherent sheaves on affine schemes have vanishing higher cohomology, which is because of the fact that standard affine covers of local rings have an element which is the whole space already.

In fact, in this case $\check{C}^\bullet(\mathcal{U}, \mathcal{F}) \simeq \mathcal{F}(X)[0]$ already in the homotopy category: indeed, we just showed that

$$K(\mathcal{U}) \simeq \mathcal{O}_{\mathcal{U}}[0] = \mathcal{O}_X[0]$$

in the homotopy category, so applying $\underline{\mathrm{Hom}}_{\mathcal{O}_X}^{\mathrm{pre}}(-, \mathcal{F})$ gives homotopy equivalences

$$\check{C}(\mathcal{U}, \mathcal{F}) = \underline{\mathrm{Hom}}_{\mathcal{O}_X}^{\mathrm{pre}}(K(\mathcal{U}), \mathcal{F}) \simeq \underline{\mathrm{Hom}}_{\mathcal{O}_X}^{\mathrm{pre}}(\mathcal{O}_X, \mathcal{F}) = \mathcal{F}(X)[0]$$

as desired.

Example 2.1.10 One also might want to know the relation between Čech complexes when we add an open set. Let $\mathcal{U}$ be an open cover of X , and $\mathcal{U} \cup V = \mathcal{U} \cup \{V\}$ be the cover obtained from adding V . Also let $\mathcal{U} \cap V = \{U \cap V : U \in \mathcal{U}\}$ be the open cover of V obtained by restricting $\mathcal{U}$. From the definition it is clear that we have a canonical inclusion $\mathcal{O}_{\mathcal{U}}^{\mathrm{pre}} \subset \mathcal{O}_{\mathcal{U} \cup V}^{\mathrm{pre}}$ of presheaves.

This data gives a natural map $\mathcal{O}_{\mathcal{U}}^{\mathrm{pre}} \oplus j_{V,!}^p \mathcal{O}_V \rightarrow \mathcal{O}_{\mathcal{U} \cup V}^{\mathrm{pre}}$, which is surjective by definition of the relevant objects, and we wish to compute the kernel. Fix an open set $W \subset X$ and consider the map on sections. The codomain is $\mathcal{O}_X(W)$ if W is contained in V or an element of $\mathcal{U}$, and zero otherwise. The domain is $\mathcal{O}_X(W)$ if W is contained in an element of $\mathcal{U}$ but not V , zero if W is contained in neither, $\mathcal{O}_X(W) \oplus \mathcal{O}_X(W)$ if W is contained in an element of $\mathcal{U}$ and V , and $\mathcal{O}_X(W)$ if it's contained in V and not an element of $\mathcal{U}$.

Based on this explicit description, it is easy to see that we have an exact sequence

$$0 \rightarrow j_{V,!}^p \mathcal{O}_{\mathcal{U} \cap V}^{\mathrm{pre}} \rightarrow \mathcal{O}_{\mathcal{U}}^{\mathrm{pre}} \oplus j_{V,!}^p \mathcal{O}_V \rightarrow \mathcal{O}_{\mathcal{U} \cup V}^{\mathrm{pre}} \rightarrow 0$$

of presheaves on X , and hence an exact triangle in the derived category. Applying $\mathrm{RHom}_{\mathcal{O}_X}^{\mathrm{pre}}(-, \mathcal{F})$ gives an exact triangle

$$\check{C}(\mathcal{U} \cup V, \mathcal{F}) \rightarrow \check{C}(\mathcal{U}, \mathcal{F}) \oplus \mathcal{F}(V)[0] \rightarrow \check{C}(\mathcal{U} \cap V, \mathcal{F}|_V) \rightarrow \check{C}(\mathcal{U} \cup V, \mathcal{F})[1]$$

in the derived category of $\mathcal{O}_X(X)$. For the second term, we used the fact that $j_{V,!}^p \mathcal{O}_V$ was a projective object, hence RHom could be computed using Hom . This is called the Čech Mayer-Vietoris sequence. Thus we get a long exact sequence which looks like this:

$$\begin{aligned} 0 \rightarrow \check{H}^0(\mathcal{U} \cup V, \mathcal{F}) \rightarrow \check{H}^0(\mathcal{U}, \mathcal{F}) \oplus \mathcal{F}(V) \rightarrow \check{H}^0(\mathcal{U} \cap V, \mathcal{F}) \rightarrow \\ \check{H}^1(\mathcal{U} \cup V, \mathcal{F}) \rightarrow \check{H}^1(\mathcal{U}, \mathcal{F}) \rightarrow \check{H}^1(\mathcal{U} \cap V, \mathcal{F}) \rightarrow \check{H}^2(\mathcal{U} \cup V, \mathcal{F}) \rightarrow \dots \end{aligned}$$

As a sample application, let's show by induction on n that a cover $\mathcal{U}$ consisting of n opens has vanishing Čech cohomology in degree $\geq n$. In the case $n = 1$ we can apply the previous result that the Čech complex only has $\check{H}^0$. Assuming the result for covers with n elements (for all spaces and presheaves), let $\mathcal{U} = \{U_1, \dots, U_n\}$, $V = U_{n+1}$, then for all $i \geq 0$ there are exact sequences

$$\check{H}^i(\mathcal{U} \cap U_{n+1}, \mathcal{F}|_{U_{n+1}}) \rightarrow \check{H}^{i+1}(\mathcal{U} \cup U_{n+1}, \mathcal{F}) \rightarrow \check{H}^{i+1}(\mathcal{U}, \mathcal{F}).$$

Since $\mathcal{U}$ and $\mathcal{U} \cap U_{n+1}$ are covers consisting of n elements, for all $i \geq n$ the left- and right-hand terms in the sequence above are zero. Thus the middle term is zero, proving the claim. We will prove shortly a much stronger result by exhibiting a homotopy equivalent complex which has only n terms in it.

We now use what we have shown to relate Čech cohomology to sheaf cohomology, which was the point of introducing it. We'll give two different methods of doing this, the latter of which will illustrate a slightly more general technique.

Theorem 2.1.11 *Let X be a ringed space and $\mathcal{U}$ an open cover of X .*

- (a) *There is a natural map $\check{C}(\mathcal{U}, -) \rightarrow R\Gamma(X, -)$ in $D^+(\mathcal{O}_X(X))$.*
 (b) *For each $\mathcal{O}_X$ -module $\mathcal{F}$, there is a bounded spectral sequence*

$$E_2^{p,q} = \check{H}^p(\mathcal{U}, \underline{H}^q(\mathcal{F})) \implies H^{p+q}(X, \mathcal{F})$$

converging to a finite filtration on the RHS, which is natural in $\mathcal{F}$.

In particular, if the restrictions of $\mathcal{F}$ to all finite intersections of elements of the open cover $\mathcal{U}$ are Γ -acyclic, then the map $\check{C}(\mathcal{U}, \mathcal{F}) \rightarrow R\Gamma(X, \mathcal{F})$ is an isomorphism in $D^+(\mathcal{O}_X(X))$.

Proof Consider the composition of functors

$$\text{Mod}(\mathcal{O}_X) \xrightarrow{i_X} \text{PMod}(\mathcal{O}_X) \xrightarrow{\check{H}^0(\mathcal{U}, -)} \text{Mod}(\mathcal{O}_X(X)).$$

First observe that $\check{H}^0(\mathcal{U}, \mathcal{F})$ is the kernel in

$$0 \rightarrow \check{H}^0(\mathcal{U}, \mathcal{F}) \rightarrow \prod_i \mathcal{F}(U_i) \rightarrow \prod_{i,j} \mathcal{F}(U_i \cap U_j)$$

where the second map is the same one as in the sheaf condition, by inspection of the Čech differential. Thus this gives a natural map $H^0(X, \mathcal{F}) \rightarrow \check{H}^0(\mathcal{U}, \mathcal{F})$ induced by restriction, which is an isomorphism iff $\mathcal{F}$ satisfies the sheaf condition for the cover $\mathcal{U}$. So, $\check{H}^0(\mathcal{U}, -) \circ i_X = \Gamma(X, -)$.

We know $\text{Mod}(\mathcal{O}_X)$ and $\text{PMod}(\mathcal{O}_X)$ have enough injectives and that i_X preserves them, so the theorem on a composition of derived functors says that $R\Gamma = R\check{H}^0(\mathcal{U}, -) \circ Ri_X$. The natural map $i_X \rightarrow Ri_X$ provides a natural transformation

$$\check{C}(\mathcal{U}, -) = R\check{H}^0(\mathcal{U}, -) \circ i_X \rightarrow R\check{H}^0(\mathcal{U}, -) \circ Ri_X = R\Gamma.$$

The spectral sequence is exactly the Grothendieck spectral sequence for a composition of right derived functors, and our explicit computation of $R^p i_X$ as cohomology presheaves. For the last remark, we have

$$E_1^{p,q} = \check{C}^p(\mathcal{U}, \underline{H}^q(\mathcal{F})),$$

so by assumption for $q > 0$ and $p \geq 0$, this is zero. Thus the spectral sequence degenerates at E_2 , with only the first row

$$E_2^{p,0} = \check{H}^p(\mathcal{U}, \underline{H}^0(\mathcal{F})) = \check{H}^p(\mathcal{U}, \mathcal{F})$$

begin nonzero. Thus $\check{H}^p(\mathcal{U}, \mathcal{F}) \rightarrow H^p(X, \mathcal{F})$ is an isomorphism, so the map of (a) is a quasi-isomorphism. We are talking about the same map because of the description of edge maps in the Grothendieck spectral sequence. $\square$

The spectral sequence in the theorem above is called the Čech-to-derived spectral sequence. We will give a different proof and slight generalization shortly. First, some other results on Čech cohomology:

Example 2.1.12 If $\mathcal{F}$ is a flasque $\mathcal{O}_X$ -module, then $\check{H}^n(\mathcal{U}, \mathcal{F}) = 0$ for $n > 0$ and any open cover $\mathcal{U}$ (and $\check{H}^0(\mathcal{U}, \mathcal{F}) = \mathcal{F}(X)$, since $\mathcal{F}$ is a sheaf). Indeed, any restriction of a flasque sheaf is flasque, so by the theorem its Čech cohomology agrees with its sheaf cohomology.

Lemma 2.1.13 *Let $\mathcal{F}'$ be an $\mathcal{O}_X$ -module such that $\check{H}^1(\mathcal{U}, \mathcal{F}') = 0$ for every cover $\mathcal{U}$ in a cofinal system of open covers of X . Then for any exact sequence $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ of $\mathcal{O}_X$ -modules,*

$$0 \rightarrow \mathcal{F}'(X) \rightarrow \mathcal{F}(X) \rightarrow \mathcal{F}''(X) \rightarrow 0$$

is an exact sequence of $\mathcal{O}_X(X)$ -modules.

In this context, a *cofinal system of open covers* is a collection of open covers such that every open cover has a refinement in the collection.

Proof We just have to check surjectivity of $\mathcal{F}(X) \rightarrow \mathcal{F}''(X)$, so let $s \in \mathcal{F}''(X)$ be a section. Then there's an open cover $\{U_i\}$ and sections $t_i \in \mathcal{F}(U_i)$ such that $t_i \mapsto s|_{U_i}$, and by restricting we may assume that this cover is in the given cofinal system. For each i, j , $t_j - t_i \mapsto s - s = 0$, so there is a unique $r_{ij} \in \mathcal{F}'(U_{ij})$ such that $r_{ij} \mapsto t_j - t_i$. Thus $r = (r_{ij})_{ij} \in \check{C}^1(\{U_i\}, \mathcal{F}')$, and we claim it is a cocycle. Indeed,

$$d(r)_{ijk} = r_{jk} - r_{ik} + r_{ij} = 0$$

holds because of injectivity of $\mathcal{F}' \rightarrow \mathcal{F}$ and

$$d(r)_{ijk} \mapsto t_k - t_j - (t_k - t_i) + t_j - t_i = 0$$

over $\mathcal{F}(U_{ijk})$. Hence by assumption there exists $q_i \in \mathcal{F}'(U_i)$ such that $q_j - q_i = r_{ij}$ for all i, j . Letting f be the map $\mathcal{F}' \rightarrow \mathcal{F}$, consider the sections $t_i - f(q_i)$. Then

$$t_j - f(q_j) - (t_i - f(q_i)) = t_j - t_i - f(r_{ij}) = 0$$

over U_{ij} , so these sections glue to an element of $\mathcal{F}(X)$. This maps to s because it does so on an open cover. $\square$

Example 2.1.14 Suppose $\mathcal{F}$ is an $\mathcal{O}_X$ -module such that for all $U \subset X$ and open covers $\mathcal{U}$ of U , $\check{H}^p(\mathcal{U}, \mathcal{F}|_U) = 0$ for $p > 0$. Then $H^p(X, \mathcal{F}) = 0$ for $p > 0$. We prove this by induction on p . Choose an exact sequence $0 \rightarrow \mathcal{F} \rightarrow \mathcal{I} \rightarrow \mathcal{G} \rightarrow 0$ with $\mathcal{I}$ injective, so that $H^i(X, \mathcal{G}) \xrightarrow{\sim} H^{i+1}(X, \mathcal{F})$ for $i > 0$. Moreover, since $H^1(X, \mathcal{I}) = 0$, [Lemma 2.1.13](#) shows that $H^1(X, \mathcal{F}) = 0$, which is the base case. For the general case, [Lemma 2.1.13](#) shows that this sequence is in fact an exact sequence of presheaves, and since Čech cohomology is a derived functor on presheaves, $\check{H}^i(\mathcal{U}, \mathcal{G}|_U) = \check{H}^{i+1}(\mathcal{U}, \mathcal{F}|_U) = 0$ for $i > 0$ since $\mathcal{I}$ is flasque. Thus by the inductive hypothesis applied to $\mathcal{G}$, we are done.

A very slight generalization of the argument above will give us affine cohomology vanishing for quasi-coherent sheaves, and gives a more practical criterion for checking cohomology vanishing.

Lemma 2.1.15 *Let X be a ringed space and $\mathcal{F}$ an $\mathcal{O}_X$ -module. Assume there exists a basis $\mathcal{B}$ such that for each $U \in \mathcal{B}$, there is a cofinal system of open coverings Cov_U of U such that:*

- (a) *For $\mathcal{U} \in \text{Cov}_U$, $U_{i_0 \dots i_p} \in \mathcal{B}$.*
- (b) *For $\mathcal{U} \in \text{Cov}_U$, $\check{H}^p(\mathcal{U}, \mathcal{F}|_U) = 0$ for $p > 0$.*

Then $H^p(U, \mathcal{F}) = 0$ for $p > 0$ and all $U \in \mathcal{B}$.

Proof We proceed by induction on $p > 0$ over all $\mathcal{F}$ simultaneously. Choose an exact sequence $0 \rightarrow \mathcal{F} \rightarrow \mathcal{I} \rightarrow \mathcal{G} \rightarrow 0$ where $\mathcal{I}$ is injective. First of all, this gives an exact sequence

$$\mathcal{I}(U) \rightarrow \mathcal{G}(U) \rightarrow H^1(U, \mathcal{F}) \rightarrow H^1(U, \mathcal{I}) = 0,$$

and since the first Čech cohomology vanishes on a cofinal system of open covers of U , we conclude $H^1(U, \mathcal{F}) = 0$ by [Lemma 2.1.13](#). This is the base case. For $p > 1$, note moreover that $H^{p-1}(U, \mathcal{G}) \xrightarrow{\sim} H^p(U, \mathcal{F})$, so it suffices to show that $H^{p-1}(U, \mathcal{G}) = 0$, for which it suffices to show that $\mathcal{G}$ satisfies the same hypotheses as $\mathcal{F}$ in the assumptions.

To check this, for each U and $\mathcal{U} \in \text{Cov}_U$, the sequence of sheaves induces a sequence

$$0 \rightarrow \check{C}^\bullet(\mathcal{U}, \mathcal{F}|_U) \rightarrow \check{C}^\bullet(\mathcal{U}, \mathcal{I}|_U) \rightarrow \check{C}^\bullet(\mathcal{U}, \mathcal{G}|_U) \rightarrow 0 \quad (*)$$

of complexes which we claim is exact. Since products of modules are exact, this reduces to showing that $0 \rightarrow \mathcal{F}(V) \rightarrow \mathcal{I}(V) \rightarrow \mathcal{G}(V) \rightarrow 0$ is exact, where V is of the form $U_{i_0 \dots i_p}$. But then $V \in \mathcal{B}$, so this follows again from [Lemma 2.1.13](#) as before.

The exact sequence $*$ of complexes gives an exact sequence

$$\check{H}^n(\mathcal{U}, \mathcal{I}|_U) \rightarrow \check{H}^n(\mathcal{U}, \mathcal{G}|_U) \rightarrow \check{H}^{n+1}(\mathcal{U}, \mathcal{F}|_U)$$

in Čech cohomology for every $n \geq 1$. The left term is 0 because $\mathcal{I}|_U$ is flasque, and the right term is zero by assumption. Therefore the middle term is zero, so $\mathcal{G}$ satisfies the same hypotheses as $\mathcal{F}$. $\square$

Now we come to the second method for relating Čech cohomology to derived-functor cohomology. The idea is to sheafify (not literally) the Čech complex into a complex of sheaves.

Continue the assumptions that X is a ringed space and $\mathcal{U}$ is an open cover of X . Let $\mathcal{F}$ be a presheaf, and define the complex of presheaves

$$\check{C}^p(\mathcal{U}, \mathcal{F}) := \prod_{i_0, \dots, i_p} j_{i_0 \dots i_p, *} \mathcal{F}|_{U_{i_0 \dots i_p}} = \prod_{\ell(I)=p} j_{I, *} \mathcal{F}|_{U_I},$$

where we are using the obvious convention for the second notation, noting that I runs over tuples of the form $I = (i_0, \dots, i_p)$ and its length ℓ is p , not $p + 1$. We define the differential component-wise as before: to give a map $\check{\mathcal{C}}^p(\mathcal{U}, \mathcal{F}) \rightarrow \check{\mathcal{C}}^{p+1}(\mathcal{U}, \mathcal{F})$ we have to give maps $\check{\mathcal{C}}^p(\mathcal{U}, \mathcal{F})|_{U_I} \rightarrow \mathcal{F}|_{U_I}$ for each I of length p , and restricting commutes with the product so we define

$$d(s)_{i_0 \dots i_{p+1}} = \sum_{k=0}^{p+1} (-1)^k s_{i_0 \dots \widehat{i_k} \dots i_{p+1}}$$

sectionwise. Moreover, this is natural in $\mathcal{F}$, so if $\mathcal{F}^\bullet$ is a complex of presheaves, we can define a bicomplex $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}^\bullet)$ by the same formula levelwise, except that we should twist the vertical differentials by $(-1)^q$ to agree with our convention for bicomplexes. Finally, note that there is no ambiguity about sheaves/presheaves because arbitrary products in the two categories agree.

Lemma 2.1.16 *With notation as above, let $\mathcal{U}$ be an open cover, $\mathcal{F}$ be an $\mathcal{O}_X$ -module, and $\mathcal{F}^\bullet$ a bounded above complex of $\mathcal{O}_X$ -modules.*

- (a) *As defined above, $\check{\mathcal{C}}(\mathcal{U}, \mathcal{F})$ is indeed a complex, and $\check{\mathcal{C}}(\mathcal{U}, \mathcal{F}^\bullet)$ is indeed a bicomplex.*
- (b) *The identities*

$$\Gamma(V, \check{\mathcal{C}}(\mathcal{U}, \mathcal{F})) = \check{\mathcal{C}}(\mathcal{U} \cap V, \mathcal{F}|_V) \quad \check{\mathcal{C}}(\mathcal{U}, \mathcal{F}^\bullet)|_V = \check{\mathcal{C}}(\mathcal{U} \cap V, \mathcal{F}^\bullet|_V)$$

both hold for every open subspace $V \subset X$.

- (c) *The natural morphism of complexes $\mathcal{F}[0] \rightarrow \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$ is a quasi-isomorphism. More generally, the natural morphism of complexes*

$$\mathcal{F}^\bullet \rightarrow \text{Tot } \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}^\bullet)$$

is a quasi-isomorphism.

In light of (c), we usually call $\check{\mathcal{C}}(\mathcal{U}, \mathcal{F})$ or $\check{\mathcal{C}}(\mathcal{U}, \mathcal{F}^\bullet)$ a Čech resolution of $\mathcal{F}$ or $\mathcal{F}^\bullet$ respectively.

Proof (a-b) Note that the first claim in (b) holds by definition of j_* . Thus the second claim of (b) holds for a single sheaf, and so its clear for a complex of sheaves. For (a), the second claim follows from the first, and the first follows from (b) applied to every open and the fact that the regular Čech complex is a complex.

- (c) The natural map $\mathcal{F} \rightarrow \prod_i j_{i,*} \mathcal{F}|_{U_i}$ is the one induced by the restriction maps. We need to show that

$$0 \rightarrow \mathcal{F} \rightarrow \check{\mathcal{C}}^0(\mathcal{U}, \mathcal{F}) \rightarrow \check{\mathcal{C}}^1(\mathcal{U}, \mathcal{F}) \rightarrow \dots$$

is an exact sequence of $\mathcal{O}_X$ -modules. It suffices to show this after restricting to an open cover, namely $\mathcal{U}$, so by the second part of (b) we may assume that $X \in \mathcal{U}$. Then, by the first part of (b), evaluation on every open V produces an

exact sequence, because the augmented Čech complex is exact when the cover contains the whole space. This proves the first part of (c).

For the second part, the natural map above extends termwise to a map $\mathcal{F}^\bullet \rightarrow \check{C}^0(\mathcal{U}, \mathcal{F}^\bullet) \hookrightarrow \text{Tot } \check{C}^\bullet(\mathcal{U}, \mathcal{F}^\bullet)$. On the other hand, we have the spectral sequence

$$E_1^{p,q} = H_v^q H_h^p(\check{C}^\bullet(\mathcal{U}, \mathcal{F}^q)) \implies H^{p+q}(\text{Tot } \check{C}^\bullet(\mathcal{U}, \mathcal{F}^\bullet)).$$

The LHS is 0 for $p > 0$, and $H^q(\mathcal{F}^\bullet)$ by the first part, so this coincides with the cohomology of the totalization, and the induced map

$$H^q(\mathcal{F}^\bullet) \rightarrow H^q(\text{Tot } \check{C}^\bullet(\mathcal{U}, \mathcal{F}^\bullet)) \cong H^q(\mathcal{F}^\bullet)$$

is an isomorphism. $\square$

This gives us a very easy way to formulate the Čech to sheaf cohomology map. Namely, we have the natural transformation $\Gamma \rightarrow R\Gamma$, and thus a commutative diagram:

$$\begin{array}{ccc} \Gamma(X, \mathcal{F}) & \longrightarrow & C^\bullet(\mathcal{U}, \mathcal{F}) \\ \downarrow & \nwarrow \text{dashed} & \downarrow \\ R\Gamma(X, \mathcal{F}) & \xrightarrow{\cong} & R\Gamma(X, \check{C}^\bullet(\mathcal{U}, \mathcal{F})) \end{array}$$

To check that this coincides with the comparison map we defined earlier, we have to show that

$$\begin{array}{ccc} \check{C}(\mathcal{U}, \mathcal{F}) & \xlongequal{\quad} & \Gamma(\check{C}(\mathcal{U}, \mathcal{F})) \\ \cong \downarrow & & \downarrow \\ R\check{H}^0(\mathcal{F}) & \longrightarrow R\check{H}^0 \circ Ri_X(\mathcal{F}) \longrightarrow & R\Gamma(\check{C}(\mathcal{U}, \mathcal{F})) \end{array}$$

commutes. We first claim that $\check{C}^\bullet(\mathcal{U}, \mathcal{F})$ is a complex of $\check{H}^0(\mathcal{U}, -)$ -acyclic objects for any presheaf $\mathcal{F}$. Indeed, since Čech cohomology commutes with products, we may show that for $j: V \rightarrow X$ an inclusion,

$$\check{H}^p(\mathcal{U}, j_{V,*}\mathcal{F}|_V) = 0 \quad p > 0$$

where V is contained in an element of $\mathcal{U}$; this is clear because $\check{C}(\mathcal{U}, j_{V,*}\mathcal{F}|_V) = \check{C}(\mathcal{U} \cap V, \mathcal{F}|_V)$, and the latter complex only has zeroth cohomology since V is an element of $\mathcal{U} \cap V$. To apply this fact in showing that the two comparison maps, we consider the following diagram:

$$\begin{array}{ccccc} \check{H}^0(\mathcal{F}) & \longrightarrow & R\check{H}^0 i_X(\mathcal{F}) & \longrightarrow & R\Gamma(X, \mathcal{F}) \\ \downarrow & & \downarrow & & \downarrow \cong \\ \check{H}^0(\check{C}(\mathcal{U}, \mathcal{F})) & \xrightarrow{\cong} & R\check{H}^0(\check{C}(\mathcal{U}, \mathcal{F})) & \longrightarrow & R\Gamma(X, \check{C}(\mathcal{U}, \mathcal{F})) \end{array}$$

Since $\mathcal{F}$ is a sheaf, the outer composites agree with the natural maps $\Gamma \rightarrow R\Gamma$ because of the uniqueness statement in the universal property of derived functors. The bottom left map is an isomorphism because we just showed that $\check{C}(\mathcal{U}, \mathcal{F})$ was a bounded below complex of $R\check{H}^0(\mathcal{U}, -)$. So, what we have to

show is that the identification of $R\check{H}^0 i_X(\mathcal{F})$ with $\check{C}(\mathcal{U}, \mathcal{F})$ under the map to $R\check{H}^0(\check{C}(\mathcal{U}, \mathcal{F}))$ agrees under the displayed isomorphism with $\check{H}^0(\check{C}(\mathcal{U}, \mathcal{F})) = \check{C}(\mathcal{U}, \mathcal{F})$. This is evident from the bifactoriality of RHom and the projective resolution of $\mathcal{O}_{\mathcal{U}}^{\mathrm{pre}}$.

The Čech resolution gives a more natural way to describe the Čech-to-derived spectral sequence, and we will state a slightly more general version of it.

Proposition 2.1.17 *Let X be a ringed space, $\mathcal{F}^\bullet \in \mathrm{Ch}^+(\mathcal{O}_X)$, and $\mathcal{U}$ an open cover of X .*

(a) *There is a natural bounded spectral sequence*

$$E_2^{p,q} = \check{H}^p(\mathcal{U}, \mathbb{H}(\mathcal{F}^\bullet)) \implies \mathbb{H}^{p+q}(X, \mathcal{F}^\bullet)$$

converging to a bounded filtration on the hypercohomology.

(b) *Suppose $f: X \rightarrow Y$ is a morphism, and that $\mathcal{U}$ is finite. Denote the composite $U_{i_0 \dots i_p} \hookrightarrow X \rightarrow Y$ by $f_{i_0 \dots i_p}$. There is a bounded spectral sequence*

$$E_1^{p,q} = \bigoplus_{i_0, \dots, i_p} H^q(Rf_{i_0 \dots i_p, *} \mathcal{F}^\bullet|_{U_{i_0 \dots i_p}}) \implies H^{p+q} Rf_* (\mathcal{F}^\bullet)$$

converging to a bounded filtration on the derived pushforwards. Here the E_1 differential is the Čech differential on the higher pushforwards.

Proof (a) This is just the spectral sequence associated to a bicomplex, where we take vertical cohomology first. Namely, choose an injective resolution $\mathcal{F}^\bullet \rightarrow \mathcal{I}^\bullet$, and consider $\check{C}^\bullet(\mathcal{U}, \mathcal{I}^\bullet)$. Since restriction to an open set is exact, the right adjoint j_* preserves injectives. Since arbitrary products of injectives are injective, it follows that $\check{C}^\bullet(\mathcal{U}, \mathcal{I}^\bullet)$ is a bicomplex of injective objects, and hence so is the totalization.

By [Lemma 2.1.16](#), this means that we have isomorphisms

$$R\Gamma(X, \mathcal{F}^\bullet) \xrightarrow{\sim} \Gamma(X, \mathcal{I}^\bullet) \xrightarrow{\sim} \Gamma(X, \mathrm{Tot} \check{C}(\mathcal{U}, \mathcal{I}^\bullet)) = \mathrm{Tot} \check{C}(\mathcal{U}, \mathcal{I}^\bullet)$$

and likewise on open subsets. Computing vertical cohomology for $\check{C}^\bullet(\mathcal{U}, \mathcal{I}^\bullet)$ first, we obtain

$$\begin{aligned} H_v^q(\check{C}^p(\mathcal{U}, \mathcal{I}^\bullet)) &= \prod_{i_0, \dots, i_p} H^q(\mathcal{I}^\bullet(U_{i_0 \dots i_p})) \\ &= \prod_{i_0, \dots, i_p} \mathbb{H}^q(U_{i_0 \dots i_p}, \mathcal{F}^\bullet), \end{aligned}$$

and so the corresponding spectral sequence will give

$$\check{H}^p(\mathcal{U}, \mathbb{H}^q(\mathcal{F}^\bullet)) \implies H^{p+q} \mathrm{Tot} \check{C}^\bullet(\mathcal{U}, \mathcal{I}^\bullet),$$

but we showed above that the RHS is $R^{p+q}\Gamma(X, \mathcal{F}^\bullet)$ as desired.

(b) We argue the same way as before, except that while product of R -modules are exact, products of $\mathcal{O}_X$ -modules aren't. Once again consider $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{I}^\bullet)$, which is a bicomplex of injectives, and therefore

$$Rf_*\mathcal{F} \xrightarrow{\sim} f_*\mathcal{I}^\bullet \xrightarrow{\sim} f_* \operatorname{Tot}(\check{\mathcal{C}}(\mathcal{U}, \mathcal{I}^\bullet)) = \operatorname{Tot} f_*\check{\mathcal{C}}(\mathcal{U}, \mathcal{I}^\bullet).$$

This will take care of the RHS of the spectral sequence. For the LHS, note that

$$f_*\check{\mathcal{C}}^p(\mathcal{U}, \mathcal{I}1) = \bigoplus_{i_0, \dots, i_p} f_{i_0, \dots, i_p, *} \mathcal{I}^q|_{U_{i_0 \dots i_p}},$$

so taking vertical cohomology and using the fact that sums commute with cohomology, we obtain the desired spectral sequence. $\square$

To conclude this section, we discuss an often-used variant of the Čech complex, which is more convenient when actually computing with small open covers. This is the ordered Čech complex, denoted $\check{\mathcal{C}}_{\text{ord}}^\bullet(\mathcal{U}, \mathcal{F})$, where $\mathcal{U} = \{U_i : i \in I\}$ is an open cover of X and I is an ordered set. Then we define

$$\check{\mathcal{C}}_{\text{ord}}^p(\mathcal{U}, \mathcal{F}) := \prod_{i_0 < \dots < i_p} \mathcal{F}(U_{i_0 \dots i_p}),$$

and so $\check{\mathcal{C}}_{\text{ord}}^p(\mathcal{U}, \mathcal{F}) = 0$ if $|\mathcal{U}| \leq p$. We use the same differential as in the standard Čech complex, which still works because if $(i_0, \dots, i_p)$ is in strictly increasing order, then $(i_0, \dots, \widehat{i_k}, \dots, i_p)$ is also in strictly increasing order. Thus we have a new chain complex, and this admits a natural map $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}) \rightarrow \check{\mathcal{C}}_{\text{ord}}^\bullet(\mathcal{U}, \mathcal{F})$ which forgets the components that aren't in strictly increasing order. The main point is:

Lemma 2.1.18 *Let X be a ringed space, $\mathcal{U}$ an ordered open cover of X , and $\mathcal{F}$ an object of $\operatorname{PMod}(\mathcal{O}_X)$. The natural map*

$$\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}) \rightarrow \check{\mathcal{C}}_{\text{ord}}^\bullet(\mathcal{U}, \mathcal{F})$$

is a homotopy equivalence.

Proof Recall from [Prop. 2.1.8](#) that we defined $\mathcal{O}_{\mathcal{U}}^{\text{pre}}$, constructed a projective resolution $K^\bullet(\mathcal{U}) \rightarrow \mathcal{O}_{\mathcal{U}}^{\text{pre}}[0]$, and saw that $\underline{\operatorname{Hom}}(K^\bullet(\mathcal{U}), \mathcal{F}) = \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$. To prove this lemma, we can make an almost identical definition

$$K_{\text{ord}}^{-n} = \bigoplus_{i_0 < \dots < i_n} j_{i_0 \dots i_n}^p \mathcal{O}_{U_{i_0 \dots i_n}},$$

and it is just as clear from earlier that $\underline{\operatorname{Hom}}_{\mathcal{O}_X}^{\text{pre}}(K_{\text{ord}}^\bullet(\mathcal{U}), \mathcal{F}) = \check{\mathcal{C}}_{\text{ord}}^\bullet(\mathcal{U}, \mathcal{F})$. It is also clear that the canonical projection as in the lemma statement is induced (after applying $\underline{\operatorname{Hom}}(-, \mathcal{F})$) by a canonical inclusion $K_{\text{ord}}^\bullet(\mathcal{U}) \hookrightarrow K^\bullet(\mathcal{U})$. It suffices to show that this map is a homotopy equivalence. Since these are complexes of projectives, we may show it is a quasi-isomorphism.

The key point is that $K^0(\mathcal{U}) = K_{\text{ord}}^0(\mathcal{U})$, and so the corresponding definition for $\mathcal{O}_{\mathcal{U}}^{\text{pre}}$ is the same in the ordered and unordered case. Therefore, we just

need to show that the augmented complex

$$\cdots \rightarrow K_{\text{ord}}^{-2}(\mathcal{U}) \rightarrow K_{\text{ord}}^{-1}(\mathcal{U}) \rightarrow K_{\text{ord}}^0(\mathcal{U}) \rightarrow \mathcal{O}_{\mathcal{U}}^{\text{pre}} \rightarrow 0$$

is exact. We now basically use the same proof as in [Prop. 2.1.8](#), whereby we fix an open $W \subset X$, and consider the set I' of indices for which $W \subset U_i$. Over W , the augmented complex becomes

$$\cdots \rightarrow \bigoplus_{i_0 < i_1 \in I'} \mathcal{O}_X(W) \rightarrow \bigoplus_{i_0 \in I'} \mathcal{O}_X(W) \rightarrow \mathcal{O}_X(W) \rightarrow 0,$$

(assuming $I' \neq \emptyset$). We can use almost the same cone contraction as earlier, except that placing a fixed index at the beginning might not retain the ordering. Thus, choose $\ell \in I'$, and define a homotopy by

$$h(s)_{j_0 \cdots j_{p+1}} = \begin{cases} (-1)^r s_{j_0 \cdots \widehat{j_r} \cdots j_{p+1}} & \exists j_r = \ell \\ 0 & \text{otherwise} \end{cases}.$$

Now, just as in the earlier proof, $hd + dh = 0$ follows from breaking the sum into terms including components with index ℓ and excluding them, and those including ℓ cancel out. $\square$

This gives another proof that if $\mathcal{U}$ is a complex with n elements, then $\check{H}^p(\mathcal{U}, \mathcal{F}) = 0$ for $p \geq n$. Note also that since the ordered Čech cohomology is identified with the unordered one, we also have properties such as a cover being ordered Čech-acyclic if one of the opens is the whole space. Using this, we can make the construction sheafy:

Remark 2.1.19 We may make a corresponding definition $\check{\mathcal{C}}_{\text{ord}}^{\bullet}(\mathcal{U}, \mathcal{F})$ for the sheaf Čech complex by only including terms with strictly increasing indices. Then the same argument we used earlier shows that the augmented complex $\mathcal{F}[0] \rightarrow \check{\mathcal{C}}_{\text{ord}}^{\bullet}(\mathcal{U}, \mathcal{F})$ is exact, provided $\mathcal{F}$ is a sheaf. Moreover, the corresponding projection map $\check{\mathcal{C}}^{\bullet}(\mathcal{U}, \mathcal{F}) \rightarrow \check{\mathcal{C}}_{\text{ord}}^{\bullet}(\mathcal{U}, \mathcal{F})$ is therefore a quasi-isomorphism.

2.2 The derived category $D(\mathcal{O}_X)$

2.2.1 The main derived functors

In the previous sections we focused on $R\Gamma$ and Čech cohomology. Now we will describe the other derived functors, such as

$$Rf_* \quad Lf^* \quad (-) \otimes_{\mathcal{O}_X}^{\mathbf{L}} (-) \quad \mathbf{R}\mathcal{H}om \quad \mathcal{E}xt$$

to name a few. We will discuss their relation to $R\Gamma$, and some basic compatibilities that they have with each other. Often, we can say a lot more if we consider certain objects in $D^+(\mathcal{O}_X)$ satisfy more hypotheses, such as ‘pseudo-coherence’ or ‘perfectness’, which we will study more in the next section. Examples are the consideration of duals, projection formulae, and so on.

The most important derived functor is Rf_* for $f: X \rightarrow Y$ a morphism of ringed spaces. We showed in [Remark 2.1.6](#) that if $f: X \rightarrow Y$ is such a morphism, then there is a commutative diagram

$$\begin{array}{ccc} D^+(\mathcal{O}_X) & \xrightarrow{Rf_*} & D^+(\mathcal{O}_Y) \\ R\Gamma \downarrow & & \downarrow R\Gamma \\ D^+(\mathcal{O}_X(X)) & \longrightarrow & D^+(\mathcal{O}_Y(Y)) \end{array}$$

where the bottom map is the restriction of scalars functor induced by $\mathcal{O}_X(X) \rightarrow \mathcal{O}_Y(Y)$. Thus, the natural map $f_* \rightarrow Rf_*$ induces a comparison map

$$R\Gamma(Y, f_* \mathcal{F}^\bullet) \rightarrow R\Gamma(Y, Rf_* \mathcal{F}^\bullet) = R\Gamma(X, \mathcal{F}^\bullet)$$

which is an isomorphism, e.g., if $\mathcal{F}^\bullet$ is a bounded above complex of f_* -acyclics. Here is a much more general statement:

Theorem 2.2.1 *Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be morphisms of ringed spaces.*

(a) *There is a canonical identification*

$$Rg_* \circ Rf_* = R(g \circ f)_*$$

of pushforwards $D^+(\mathcal{O}_X) \rightarrow D^+(\mathcal{O}_Z)$.

(b) *For $A \in D^+(\mathcal{O}_X)$, there is a bounded spectral sequence*

$$E_2^{p,q} = H^p(Rg_* H^q(Rf_* A)) \implies H^{p+q}(R(g \circ f)_* A)$$

converging to a finite filtration on the RHS. In particular, if $\mathcal{F}$ is an $\mathcal{O}_X$ -module, then we have a spectral sequence

$$R^p g_* (R^q f_* \mathcal{F}) \implies R^{p+q} (gf)_* \mathcal{F}.$$

This is called the Leray spectral sequence.

Proof This is a composition of derived functors and the corresponding Grothendieck spectral sequence. We have to check that f_* sends e.g. injectives to g_* -acyclic sheaves. Indeed, injectives are flasque, g_* preserves flasqueness, and flasque sheaves are g_* acyclic. $\square$

Sheaf cohomology is a special case of derived pushforward. If X is a ringed space, the unique map $p: X \rightarrow *$ extends to a map of ringed spaces by viewing $(*, \mathcal{O}_X(X))$ as a ringed space. Then $\text{Mod}(\mathcal{O}_*) = \text{Mod}(\mathcal{O}_X(X))$, and the functor p_* coincides with Γ .

Other useful results are the ‘hypercohomology spectral sequences’, for which we apply [Thm. 1.3.23](#). If $\mathcal{F}^\bullet$ is a bounded below complex of $\mathcal{O}_X$ -modules, this gives us bounded spectral sequences

$$E_1^{p,q} = R^q f_* \mathcal{F}^p \implies H^{p+q} Rf_*(\mathcal{F}^\bullet)$$

and, for $R\Gamma$,

$$E_1^{p,q} = H^q(X, \mathcal{F}^p) \implies \mathbb{H}^{p+q}(X, \mathcal{F}^\bullet).$$

This gives a way to convert results about cohomology of a single sheaf, to results about arbitrary objects of the derived category. We also have the corresponding secondary sequences on cohomology:

$$E_2^{p,q} = R^p f_*(H^q A) \implies H^{p+q} Rf_*(A)$$

and

$$E_2^{p,q} = H^p(X, H^q(A)) \implies \mathbb{H}^{p+q}(X, A)$$

for $A \in D^+(\mathcal{O}_X)$. In applications to algebraic geometry, we will often want to argue about complexes which have quasicohherent or coherent cohomology, and this spectral sequence lets us translate sheaf level statements about such things to bounded below statements.

The following result, called the *Mayer-Vietoris sequence*, is a good demonstration of how to formulate results in the derived category:

Proposition 2.2.2 *Let $f: X \rightarrow Y$ be a morphism of ringed spaces, and $A \in D^+(\mathcal{O}_X)$. Let $U, V \subset X$ be open subsets, and denote by f_U, f_V , and $f_{U \cap V}$ the composites of f with the inclusions of U, V , and $U \cap V$ into X respectively. Then there is a distinguished triangle*

$$Rf_* A \xrightarrow{(1 \ 1)^T} Rf_{U,*} A|_U \oplus Rf_{V,*} A|_V \xrightarrow{(1 \ -1)} Rf_{U \cap V,*} A|_{U \cap V} \rightarrow Rf_* A[1]$$

in $D^+(\mathcal{O}_Y)$ which is natural in A .

Proof Choose a quasi-isomorphism $A \xrightarrow{\sim} \mathcal{I}^\bullet$ with $\mathcal{I}^\bullet \in K^+(\mathcal{I}_{\mathcal{O}_X})$, then the (augmented) ordered Čech complex for $\mathcal{I}^\bullet$ with respect to U, V , is the exact sequence of complexes:

$$0 \rightarrow \mathcal{I}^\bullet \xrightarrow{(1 \ 1)^T} j_{U,*} \mathcal{I}^\bullet|_U \oplus j_{V,*} \mathcal{I}^\bullet|_V \xrightarrow{(1 \ -1)} j_{U \cap V,*} \mathcal{I}^\bullet|_{U \cap V} \rightarrow 0$$

This gives an exact triangle

$$A \rightarrow Rj_{U,*} A|_U \oplus Rj_{V,*} A|_V \rightarrow Rj_{U \cap V,*} A \rightarrow A[1]$$

in $D^+(\mathcal{O}_X)$. Now apply Rf_* to this. □

The following observation justifies the name: take f to be the identity, apply $R\Gamma$, and then the long exact sequence in cohomology is

$$\cdots \rightarrow \mathbb{H}^i(X, A) \rightarrow \mathbb{H}^i(U, A) \oplus \mathbb{H}^i(V, A) \rightarrow \mathbb{H}^i(U \cap V, A) \rightarrow \mathbb{H}^{i+1}(X, A) \rightarrow \cdots$$

mirroring the Mayer-Vietoris sequence in topology.

We next discuss the locality of derived pushforward, and a fundamental computation which expresses pushforward in terms of cohomology. First, let $f: X \rightarrow Y$ be a map and $U \subset Y$ an open subset. Let $f|_{f^{-1}U}$ denote the restricted map $f^{-1}(U) \rightarrow U$. Then by definition we have an equality of functors

$$(-)|_U \circ f_* = (f|_{f^{-1}U})_* \circ (-)|_{f^{-1}U}.$$

Now restriction is exact, and we showed that it preserves injectives. It follows that we have a canonical identification

$$(Rf_*A)|_U = R(f|_{f^{-1}U})_*(A|_{f^{-1}U}),$$

and therefore

$$(R^p f_* \mathcal{F})|_U = R^p (f|_{f^{-1}U})_*(\mathcal{F}|_{f^{-1}U}) \quad p \geq 0$$

for $\mathcal{F}$ an $\mathcal{O}_X$ -module. This means that whenever we are checking a property which is target local, we may always check on an open cover. For example, if $U \subset Y$ we see that

$$R\Gamma(U, Rf_*A) = R\Gamma(U, R(f|_{f^{-1}U})_*A) = R\Gamma(f^{-1}(U), A),$$

viewed in the derived category of the ring $\mathcal{O}_X(f^{-1}(U))$ by restriction of scalars. Next let us consider a seemingly stupid composition of functors:

$$\mathrm{Mod}(\mathcal{O}_X) \xrightarrow{f_*} \mathrm{Mod}(\mathcal{O}_Y) \xrightarrow{i_Y} \mathrm{PMod}(\mathcal{O}_Y) \xrightarrow{\mathbf{a}} \mathrm{Mod}(\mathcal{O}_Y)$$

Sheafification $\mathbf{a}$ is exact, and the composite of the last two functors is the identity. Thus the whole composite equals f_* . Since everything is $\mathbf{a}$ -acyclic, this implies that

$$Rf_* = \mathbf{a} \circ R(i_Y \circ f_*) = \mathbf{a} \circ Ri_Y \circ Rf_*,$$

where in the second equality we have used the fact that a flasque sheaf is i_Y -acyclic. That is the case because $R^p i_Y(\mathcal{G}) = \underline{H}^p(\mathcal{G}) = 0$ as $H^p(U, \mathcal{G}) = 0$ if $\mathcal{G}$ is flasque. Therefore, for $A \in D^+(\mathcal{O}_X)$, we have

$$\begin{aligned} H^p(Rf_*A) &= \mathbf{a}H^p(Ri_Y(Rf_*A)) = \mathbf{a}\underline{H}^p(Rf_*A) \\ &= \mathbf{a}(U \mapsto \mathbb{H}^p(U, Rf_*A)) \\ &= \mathbf{a}(U \mapsto \mathbb{H}^p(f^{-1}(U), A)). \end{aligned}$$

Once again, we are implicitly viewing $U \mapsto \underline{H}^p(f^{-1}(U), A)$ as a presheaf of $\mathcal{O}_Y$ -modules via restriction of scalars.

Example 2.2.3 Take $A = \mathcal{F}[0]$ and $f = 1_X$. Then this says that $\mathcal{F} = \mathbf{a}(U \mapsto H^0(U, A))$, which we already know, and

$$0 = \mathbf{a}(U \mapsto H^p(U, \mathcal{F})) = \mathbf{a}\underline{H}^p(\mathcal{F})$$

for $p > 0$, which is new. In other words, each cohomology class in $H^p(X, \mathcal{F})$ restricts to zero in a nbhd of every point. This is a reincarnation of the fact that taking sections is locally exact, but globally not right exact.

(Perhaps reorganize to add something more apt here.)

We now discuss a useful criterion for derived pushforward and cohomology commuting with filtered colimits. First of all, exactness of a sequence sheaves is checked on stalks, which is a certain colimit, and filtered colimits commute with colimits; therefore taking filtered colimits of $\mathcal{O}_X$ -modules is an exact. It follows that on each of the derived categories we obtain a functor

$$\varinjlim_I^D : D^*(\text{Mod}(\mathcal{O}_X)^I) \rightarrow D^*(\mathcal{O}_X),$$

where we use the superscript D to distinguish this from some kind of colimit inside $D^*(\mathcal{O}_X)$. If we restrict to the bounded above (or below) case, we know that this can be computed by any preadapted class in $K^+(\text{Mod}(\mathcal{O}_X)^I)$, where I . We usually take this class to be the pointwise injectives. Every object of $\text{Mod}(\mathcal{O}_X)^I$ admits a monomorphism into a pointwise injective object because we have functorial injective embeddings in $\text{Mod}(\mathcal{O}_X)$.

This preadapted class has good properties. Even if its elements are not itself injective objects of the functor category, we note that exactness in a functor category is checked pointwise. So, given a left exact functor $F: \text{Mod}(\mathcal{O}_X) \rightarrow \mathcal{A}$, the induced functor $F: \text{Mod}(\mathcal{O}_X)^I \rightarrow \mathcal{A}^I$ is also left exact, and given an acyclic complex of pointwise injective objects A , $F(A)$ is acyclic, because $F(A)(i) = F(A(i))$ is acyclic as $A(i)$ is an acyclic complex of injectives. This means that we can define the right derived functor $RF: D^+(\text{Mod}(\mathcal{O}_X)) \rightarrow D^+(\mathcal{A}^I)$.

Remark 2.2.4 More generally, here is what we have just said: If $\mathcal{A}$ is an abelian category and $\mathcal{R}$ is a preadapted class with *functorial embeddings*, then for any small category C , the class $\mathcal{R}^C$ forms a preadapted class in $\mathcal{A}^C$. Next, if $\mathcal{R}$ is adapted to a left exact functor $F: \mathcal{A} \rightarrow \mathcal{B}$, then $\mathcal{R}^C$ is adapted to $F^C: \mathcal{A}^C \rightarrow \mathcal{B}^C$ so that we can define the derived functor $RF^C: D^+(\mathcal{A}^C) \rightarrow D^+(\mathcal{B}^C)$. Lastly, suppose $A^\bullet \in \text{Ch}^+(\mathcal{A}^C)$, and choose a resolution $A^\bullet \rightarrow X^\bullet$ with $X^\bullet(c) \in K^+(\mathcal{R})$ for each $c \in C$. Then $RF^C(A) = F(X)$ pointwise and termwise, i.e. $RF^C(A)(c) = F(X(c)^\bullet)$. Since the evaluation functors are exact, $A^\bullet(c) \rightarrow X^\bullet(c)$ is also a quasi-isomorphism, and $RF(A^\bullet(c)) = F(X(c)^\bullet)$. Therefore, $RF(A)$ is represented the object $c \mapsto RF(A(c))$, where if $c \rightarrow c'$ is a map in C , then the corresponding map $RF(A(c)) \rightarrow RF(A(c'))$ is given by functoriality of RF and the fact that $A(c) \rightarrow A(c')$ is a chain map. Another way to say this last part is that the derived functor RF^C factors through

$$D^+(\mathcal{A}^C) \rightarrow D^+(\mathcal{A})^C \dashrightarrow D^+(\mathcal{B}^I)$$

where the first map is induced by the exact evaluation functors.

We can combine these two observations above to discuss how filtered colimits interact with sheaf cohomology.

Proposition 2.2.5 *Let Y have a basis of qcqs opens, X have a basis of quasicompact opens, and $f: X \rightarrow Y$ be a qcqs morphism of ringed spaces. For any filtered category I , the diagram*

$$\begin{array}{ccc} D^+(\mathrm{Mod}(\mathcal{O}_X)^I) & \xrightarrow{\lim_I^D} & D^+(\mathcal{O}_X) \\ Rf_*^I \downarrow & & Rf_* \downarrow \\ D^+(\mathrm{Mod}(\mathcal{O}_Y)^I) & \xrightarrow{\lim_I^D} & D^+(\mathcal{O}_Y) \end{array}$$

commutes up to canonical isomorphism. In particular, if $\mathcal{F}(-)$ is a filtered diagram of $\mathcal{O}_X$ -modules, then:

(a) *With the assumptions on f as above, the natural map*

$$\varinjlim_{i \in I} R^p f_* \mathcal{F}(i) \rightarrow R^p f_* \left(\varinjlim_{i \in I} \mathcal{F}(i) \right)$$

is an isomorphism for all $p \geq 0$.

(b) *If X is qcqs and has a basis of quasicompact opens, then the natural map*

$$\varinjlim_{i \in I} H^p(X, \mathcal{F}(i)) \rightarrow H^p\left(X, \varinjlim_{i \in I} \mathcal{F}(i)\right)$$

is an isomorphism for all $p \geq 0$.

Therefore, in the circumstances described above, the functors $R^p f_$ and $H^p(X, -)$ preserve arbitrary coproducts for all $p \geq 0$.*

Proof Consequence (a) follows directly from the diagram and the fact that Rf_*^I is computed pointwise, which was discussed above. If we take $Y = (*, \mathcal{O}_X(X))$, then f qcqs means that X is qcqs, and we get (b). The fact about arbitrary coproducts holds because a coproduct is a filtered colimit of biproducts (or empty, in which case we already know the functors preserve 0).

Now the main result. First let us show that under the assumptions of the proposition, $\varinjlim_I f_*(\mathcal{F}(i)) \xrightarrow{\sim} f_*(\varinjlim_i \mathcal{F}(i))$. Since Y has a basis of qcqs opens and f is qcqs, we reduce immediately to showing that $\varinjlim_i H^0(X, \mathcal{F}(i)) \xrightarrow{\sim} H^0(X, \varinjlim_i \mathcal{F}(i))$ is an isomorphism whenever X is qcqs and has a basis of qc opens. So, consider each $\mathcal{F}(i)$ as a sheaf on the basis of all qc opens, which is closed under finite intersection by assumption.

We claim that the presheaf filtered colimit $U \mapsto \varinjlim_i \mathcal{F}(i)(U)$ (for U quasicompact) is already a sheaf on the basis, which completes the proof since X is quasicompact, and the sheaf colimit is the sheafification of the presheaf colimit. To demonstrate the sheaf condition for an open cover $\{U_j\}$ of some qc

open U , it suffices to show the sheaf condition on any subcover, and hence we may assume $\{U_j\}$ is finite. Thus the sheaf condition for each $\mathcal{F}(i)$ is the exact sequence

$$0 \rightarrow \mathcal{F}(i)(U) \rightarrow \bigoplus_j \mathcal{F}(i)(U_j) \rightarrow \bigoplus_{j,k} \mathcal{F}(i)(U_j \cap U_k),$$

and taking filtered colimits gives the sheaf condition for $(\varinjlim_i \mathcal{F}(i))(U)$.

This shows that f_* commutes with filtered colimits, and so the underived version of the diagram in the proposition commutes. To conclude, we apply composition of derived functors. By the discussion above, it suffices to show that if $\mathcal{I}(-)$ is a filtered diagram of injective objects, then $\varinjlim_i \mathcal{I}(i)$ is f_* -acyclic. By the description of $R^p f_*$ as a sheafification, we may show that $H^p(f^{-1}(U), \varinjlim_i \mathcal{I}(i)) = 0$ for all $p > 0$ and $U \subset Y$ qcqs. Then $f^{-1}(U)$ is qcqs, so we reduce to showing that $H^p(X, \varinjlim_i \mathcal{I}(i)) = 0$ for $p > 0$ whenever X is qcqs and has a basis of quasicompact opens, and $\mathcal{I}(-)$ is a filtered diagram of injectives.

We show this using Čech cohomology. Let $\mathcal{U}$ be a finite open cover of a quasicompact open $U \subset X$ by qc opens. Since $U_{i_0 \dots i_p}$ is still qcqs and has a basis of qc opens, we have

$$\varinjlim_{i \in I} H^0(U_{i_0 \dots i_p}, \mathcal{F}(i)) \xrightarrow{\sim} H^0(U_{i_0 \dots i_p}, \varinjlim_{i \in I} \mathcal{F}(i)),$$

so $\varinjlim_i \check{C}^p(\mathcal{U}, \mathcal{I}(i)|_U) \xrightarrow{\sim} \check{C}^p(\mathcal{U}, \varinjlim_i \mathcal{I}(i)|_U)$ by finiteness of $\mathcal{U}$. Since filtered colimits are exact, this statement holds on the level of Čech cohomology. Since each $\mathcal{I}(i)$ is injective, the cohomology on the LHS vanishes, which proves that

$$\check{H}^p(\mathcal{U}, \varinjlim_i \mathcal{I}(i)|_U) = 0 \quad p > 0$$

for every finite open cover $\mathcal{U}$ of U by qc opens. Since such coverings form a cofinal system, we conclude by [Lemma 2.1.15](#) that higher cohomology of $\varinjlim_i \mathcal{I}(i)$ vanishes for all qc opens in X . $\square$

Our next topic is $\mathbf{R}\mathcal{H}om$, which we will define in the exact same way as $\mathbf{R}Hom$. We will see however that to prove the relevant properties of $\mathbf{R}\mathcal{H}om$, many specific properties of sheaves were used; namely, what follows is *not* just an obvious generalization of $\mathbf{R}Hom$.

Let $\mathcal{F}^\bullet, \mathcal{G}^\bullet$ be (possibly unbounded) complexes of $\mathcal{O}_X$ -modules, and define

$$\underline{Hom}_{\mathcal{O}_X}^n(\mathcal{F}^\bullet, \mathcal{G}^\bullet) := \prod_{p \in \mathbb{Z}} Hom_{\mathcal{O}_X}(\mathcal{F}^p, \mathcal{G}^{n+p}),$$

with differential given by the same formula $df - (-1)^n fd$, where this formula makes sense on every open and hence as sheaves. So,

$$H^0(U, \underline{Hom}_{\mathcal{O}_X}^\bullet(\mathcal{F}^\bullet, \mathcal{G}^\bullet)) = \underline{Hom}_{\mathcal{O}_U}(\mathcal{F}^\bullet|_U, \mathcal{G}^\bullet|_U)$$

as complexes of $\mathcal{O}_X(U)$ -modules since products commute with sections, from which we immediately deduce e.g. that the definition above actually gives a complex. More generally, most facts that we prove about the sheaf hom complex will pass through the formula above. For example, we get for free that

$$\underline{\mathrm{Hom}}_{\mathcal{O}_X}^{\bullet}(\mathcal{F}^{\bullet}, \mathcal{G}^{\bullet}[k]) = \underline{\mathrm{Hom}}_{\mathcal{O}_X}^{\bullet}(\mathcal{F}^{\bullet}[-k], \mathcal{G}^{\bullet}) = \underline{\mathrm{Hom}}_{\mathcal{O}_X}^{\bullet}(\mathcal{F}^{\bullet}, \mathcal{G}^{\bullet})[k]$$

because it is true on every open.

Lemma 2.2.6 *Let X be a ringed space. In what follows,*

$$\mathcal{F}^{\bullet}, \mathcal{F}_1^{\bullet}, \mathcal{F}_2^{\bullet}, \mathcal{G}^{\bullet}, \mathcal{G}_1^{\bullet}, \mathcal{G}_2^{\bullet}$$

are all possibly unbounded complexes of $\mathcal{O}_X$ -modules.

(a) *$\underline{\mathrm{Hom}}$ naturally defines a functor*

$$\mathrm{Ch}(\mathcal{O}_X)^{\mathrm{op}} \times \mathrm{Ch}(\mathcal{O}_X) \rightarrow \mathrm{Ch}(\mathcal{O}_X).$$

This restricts to $\mathrm{Ch}^{-}(\mathcal{O}_X)^{\mathrm{op}} \times \mathrm{Ch}^{+}(\mathcal{O}_X) \rightarrow \mathrm{Ch}^{+}(\mathcal{O}_X)$. Moreover, these are homotopy invariant, meaning that both restrict to functors

$$\mathrm{K}(\mathcal{O}_X)^{\mathrm{op}} \times \mathrm{K}(\mathcal{O}_X) \rightarrow \mathrm{K}(\mathcal{O}_X) \quad \mathrm{K}^{-}(\mathcal{O}_X)^{\mathrm{op}} \times \mathrm{K}^{+}(\mathcal{O}_X) \rightarrow \mathrm{K}^{+}(\mathcal{O}_X)$$

on the homotopy categories.

(b) *For any map $\phi: \mathcal{G}_1^{\bullet} \rightarrow \mathcal{G}_2^{\bullet}$, there is a canonical identification*

$$\underline{\mathrm{Hom}}_{\mathcal{O}_X}^{\bullet}(\mathcal{F}^{\bullet}, \mathrm{Cone}(\phi)) = \mathrm{Cone}\left(\underline{\mathrm{Hom}}_{\mathcal{O}_X}^{\bullet}(\mathcal{F}^{\bullet}, \phi)\right).$$

For any map $\phi: \mathcal{F}_1^{\bullet} \rightarrow \mathcal{F}_2^{\bullet}$, there is a canonical identification

$$\underline{\mathrm{Hom}}_{\mathcal{O}_X}^{\bullet}(\mathrm{Cone}(\phi), \mathcal{G}^{\bullet}) = \mathrm{Cone}\left(\underline{\mathrm{Hom}}_{\mathcal{O}_X}^{\bullet}(\phi, \mathcal{G}^{\bullet})\right)$$

as well.

(c) *If $\mathcal{I}^{\bullet} \in \mathrm{K}^{+}(\mathcal{I}_{\mathcal{O}_X})$ and $\phi: \mathcal{F}_1^{\bullet} \rightarrow \mathcal{F}_2^{\bullet}$ is a quasi-isomorphism, then*

$$\underline{\mathrm{Hom}}_{\mathcal{O}_X}^{\bullet}(\mathcal{F}_1^{\bullet}, \mathcal{I}^{\bullet}) \rightarrow \underline{\mathrm{Hom}}_{\mathcal{O}_X}^{\bullet}(\mathcal{F}_2^{\bullet}, \mathcal{I}^{\bullet})$$

is a quasi-isomorphism.

(d) *If $\mathcal{I}^{\bullet} \in \mathrm{K}(\mathcal{I}_{\mathcal{O}_X})$, then $\underline{\mathrm{Hom}}_{\mathcal{O}_X}^{\bullet}(\mathcal{F}^{\bullet}, \mathcal{I}^{\bullet})$ is a complex of flasque $\mathcal{O}_X$ -modules. If in addition $\mathcal{F}^{\bullet}$ is a complex of flat $\mathcal{O}_X$ -modules, then $\underline{\mathrm{Hom}}_{\mathcal{O}_X}^{\bullet}(\mathcal{F}^{\bullet}, \mathcal{I}^{\bullet})$ is a complex of injectives.*

Proof (a) The first part is from functoriality of sheaf hom and of products. Note that if the first argument is zero after degree q and the second is zero below degree r , then whenever $n < -q + r$, either $p > q$ or $n + p \leq n + q < r$, so $\mathrm{Hom}(\mathcal{F}^p, \mathcal{G}^{n+p}) = 0$ either way. For homotopy invariance, note e.g. that a homotopy $\mathcal{G}_1^{\bullet} \rightarrow \mathcal{G}_2^{\bullet-1}$ induces a map $\mathrm{Hom}(\mathcal{F}^p, \mathcal{G}_1^{n+p}) \rightarrow \mathrm{Hom}(\mathcal{F}^p, \mathcal{G}_2^{n+p-1})$, and hence a map

$$\underline{\mathrm{Hom}}_{\mathcal{O}_X}^n(\mathcal{F}^{\bullet}, \mathcal{G}_1^{\bullet}) \rightarrow \underline{\mathrm{Hom}}_{\mathcal{O}_X}^{n-1}(\mathcal{F}^{\bullet}, \mathcal{G}_2^{\bullet}) \quad n \in \mathbb{Z}$$

which is a homotopy by functoriality. Likewise for the first argument.

(b) This is true after taking sections over every open set, and the identification is compatible with restriction. Here we are using the fact that $\mathcal{H}om$ preserves finite biproducts in either variable.

(c) It suffices by (b) to prove that if $\mathcal{F}^\bullet$ is acyclic, then $\mathcal{H}om_{\mathcal{O}_X}^\bullet(\mathcal{G}^\bullet, \mathcal{I}^\bullet)$ is acyclic. But to show exactness we may show exactness on sections over every open set, and indeed $\underline{\mathcal{H}om}_{\mathcal{O}_U}(\mathcal{F}^\bullet|_U, \mathcal{I}^\bullet|_U)$ is acyclic for all U because $\mathcal{I}^\bullet|_U$ is a bounded above complex of injectives.

(d) Note that a product of flasque sheaves is flasque, because products commute with sections, and products are exact for modules over a ring. Thus we may show that $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{I})$ is flasque for $\mathcal{I}$ an injective sheaf. Indeed, for an open inclusion $j: U \rightarrow X$, applying $\mathcal{H}om_{\mathcal{O}_X}(-, \mathcal{I})$ to the injective morphism $j_!\mathcal{F}|_U \hookrightarrow \mathcal{F}$ gives the exact sequence

$$\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{I}) \rightarrow \mathcal{H}om_{\mathcal{O}_X}(j_!\mathcal{F}|_U, \mathcal{I}) = \mathcal{H}om_{\mathcal{O}_U}(\mathcal{F}|_U, \mathcal{I}|_U) \rightarrow 0$$

by adjunction. This shows flasqueness.

For the second statement, we once again reduce to showing that $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{I})$ is injective if $\mathcal{F}$ is flat and $\mathcal{I}$ is injective, because products preserve injectives. But by the tensor-hom adjunction,

$$\mathcal{H}om_{\mathcal{O}_X}(-, \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{I})) \cong \mathcal{H}om_{\mathcal{O}_X}((-) \otimes_{\mathcal{O}_X} \mathcal{F}, \mathcal{I}),$$

so the LHS is a composition of two exact functors. $\square$

By part (a) of the lemma, the sheaf hom complex gives a functor $K(\mathcal{O}_X)^{\text{op}} \times K^+(\mathcal{I}_{\mathcal{O}_X}) \rightarrow K(\mathcal{O}_X)$, which by (c) descends to a functor $D(\mathcal{O}_X)^{\text{op}} \times K^+(\mathcal{O}_X) \rightarrow D(\mathcal{O}_X)$. Via the equivalence $K^+(\mathcal{I}_{\mathcal{O}_X}) \rightarrow D^+(\mathcal{O}_X)$, we get a functor

$$\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}: D(\mathcal{O}_X)^{\text{op}} \times D^+(\mathcal{O}_X) \rightarrow D(\mathcal{O}_X).$$

By part (c) of the lemma, the functors

$$\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A, -) \quad \mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(-, B)$$

are both triangulated for any $A \in D(\mathcal{O}_X)$ and $B \in D^+(\mathcal{O}_X)$. Note that by (a), if $A \in D^-(\mathcal{O}_X)$, then this lands in $D^+(\mathcal{O}_X)$. Lastly, the choice of an inverse equivalence $D^+(\mathcal{O}_X) \rightarrow K^+(\mathcal{I}_{\mathcal{O}_X})$ comes with a natural transformation which induces a natural morphism

$$\underline{\mathcal{H}om} \rightarrow \mathbf{R}\mathcal{H}om$$

of bifunctors, which on the bounded below complexes of injectives in the second variable is an isomorphism.

Proposition 2.2.7 *Let X be a ringed space.*

(a) *If $U \subset X$, then*

$$\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A, B)|_U = \mathbf{R}\mathcal{H}om_{\mathcal{O}_U}(A|_U, B|_U)$$

for any $A \in D(\mathcal{O}_X)$ and $B \in D^+(\mathcal{O}_X)$.

(b) For $\mathcal{F} \in \text{Mod}(\mathcal{O}_X)$, there is a canonical identification

$$R\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, -) = \mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}[0], -)$$

of the right derived functor with sheaf $R\mathcal{H}om$.

(c) There is a canonical identification

$$R\Gamma\left(X, \mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A, B)\right) = R\mathcal{H}om_{\mathcal{O}_X}(A, B)$$

in $D^+(\mathcal{O}_X(X))$ for $A \in D^-(\mathcal{O}_X)$ and $B \in D^+(\mathcal{O}_X)$.

Proof (a) Replace B by a bounded below complex of injectives, then we conclude by exactness of restriction and the fact that the restriction of a bounded below complex of injectives still has the same property.

(b) Since $\mathbf{R}\mathcal{H}om(\mathcal{F}[0], -)$ is triangulated, and due to the natural map from $\underline{\mathcal{H}om}(\mathcal{F}[0], -) = \mathcal{H}om(\mathcal{F}, -)$, there is a natural map in the direction specified by the universal property of derived functors, making the obvious diagram commute. This is an isomorphism on $K^+(\mathcal{I}_{\mathcal{O}_X})$, and hence on $D^+(\mathcal{O}_X)$.

(c) Note that $\Gamma(X, \underline{\mathcal{H}om}(A, B)) = \underline{\mathcal{H}om}(A, B)$, so we have some natural maps

$$R\mathcal{H}om(A, -) \leftarrow \underline{\mathcal{H}om}(A, -) \rightarrow R\Gamma(\underline{\mathcal{H}om}(A, -)) \rightarrow R\Gamma(\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A, -))$$

in the derived category of $\mathcal{O}_X(X)$. Testing on $K^+(\mathcal{I}_{\mathcal{O}_X})$, the left and right hand maps are isomorphisms, and the remaining map is an isomorphism as well because $\underline{\mathcal{H}om}(A, \mathcal{I}^\bullet)$ is flasque, by [Lemma 2.2.6](#). $\square$

In light of (b) above, we write

$$\mathcal{E}xt_{\mathcal{O}_X}^p(-, -) := H^p(\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}((-)[0], (-)[0]))$$

for the classical right derived functors of $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, -)$, so these form bifunctors

$$\text{Mod}(\mathcal{O}_X)^{\text{op}} \times \text{Mod}(\mathcal{O}_X) \rightarrow \text{Mod}(\mathcal{O}_X).$$

Then by (c), or simply the fact that $\mathcal{H}om(\mathcal{F}, \mathcal{I})$ is flasque if $\mathcal{I}$ is injective, we have the following spectral sequence:

$$H^p(X, \mathcal{E}xt_{\mathcal{O}_X}^q(\mathcal{F}, \mathcal{G})) \implies \text{Ext}_{\mathcal{O}_X}^{p+q}(\mathcal{F}, \mathcal{G})$$

This is called the *local-to-global* spectral sequence. More generally, instead of applying the Grothendieck spectral sequence, we can use the main ingredient in its proof, which gives a spectral sequence

$$H^p\left(X, H^q(\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A, B))\right) \implies \text{Hom}_{D(\mathcal{O}_X)}(A, B)$$

of $\mathcal{O}_X(X)$ -modules, for $A \in D^-(\mathcal{O}_X)$ and $B \in D^+(\mathcal{O}_X)$. The point of introducing $\mathbf{R}\mathcal{H}om$ is that it is in fact usually much easier to compute, and as we will see it behaves a lot more like classical $R\mathcal{H}om$'s for modules.

For example, we know that $\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(\mathcal{O}_X, -)$ is the identity functor while $\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(\mathcal{O}_X, -)$ is $R\Gamma$. Thus the spectral sequence above is a generalization of the hypercohomology spectral sequence.

Another reason that $\mathbf{R}\mathcal{H}om$ is more similar to the module case is because we can often compute it using resolutions in the first variable. Let us temporarily call an $\mathcal{O}_X$ -module *internally projective* if $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, -)$ is exact. Thus in this case $\mathcal{E}xt^p(\mathcal{F}, -) = 0$ for $p > 0$, and conversely if $\mathcal{E}xt^p(\mathcal{F}, -) = 0$ for $p > 0$, then an exact sequence $0 \rightarrow \mathcal{G}_1 \rightarrow \mathcal{G}_2 \rightarrow \mathcal{G}_3 \rightarrow 0$ induces a long exact sequence

$$0 \rightarrow \mathcal{H}om(\mathcal{F}, \mathcal{G}_1) \rightarrow \mathcal{H}om(\mathcal{F}, \mathcal{G}_2) \rightarrow \mathcal{H}om(\mathcal{F}, \mathcal{G}_3) \rightarrow \mathcal{E}xt^1(\mathcal{F}, \mathcal{G}_1) = 0$$

by exactness of $\mathbf{R}\mathcal{H}om$. So being internally projective is equivalent to having $\mathcal{E}xt^p(\mathcal{F}, -) = 0$ for $p > 0$. For example, a finite locally free sheaf is internally projective.

Example 2.2.8 If $\mathcal{F}$ is an internally projective module, then the local-to-global spectral sequence for Exts immediately gives a natural identification $H^p(X, \mathcal{H}om(\mathcal{F}, -)) = \text{Ext}_{\mathcal{O}_X}^p(\mathcal{F}, -)$. For instance if $\mathcal{F} = \mathcal{E}$ is finite locally free, then $H^p(X, \mathcal{E}^\vee \otimes_{\mathcal{O}_X} \mathcal{G}) = \text{Ext}_{\mathcal{O}_X}^p(\mathcal{E}, \mathcal{G})$.

Example 2.2.9 Just as in the case where the target is modules, $\underline{\mathcal{H}om}(\mathcal{F}^\bullet, \mathcal{G}^\bullet)$ is the product totalization of the bicomplex $C^{p,q} = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}^{-p}, \mathcal{G}^q)$, so there is an associated spectral sequence. However this doesn't automatically converge to what we expect since products aren't exact; so, assume either that $\mathcal{F}^\bullet \in \text{Ch}^b(\mathcal{O}_X)$ and $\mathcal{G}^\bullet$ is arbitrary, or $\mathcal{F}^\bullet \in \text{Ch}^-(\mathcal{O}_X)$ and $\mathcal{G}^\bullet \in \text{Ch}^+(\mathcal{O}_X)$. Both of these ensure that the products taken are finite, so we get spectral sequences

$$H_h^q(H_v^p(\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}^\bullet, \mathcal{G}^\bullet))) \implies H^{p+q}(\underline{\mathcal{H}om}_{\mathcal{O}_X}(\mathcal{F}^\bullet, \mathcal{G}^\bullet))$$

and

$$H_v^p(H_h^q(\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}^\bullet, \mathcal{G}^\bullet))) \implies H^{p+q}(\underline{\mathcal{H}om}_{\mathcal{O}_X}(\mathcal{F}^\bullet, \mathcal{G}^\bullet)).$$

So, for example, if $\mathcal{F}^\bullet$ is a bounded above complex of internally projective sheaves, then for $\mathcal{G}^\bullet$ a bounded below acyclic complex, $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}^i, \mathcal{G}^\bullet)$ is acyclic. Thus the first spectral sequence degenerates to 0 and the total complex $\underline{\mathcal{H}om}_{\mathcal{O}_X}(\mathcal{F}^\bullet, \mathcal{G}^\bullet)$ is acyclic.

Lemma 2.2.10 *Let $\mathcal{F}^\bullet$ be a bounded above complex of internally projective modules. Then the natural map*

$$\underline{\mathcal{H}om}_{\mathcal{O}_X}(\mathcal{F}^\bullet, -) \rightarrow \mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}^\bullet, -)$$

is an isomorphism of functors $\mathbf{K}^+(\mathcal{O}_X) \rightarrow D^+(\mathcal{O}_X)$.

Proof Let $\mathcal{G}^\bullet \in \mathbf{K}^+(\mathcal{O}_X)$ and $\mathcal{G}^\bullet \rightarrow \mathcal{I}^\bullet$ be a qis with $\mathcal{I}^\bullet \in \mathbf{K}^+(\mathcal{I}_{\mathcal{O}_X})$. Then the diagram

$$\begin{array}{ccc} \underline{\mathcal{H}om}_{\mathcal{O}_X}(\mathcal{F}^\bullet, \mathcal{G}^\bullet) & \longrightarrow & \mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}^\bullet, \mathcal{G}^\bullet) \\ \downarrow & & \downarrow \cong \\ \underline{\mathcal{H}om}_{\mathcal{O}_X}(\mathcal{F}^\bullet, \mathcal{I}^\bullet) & \xrightarrow{\cong} & \mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}^\bullet, \mathcal{I}^\bullet) \end{array}$$

commutes by naturality in both variables, and we already know that the right and bottom maps are isomorphisms. By [Lemma 2.2.6\(b\)](#) and the fact that $\underline{\mathcal{H}om}_{\mathcal{O}_X}(\mathcal{F}^\bullet, -)$ preserves acyclicity, the left map is also an isomorphism. $\square$

Here's how this occurs in practice: we often care geometrically about computing $\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$, e.g. where $\mathcal{F}$ is some coherent sheaf, and we have a finite resolution of $\mathcal{F}$ by vector bundles. So, we can first compute $\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$ which is just $\underline{\mathcal{H}om}$ applied to the resolution, and then compute cohomology and use the local-to-global spectral sequence. We will return to internally projective sheaves in the next section.

Now let's discuss Lf^* and the derived tensor product. First, the latter; we know that $\mathrm{Mod}(\mathcal{O}_X)$ is closed symmetric monoidal and has exact arbitrary sums, and enough flat objects, so the entire discussion about $(-) \otimes^{\mathbf{L}} (-)$ already applies. By convention, we denote the corresponding Tor sheaves as $\mathrm{Tor}_{\mathcal{O}_X}^n(\mathcal{F}, \mathcal{G})$, instead of $\mathrm{Tor}_{\mathrm{Mod}(\mathcal{O}_X)}^n(\mathcal{F}, \mathcal{G})$, though they mean the same thing.

Example 2.2.11 Let $A, B \in D^-(\mathcal{O}_X)$, and represent them by complexes $\mathcal{F}^\bullet \rightarrow A, \mathcal{G}^\bullet = B$ where $\mathcal{F}^\bullet$ is a bounded above complex of flats. Then we defined the complex $\mathcal{F}^\bullet \otimes \mathcal{G}^\bullet$ as the (sum) totalization of the natural bicomplex. Since this is a shifted third quadrant bicomplex, we have the associated spectral sequence

$$H_h^p H_v^q(\mathcal{F}^\bullet \otimes \mathcal{G}^\bullet) \implies H^{p+q}(\mathcal{F}^\bullet \otimes \mathcal{G}^\bullet).$$

Since each $\mathcal{F}^i$ is flat, we have $H_v^q(\mathcal{F}^\bullet \otimes \mathcal{G}^\bullet) = \mathcal{F}^\bullet \otimes H^q(\mathcal{G}^\bullet)$. Thus the LHS above is $H^p(\mathcal{F}^\bullet \otimes H^q(\mathcal{G}^\bullet))$. Therefore, this gives a bounded spectral sequence

$$H^p(A \otimes_{\mathcal{O}_X}^{\mathbf{L}} H^q(B)) \implies H^{p+q}(A \otimes_{\mathcal{O}_X}^{\mathbf{L}} B),$$

which is called the *hyper-Tor spectral sequence*. In particular, if $A = \mathcal{F}$ is represented by a single sheaf, then

$$\mathrm{Tor}^p(\mathcal{F}, H^q(B)) \implies H^{p+q}(\mathcal{F} \otimes_{\mathcal{O}_X}^{\mathbf{L}} B).$$

The general version is useful because it relates the cohomology of $A \otimes_{\mathcal{O}_X}^{\mathbf{L}} B$, which isn't a derived functor of anything, to the cohomology of something which is a derived functor, since

$$A \otimes_{\mathcal{O}_X}^{\mathbf{L}} H^q(B) = L\left((-) \otimes_{\mathcal{O}_X} H^q(B)\right)(A)$$

as $H^q(B)$ is just an $\mathcal{O}_X$ -module.

Example 2.2.12 We can repeat the same trick as above for $\mathbf{R}\mathcal{H}om$, but we have to make sure the first argument is bounded above as explained earlier. So if $A \in D^-(\mathcal{O}_X)$ and $B \in D^+(\mathcal{O}_X)$, then by replacing B with a bounded below complex of injectives we obtain a spectral sequence

$$H^p(\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(H^q(A), B)) \implies H^{p+q}(\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A, B)),$$

so in particular

$$\mathcal{E}xt_{\mathcal{O}_X}^p(H^q(A), \mathcal{G}) \implies H^{p+q}(\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A, \mathcal{G})).$$

The same thing of course works for regular $\mathbf{R}\mathcal{H}om$, so we have

$$H^p(\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(H^q(A), B)) \implies H^{p+q}(\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A, B))$$

and likewise an Ext version as above.

The nice property of the derived tensor product is that it behaves well with respect to stalks. First of all, since the stalk functor $(-)_x: \text{Mod}(\mathcal{O}_X) \rightarrow \text{Mod}(\mathcal{O}_{X,x})$ is exact, it induces a termwise map on the derived categories $(-)_x: D^*(\mathcal{O}_X) \rightarrow D^*(\mathcal{O}_{X,x})$. Then, here's a simple lemma about flat sheaves.

Lemma 2.2.13 *Let X be a ringed space. An $\mathcal{O}_X$ -module $\mathcal{F}$ is flat if and only if $\mathcal{F}_x$ is flat over $\mathcal{O}_{X,x}$.*

Proof We already know the $\Leftarrow$ implication. For the other direction, we just need to know that for any $\mathcal{O}_{X,x}$ module M there is an $\mathcal{O}_X$ -module with stalk M at x . So take the skyscraper sheaf corresponding to M at x . $\square$

Remark 2.2.14 In the tensor product case, we want to (as before) assert the existence of a map

$$(-) \otimes_{\mathcal{O}_X}^{\mathbf{L}} (-) \rightarrow (-) \otimes_{\mathcal{O}_X} (-)$$

natural in both variables, of functors $K^-(\mathcal{O}_X) \times K^-(\mathcal{O}_X) \rightarrow D^-(\mathcal{O}_X)$. This is more delicate than the $\mathbf{R}\mathcal{H}om$ case because we don't have enough projectives (actually, that case was a bit delicate too, so I should go back and fix it); so, if $\mathcal{R}$ is the class of flats, while we have a map $K^-(\mathcal{O}_X) \rightarrow K^-(\mathcal{R})[S_{\mathcal{R}}^{-1}]$, we don't automatically have a map $K^-(\mathcal{O}_X) \rightarrow K^-(\mathcal{R})$ coming from a choice of inverse.

So here's what we can do instead: By construction, for each $\mathcal{F}$, the functor $\mathcal{F} \otimes_{\mathcal{O}_X}^{\mathbf{L}} (-)$ as $D^-(\mathcal{O}_X) \rightarrow D^-(\mathcal{O}_X)$ is a left derived functor of $\mathcal{F} \otimes (-)$ using an adapted class, so we have a natural map $\mathcal{F} \otimes_{\mathcal{O}_X}^{\mathbf{L}} (-) \rightarrow \mathcal{F} \otimes_{\mathcal{O}_X} (-)$ of functors $K^-(\mathcal{O}_X) \rightarrow D^-(\mathcal{O}_X)$. Since a morphism $\mathcal{F} \rightarrow \mathcal{F}'$ gives a natural transformation $\mathcal{F} \otimes (-) \rightarrow \mathcal{F}' \otimes (-)$, there is an induced morphism of left derived functors making the diagram

$$\begin{array}{ccc} \mathcal{F} \otimes_{\mathcal{O}_X}^{\mathbf{L}} (-) & \longrightarrow & \mathcal{F} \otimes_{\mathcal{O}_X} (-) \\ \downarrow & & \downarrow \\ \mathcal{F}' \otimes_{\mathcal{O}_X}^{\mathbf{L}} (-) & \longrightarrow & \mathcal{F}' \otimes_{\mathcal{O}_X} (-) \end{array}$$

commute. We know that if $\mathcal{G}^\bullet \rightarrow B$ is a flat resolution, then the comparison map is induced by $\mathcal{F} \otimes_{\mathcal{O}_X}^{\mathbf{L}} B = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}^\bullet \rightarrow \mathcal{F} \otimes_{\mathcal{O}_X} B$; now if $\mathcal{F}^\bullet$ is any bounded above complex, the differentials induce commutative diagrams like the one above, and by taking direct sums we get a map of totalizations

$$\mathcal{F}^\bullet \otimes_{\mathcal{O}_X}^{\mathbf{L}} B = \text{Tot}(\mathcal{F}^\bullet \otimes_{\mathcal{O}_X} \mathcal{G}^\bullet) \rightarrow \mathcal{F}^\bullet \otimes_{\mathcal{O}_X} B,$$

This is exactly the desired map of complexes. If B is a bounded above complex of flats, then this is obviously an isomorphism, and $\mathcal{F}^\bullet$ is a bounded above complex of flats, then tensoring with it preserves quasi-isomorphisms.

Remark 2.2.15 Here is another thing we are using to construct natural maps when comparing derived functors. If $\mathcal{R}$ is a (say left) preadapted class, then we know that the natural map $Q: K^-(\mathcal{R}) \rightarrow D^-(\mathcal{A})$ is a localization by the quasi-isomorphisms. This means that restriction along Q induces an equivalence

$$Q^*: \text{Fun}(D^-(\mathcal{A}), \mathcal{C}) \rightarrow \text{Fun}^S(K^-(\mathcal{R}), \mathcal{C}),$$

where Fun^S denotes the full subcategory of functors which send quasi-isomorphisms in $K^-(\mathcal{R})$ to isomorphisms. By currying/product-hom adjunction, this also induces an equivalence

$$\text{Fun}(D^-(\mathcal{A}) \times \mathcal{C}, \mathcal{D}) \rightarrow \text{Fun}^{S \times 1}(K^-(\mathcal{R}) \times \mathcal{C}, \mathcal{D}),$$

where the RHS denotes the full subcategory of bifunctors G for which $G(-, C)$ sends quasi-isomorphisms to isomorphisms for every $C \in \mathcal{C}$. Therefore, to provide natural transformations between functors in the LHS, we can precompose with Q and then provide natural transformations there.

The natural map $(-) \otimes_{\mathcal{O}_X}^{\mathbf{L}} (-) \rightarrow (-) \otimes_{\mathcal{O}_X} (-)$ gives rise to a zig-zag of maps

$$\left((-) \otimes_{\mathcal{O}_X}^{\mathbf{L}} (-) \right)_x \rightarrow (-)_x \otimes_{\mathcal{O}_{X,x}} (-)_x \leftarrow (-)_x \otimes_{\mathcal{O}_{X,x}}^{\mathbf{L}} (-)_x$$

because regular tensor product commutes with stalks, all landing in $D^-(\mathcal{O}_{X,x})$. Now in the first variable, place a bounded above complex $\mathcal{F}^\bullet$ of flat sheaves. The left hand map becomes an isomorphism because the same was true before taking stalks. The RHS is an isomorphism because $\mathcal{F}_x^\bullet$ is a bounded above complex of flat $\mathcal{O}_{X,x}$ -modules. Therefore, as functors $K^-(\mathcal{R}) \times D^-(\mathcal{O}_X) \rightarrow D^-(\mathcal{O}_{X,x})$, the LHS and RHS are isomorphic, so the same is true on $D^-(\mathcal{O}_X) \times D^-(\mathcal{O}_X)$ by the remark above. We have shown:

Lemma 2.2.16 *Let $A, B \in D^-(\mathcal{O}_X)$. For all $x \in X$, there is a canonical identification*

$(A \otimes_{\mathcal{O}_X}^{\mathbf{L}} B)_x = A_x \otimes_{\mathcal{O}_{X,x}}^{\mathbf{L}} B_x$

in $D^-(\mathcal{O}_{X,x})$. In particular, for $\mathcal{F}, \mathcal{G} \in \text{Mod}(\mathcal{O}_X)$, the natural map

$$\text{Tor}_n^{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})_x \rightarrow \text{Tor}_n^{\mathcal{O}_{X,x}}(\mathcal{F}_x, \mathcal{G}_x)$$

for every $x \in X$ and $n \in \mathbb{Z}$.

Now we discuss derived pullback. Let $f: X \rightarrow Y$ be a morphism of ringed spaces. Consider the class $\mathcal{R}$ of flat $\mathcal{O}_Y$ -modules, which we already know is preadapted. We claim it is adapted to the pullback functor f^* . If $\mathcal{F}^\bullet$ is a bounded above acyclic complex of flat $\mathcal{O}_X$ -modules, then apply f^* , and consider the resulting sequence on stalks at a point $x \in X$:

$$\cdots \rightarrow \mathcal{F}_{f(x)}^{n-1} \otimes_{\mathcal{O}_{Y,f(x)}} \mathcal{O}_{X,x} \rightarrow \mathcal{F}_{f(x)}^n \otimes_{\mathcal{O}_{Y,f(x)}} \mathcal{O}_{X,x} \rightarrow \cdots$$

Since each $\mathcal{F}_{f(x)}^n$ is flat over $\mathcal{O}_{Y,f(x)}$ and the sequence is bounded above, the sequence above is in fact exact (by a result early on in the discussion of derived tensor product). Therefore $f^*\mathcal{F}^\bullet$ is an acyclic complex, and we deduce that $\mathcal{R}$ is an adapted class, so the derived functor $Lf^*: D^-(\mathcal{O}_Y) \rightarrow D^-(\mathcal{O}_X)$ exists. Here are the very basic properties:

Proposition 2.2.17 *Let $f: X \rightarrow Y$ be a morphism of ringed spaces.*

(a) *If $g: Y \rightarrow Z$ is a morphism of ringed spaces, the natural map*

$$L(g \circ f)^* \rightarrow Lf^* \circ Lg^*$$

is an isomorphism of functors $D^-(\mathcal{O}_Z) \rightarrow D^-(\mathcal{O}_X)$.

(b) *Let $U \subset Y$ be an open subset, and $f|_{f^{-1}(U)}$ denote the restricted morphism $f^{-1}(U) \rightarrow U$. Then there is a canonical identification*

$$Lf^*(A)|_{f^{-1}(U)} = L(f|_{f^{-1}(U)})^*(A|_U)$$

in $D^-(\mathcal{O}_X|_{f^{-1}(U)})$ for $A \in D^-(\mathcal{O}_Y)$.

(c) *For any $A \in D^-(\mathcal{O}_Y)$ and $x \in X$, the natural map*

$$Lf^*(A)_x \rightarrow A_{f(x)} \otimes_{\mathcal{O}_{Y,f(x)}}^{\mathbf{L}} \mathcal{O}_{X,x}$$

is an isomorphism, where on the RHS we mean the derived functor of the base change by $\mathcal{O}_{X,x}$ defined previously.

(d) *For any $A, B \in D^-(\mathcal{O}_Y)$, the natural map*

$$Lf^*(A \otimes_{\mathcal{O}_Y}^{\mathbf{L}} B) \rightarrow Lf^*(A) \otimes_{\mathcal{O}_X}^{\mathbf{L}} Lf^*(B)$$

is an isomorphism.

Proof (a) If $\mathcal{F}$ is flat over $\mathcal{O}_Y$, then $f^*\mathcal{F}$ is flat over $\mathcal{O}_X$, because its stalk at x is the base change $\mathcal{F}_{f(x)} \otimes_{\mathcal{O}_{Y,f(x)}} \mathcal{O}_{X,x}$ and flatness is stable under base change.

(b) This follows from the commutative diagram

$$\begin{array}{ccc} f^{-1}(U) & \xrightarrow{f|_{f^{-1}(U)}} & U \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

of ringed spaces, (a), and the fact that pulling back along an open inclusion is just restriction.

(c) First of all, recall that we justified the notation on the RHS because the restriction of scalars (which is exact) of the derived base change is indeed the derived tensor product. The statement is derived functors of the composite in

the commutative square:

$$\begin{array}{ccc} \mathrm{Mod}(\mathcal{O}_Y) & \xrightarrow{(-)^{f(x)}} & \mathrm{Mod}(\mathcal{O}_{Y,f(x)}) \\ f^* \downarrow & & \downarrow \otimes_{\mathcal{O}_{Y,f(x)}} \mathcal{O}_{X,x} \\ \mathrm{Mod}(\mathcal{O}_X) & \xrightarrow{(-)_x} & \mathrm{Mod}(\mathcal{O}_{X,x}) \end{array}$$

Taking stalks is exact, so the only thing to do is find a preadapted class on $\mathrm{Mod}(\mathcal{O}_Y)$ whose stalks land in an adapted class for base change to $\mathcal{O}_{X,x}$, for which we can take the flat $\mathcal{O}_Y$ -modules.

(d) We have maps

$$Lf^*(A \otimes_{\mathcal{O}_Y}^{\mathbf{L}} B) \rightarrow Lf^*(A \otimes_{\mathcal{O}_Y} B) \rightarrow f^*(A \otimes_{\mathcal{O}_Y} B)$$

and maps

$$Lf^*(A) \otimes_{\mathcal{O}_X}^{\mathbf{L}} Lf^*(B) \rightarrow f^*(A) \otimes_{\mathcal{O}_X}^{\mathbf{L}} f^*(B) \rightarrow f^*(A) \otimes_{\mathcal{O}_X} f^*(B),$$

all of which are natural in A and in B . The two RHS's are naturally isomorphic in A and B . If we take A and B to be bounded above complexes of flat sheaves, then the first maps in the bottom and top row are isomorphisms. The second in the top row is also an isomorphism because the tensor product of flat sheaves is flat, and the second in the bottom row is an isomorphism because the pullback of flat modules is flat. Thus we obtain a canonical isomorphism on $D^-(\mathcal{O}_Y) \times D^-(\mathcal{O}_Y)$. $\square$

Remark 2.2.18 Let $A \rightarrow B$ be a morphism in $D(\mathcal{O}_X)$, and suppose that $A_x \rightarrow B_x$ is an isomorphism in $D(\mathcal{O}_{X,x})$ for every $x \in X$. Since taking stalks is exact and termwise, this implies that $H^n(A) \rightarrow H^n(B)$ is a stalkwise isomorphism, and hence an isomorphism of sheaves. Thus $A \rightarrow B$ is an isomorphism. Therefore we can often argue on stalks even in the derived category.

Remark 2.2.19 Let $A \rightarrow B$ be a morphism in $D^+(\mathcal{O}_X)$, and suppose that $R\Gamma(U, A) \rightarrow R\Gamma(U, B)$ is an isomorphism in $D^+(\mathcal{O}_X(X))$ (or $D^+(\mathcal{O}_X(U))$), or even $D^+(\mathbb{Z})$ for every open $U \subset X$. We claim $A \rightarrow B$ is an isomorphism. We may assume $A = \mathcal{I}_1^\bullet$ and $B = \mathcal{I}_2^\bullet$ are in $K^+(\mathcal{I}_{\mathcal{O}_X})$, and then since $\mathcal{I}_2^\bullet$ is a bounded below complex of injectives, the map $\mathcal{I}_1^\bullet \rightarrow \mathcal{I}_2^\bullet$ comes from a chain map. Then for each n , $\mathcal{I}_1^n(U) \rightarrow \mathcal{I}_2^n(U)$ is an isomorphism for every U , so $\mathcal{I}_1 \xrightarrow{\sim} \mathcal{I}_2$ in $K^+(\mathcal{O}_X)$, which implies the result.

Note that for this result, we only need to check on all opens U in a basis for X , because that is all we used in the last step of the proof. Moreover, it is of course enough to show that $\mathbb{H}^n(U, A) \rightarrow \mathbb{H}^n(U, B)$ is an isomorphism for all n and all U in a basis, because that implies the condition in the statement.

Remark 2.2.20 Let $\varphi: \mathcal{O}_1 \rightarrow \mathcal{O}_2$ be a morphism of sheaves of rings on X , so that we get an associated morphism $f: (X, \mathcal{O}_2) \rightarrow (X, \mathcal{O}_1)$ of ringed spaces. We have already discussed the fact that f_* is exact in this case and is just restriction of scalars. Since $f^{-1} = \mathrm{Id}$, $f^*(\mathcal{F}) = \mathcal{F} \otimes_{\mathcal{O}_1} \mathcal{O}_2$ is extension of

scalars. Then we have a canonical identification $f_*(\mathcal{F} \otimes_{\mathcal{O}_1} \mathcal{O}_2) = \mathcal{F} \otimes_{\mathcal{O}_1} \mathcal{O}_2$, where on the RHS we mean as $\mathcal{O}_1$ -modules. Since f_* is exact, this implies that $f_* Lf^*(A) = A \otimes_{\mathcal{O}_1}^{\mathbf{L}} \mathcal{O}_2$, so we often write $(-) \otimes_{\mathcal{O}_1}^{\mathbf{L}} \mathcal{O}_2$, for derived change of scalars.

Now let $f: X \rightarrow Y$ be a morphism, and observe that the functor f^{-1} is exact, so descends to the derived categories. We may write f as the composite

$$(X, \mathcal{O}_X) \rightarrow (X, f^{-1}(\mathcal{O}_Y)) \rightarrow (Y, \mathcal{O}_Y)$$

where the first map is induced by $f^{-1}\mathcal{O}_Y \rightarrow \mathcal{O}_X$, and the second is given on spaces by f and the morphism of sheaves $f^{-1}\mathcal{O}_Y \rightarrow f^{-1}\mathcal{O}_Y$ is the identity. The second map is flat, so we are justified in writing

$$Lf^*(A) = f^{-1}(A) \otimes_{f^{-1}\mathcal{O}_Y}^{\mathbf{L}} \mathcal{O}_X$$

according to the convention above.

A simple and practical example of left derived functor is the closed-embedding projection formula. Let $i: Z \rightarrow X$ be a morphism of ringed spaces which is a closed embedding on the underlying topological spaces. If $\mathcal{F}$ is a sheaf on Z and $\mathcal{G}$ is a sheaf on Y , then since $i^*(i_*\mathcal{F} \otimes \mathcal{G}) = i^*i_*\mathcal{F} \otimes i^*\mathcal{G}$, the counit map induces a morphism $i^*(i_*\mathcal{F} \otimes \mathcal{G}) \rightarrow \mathcal{F} \otimes i^*\mathcal{G}$, and thus a morphism $i_*\mathcal{F} \otimes \mathcal{G} \rightarrow i_*(\mathcal{F} \otimes i^*\mathcal{G})$ by adjunction.

Consider the stalk at a point $x \notin i(Z)$: the RHS is 0, and the LHS is $0 \otimes \mathcal{G} = 0$. On the other hand, if $x = i(z)$, then the stalk at x on either side is

$$\mathcal{F}_z \otimes_{\mathcal{O}_{X, i(z)}} \mathcal{G}_{i(z)} \rightarrow \mathcal{F}_z \otimes_{\mathcal{O}_{Z, z}} (\mathcal{G}_z \otimes_{\mathcal{O}_{X, i(z)}} \mathcal{O}_{Z, z}),$$

which is an isomorphism. Thus the natural projection map is an isomorphism. Since i_* is exact, this likewise induces natural morphisms

$$i_*(A) \otimes_{\mathcal{O}_X}^{\mathbf{L}} B \rightarrow i_*A \otimes_{\mathcal{O}_X} B \cong i_*(A \otimes_{\mathcal{O}_Z} i^*B) \leftarrow i_*(A \otimes_{\mathcal{O}_Z}^{\mathbf{L}} Li^*B).$$

If we replace B with a bounded above complex of flat modules, the left map is an isomorphism, and i^*B is a bounded above complex of flat $\mathcal{O}_Z$ -modules, so the right-hand map is also an isomorphism. Thus we deduce the derived projection formula:

Lemma 2.2.21 *Let $i: Z \rightarrow X$ be a closed topological embedding, and $A \in D^-(\mathcal{O}_Z)$, $B \in D^-(\mathcal{O}_X)$. Then the natural map*

$$i_*A \otimes_{\mathcal{O}_X}^{\mathbf{L}} B \rightarrow i_*(A \otimes_{\mathcal{O}_Z}^{\mathbf{L}} Li^*B)$$

is an isomorphism.

Because of this result, combined with the fact that hypercohomology is invariant under pushing forward by closed embeddings, we often drop the i_* and just implicitly view an object $A \in D^-(\mathcal{O}_Z)$ as an object of $D^-(\mathcal{O}_X)$. It is separately also true that $i_*: D^-(\mathcal{O}_Z) \rightarrow D^-(\mathcal{O}_X)$ is a fully faithful embedding, but we won't discuss this yet.

Example 2.2.22 Since i_* is exact, it commutes with cohomology (of complexes), and so

$$H^p(i_*A \otimes_{\mathcal{O}_X}^{\mathbf{L}} B) = i_*H^p(A \otimes_{\mathcal{O}_Z}^{\mathbf{L}} Li^*B).$$

In particular, for sheaves $\mathcal{F}$ and $\mathcal{G}$ we have

$$\mathrm{Tor}_p^{\mathcal{O}_X}(i_*\mathcal{F}, \mathcal{G}) = i_*\mathrm{Tor}_p^{\mathcal{O}_Z}(\mathcal{F}, Li^*\mathcal{G})$$

for all p . Now on the other hand, we have shown that i_* actually commutes with $R\Gamma$. So, suppose that $A \otimes_{\mathcal{O}_Z}^{\mathbf{L}} Li^*B \in D^b(\mathcal{O}_Z)$. Thus when we apply i_* we land in $D^b(\mathcal{O}_X)$, and therefore

$$R\Gamma(X, i_*A \otimes_{\mathcal{O}_X}^{\mathbf{L}} B) = R\Gamma(Z, A \otimes_{\mathcal{O}_Z}^{\mathbf{L}} Li^*B)$$

viewed in $D^+(\mathcal{O}_X(X))$.

Example 2.2.23 Suppose $i_1: Z_1 \rightarrow X$ and $i_2: Z_2 \rightarrow X$ are two closed embeddings, and consider the structure sheaves $i_{1,*}\mathcal{O}_{Z_1}$ and $i_{2,*}\mathcal{O}_{Z_2}$ as sheaves on X . In scheme theory, we often want to think of their derived tensor product as a derived intersection product, which generalizes the fact that their ordinary product is the structure sheaf of their scheme-theoretic intersection. Then the projection formula shows that:

$$\begin{aligned} i_{1,*}\mathcal{O}_{Z_1} \otimes_{\mathcal{O}_X}^{\mathbf{L}} i_{2,*}\mathcal{O}_{Z_2} &= i_{1,*}(\mathcal{O}_{Z_1} \otimes_{\mathcal{O}_{Z_1}}^{\mathbf{L}} Li_1^*i_{2,*}\mathcal{O}_{Z_2}) \\ &= i_{1,*}(Li_1^*i_{2,*}\mathcal{O}_{Z_2}) \\ &= i_{2,*}(Li_2^*i_{1,*}\mathcal{O}_{Z_1}) \end{aligned}$$

In particular, we can compute the derived intersection by pulling back $\mathcal{O}_{Z_2}$ to Z_1 or pulling back $\mathcal{O}_{Z_1}$ to Z_2 , and we get the same answer.

For example, suppose Z_1 and Z_2 are effective divisors on a scheme X . Then we have two term resolutions

$$i_{1,*}\mathcal{O}_{Z_1} \simeq [\mathcal{O}_X(-Z_1) \hookrightarrow \mathcal{O}_X][1] \quad i_{2,*}\mathcal{O}_{Z_2} \simeq [\mathcal{O}_X(-Z_2) \hookrightarrow \mathcal{O}_X][1]$$

by line bundles, which are in particular flat resolutions. So we have a few options for computing the derived intersection. If we replace both sides in the definition by resolutions, then we will obtain a three term complex

$$[\mathcal{O}_X(-Z_1 - Z_2) \rightarrow \mathcal{O}_X(Z_1) \oplus \mathcal{O}_X(Z_2) \rightarrow \mathcal{O}_X][2]$$

by definition of the tensor product. This is most ideal if we need to compute something like $\mathbf{R}\mathcal{H}om$ or further pullback, because it is a complex of locally frees. Alternatively, we can directly apply e.g. Li_1^* to this, which gives the complex

$$[\mathcal{O}_X(-Z_2)|_{Z_1} \rightarrow \mathcal{O}_{Z_1}][1],$$

which is best suited for directly computing cohomology.

The most interesting statements usually arise from relating left and right derived functors. This often happens through adjunctions. For example, let

$f: X \rightarrow Y$ be a morphism, then we have the standard adjunction $f^* \dashv f_*$. Therefore, by our results on derived adjunctions, for $A \in D^-(\mathcal{O}_Y)$ and $B \in D^+(\mathcal{O}_X)$ we have a canonical isomorphism

$$\mathrm{RHom}_{\mathcal{O}_X}(Lf^*A, B) \cong \mathrm{RHom}_{\mathcal{O}_Y}(A, Rf_*B)$$

in $D^+(\mathcal{O}_Y(Y))$, where the LHS is viewed via restriction of scalars along $\mathcal{O}_Y(Y) \rightarrow \mathcal{O}_X(X)$. We would like to generalize this statement to something on the level of sheaves.

To do this, first consider the sheaf-level adjunction; given a morphism $f: X \rightarrow Y$, applying the sheaf adjunction to $f^{-1}(U) \rightarrow U$ and recalling the behavior of pullback and pushforward with respect to this, we deduce a natural isomorphism

$$\mathrm{Hom}_{\mathcal{O}_{f^{-1}(U)}}(f^*\mathcal{F}|_{f^{-1}(U)}, \mathcal{G}|_{f^{-1}(U)}) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{O}_U}(\mathcal{F}|_U, f_*\mathcal{G}|_U).$$

This commutes with the restriction maps (i.e. is natural in U as well), so it induces an isomorphism

$$f_* \mathrm{Hom}_{\mathcal{O}_X}(f^*\mathcal{F}, \mathcal{G}) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{F}, f_*\mathcal{G})$$

of $\mathcal{O}_Y$ -modules, by definition of the relevant objects, again natural in $\mathcal{F}$ and $\mathcal{G}$. By taking direct sums of both sides, this immediately generalizes to the case where $\mathcal{F}^\bullet$ is in $K^-(\mathcal{O}_Y)$ and $\mathcal{G}^\bullet \in K^+(\mathcal{O}_X)$. Now, we have natural morphisms

$$\begin{array}{ccc} f_* \mathrm{Hom}_{\mathcal{O}_X}(f^*\mathcal{F}^\bullet, \mathcal{G}^\bullet) & \xrightarrow{\cong} & \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{F}^\bullet, f_*\mathcal{G}^\bullet) \\ \downarrow & & \downarrow \\ Rf_* \mathrm{Hom}_{\mathcal{O}_X}(f^*\mathcal{F}^\bullet, \mathcal{G}^\bullet) & & \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{F}^\bullet, Rf_*\mathcal{G}^\bullet) \\ \downarrow & & \downarrow \\ Rf_* \mathbf{RHom}_{\mathcal{O}_X}(f^*\mathcal{F}^\bullet, \mathcal{G}^\bullet) & & \mathbf{RHom}_{\mathcal{O}_Y}(\mathcal{F}^\bullet, Rf_*\mathcal{G}^\bullet) \\ \downarrow & & \\ Rf_* \mathbf{RHom}_{\mathcal{O}_X}(Lf^*\mathcal{F}^\bullet, \mathcal{G}^\bullet) & & \end{array}$$

as functors $K^-(\mathcal{O}_Y) \times K^+(\mathcal{O}_X) \rightarrow D^+(\mathcal{O}_Y)$. If we take $\mathcal{G}^\bullet$ to be a complex of injectives, then the two upper-most maps on the left are isomorphisms, the first one because $\mathrm{Hom}_{\mathcal{O}_X}(-, \mathcal{G}^\bullet)$ is flasque by [Lemma 2.2.6](#). If we also take $\mathcal{F}^\bullet$ to be a bounded above complex of flats, then $Lf^*\mathcal{F}^\bullet \rightarrow f^*\mathcal{F}^\bullet$ is an isomorphism, so the bottom map on the left is an isomorphism. Therefore, this induces a natural map

$$Rf_* \mathbf{RHom}_{\mathcal{O}_X}(Lf^*A, B) \rightarrow \mathbf{RHom}_{\mathcal{O}_Y}(A, Rf_*B) \quad (*)$$

in $D^+(\mathcal{O}_Y)$, natural in $A \in D^-(\mathcal{O}_Y)$ and $B \in D^+(\mathcal{O}_X)$. We note at this stage that it does not obviously follow from the diagram that this is an isomorphism; a bounded above complex of flats in the first variable doesn't necessarily compute $\mathbf{RHom}$, and it is not clear if the pushforward of a bounded below complex of injectives (which is only complex of flasques) computes $\mathbf{RHom}$. Yet nonetheless we still can prove the following theorem by indirect means:

Proposition 2.2.24 *For $A \in D^-(\mathcal{O}_Y)$ and $B \in D^+(\mathcal{O}_X)$, the map $(*)$ is an isomorphism*

$$Rf_* \mathbf{RHom}_{\mathcal{O}_X}(Lf^*A, B) \xrightarrow{\sim} \mathbf{RHom}_{\mathcal{O}_Y}(A, Rf_*B)$$

in $D^+(\mathcal{O}_Y)$, natural in A and B .

Proof We are going to apply [Remark 2.2.19](#), which allows us to check this after applying $R\Gamma(U, -)$ for every $U \subset Y$. But because of the properties we have proven regarding the relation of pushforward and pullback with restriction, doing this gives a map

$$\mathbf{RHom}_{f^{-1}(U)}(Lg^*(A|_U), B|_{f^{-1}(U)}) \rightarrow \mathbf{RHom}_{\mathcal{O}_U}(A|_U, Rg_*(B|_{f^{-1}(U)}))$$

where g is the restricted map $f^{-1}(U) \rightarrow U$. This is the same map as the derived adjunction applied to g , because both were defined using the adjunction $g^* \dashv g_*$, as one can check easily. $\square$

We note that this is a nice example of using derived categories, because its statement was not trivial after replacing with a complex of injectives or flats. We also note that, because we have ignored unbounded derived categories, the isomorphism is not induced by any unit or counit map, but instead we describe it as the unique extension of the underived formula upon restricting to injectives and flats.

Example 2.2.25 Let $A = \mathcal{F}$ be a sheaf, and take the map f to be the inclusion $x \hookrightarrow X$ of a point where the domain has structure sheaf $\mathcal{O}_{X,x}$. Then f^* is the functor $(-)_x$ which is exact, and f_* is the skyscraper sheaf functor, which is also exact. So, both commute with cohomology, and since sheaf ext on a point is just module ext, we get

$$f_* \mathrm{Ext}_{\mathcal{O}_{X,x}}^p(\mathcal{F}_x, M) = \mathrm{Ext}_{\mathcal{O}_X}^p(\mathcal{F}, f_*M)$$

for any $\mathcal{O}_{X,x}$ -module M , so $\mathrm{Ext}_{\mathcal{O}_{X,x}}^p(\mathcal{F}_x, M) = \mathrm{Ext}_{\mathcal{O}_X}^p(\mathcal{F}, f_*M)_x$. In particular, if $\mathcal{F}$ is internally projective, then $\mathcal{F}_x$ is projective for every $x \in X$. E.g. if $\mathcal{F}$ is also of finite presentation and X is a locally ringed space, then $\mathcal{F}$ is a locally free sheaf of finite rank. So, being internally projective is usually equivalent to being a vector bundle.

Example 2.2.26 Let us record the fact from the example above that if we denote the skyscraper sheaf functor by Sk_x , which descends to the derived category, then

$$\mathrm{Sk}_x \mathbf{RHom}_{\mathcal{O}_{X,x}}(A_x, B) = \mathbf{RHom}_{\mathcal{O}_X}(A, \mathrm{Sk}_x B)$$

for $B \in D^+(\mathcal{O}_{X,x})$ and $A \in D^-(\mathcal{O}_X)$. In particular, suppose A is itself of the form $\mathrm{Sk}_x A$ for $A \in D^-(\mathcal{O}_{X,x})$, then

$$\mathrm{Sk}_x \mathbf{RHom}_{\mathcal{O}_{X,x}}(A, B) = \mathbf{RHom}_{\mathcal{O}_X}(\mathrm{Sk}_x A, \mathrm{Sk}_x B).$$

As an aside, we note that the usual derived adjunction formula tells us that $\mathbf{R}\mathrm{Hom}_{\mathcal{O}_{X,x}}(A_x, B) = \mathbf{R}\mathrm{Hom}_{\mathcal{O}_X}(A, \mathrm{Sk}_x B)$, which is an incarnation of the fact that skyscraper sheaves are (clearly) flasque.

Remark 2.2.27 Let A be a bounded above complex of flats, and B a bounded below complex of injectives. Now we can verify that

$$\underline{\mathrm{Hom}}_{\mathcal{O}_Y}(A, f_* B) \rightarrow \mathbf{R}\mathrm{Hom}_{\mathcal{O}_Y}(A, Rf_* B)$$

is in fact an isomorphism. Indeed by [Prop. 2.2.24](#), it suffices to check that

$$\underline{\mathrm{Hom}}_{\mathcal{O}_Y}(f^* A, B) \rightarrow Rf_* \mathbf{R}\mathrm{Hom}_{\mathcal{O}_Y}(Lf^* A, B)$$

is an isomorphism, which we already showed in this case.

For the last result in this section, let us show the easier proposition that the tensor-hom adjunction derives well. Recall that from the adjunction, one immediately proves

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}_1 \otimes_{\mathcal{O}_X} \mathcal{F}_2, \mathcal{F}_3) \cong \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}_1, \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}_2, \mathcal{F}_3))$$

Now suppose $\mathcal{F}_1^\bullet$, $\mathcal{F}_2^\bullet$, and $\mathcal{F}_3^\bullet$ are (arbitrary) complexes of $\mathcal{O}_X$ -modules. Then

$$\begin{aligned} \mathrm{Hom}_{\mathcal{O}_X}\left(\bigoplus_{p+q=n} \mathcal{F}_1^p \otimes_{\mathcal{O}_X} \mathcal{F}_2^q, \mathcal{F}_3^r\right) &\cong \prod_{p+q=n} \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}_1^p \otimes_{\mathcal{O}_X} \mathcal{F}_2^q, \mathcal{F}_3^r) \\ &\cong \prod_{p+q=n} \mathrm{Hom}_{\mathcal{O}_X}\left(\mathcal{F}_1^p, \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}_2^q, \mathcal{F}_3^r)\right). \end{aligned}$$

Now take the product over all n , and set $r = n + m$; thus we will get

$$\begin{aligned} \underline{\mathrm{Hom}}_{\mathcal{O}_X}^m(\mathcal{F}_1^\bullet \otimes_{\mathcal{O}_X} \mathcal{F}_2^\bullet, \mathcal{F}_3^\bullet) &\cong \prod_{p,q} \mathrm{Hom}_{\mathcal{O}_X}\left(\mathcal{F}_1^p, \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}_2^q, \mathcal{F}_3^{p+q+m})\right) \\ &\cong \prod_p \mathrm{Hom}_{\mathcal{O}_X}\left(\mathcal{F}_1^p, \underline{\mathrm{Hom}}_{\mathcal{O}_X}^{p+m}(\mathcal{F}_2^\bullet, \mathcal{F}_3^\bullet)\right) \\ &\cong \underline{\mathrm{Hom}}_{\mathcal{O}_X}^m\left(\mathcal{F}_1^\bullet, \underline{\mathrm{Hom}}_{\mathcal{O}_X}(\mathcal{F}_2^\bullet, \mathcal{F}_3^\bullet)\right), \end{aligned}$$

which gives a tensor-hom adjunction in the category $\mathrm{Ch}(\mathcal{O}_X)$. Now we bring this to the derived category.

Proposition 2.2.28 *Let X be a ringed space. There is a canonical isomorphism*

$$\mathbf{R}\mathrm{Hom}_{\mathcal{O}_X}(A \otimes_{\mathcal{O}_X}^{\mathbf{L}} B, C) \cong \mathbf{R}\mathrm{Hom}_{\mathcal{O}_X}\left(A, \mathbf{R}\mathrm{Hom}_{\mathcal{O}_X}(B, C)\right)$$

in $D^+(\mathcal{O}_X)$ natural in $A, B \in D^-(\mathcal{O}_X)$ and $C \in D^+(\mathcal{O}_X)$.

Proof We will do this more quickly than the previous results. Choosing representatives, the natural maps $A \otimes_{\mathcal{O}_X}^{\mathbf{L}} B \rightarrow A \otimes_{\mathcal{O}_X} B$ and the natural maps

from $\underline{\mathcal{H}om}$ to $\mathbf{R}\mathcal{H}om$, induce maps

$$\underline{\mathcal{H}om}_{\mathcal{O}_X}(A \otimes_{\mathcal{O}_X} B, C) \rightarrow \mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A \otimes_{\mathcal{O}_X}^{\mathbf{L}} B, C)$$

and

$$\underline{\mathcal{H}om}_{\mathcal{O}_X}\left(A, \underline{\mathcal{H}om}_{\mathcal{O}_X}(B, C)\right) \rightarrow \mathbf{R}\mathcal{H}om_{\mathcal{O}_X}\left(A, \mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(B, C)\right).$$

By the result in $\mathrm{Ch}(\mathcal{O}_X)$, the two LHS's are identified. Now, take B to be a bounded above complex of flat $\mathcal{O}_X$ -modules, and C to be a bounded below complex of injectives. Then the top map is an isomorphism. Moreover, by [Lemma 2.2.6\(d\)](#), each $\underline{\mathcal{H}om}_{\mathcal{O}_X}(B, C)$ is a bounded below complex of injectives, so the bottom map is an isomorphism too. $\square$

In particular, under the same assumptions for A, B, C , we have

$$\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A \otimes_{\mathcal{O}_X}^{\mathbf{L}} B, C) \cong \mathbf{R}\mathcal{H}om_{\mathcal{O}_X}\left(A, \mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(B, C)\right).$$

This is a result that we already knew if $B \in \mathrm{Mod}(\mathcal{O}_X)$, so this is a direct generalization of that.

2.2.2 Appendix: Cohomology of a Noetherian space

The goal of this short section is to prove the following theorem:

Theorem 2.2.29 *Let X be a ringed space whose underlying space is Noetherian, and suppose $\dim(X) = d$. Then $H^p(X, \mathcal{F}) = 0$ for all $p > d$ and $\mathcal{F} \in \mathrm{Mod}(\mathcal{O}_X)$.*

Corollary 2.2.30 *Let X be a Noetherian ringed space of dimension d .*

- (a) *For $A \in D^{[a,b]}(\mathcal{O}_X)$, $R\Gamma(X, A) \in D^{[a,b+d]}(\mathcal{O}_X(X))$. In particular $R\Gamma$ restricts to a functor $D^b(\mathcal{O}_X) \rightarrow D^b(\mathcal{O}_X(X))$.*
- (b) *Suppose $f: X \rightarrow Y$ is a morphism. For $A \in D^{[a,b]}(\mathcal{O}_X)$, $Rf_* A \in D^{[a,b+d]}(\mathcal{O}_Y)$. In particular Rf_* restricts to a functor $D^b(\mathcal{O}_X) \rightarrow D^b(\mathcal{O}_Y)$.*

Proof (a) is a special case of (b). For each $\mathcal{F}$, $R^p f_*(\mathcal{F})$ is the sheafification of $U \mapsto H^p(f^{-1}(U), \mathcal{F})$, which for $p > d$ is zero since $f^{-1}(U)$ is a Noetherian space of dimension $\leq d$. Thus the result follows from [Example 1.2.29](#). $\square$

The proof of the main result is by some trickery. First, to simplify notation, we may assume that $\mathcal{O}_X = \mathbb{Z}$, so that all sheaves considered are abelian (because cohomology commutes with change of ring). Thus all sheaves and ringed spaces in this section are considered abelian. Also, the number in the theorem (namely $\dim(X)$) is definitely achieved, as e.g. the example of $\mathbb{P}^n$ will show.

For this section, we will use the following special notation: for an open set $U \subset X$, we write $\underline{\mathbb{Z}}_U := j_* \mathbb{Z}$ where $j: U \hookrightarrow X$. If $\mathcal{F}$ is an abelian sheaf and $\{s_i\}$

is a collection of local sections with $s_i \in \mathcal{F}(U_i)$, we say that $\mathcal{F}$ is generated by $\{s_i\}$ if the induced map

$$\bigoplus_i \mathbb{Z}_{U_i} \rightarrow \mathcal{F} \rightarrow 0$$

is surjective. Warning: a sheaf generated by finitely many local sections need not be of finite type (e.g. $\mathbb{Z}_U$ itself has support U , so on a connected space it can't be of finite type).

But, what is true is that every sheaf $\mathcal{F}$ is a filtered colimit of sheaves generated by finitely many sections. Indeed, let $\{\mathcal{F}_\alpha\}$ be the collection of all subsheaves generated by finitely many sections of $\mathcal{F}$, considered as a filtered system with inclusion maps. Then the natural map $\varinjlim_\alpha \mathcal{F}_\alpha \rightarrow \mathcal{F}$ is an isomorphism. Indeed, its injective, and surjectivity can be checked on stalks; but this is clear because a local section $s \in \mathcal{F}(U)$ generates a subsheaf of $\mathcal{F}$ which is generated by finitely many sections.

The main lemma which actually does real ‘work’ is somehow the following simple one. We will explain how to prove the theorem after stating the lemma, and then prove the lemma at the end.

Lemma 2.2.31 *Let X be an irreducible Noetherian space and $\mathcal{F} \subset \mathbb{Z}$ an abelian subsheaf generated by finitely many local sections. Then there is filtration*

$$0 = \mathcal{F}_0 \subset \cdots \subset \mathcal{F}_n = \mathcal{F}$$

such that for $0 < i \leq n$, there are open sets $U, V \subset X$ and an exact sequence of the form $0 \rightarrow \mathbb{Z}_U \rightarrow \mathbb{Z}_V \rightarrow \mathcal{F}_i/\mathcal{F}_{i-1} \rightarrow 0$.

Intuitively, this lemma says that we can finitely filter the sheaf $\mathcal{F}$ so that the filtration quotients are generated by one local section with one local relation. As mentioned let's prove the main theorem. The main trick we'll apply aside from the lemma and the previous remark is that for a Noetherian space, as we showed, cohomology over all opens commutes with filtered colimits of sheaves.

Proof (Thm. 2.2.29) The proof is a sequence of a bunch of reductions. We prove by induction on d that if $\dim(X) \leq d$, then all sheaves $\mathcal{F}$ satisfy $H^p(X, \mathcal{F}) = 0$ for $p > d$.

1. We may assume X is irreducible. Indeed, induct on the (finite) number of components and assume the the base case is done. Let $X_1 \cup \cdots \cup X_k = X$ be the components and write $U = X \setminus X_1$, then we have the extension-by-zero sequence

$$0 \rightarrow j_{U,!}\mathcal{F}|_U \rightarrow \mathcal{F} \rightarrow i_{X_1,*}\mathcal{F}|_{X_1} \rightarrow 0.$$

Then $j_{U,!}\mathcal{F}|_U$ is supported in $U \subset X_2 \cup \cdots \cup X_k$ and $i_{X_1,*}\mathcal{F}|_{X_1}$ is supported in X_1 , so both are pushed forward from a Noetherian space of lower or equal to dimension, and with strictly less irreducible components.

2. Now if $\dim(X) = 0$, then since X is irreducible it is the only nonempty closed set: if there were another one Z , then it would contain some nonempty irreducible component Z_1 , and then $Z_1 \subset X$ would contradict

dimension 0 unless $Z = \emptyset$. But then X has only one nonempty open, so every sheaf on X is flasque.

3. For the remainder of the proof, assume X is irreducible and $d > 0$. Since cohomology commutes with filtered colimits, we may assume $\mathcal{F}$ is generated by finitely many local sections, say $\{s_1, \dots, s_k\}$. If we let $\mathcal{G} \subset \mathcal{F}$ be the subsheaf generated by $s_1, \dots, s_{k-1}$, then in the exact sequence $0 \rightarrow \mathcal{G} \rightarrow \mathcal{F} \rightarrow \mathcal{F}/\mathcal{G} \rightarrow 0$ the left and right sheaves are generated by strictly less than k local sections. Thus inductively we may assume $\mathcal{F}$ is generated by one local section.
4. We claim that we can actually assume that $\mathcal{F}$ is of the form $\mathbb{Z}_V$. Indeed, by (3) there is an exact sequence $0 \rightarrow \mathcal{G} \rightarrow \mathbb{Z}_U \rightarrow \mathcal{F} \rightarrow 0$ for some open set $U \subset X$, so under such an assumption $H^p(X, \mathcal{F}) \xrightarrow{\sim} H^{p+1}(X, \mathcal{G})$ for $p > d$, and hence by replacing $\mathcal{F}$ with $\mathcal{G}$ we may assume $\mathcal{F} \subset \mathbb{Z}_U \subset \mathbb{Z}$. Once again by writing $\mathcal{F}$ as filtered colimit of its subsheaves generated by finitely many sections, we may assume $\mathcal{F} \subset \mathbb{Z}$ is generated by finitely many sections. Thus we have a filtration $0 \subset \mathcal{F}_1 \subset \dots \subset \mathcal{F}_n = \mathcal{F}$ where the lemma and claimed assumption show that $H^p(X, \mathcal{F}_i/\mathcal{F}_{i-1}) = 0$ for $p > d$. Thus the sequence $0 \rightarrow \mathcal{F}_{i-1} \rightarrow \mathcal{F}_i \rightarrow \mathcal{F}_i/\mathcal{F}_{i-1} \rightarrow 0$ shows inductively that $H^p(X, \mathcal{F}_i) = 0$ for $p > d$, which includes the result.
5. Thus, let $V \subset X$ be a nonempty open and consider the extension-by-zero sequence $0 \rightarrow \mathbb{Z}_V \rightarrow \mathbb{Z} \rightarrow \mathcal{G} \rightarrow 0$ where $\mathcal{G}$ is supported on the closed subset $X \setminus V$. Since X is irreducible, and V is nonempty, $\dim(X \setminus V) \leq d-1$, and hence $H^p(X, \mathcal{G}) = 0$ for $p \geq d$. Thus, $H^p(X, \mathbb{Z}_V) \xrightarrow{\sim} H^p(X, \mathbb{Z}) = 0$ for $p > d$, however $\mathbb{Z}$ is flasque because every open set of X is irreducible and hence connected.

This proves the induction step as desired. $\square$

We will see a few arguments of this form in the sequel, where we argue by various inductions and sub-inductions, and keep reducing cohomology statements via exact sequences until we get to something trivial. Finally, the lemma:

Proof (of Lemma 2.2.31) Since X is irreducible, $\mathcal{F}(U) \subset \mathbb{Z}$ for all U , so $\mathcal{F}$ is generated by $(U_1, n_1), \dots, (U_k, n_k)$, where we may assume $n_i > 0$ for all i . Note that for any subcollection $\{(U_i, n_i) : i \in I\}$ of these generators, $\mathcal{F}(\bigcap_i U_i)$ contains the ideal $(n_i : i \in I) = (d)$ where $d = \gcd(n_i : i \in I)$; so by adding all such opens and gcds over all $I \subset \{1, \dots, k\}$, we may assume that for each $x \in X$, $\gcd(n_i : i \in \{i : x \in U_i\})$ is attained by some (U_i, n_i) containing x .

Let $\mathcal{F}_0 = 0$ and for each n let $\mathcal{F}_n$ be generated by $\{(U_i, n_i) : n_i \leq n\}$. Thus we get an ascending filtration as in the lemma statement which ends at $\mathcal{F}$. Write $U_{\leq n} = \bigcup_{i, n_i \leq n} U_i$, and note that

$$(\mathcal{F}_n)_x = (n_i : \{i \leq n : x \in U_i\}) = \gcd(n_i : \{i \leq n : x \in U_i\})\mathbb{Z}.$$

$(\mathcal{F}_n)_x = (\mathcal{F}_{n-1})_x = 0$ if $x \notin U_{\leq n}$, and $(\mathcal{F}_n)_x = (\mathcal{F}_{n-1})_x$ if $x \in U_{\leq n-1}$, because in the latter case the gcd already occurs as a generator at x with value $\leq n-1$.

In the remaining case $(\mathcal{F}_n)_x = n\mathbb{Z}$ but $(\mathcal{F}_{n-1})_x = 0$. Thus, the section n over $U_{\leq n}$ generates $\mathcal{F}_n/\mathcal{F}_{n-1}$, and the kernel of $\underline{\mathbb{Z}}_{U_{\leq n}} \rightarrow \mathcal{F}_n/\mathcal{F}_{n-1} \rightarrow 0$ is exactly the subsheaf $\underline{\mathbb{Z}}_{U_{\leq n-1}}$. $\square$

2.2.3 Pseudocoherent and perfect complexes

In this section, we start to focus on more advanced notions regarding the derived category $D(\mathcal{O}_X)$ where X is a ringed space. Pseudocoherent objects of the derived category will be a generalization of finitely presented $\mathcal{O}_X$ -modules, and perfect objects will be a generalization of finite locally free $\mathcal{O}_X$ -modules. In practice, pseudocoherence will not be very different from a finite type assumption because we mostly care about Noetherian schemes. Perfect complexes will not be very different from bounded complexes of vector bundles.

Before moving forward, let us also make the following convention: so far D^+ , D^- and D^b have referred to the localizations of the respective homotopy categories. We now will use these to refer to the essential images of these subcategories in $D(\mathcal{O}_X)$. In other words, these are the subcategories with cohomology bounded below, above, or in both directions. Indeed, one direction is clear, and for the other observe that the map from/to the smart truncation τ will be a quasi-isomorphism. We may also use the notation $D^{[a,b]}(\mathcal{O}_X)$ for objects whose cohomology is concentrated in the interval $[a, b]$. We want to start to view the derived category $D(\mathcal{O}_X)$ more abstractly, and think of its objects as more than just complexes. So, we use letters $A, B \in D(\mathcal{O}_X)$ when talking abstractly about its objects, and $\mathcal{F}^\bullet, \mathcal{G}^\bullet \in K(\mathcal{O}_X)$ when thinking about chain complexes of $\mathcal{O}_X$ -modules.

Here are the major definitions of this section.

Definition 2.2.32 An $\mathcal{O}_X$ -module $\mathcal{E}$ is *strictly perfect* if it is a direct summand of a finite free $\mathcal{O}_X$ -module. An object $\mathcal{E}^\bullet \in \text{Ch}(\mathcal{O}_X)$ is *strictly perfect* if it is bounded, and $\mathcal{E}^n$ is strictly perfect for all $n \in \mathbb{Z}$.

Definition 2.2.33 Let X be a ringed space.

1. A complex $\mathcal{F}^\bullet \in K(\mathcal{O}_X)$ is *perfect* if there is an open cover $\{U_i\}$ of X such that for each i , there exists a quasi-isomorphism $\mathcal{E}^\bullet \rightarrow \mathcal{F}|_{U_i}$ in $\text{Ch}(\mathcal{O}_{U_i})$ such that $\mathcal{E}$ is strictly perfect.
2. An object $A \in D(\mathcal{O}_X)$ is *perfect* if it is represented by a perfect complex. We denote by $\text{Perf}^*(\mathcal{O}_X) \subset D^*(\mathcal{O}_X)$ the full subcategory of perfect objects.

Definition 2.2.34 Let X be a ringed space and $n \in \mathbb{Z}$.

1. A complex $\mathcal{F}^\bullet \in K(\mathcal{O}_X)$ is *n-pseudocoherent* if there is an open cover $\{U_i\}$ of X such that for each i , there exists a morphism $\mathcal{E}^\bullet \rightarrow \mathcal{F}|_{U_i}$ in $\text{Ch}(\mathcal{O}_X)$ with $\mathcal{E}$ strictly perfect, such that $H^p(\mathcal{E}^\bullet) \rightarrow H^p(\mathcal{F}^\bullet|_{U_i})$ is an isomorphism for $j > m$ and an epimorphism for $j = m$.
2. A complex $\mathcal{F}^\bullet \in K(\mathcal{O}_X)$ is $(-\infty)$ -pseudocoherent, or just pseudocoherent, if it is n -pseudocoherent for all $n \in \mathbb{Z}$.
3. An object $A \in D(\mathcal{O}_X)$ is *n-pseudocoherent* if it is represented by an n -pseudocoherent complex for $n \in [-\infty, \infty)$.

Remark 2.2.35 Here are some basic observations about these definitions. First, a strictly perfect sheaf is clearly finitely presented, and thus over a locally ringed space it is finite locally free. In that case, it doesn't change anything if we define perfectness and pseudocoherence using finite free sheaves, or finite locally free sheaves. Second, if X is quasicompact, then a perfect object A is in $D^b(\mathcal{O}_X)$, because there is a finite open cover $\{U_i\}$ such that $H^p(A)|_{U_i} = 0$ for $p \gg 0$ and $p \ll 0$. Similarly, if X is quasicompact then a pseudocoherent object A is in $D^-(\mathcal{O}_X)$.

Here are the basic properties of strictly perfect complexes.

Lemma 2.2.36 *Let X be a ringed space.*

1. *If $\phi: \mathcal{E}_1^\bullet \rightarrow \mathcal{E}_2^\bullet$ is a morphism of strictly perfect complexes, then $\text{Cone}(\phi)$ is a strictly perfect complex.*
2. *If $\mathcal{E}_1^\bullet, \mathcal{E}_2^\bullet$ are strictly perfect complexes, then $\mathcal{E}_1^\bullet \otimes \mathcal{E}_2^\bullet$ is strictly perfect.*
3. *If $f: X \rightarrow Y$ is a morphism of ringed spaces and $\mathcal{E}^\bullet$ is strictly perfect on Y , then $f^*\mathcal{E}^\bullet$ is strictly perfect on X . In particular if $\mathcal{E}^\bullet$ is strictly perfect, then $\mathcal{E}^\bullet|_U$ is strictly perfect for any $U \subset X$.*
4. *A strictly perfect complex is a bounded complex of internally projective sheaves. In particular, if $\mathcal{E}^\bullet$ is perfect, then the natural map $\underline{\text{Hom}}_{\mathcal{O}_X}(\mathcal{E}^\bullet, -) \rightarrow \text{RHom}_{\mathcal{O}_X}(\mathcal{E}^\bullet, -)$ is an isomorphism in $D(\mathcal{O}_X)$.*

Proof It is clear from the definition that the direct sum of two strictly perfect $\mathcal{O}_X$ -modules is strictly perfect. Therefore (1) is trivial. This also reduces (2) to showing that if $\mathcal{F}, \mathcal{G}$ are strictly perfect then $\mathcal{F} \otimes \mathcal{G}$ is. Indeed if $\mathcal{F} \oplus \mathcal{F}' = \mathcal{O}_X^n$ and $\mathcal{G} \oplus \mathcal{G}' = \mathcal{O}_X^m$, then taking the tensor product of both sides and distributing has one summand as $\mathcal{F} \otimes \mathcal{G}$, and on the RHS we have $\mathcal{O}_X^{mn}$. (3) is clear because f^* takes $\mathcal{O}_Y^n$ to $\mathcal{O}_X^n$, and preserves direct sums. For (4), note that if $\mathcal{F} \oplus \mathcal{F}'$ is finite free, then $\text{Ext}^p(\mathcal{F}, -) \oplus \text{Ext}^p(\mathcal{F}', -) = 0$ for $p > 0$. $\square$

Next we have some results that we aim to apply to the strictly perfect case, but we can prove in more generality about internally projective objects.

Lemma 2.2.37 *Let $\phi: \mathcal{E}^\bullet \rightarrow \mathcal{F}^\bullet$ be a morphism of complexes with $\mathcal{E}^\bullet$ a bounded above complex of internally projective objects. Assume we're in one of the following situations:*

- (a) *$a \in \mathbb{Z}$ is such that $\mathcal{E}^p = 0$ for $p < a$, and $H^p(\mathcal{F}^\bullet) = 0$ for $p \geq a$.*
- (b) *$\mathcal{E}^\bullet$ is bounded, and $\mathcal{F}^\bullet$ is acyclic.*
- (c) *$\mathcal{F}^\bullet$ is acyclic and bounded below.*

Then there is an open cover $\{U_i\}$ of X such that $\phi|_{U_i}$ is nullhomotopic.

Proof We claim that in any of the cases, $H^0 \underline{\text{Hom}}_{\mathcal{O}_X}(\mathcal{E}^\bullet, \mathcal{F}^\bullet) = 0$. For cases (b) and (c), we showed this in [Example 2.2.9](#). For (a), just use the same spectral

sequence of [Example 2.2.9](#) slightly more carefully, which converges as expected because $\mathcal{E}^\bullet$ is bounded.

Now, observe that $H^0 \underline{\text{Hom}}_{\mathcal{O}_X}(\mathcal{E}^\bullet, \mathcal{F}^\bullet)$ is the sheafification of the presheaf

$$U \mapsto H^0 \underline{\text{Hom}}_{\mathcal{O}_U}(\mathcal{E}^\bullet|_U, \mathcal{F}^\bullet|_U) = \text{Hom}_{K(\mathcal{O}_U)}(\mathcal{E}^\bullet|_U, \mathcal{F}^\bullet|_U),$$

so the fact that this is zero means that every global section of this presheaf is zero on an open cover. $\square$

Corollary 2.2.38 *Let $\mathcal{E}^\bullet$ be a bounded complex of internally projective objects and $\mathcal{F}^\bullet$ a complex.*

1. *Suppose $\mathcal{E}^\bullet \in K^{[a, \infty)}(\mathcal{O}_X)$, and let $\phi: \mathcal{G}^\bullet \rightarrow \mathcal{F}^\bullet$ be a morphism of complexes for which $H^p(\phi)$ is an isomorphism for $p > a$, and an epimorphism for $p = a$. Then for any map $\alpha: \mathcal{E}^\bullet \rightarrow \mathcal{F}^\bullet$, there is an open cover $\{U_i\}$ of X and maps $\beta_i: \mathcal{E}^\bullet|_{U_i} \rightarrow \mathcal{G}^\bullet|_{U_i}$ such that $\phi|_{U_i} \beta_i = \alpha$ in $K(\mathcal{O}_U)$.*
2. *For any $\alpha \in \text{Hom}_{D(\mathcal{O}_X)}(\mathcal{E}^\bullet, \mathcal{F}^\bullet)$, there exists an open cover $\{U_i\}$ of X such that $\alpha|_{U_i}$ is given by a morphism of complexes $\mathcal{E}^\bullet|_{U_i} \rightarrow \mathcal{F}^\bullet|_{U_i}$.*
3. *If $\alpha: \mathcal{E}^\bullet \rightarrow \mathcal{F}^\bullet$ is a morphism of complexes which becomes 0 in $D(\mathcal{O}_X)$, then it is locally nullhomotopic.*

Proof (1) The distinguished triangle $\mathcal{G}^\bullet \rightarrow \mathcal{F}^\bullet \rightarrow \text{Cone}(\phi) \rightarrow \mathcal{G}^\bullet[1]$ shows that $H^p(\text{Cone}(\phi)) = 0$ for $p \geq a$. This also induces an exact sequence

$$\text{Hom}_{K(\mathcal{O}_X)}(\mathcal{E}^\bullet|_U, \mathcal{G}^\bullet|_U) \rightarrow \text{Hom}_K(\mathcal{E}^\bullet|_U, \mathcal{F}^\bullet|_U) \rightarrow \text{Hom}_K(\mathcal{E}^\bullet|_U, \text{Cone}(\phi)|_U)$$

because restriction is exact. By the lemma part (a), the composite map $\mathcal{E}^\bullet \rightarrow \text{Cone}(\phi)$ is locally 0, so on any such open cover, the exact sequence says that we can lift it to a map $\mathcal{E}^\bullet|_U \rightarrow \mathcal{G}^\bullet|_U$.

(2) Write α as a roof $\mathcal{E}^\bullet \rightarrow \mathcal{G}^\bullet \xleftarrow{q} \mathcal{F}^\bullet$ where q is a qis. Applying (1) to q , we can locally lift against q to get $\mathcal{E}^\bullet|_U \rightarrow \mathcal{F}^\bullet|_U$ making the diagram commute, which is therefore a representative for α .

(3) A map becomes zero iff there is a roof $\mathcal{E}^\bullet \rightarrow \mathcal{G}^\bullet \xleftarrow{q} \mathcal{F}^\bullet$ with the first map being zero, commuting with α . Again by (a) we can locally lift against q , which means that α has to locally be 0. $\square$

Now we are going to apply these results to m -pseudocoherent and perfect complexes. We'll first focus on m -pseudocoherence. First of all, as we have already explained, the condition in the definition of m -pseudocoherent on the map $\mathcal{E}^\bullet \rightarrow \mathcal{F}|_U$ is equivalent to the cone having no cohomology in degrees $\geq m$. We'll use this all the time.

Lemma 2.2.39 *Let X be a ringed space and $E \in D(\mathcal{O}_X)$.*

- (a) *Suppose that for each U in an open cover of X , there is a strictly perfect complex $\mathcal{E}^\bullet \in K(\mathcal{O}_U)$ and a morphism $\mathcal{E}^\bullet \rightarrow E|_U$ whose cone*

is in $D^{<m}(\mathcal{O}_U)$. Then E is m -pseudocoherent. Moreover, any complex representing E is m -pseudocoherent.

- (b) Suppose that for each U in an open cover of X , there is a strictly perfect complex $\mathcal{E}^\bullet \in K(\mathcal{O}_U)$ such that $\mathcal{E}|_U$ is isomorphic in $D(\mathcal{O}_U)$ to $\mathcal{E}^\bullet$. Then E is perfect. Moreover, any complex representing E is perfect.

Proof In either case, choose a representative $\mathcal{F}^\bullet$ for E , and consider the triangle $\mathcal{E}^\bullet \rightarrow \mathcal{F}^\bullet|_U$ in $D(\mathcal{O}_U)$. It suffices to show that $\mathcal{F}^\bullet$ is m -pseudocoherent or perfect. By Cor. 2.2.38, we may shrink U and assume $\mathcal{E}^\bullet \rightarrow \mathcal{F}^\bullet|_U$ is represented by a morphism of complexes, with the same hypothesis that either the cone is in $D^{<m}(\mathcal{O}_U)$ or it is 0. This is the definition of m -pseudocoherent or perfect, and the representative was arbitrary. $\square$

Next, there are some obvious properties of pseudocoherence which we write down.

Lemma 2.2.40 *Let X be a ringed space and $E \in D(\mathcal{O}_X)$.*

- (a) *If E is m -pseudocoherent, then it is k -pseudocoherent for all $k \geq m$.*
- (b) *If E is m -pseudocoherent, then $E[n]$ is $(m - n)$ -pseudocoherent.*
- (c) *If $E \in D^{<m}(\mathcal{O}_X)$, then E is m -pseudocoherent.*
- (d) *If E is perfect, then it is pseudocoherent.*

Proof (a) and (d) are by definition. (b) Locally $\mathcal{E}^\bullet[n] \rightarrow \mathcal{E}|_U[n] \rightarrow C[n] \rightarrow$ has $C[n] \in D^{<m-n}$ since $C \in D^{<m}$. (c) The triangle $0 \rightarrow E \rightarrow E \rightarrow 0$ has $E \in D^{<m}$. $\square$

Remark 2.2.41 In the following discussion, we will use the following facts (that we have essentially already discussed) often. First, a left derived functor LF (in the presence of an adapted class) sends $D^{\leq m}(\mathcal{A}) \rightarrow D^{\leq m}(\mathcal{B})$. The proof is that if we choose a representative in $K^{\leq m}(\mathcal{A})$, then we can choose the resolution in the adapted class to be in $K^{\leq m}$ as well. Next, specializing to $\text{Mod}(\mathcal{O}_X)$, the derived tensor product and $\mathbf{R}\mathcal{H}om$ preserve the following categories.

$$(-) \otimes_{\mathcal{O}_X}^{\mathbf{L}} (-): D^{\leq a}(\mathcal{O}_X) \times D^{\leq b}(\mathcal{O}_X) \rightarrow D^{\leq a+b}(\mathcal{O}_X).$$

Indeed, we can choose a bounded above complex of flats in $K^{\leq a}(\mathcal{O}_X)$ for the first variable, and a representative in $K^{\leq b}$ for the second, then the tensor product lands in $K^{\leq a+b}$. Similarly,

$$\mathbf{R}\mathcal{H}om: D^{\leq a}(\mathcal{O}_X)^{\text{op}} \times D^{\geq b}(\mathcal{O}_X) \rightarrow D^{-a+b}(\mathcal{O}_X),$$

because choosing a bounded below complex of injectives in $K^{\geq b}(\mathcal{O}_X)$ for the second variable and a representative in $K^{\leq a}$ for the first, any $m > b - a$ satisfies either $p > a$ or $p < m + b$.

Lemma 2.2.42 *Let $A \rightarrow B \rightarrow C \rightarrow A[1]$ be a distinguished triangle in $D(\mathcal{O}_X)$.*

- (a) *If A is $(m+1)$ -pseudocoherent and B is m -pseudocoherent, then C is m -pseudocoherent.*
- (b) *If A and C are m -pseudocoherent, then B is m -pseudocoherent.*
- (c) *If B is m -pseudocoherent and C is $(m-1)$ -pseudocoherent, then A is m -pseudocoherent.*

In particular, if any two of A , B , and C are pseudocoherent, then the third is pseudocoherent as well.

Proof The conclusion follows immediately from the three statements, and observe that (b) and (c) follow from (a) by rotation and the fact that $A[1]$ is m -pseudocoherent iff A is $(m+1)$ -pseudocoherent.

For (a), fix $x \in X$ so that on a nbhd U we have maps as in the solid part of the following diagram:

$$\begin{array}{ccccccc} \mathcal{F}^\bullet & \dashrightarrow & \mathcal{G}^\bullet & \dashrightarrow & \mathcal{H}^\bullet & \dashrightarrow & \mathcal{F}^\bullet[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ A|_U & \longrightarrow & B|_U & \longrightarrow & C|_U & \longrightarrow & A|_U[1] \end{array}$$

So, here $\mathcal{F}^\bullet, \mathcal{G}^\bullet$ are strictly perfect, and the maps to $A|_U$ and $B|_U$ preserve cohomology in degrees above $(m+1)$ and m respectively, and are surjections on those degrees. By throwing out lower degree terms we may assume $\mathcal{F}^\bullet \in K^{\geq m+1}$. Now represent B by a complex (but we use the same letter).

By further shrinking U , we may assume that the composite $\mathcal{F}^\bullet \rightarrow B|_U$ is represented by a morphism of complexes (Cor. 2.2.38(2)). Likewise, we may assume that $\mathcal{G}^\bullet \rightarrow B|_U$ is represented by a morphism of complexes by shrinking again. By the condition on $\mathcal{G}^\bullet \rightarrow B|_U$, the composite $\mathcal{F}^\bullet \rightarrow B|_U$ lifts to a map $\mathcal{F} \dashrightarrow \mathcal{G}$ by again shrinking U , by Cor. 2.2.38(1).

By the triangulated structure, this data can be completed to a morphism of triangles as shown in the diagram, where $\mathcal{H}^\bullet$ is strictly perfect as a cone of morphisms of strictly perfect complexes. Now consider the diagram of long exact sequences: the epimorphism 4-lemma shows that $H^m(\mathcal{H}^\bullet) \rightarrow H^m(C|_U)$ is surjective, and the 5-lemma shows that $H^p(\mathcal{H}^\bullet) \rightarrow H^p(C|_U)$ is an isomorphism for $p > m$. $\square$

This is a useful lemma because often we will try prove things by inducting along smart truncation triangles. Also, note that in particular, the pseudocoherent complexes form a full triangulated subcategory of $D(\mathcal{O}_X)$. The same is true for $\text{Perf}^*(\mathcal{O}_X)$, though this is much easier; we reduce immediately to the case of strictly perfect complexes, and we already know they are closed under cones. Here is one application:

Lemma 2.2.43 *For $m \leq 0$, an $\mathcal{O}_X$ -module $\mathcal{F}$ is m -pseudocoherent if and only for each point there is a nbhd U and an exact sequence*

$$\mathcal{O}_U^{n_m} \rightarrow \mathcal{O}_U^{n_{m+1}} \rightarrow \cdots \rightarrow \mathcal{O}_U^{n_{-1}} \rightarrow \mathcal{O}_U^{n_0} \rightarrow \mathcal{F}|_U \rightarrow 0$$

of finite free modules.

Proof Since a finite free module is strictly perfect, and such local resolution implies m -pseudocoherence. We remark that the only reason this isn't trivial is because we may not be in the locally ringed case. Conversely, note that if $\mathcal{F}$ is 0-pseudocoherent then it admits a surjection from a strictly perfect (so finite type) module, hence is finite type. So 0-pseudocoherent is equivalent to finite type.

To complete the proof, suppose that there is a resolution as in the lemma statement of length m , and assume $\mathcal{F}$ is $(m-1)$ -pseudocoherent. Split up into short exact sequences

$$0 \rightarrow \mathcal{G}^{i-1} \rightarrow \mathcal{O}_U^{n_i} \rightarrow \mathcal{G}^i \rightarrow 0$$

where $\mathcal{G}^0 = \mathcal{F}|_U$. It suffices to show that $\mathcal{G}^{m-1}$ is finite type, i.e. 0-pseudocoherent. Each sequence corresponds to a triangle in $D(\mathcal{O}_X)$, each $\mathcal{O}_U^{n_i}$ is $(-\infty)$ -pseudocoherent, and $\mathcal{G}^0$ is $(m-1)$ -pseudocoherent. Thus $\mathcal{G}^1$ is m -pseudocoherent, $\mathcal{G}^2$ is $(m+1)$ -pseudocoherent, and so on, so $\mathcal{G}^{m-1}$ is 0-pseudocoherent. $\square$

So, in particular, for modules 0-pseudocoherent is finite type, and (-1) -pseudocoherent is finite presentation.

Proposition 2.2.44 *Let $A \in D^{\leq a}(\mathcal{O}_X)$ and $B \in D^{\leq b}(\mathcal{O}_X)$.*

- (a) *If A and B are perfect, then $A \otimes_{\mathcal{O}_X}^{\mathbf{L}} B$ is perfect.*
- (b) *If A is m -pseudocoherent and B is n -pseudocoherent, then $A \otimes_{\mathcal{O}_X}^{\mathbf{L}} B$ is $\max(a+n, b+m)$ -pseudocoherent.*
- (c) *Now let $f: X \rightarrow Y$ be a morphism of ringed spaces and $A \in D^-(\mathcal{O}_Y)$. If A is m -pseudocoherent (resp. perfect), then $Lf^*(A)$ is m -pseudocoherent (resp. perfect).*

Proof For (a), we may choose U in among an open cover so that $A|_U$ and $B|_U$ are isomorphic in $D(\mathcal{O}_X)$ to strictly perfect complexes. Then by locality of tensor product, flatness of strictly perfect complexes, and the fact that strictly perfect complexes are preserved under tensor product (Lemma 2.2.36), we are done. For (c) in the perfect case, the same thing works as f^* preserves strictly perfect complexes.

For (c) in the m -pseudocoherent case, let $\mathcal{F}^\bullet$ be strictly perfect on U so that we have a triangle $\mathcal{F}^\bullet \rightarrow A|_U \rightarrow C \rightarrow \mathcal{F}^\bullet[1]$ with $C \in D^{< m}(\mathcal{O}_U)$. Applying Lf^* gives the distinguished triangle

$$f^* \mathcal{F}^\bullet \rightarrow Lf^*(A)|_U \rightarrow Lf^* C \rightarrow$$

and we conclude because $f^*\mathcal{F}^\bullet$ is strictly perfect on $f^{-1}(U)$, $Lf^*(C) \in D^{<m}(\mathcal{O}_{f^{-1}(U)})$.

Now do (b): we'll suppress open sets. We may also assume that $n \leq b$; if not, then $a+n > a+b$, and since $A \otimes_{\mathcal{O}_X}^{\mathbf{L}} B \in D^{\leq a+b}$, it is automatically $(a+n)$ (hence $\max(a+n, b+m)$)-pseudocoherent. Locally on X , we have distinguished triangles $\mathcal{E}_1^\bullet \rightarrow B \rightarrow C_1 \rightarrow$ and $\mathcal{E}_2^\bullet \rightarrow A \rightarrow C_2 \rightarrow$ with $\mathcal{E}_1^\bullet$ and $\mathcal{E}_2^\bullet$ strictly perfect, and $C_1 \in D^{<m}$, $C_2 \in D^{<n}$. Since $n \leq b$, $C_2 \in D^{<b}$, so $\mathcal{E}_2^\bullet \in D^{\leq b}$.

By exactness of $\otimes^{\mathbf{L}}$, we get three distinguished triangles:

$$\begin{aligned} \mathcal{E}_1^\bullet \otimes_{\mathcal{O}_X}^{\mathbf{L}} \mathcal{E}_2^\bullet &\rightarrow A \otimes_{\mathcal{O}_X}^{\mathbf{L}} \mathcal{E}_2^\bullet \rightarrow C_1 \otimes_{\mathcal{O}_X}^{\mathbf{L}} \mathcal{E}_2^\bullet \rightarrow \\ A \otimes_{\mathcal{O}_X}^{\mathbf{L}} \mathcal{E}_2^\bullet &\rightarrow A \otimes_{\mathcal{O}_X}^{\mathbf{L}} B \rightarrow A \otimes_{\mathcal{O}_X}^{\mathbf{L}} C_2 \rightarrow \\ \mathcal{E}_1^\bullet \otimes_{\mathcal{O}_X}^{\mathbf{L}} \mathcal{E}_2^\bullet &\rightarrow A \otimes_{\mathcal{O}_X}^{\mathbf{L}} B \rightarrow C_3 \rightarrow \end{aligned}$$

Here, $C_1 \otimes_{\mathcal{O}_X}^{\mathbf{L}} \mathcal{E}_2^\bullet \in D^{<m+b}$ and $A \otimes_{\mathcal{O}_X}^{\mathbf{L}} C_2 \in D^{<a+n}$, so both are in $D^{<\max(a+n, m+b)}$. By the octahedral axiom, there is a distinguished triangle

$$C_1 \otimes_{\mathcal{O}_X}^{\mathbf{L}} \mathcal{E}_2^\bullet \rightarrow C_3 \rightarrow A \otimes_{\mathcal{O}_X}^{\mathbf{L}} C_2 \rightarrow,$$

so $C_3 \in D^{<\max(a+n, m+b)}$. But $\mathcal{E}_1^\bullet \otimes_{\mathcal{O}_X}^{\mathbf{L}} \mathcal{E}_2^\bullet = \mathcal{E}_1^\bullet \otimes \mathcal{E}_2^\bullet$ is strictly perfect by [Lemma 2.2.36](#), so we are done. $\square$

Now let us prove a lemma which will be the key method in which we actually use pseudocoherent and perfect objects of the derived category.

Lemma 2.2.45 *Let $\mathcal{F}^\bullet, \mathcal{G}^\bullet, \mathcal{H}^\bullet \in \text{Ch}(\mathcal{O}_Y)$, and let $f: X \rightarrow Y$ be a morphism of ringed spaces.*

(a) *There are canonical natural maps*

$$\begin{aligned} \text{ev}: \underline{\text{Hom}}_{\mathcal{O}_Y}(\mathcal{F}^\bullet, \mathcal{G}^\bullet) \otimes \mathcal{F}^\bullet &\rightarrow \mathcal{G}^\bullet \\ \underline{\text{Hom}}_{\mathcal{O}_Y}(\mathcal{F}^\bullet, \mathcal{G}^\bullet) \otimes \mathcal{H}^\bullet &\rightarrow \underline{\text{Hom}}_{\mathcal{O}_Y}(\mathcal{F}^\bullet, \mathcal{G}^\bullet \otimes \mathcal{H}^\bullet) \\ \mathcal{F}^\bullet \otimes \mathcal{H}^\bullet &\rightarrow \underline{\text{Hom}}(\underline{\text{Hom}}(\mathcal{F}, \mathcal{O}_Y), \mathcal{H}) \\ f^* \underline{\text{Hom}}_{\mathcal{O}_Y}(\mathcal{F}^\bullet, \mathcal{G}^\bullet) &\rightarrow \underline{\text{Hom}}_{\mathcal{O}_X}(f^* \mathcal{F}^\bullet, f^* \mathcal{G}^\bullet) \end{aligned}$$

all induced by the relevant adjunctions.

(b) *If $\mathcal{G}_{(-)}^\bullet$ is a complex of filtered diagrams of sheaves, then there is a canonical map*

$$\varinjlim_{i \in I} \underline{\text{Hom}}_{\mathcal{O}_Y}(\mathcal{F}^\bullet, \mathcal{G}_i^\bullet) \rightarrow \underline{\text{Hom}}_{\mathcal{O}_Y}(\mathcal{F}^\bullet, \varinjlim_{i \in I} \mathcal{G}_i^\bullet)$$

where on either side we mean the termwise filtered colimit.

(c) *Assume that $\mathcal{F}^\bullet$ is a bounded complex of finitely presented modules. Then the map of (b) is an isomorphism. If $\mathcal{H}^\bullet$ is a complex of flat sheaves, then second map of (a) is an isomorphism. If f is a flat morphism, then the*

fourth map of (a) is an isomorphism. In particular, the canonical map

$$\underline{\mathcal{H}om}_{\mathcal{O}_Y}(\mathcal{F}^\bullet, \mathcal{G}^\bullet)_y \rightarrow \underline{\mathcal{H}om}_{\mathcal{O}_{Y,y}}(\mathcal{F}_y^\bullet, \mathcal{G}_y^\bullet)$$

is an isomorphism for all $y \in Y$.

- (d) If $\mathcal{F}^\bullet$ is strictly perfect, the second, third, and fourth maps of (a) are isomorphisms. In particular, $\mathcal{F}^\bullet$ is dualizable, meaning that the natural map

$$\mathcal{F}^\bullet \rightarrow \underline{\mathcal{H}om}(\underline{\mathcal{H}om}(\mathcal{F}^\bullet, \mathcal{O}_Y), \mathcal{O}_Y)$$

is an isomorphism.

- (e) If $\mathcal{F}^\bullet$ is bounded, and $\mathcal{H}^\bullet$ is strictly perfect, then the second map of (a) is an isomorphism.

Proof (a) Recall that we have already shown the internal tensor hom adjunction for complexes. The external version of this (i.e. the actual adjunction) is obtained by taking zero-cycles. The counit of this adjunction is the first map (and it is also natural in $\mathcal{F}^\bullet$). Now take the first map and tensor both sides with $\mathcal{H}^\bullet$. Then the tensor-hom adjunction produces the second map. For the third map, we have a natural map

$$\underline{\mathcal{H}om}_{\mathcal{O}_X}(\mathcal{F}^\bullet, \mathcal{G}^\bullet) \rightarrow \underline{\mathcal{H}om}_{\mathcal{O}_X}(\mathcal{F}^\bullet, f_* f^* \mathcal{G}^\bullet) \cong f_* \underline{\mathcal{H}om}_{\mathcal{O}_X}(f^* \mathcal{F}^\bullet, f^* \mathcal{G}^\bullet)$$

where the first one is induced by the unit map $\mathcal{G}^\bullet \rightarrow f_* f^* \mathcal{G}^\bullet$ of the standard pull-push adjunction on complexes. Therefore by the same adjunction this corresponds to the desired map. (b) just follows from the universal property applied termwise.

(c-e) Let us first prove all of these statements in the case of single sheaves, and then we'll generalize this to complexes. Throughout, note that if we have a natural transformation α of additive functors and $\alpha_{\mathcal{F} \oplus \mathcal{G}}$ is an isomorphism, then $\alpha_{\mathcal{F}}$ is an isomorphism. This is because e.g. we can consider the kernel and cokernel. Therefore, to prove the strictly perfect case for sheaves we usually only need to consider the finite free case.

First, consider the natural map

$$\mathcal{H}om(\mathcal{F}, \mathcal{G}) \otimes \mathcal{H} \rightarrow \mathcal{H}om(\mathcal{F}, \mathcal{G} \otimes \mathcal{H}).$$

If $\mathcal{F} \cong \mathcal{O}_Y^n$, then this reduces to $\mathcal{G}^n \otimes \mathcal{H} \rightarrow (\mathcal{G} \otimes \mathcal{H})^n$ which is an isomorphism. Hence this works if $\mathcal{F}$ is strictly perfect too. Likewise, if $\mathcal{H} \cong \mathcal{O}_Y^n$, then the map reduces to $\mathcal{H}om(\mathcal{F}, \mathcal{G})^n \rightarrow \mathcal{H}om(\mathcal{F}, \mathcal{G}^n)$ which is also an isomorphism. So we're done if $\mathcal{H}$ is strictly perfect as well. Finally, let $\mathcal{F}$ be finitely presented and $\mathcal{H}$ be flat. Since the question is local, we may assume there is a presentation $\mathcal{O}_X^m \rightarrow \mathcal{O}_X^n \rightarrow \mathcal{F} \rightarrow 0$, then we obtain a commutative diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{H}om(\mathcal{F}, \mathcal{G}) \otimes \mathcal{H} & \longrightarrow & \mathcal{G}^m \otimes \mathcal{H} & \longrightarrow & \mathcal{G}^n \otimes \mathcal{H} \\ & & \downarrow & & \downarrow \cong & & \downarrow \cong \\ 0 & \longrightarrow & \mathcal{H}om(\mathcal{F}, \mathcal{G} \otimes \mathcal{H}) & \longrightarrow & (\mathcal{G} \otimes \mathcal{H})^m & \longrightarrow & (\mathcal{G} \otimes \mathcal{H})^n \end{array}$$

The two right-hand maps are isomorphisms because tensor products commute with sums, so the left-hand one is.

Next consider the single sheaf case the fourth map of (a). If $\mathcal{F} \cong \mathcal{O}_Y^n$, then this becomes the map $f^*(\mathcal{G}^n) \rightarrow (f^*\mathcal{G})^n$. Thus the statement when $\mathcal{F}$ is strictly perfect. Now assume f is flat and $\mathcal{F}$ is finitely presented. Just as above, we may assume that we have $\mathcal{O}_Y^n \rightarrow \mathcal{O}_Y^m \rightarrow \mathcal{F} \rightarrow 0$, and then applying either $f^* \mathcal{H}om(-, \mathcal{G}^\bullet)$ or $\mathcal{H}om(f^*(-), f^*\mathcal{G})$ preserve exactness by flatness, and gives the result by the finite free case. In particular, if we take $f = \{y\} \hookrightarrow Y$, which is flat because taking stalks is exact, we get that $\mathcal{H}om(\mathcal{F}, \mathcal{G})_y = \mathcal{H}om(\mathcal{F}_x, \mathcal{G}_x)$ when $\mathcal{F}$ is finitely presented.

This last fact proves that (b) is an isomorphism in the case of sheaves with $\mathcal{F}$ finitely presented, because stalks commute with filtered colimits of sheaves, so we are reduced to the case of rings. But we have shown previously that finitely presented modules over a ring are compact. Note that this implies that $\mathcal{H}om(\mathcal{F}, -)$ commutes with arbitrary direct sums if $\mathcal{F}$ is finitely presented.

Finally, consider the third map of (a) in the sheaf case and assume $\mathcal{F}$ is finite free. Then it is just the map

$$\mathcal{H}^n \rightarrow \mathcal{H}om(\mathcal{O}_Y^n, \mathcal{H}) = \mathcal{H}om(\mathcal{O}_Y, \mathcal{H})^n$$

which is an isomorphism. Thus the isomorphism holds in the strictly perfect case, which completes all proofs in the case of individual sheaves.

It remains to reduce all claims to the sheaf case. Let's start with the second map of (a), and note that in all case we are assuming $\mathcal{F}^\bullet$ is bounded, so the products in the the hom complexes become sums. Consider the chain of isomorphisms:

$$\begin{aligned} (\mathcal{H}om_{\mathcal{O}_Y}(\mathcal{F}^\bullet, \mathcal{G}^\bullet) \otimes \mathcal{H}^\bullet)^n &\cong \bigoplus_p \mathcal{H}om_{\mathcal{O}_Y}^p(\mathcal{F}^\bullet, \mathcal{G}^\bullet) \otimes \mathcal{H}^{n-p} \\ &\cong \bigoplus_{r,p} \mathcal{H}om(\mathcal{F}^r, \mathcal{G}^{r+p}) \otimes \mathcal{H}^{n-p} \\ &\cong \bigoplus_{r,p} \mathcal{H}om(\mathcal{F}^r, \mathcal{G}^{r+p} \otimes \mathcal{H}^{n-p}) \\ &\cong \bigoplus_r \mathcal{H}om(\mathcal{F}^r, \bigoplus_p \mathcal{G}^{r+p} \otimes \mathcal{H}^{n-p}) \\ &= \bigoplus_r \mathcal{H}om(\mathcal{F}^r, (\mathcal{G}^\bullet \otimes \mathcal{H}^\bullet)^{r+n}) \\ &= \mathcal{H}om_{\mathcal{O}_Y}^n(\mathcal{F}^\bullet, \mathcal{G}^\bullet \otimes \mathcal{H}^\bullet). \end{aligned}$$

Here, in any of the three cases we are considering, the third step follows from the individual sheaf case. Now the fourth step: in two of the cases, $\mathcal{F}^r$ is finitely presented, so $\mathcal{H}om(\mathcal{F}^r, -)$ preserves arbitrary sums as explained. In the remaining case, $\mathcal{H}^\bullet$ is bounded, so there are only finitely many p for which $\mathcal{G}^{r+p} \otimes \mathcal{H}^{n-p}$ is nonzero.

Next consider the fourth map of (a). Again in the cases we consider, $\mathcal{F}^\bullet$ is bounded so the products turn into sums, and f^* commutes with sums. So termwise, we have

$$\begin{aligned} f^* \underline{\mathcal{H}om}^n(\mathcal{F}^\bullet, \mathcal{G}^\bullet) &= \bigoplus_p f^* \mathcal{H}om(\mathcal{F}^p, \mathcal{G}^{n+p}) \\ &\cong \bigoplus_p \mathcal{H}om(f^* \mathcal{F}^p, f^* \mathcal{G}^{n+p}) \\ &\cong \underline{\mathcal{H}om}^n(f^* \mathcal{F}^\bullet, f^* \mathcal{G}^\bullet), \end{aligned}$$

as desired, by the sheaf cases. The stalkwise consequence follows by taking f to be the inclusion of a point, as before.

For the map of (b), note that we reduce immediately to the sheaf case because finite direct sums commute with filtered colimits, so the $\mathcal{H}om$ complex doesn't affect anything. All that remains is the third map of (a) when $\mathcal{F}^\bullet$ is strictly perfect. Indeed,

$$\begin{aligned} (\mathcal{F}^\bullet \otimes \mathcal{H}^\bullet)^n &= \bigoplus_p \mathcal{F}^p \otimes \mathcal{H}^{n-p} \\ &\cong \bigoplus_p \mathcal{H}om(\mathcal{H}om(\mathcal{F}^p, \mathcal{O}_Y), \mathcal{H}^{n-p}) \\ &= \bigoplus_p \mathcal{H}om(\underline{\mathcal{H}om}^{-p}(\mathcal{F}^\bullet, \mathcal{O}_Y), \mathcal{H}^{n-p}) \\ &= \bigoplus_p \mathcal{H}om(\underline{\mathcal{H}om}^p(\mathcal{F}^\bullet, \mathcal{O}_Y), \mathcal{H}^{n+p}) \\ &= \underline{\mathcal{H}om}^n(\underline{\mathcal{H}om}(\mathcal{F}^\bullet, \mathcal{O}_Y), \mathcal{H}^\bullet), \end{aligned}$$

as desired, where in the fourth step we just replaced p with $-p$. $\square$

It is a good moment to quickly discuss the relationship between some of the conditions on sheaves we have been discussing. The upshot is that, in the algebro-geometric setting, considering objects which are locally isomorphic in the derived category to bounded complexes of finite locally free sheaves is really the most general nice situation we can get for computing $\mathbf{R}\mathcal{H}om$ in the first variable. Thus, hereafter, we will forget about the notion of internally projective sheaves.

Proposition 2.2.46 *Let X be a ringed space and $\mathcal{F}$ an $\mathcal{O}_X$ -module. We have the following implications: $\mathcal{F}$ locally strictly perfect $\implies \mathcal{F}$ internally projective $\implies \mathcal{F}$ stalkwise projective $\implies \mathcal{F}$ flat. Moreover:*

- (a) *If $\mathcal{F}$ is of finite type, then the first implication may be reversed.*
- (b) *If $\mathcal{F}$ is of finite presentation, then the second and third implications may be reversed as well.*

Now suppose X is a locally ringed space. Then $\mathcal{F}$ finite locally free $\iff \mathcal{F}$ locally strictly perfect. Moreover:

- (a) If $\mathcal{F}$ is of finite type, then $\mathcal{F}$ stalkwise projective implies $\mathcal{F}$ stalkwise free.
- (b) If $\mathcal{F}$ is of finite presentation, then $\mathcal{F}$ flat implies $\mathcal{F}$ locally free, i.e. all assertions are equivalent.

Proof In the first part, we have proven all of the starting implications. If $\mathcal{F}$ is finite type and internally projective, then we can find a surjection $\mathcal{O}_U^n \rightarrow \mathcal{F} \rightarrow 0$ and thus an exact sequence

$$\mathrm{Hom}_{\mathcal{O}_U}(\mathcal{F}|_U, \mathcal{O}_U^n) \rightarrow \mathrm{Hom}_{\mathcal{O}_U}(\mathcal{F}|_U, \mathcal{F}|_U) \rightarrow 0.$$

Thus by further restricting U we can lift the identity section, which gives a local splitting. Next, suppose $\mathcal{F}$ is of finite presentation. Then since $\mathrm{Hom}(\mathcal{F}, (-))_x \cong \mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{F}_x, (-)_x)$, the RHS being an exact functor for all x implies that $\mathrm{Hom}(\mathcal{F}, -)$ is an exact functor. If $\mathcal{F}$ is flat and finitely presented, then the same is true of $\mathcal{F}_x$ over $\mathcal{O}_{X,x}$ for each x , which implies that $\mathcal{F}_x$ is projective.

Now let X be locally ringed. If $\mathcal{F}$ is finite locally free, then clearly it is locally strictly perfect. Since being locally strictly perfect implies being of finite presentation, proving (b) will prove the other direction. Note that (a) holds because $\mathcal{F}_x$ is a finite projective module over a local ring for each x . For (b), we may assume $\mathcal{F}$ is stalkwise free.

Fix $x \in X$; choose a basis for $\mathcal{F}_x$ over $\mathcal{O}_{X,x}$, then by finiteness we can lift this to a map $\mathcal{O}_U^n \rightarrow \mathcal{F}|_U$ whose stalk at x is an isomorphism. Considering the cokernel $\mathcal{O}_U^n \rightarrow \mathcal{F}|_U \rightarrow \mathcal{G} \rightarrow 0$, $\mathcal{G}$ is of finite type so its support is closed, and $\mathcal{G}_x = 0$, so by shrinking U we may assume $\mathcal{O}_U^n \rightarrow \mathcal{F}|_U \rightarrow 0$ is surjective. Consider the kernel $\mathcal{K}$, which is of finite type since $\mathcal{F}$ is of finite presentation, hence its support is closed. Again $\mathcal{K}_x = 0$, so by shrinking U again $\mathcal{O}_U^n \rightarrow \mathcal{F}|_U$ becomes an isomorphism. $\square$

Now we come to some applications of pseudocoherence and perfectness to derived notions. We will not do things in full generality right now, and instead will prefer to discuss perfectness in more detail until after we have talked about tor-dimension.

Proposition 2.2.47 *Let Y be a ringed space with $A \in D^-(\mathcal{O}_Y)$ and $B \in D^{\geq b}(\mathcal{O}_Y)$.*

- (a) *Assume that $\mathcal{F}^\bullet \in \mathrm{Ch}^{\geq a}(\mathcal{O}_Y)$ is a bounded complex of flat sheaves. There is a morphism*

$$\mathbf{R}\mathrm{Hom}_{\mathcal{O}_Y}(A, B) \otimes \mathcal{F}^\bullet \rightarrow \mathbf{R}\mathrm{Hom}_{\mathcal{O}_Y}(A, B \otimes_{\mathcal{O}_Y} \mathcal{F}^\bullet)$$

in $D(\mathcal{O}_Y)$ natural in A and B .

- (b) *Assume that $f: X \rightarrow Y$ is a flat morphism. Then there is a morphism*

$$f^* \mathbf{R}\mathrm{Hom}_{\mathcal{O}_Y}(A, B) \rightarrow \mathbf{R}\mathrm{Hom}_{\mathcal{O}_X}(f^* A, f^* B)$$

natural in A and B .

(c) Let I be a filtered category. Then there is a morphism

$$\varinjlim_{i \in I}^D \mathbf{R}\mathcal{H}om_{\mathcal{O}_Y}(A, B_i) \rightarrow \mathbf{R}\mathcal{H}om_{\mathcal{O}_Y}(A, \varinjlim_I^D B_\bullet)$$

natural in A and $B_\bullet \in D^+(\mathrm{Mod}(\mathcal{O}_Y)^I)$.

(d) Let A be m -pseudocoherent. Then the morphism of (a) induces an isomorphism on H^p for $p > m + a + b$ and a monomorphism on H^{m+a+b} . The morphism of (b) induces an isomorphism on H^p for $p > m + b$ and a monomorphism on H^{m+b} . In particular, if A is pseudocoherent both are isomorphisms.

Remark 2.2.48 We are going to ignore (c) for right now, but it's easy to prove with the same arguments as in (a) and (b). It is interesting to note that it implies, in the case e.g. of a qcqs schemes, that $\mathbf{R}\mathcal{H}om_{\mathcal{O}_Y}(A, -)$ commutes with filtered colimits for A pseudocoherent. So for example in that case A is 'derived-compact'.

Proof (a-b) Note that since $\mathcal{F}^\bullet$ is a bounded complex of flat sheaves, $(-) \otimes \mathcal{F}^\bullet$ preserves arbitrary quasi-isomorphisms. Likewise, the same holds for f^* as it is exact. We have a natural map

$$\mathbf{R}\mathcal{H}om(A, B) \otimes \mathcal{F}^\bullet \leftarrow \underline{\mathcal{H}om}(A, B) \otimes \mathcal{F}^\bullet \rightarrow \mathbf{R}\mathcal{H}om(A, B \otimes \mathcal{F}^\bullet)$$

where the second map factors through $\underline{\mathcal{H}om}(A, B \otimes \mathcal{F}^\bullet)$. When B is a complex of injectives the left map is an isomorphism, so we obtain the desired map. Similarly we have

$$f^* \mathbf{R}\mathcal{H}om(A, B) \leftarrow f^* \underline{\mathcal{H}om}(A, B) \rightarrow \underline{\mathcal{H}om}(f^* A, f^* B) \rightarrow \mathbf{R}\mathcal{H}om(f^* A, f^* B),$$

and the left map is an isomorphism when $B \in K^+(\mathcal{O}_Y)$

(d) Let's do e.g. (b), and the proof of (a) will be the same. The statement is local on Y , so by replacing f with $f^{-1}(U) \rightarrow U$, we may assume there is a triangle $\mathcal{E}^\bullet \rightarrow A \rightarrow C \rightarrow \mathcal{E}^\bullet[1]$ with $C \in D^{<m}(\mathcal{O}_Y)$ and $\mathcal{E}^\bullet$ is strictly perfect. By naturality, we get a morphism of distinguished triangles:

$$\begin{array}{ccccccc} \longrightarrow & f^* \mathbf{R}\mathcal{H}om(C, B) & \longrightarrow & f^* \mathbf{R}\mathcal{H}om(A, B) & \longrightarrow & f^* \underline{\mathcal{H}om}(\mathcal{E}^\bullet, B) \\ & \downarrow & & \downarrow & & \downarrow \\ \longrightarrow & \mathbf{R}\mathcal{H}om(f^* C, f^* B) & \longrightarrow & \mathbf{R}\mathcal{H}om(f^* A, f^* B) & \longrightarrow & \underline{\mathcal{H}om}(f^* \mathcal{E}^\bullet, f^* B) \end{array}$$

Here we are using the fact that the pullback of strictly perfect complex is strictly perfect, and that a strictly perfect complex computes $\mathbf{R}\mathcal{H}om$ in the first variable. The right-hand map is an isomorphism by [Lemma 2.2.45](#) for two separate reasons, and both of the objects in the left-hand map lie in $D^{<b+m}$. Therefore the 5-lemma and the monomorphism 4-lemma provide the result regarding the middle map. $\square$

We record the important consequence that if A is m -pseudocoherent and $B \in D^{\geq b}$, then we get a natural isomorphism

$$H^p(\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A, B))_x \xrightarrow{\sim} H^p(\mathbf{R}\mathcal{H}om_{\mathcal{O}_{X,x}}(A_x, B_x))$$

for $p > m + b$. So $\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A, B)_x = \mathbf{R}\mathcal{H}om_{\mathcal{O}_{X,x}}(A_x, B_x)$ in the pseudocoherent case, which is important because on $\mathcal{O}_{X,x}$ we have enough projectives, so we can use alternative techniques to compute things there. Note in particular that for individual sheaves $\mathcal{F}, \mathcal{G}$, we have

$$\mathrm{Ext}_{\mathcal{O}_X}^p(\mathcal{F}, \mathcal{G})_x \xrightarrow{\sim} \mathrm{Ext}_{\mathcal{O}_{X,x}}^p(\mathcal{F}_x, \mathcal{G}_x)$$

for $p > m$, where $\mathcal{F}$ is m -pseudocoherent.

Before moving to tor-dimension, let us prove a few more results which allow us to relate pseudocoherence to having pseudocoherent terms or pseudocoherent cohomology.

Lemma 2.2.49 *A direct summand of an m -pseudocoherent object of $D(\mathcal{O}_X)$ is m -pseudocoherent.*

Remark 2.2.50 Here we will use that the direct sum of exact triangles is exact. This is true in any triangulated category, but in $K^*(\mathcal{A})$ or $D^*(\mathcal{A})$ for an abelian category $\mathcal{A}$, this is especially easy because cones commute with direct sums of morphisms.

Proof Suppose $A \oplus B = C$ is m -pseudocoherent. We have distinguished triangles $A \rightarrow A \rightarrow 0 \rightarrow A[1]$ and $B \xrightarrow{0} B \rightarrow B \oplus B[1] \rightarrow B[1]$, which add up to the triangle:

$$C \rightarrow C \rightarrow B \oplus B[1] \rightarrow C[1].$$

Thus $B \oplus B[1]$ is m -pseudocoherent. Applying this same result inductively to $B \oplus B[1]$ shows that $B[n] \oplus B[n+1]$ is m -pseudocoherent. Now consider the triangles

$$B[n] \rightarrow B[n] \oplus B[n-1] \rightarrow B[n-1] \rightarrow B[n+1]$$

coming from the corresponding split short exact sequence. Since this is local we may assume B is bounded above (since C is). Thus for $n \gg 0$, $B[n]$ is in $D^{< m}$ and hence is m -pseudocoherent. Thus $B[n-1]$ is m -pseudocoherent, and so inductively B is m -pseudocoherent. $\square$

Lemma 2.2.51 *Let X be a ringed space.*

- (a) *If $\mathcal{F}^\bullet \in K^-(\mathcal{O}_X)$ and $\mathcal{F}^i$ is $(m-i)$ -pseudocoherent for all i , then $\mathcal{F}^\bullet$ is m -pseudocoherent.*
- (b) *If $A \in D^-(\mathcal{O}_X)$ and $H^i(A)$ is $(m-i)$ -pseudocoherent for all i , then A is m -pseudocoherent.*

Proof We prove (b), and (a) follows from the same trick but using stupid truncations instead of smart ones. For every $p \in \mathbb{Z}$, use the canonical truncation triangle

$$\tau_{\leq p-1}A \rightarrow \tau_{\leq p}A \rightarrow H^p(A)[-p] \rightarrow \tau_{\leq p-1}A[1].$$

By assumption, $H^p(A)[-p]$ is m -pseudocoherent for all p . Also, for $p \ll 0$, $\tau_{\leq p-1}A$ will also be m -pseudocoherent (e.g. for $p < m$). Thus at that point $\tau_{\leq p}A$ is m -pseudocoherent, and $\tau_{\leq p+1}A$ is as well, and so on. Now for $p \gg 0$, $A = \tau_{\leq p}A$ so we're done. $\square$

Now we will discuss a nice condition on complexes which is that of having finite tor-dimension or having bounded tor amplitude. An object $A \in D(\mathcal{O}_X)$ has *tor amplitude* in $[a, b]$ if it is bounded above and $A \otimes_{\mathcal{O}_X}^{\mathbf{L}} \mathcal{F} \in D^{[a, b]}(\mathcal{O}_X)$ for each $\mathcal{O}_X$ -module $\mathcal{F}$. In particular, it must in fact be bounded. An object A has *finite tor dimension* if it has tor amplitude in $[a, b]$ for some a, b .

The good behavior of tor dimension comes from the following result.

Proposition 2.2.52 *Let X be a ringed space.*

- (a) *Let $\mathcal{F}^\bullet$ be a bounded above complex of flat $\mathcal{O}_X$ -modules. If $\mathcal{F}^\bullet$ has tor-amplitude in $[a, b]$, then $\text{cok}(\mathcal{F}^{a-1} \rightarrow \mathcal{F}^a)$ is flat. In particular, $\mathcal{F}^\bullet \rightarrow \tau_{\geq a}\mathcal{F}^\bullet$ is a quasi-isomorphism and $\tau_{\geq a}\mathcal{F}^\bullet$ is a bounded complex of flat modules.*
- (b) *An object $E \in D(\mathcal{O}_X)$ has tor amplitude in $[a, b]$ if and only if it is represented by complex of flat $\mathcal{O}_X$ -modules in $\text{Ch}^{[a, b]}(\mathcal{O}_X)$.*
- (c) *Let $\mathcal{R}_{\text{flat}} \subset \text{Mod}(\mathcal{O}_X)$ denote the preadapted class of flat $\mathcal{O}_X$ -modules. Let $D_{\text{tor}}^{[a, b]}(\mathcal{O}_X)$ and $D_{\text{tor}}^b(\mathcal{O}_X)$ be the full subcategories of $D(\mathcal{O}_X)$ of objects with tor amplitude in $[a, b]$, and finite tor dimension respectively. Then the natural maps*

$$K^{[a, b]}(\mathcal{R}_{\text{flat}}) \rightarrow D_{\text{tor}}^{[a, b]}(\mathcal{O}_X) \quad K^b(\mathcal{R}_{\text{flat}}) \rightarrow D_{\text{tor}}^b(\mathcal{O}_X)$$

are triangulated localizations by the quasi-isomorphisms.

Proof By assumption, $\mathcal{F}^\bullet$ is exact in degrees $< a$. Therefore

$$\dots \rightarrow \mathcal{F}^{a-2} \rightarrow \mathcal{F}^{a-1} \rightarrow \mathcal{F}^a \rightarrow \text{cok}(d^{a-1}) \rightarrow 0$$

is a flat resolution of the cokernel, so $\text{Tor}_{\mathcal{O}_X}^{-1}(\text{cok}(d^{a-1}), \mathcal{G})$ is isomorphic to the cohomology of

$$\mathcal{F}^{a-2} \otimes \mathcal{G} \rightarrow \mathcal{F}^{a-1} \otimes \mathcal{G} \rightarrow \mathcal{F}^a \otimes \mathcal{G}.$$

in the middle (for any $\mathcal{G}$). But

$$H^{a-1}(\mathcal{F}^\bullet \otimes \mathcal{G}) = H^{a-1}(\mathcal{F}^\bullet \otimes_{\mathcal{O}_X}^{\mathbf{L}} \mathcal{G}) = 0$$

since $\mathcal{F}^\bullet$ has tor amplitude in $[a, b]$, so $\text{Tor}_{\mathcal{O}_X}^{-1}(\text{cok}(d^{a-1}), \mathcal{G}) = 0$ for every $\mathcal{G}$. For the second statement, the map is a quasi-isomorphism because $\mathcal{F}^\bullet$ only has cohomology in degrees $\geq a$.

- (b) One direction is obvious because a bounded complex of flats computes $\otimes_{\mathcal{O}_X}^{\mathbf{L}}$. For the other direction, by (a) we may choose a complex of flats $\mathcal{F}^\bullet$

concentrated in $[a, \infty)$. Since it can't have cohomology in degrees $\geq b$, the sequence

$$0 \rightarrow \ker d^b \rightarrow \mathcal{F}^b \rightarrow \mathcal{F}^{b+1} \rightarrow 0$$

is exact, and thus $\ker d^b$ is flat. Thus $\tau_{\leq b} \mathcal{F}^\bullet \rightarrow \mathcal{F}^\bullet$ is a quasi-isomorphism and $\tau_{\leq b} \mathcal{F}$ has the desired properties.

(c) Let us prove the bounded case, which is the one that we'll actually use. Consider the functor $K^b(\mathcal{R}_{\text{flat}}) \rightarrow D^-(\mathcal{O}_X)$, which has essential image D_{tor}^b by (b). Thus we have to show it is fully faithful. Since $\mathcal{R}$ is preadapted, we already know that quasi-isomorphisms form a localizing class (compatible with the triangulation) in the source category, so we apply the criterion of [Prop. 1.1.21](#).

Namely, we want to show that for any quasi-isomorphism $\mathcal{F}_1^\bullet \rightarrow \mathcal{G}^\bullet$ in $K^-(\mathcal{R}_{\text{flat}})$ with $\mathcal{F}_1^\bullet$ in $K^b(\mathcal{R}_{\text{flat}})$, there is a map $\mathcal{G}^\bullet \rightarrow \mathcal{F}_2^\bullet$ with $\mathcal{F}_2^\bullet \in K^b(\mathcal{R}_{\text{flat}})$ such that the composite is a quasi-isomorphism. Assume $\mathcal{F}_1^\bullet$ is concentrated in degrees $\geq a$, then the map $\mathcal{G}^\bullet \rightarrow \tau_{\geq a} \mathcal{G}^\bullet$ satisfies these hypotheses by (a). $\square$

Remark 2.2.53 (experimental) Take the unbounded tensor product functor $K(\mathcal{O}_X) \times K(\mathcal{O}_X) \rightarrow K(\mathcal{O}_X)$, and restrict it to the two subcategories

$$K(\mathcal{O}_X) \times K^b(\mathcal{R}_{\text{flat}}) \rightarrow K(\mathcal{O}_X) \quad K^b(\mathcal{R}_{\text{flat}}) \times K(\mathcal{O}_X) \rightarrow K(\mathcal{O}_X).$$

Now if $\mathcal{F}^\bullet \in K^b(\mathcal{R}_{\text{flat}})$, then we know that $(-) \otimes \mathcal{F}^\bullet$ preserves quasi-isomorphisms of unbounded complexes, because in fact we showed this for all bounded above complexes of flat $\mathcal{O}_X$ -modules. But now suppose $\mathcal{G}^\bullet$ is an unbounded complex, then we claim $\mathcal{G}^\bullet \otimes (-)$ preserves quasi-isomorphisms in $K^b(\mathcal{R}_{\text{flat}})$. Note that it is closed under cones, so we may show it preserves acyclics. But both spectral sequences for a totalization apply here, because the corresponding bicomplex looks like either a vertical or horizontal strip. Thus by taking cohomology of $\mathcal{G}^\bullet \otimes \mathcal{F}^n$ first shows that the totaliation is acyclic. Therefore this descends to a bifunctor

$$(-) \otimes_{\mathcal{O}_X}^{\mathbf{L}} (-): D(\mathcal{O}_X) \times D_{\text{tor}}^b(\mathcal{O}_X) \rightarrow D(\mathcal{O}_X),$$

and likewise for the one in the other direction.

Thus, we can take derived tensor products when one of the factors has finite tor dimension; but we should be very careful using this because we haven't proven many properties of it. However, in general, almost everything carries over with similar proofs except as long as two unbounded complexes are not interacting with each other. For instance, it is easy to see from the corresponding properties that $A \otimes_{\mathcal{O}_X}^{\mathbf{L}} B \cong B \otimes_{\mathcal{O}_X}^{\mathbf{L}} A$ if A has finite tor dimension (in the sense of the two functors above). Similarly,

$$(A \otimes_{\mathcal{O}_X}^{\mathbf{L}} B) \otimes_{\mathcal{O}_X}^{\mathbf{L}} C \cong A \otimes_{\mathcal{O}_X}^{\mathbf{L}} (B \otimes_{\mathcal{O}_X}^{\mathbf{L}} C)$$

if e.g. both B and C have finite tor dimension, so that both sides above make sense.

Remark 2.2.54 We note that in the real world (i.e. under Noetherianity hypotheses), pseudocoherence is usually not a real condition beyond finite-type/coherence properties. But having finite tor dimension *is* a subtler condition. For example, let $X = \operatorname{Spec} k[\epsilon]/(\epsilon)^2$ and $Z = V(\epsilon) \subset X$. We essentially showed earlier that $\mathcal{O}_Z$ has infinite tor dimension on X ; let us demonstrate this again. Indeed, we have an infinite resolution

$$\cdots \rightarrow \mathcal{O}_X \xrightarrow{\epsilon} \mathcal{O}_X \rightarrow \mathcal{O}_Z \rightarrow 0$$

where each map corresponds to multiplication by ϵ and the final map is the canonical quotient. The section ϵ is zero on Z , so

$$\begin{aligned} \mathcal{O}_Z \otimes_{\mathcal{O}_X}^{\mathbf{L}} \mathcal{O}_X &\simeq [\cdots \rightarrow \mathcal{O}_Z \xrightarrow{0} \mathcal{O}_Z \xrightarrow{0} \mathcal{O}_Z \rightarrow 0] \\ &= \bigoplus_{n \geq 0} \mathcal{O}_Z[n], \end{aligned}$$

i.e. $\operatorname{Tor}_{\mathcal{O}_X}^{-n}(\mathcal{O}_Z, \mathcal{O}_Z) = \mathcal{O}_Z$ for all $n \geq 0$. Here, the point is that infinite tor dimension is a detector of singularities. We will see later that sheaves supported in the regular locus of a scheme tend to have finite tor dimension, and those whose support intersects the singular locus have infinite tor dimension. Thus, in general, we cannot expect sheaves to automatically have finite tor dimension unless we are working over a regular scheme.

Lemma 2.2.55 *Let $A \in D(\mathcal{O}_Y)$ have tor-amplitude in $[a, b]$.*

- (a) *For any $f: X \rightarrow Y$, $Lf^*(A)$ has tor-amplitude in $[a, b]$.*
- (b) *For any $B \in D(\mathcal{O}_Y)$ with tor-amplitude in $[c, d]$, $A \otimes_{\mathcal{O}_Y}^{\mathbf{L}} B$ has tor-amplitude in $[a + c, b + d]$.*

Proof Represent A by a complex $\mathcal{E}^\bullet \in \operatorname{Ch}^{[a, b]}$ of flats. (a) Then $Lf^*(A)$ is represented by $f^*\mathcal{E}^\bullet$, which is still a complex of flats in $[a, b]$. (b) Represent B by a complex of flats $\mathcal{F}^\bullet$ in $[c, d]$, then $\mathcal{E}^\bullet \otimes \mathcal{F}^\bullet$ represents the derived tensor product, the totalization is concentrated in $[a + c, b + d]$ by definition, and the tensor product and direct sum of flat sheaves is flat. $\square$

Let $f: X \rightarrow Y$ be a morphism of ringed spaces. We say f has tor-dimension $\leq n$ if $Lf^*(\mathcal{F}) \in D^{[-n, 0]}(\mathcal{O}_X)$ for every $\mathcal{F} \in \operatorname{Mod}(\mathcal{O}_Y)$.

Lemma 2.2.56 *Let $A \in D(\mathcal{O}_X)$.*

- (a) *A has tor-amplitude in $[a, b]$ if and only if $A_x \in D(\mathcal{O}_{X, x})$ has tor-amplitude in $[a, b]$ for all x .*
- (b) *If $f: X \rightarrow Y$ is a morphism, then f has tor-dimension $\leq n$ if and only if $\mathcal{O}_X$ has tor-amplitude in $[-n, 0]$ as an object of $D(f^{-1}(\mathcal{O}_Y))$, if and only if $\mathcal{O}_{X, x}$ has tor-amplitude in $[-n, 0]$ as an object of $D(\mathcal{O}_{Y, f(x)})$ for x .*

Proof (a) One direction follows from (a) of the previous proposition. For the other, note that an object being in $D^{[a,b]}(\mathcal{O}_X)$ can be checked on stalks, and $\otimes_{\mathcal{O}_X}^{\mathbf{L}}$ commutes with taking stalks.

(b) The same proof as in (a) shows that f has tor dimension n iff $(Lf^*\mathcal{F})_x = \mathcal{F}_{f(x)} \otimes_{\mathcal{O}_{Y,f(x)}}^{\mathbf{L}} \mathcal{O}_{X,x}$ is in $D^{[-n,0]}$ for every x and $\mathcal{F}$. The other condition in (b) says that $M \otimes_{\mathcal{O}_{Y,f(x)}}^{\mathbf{L}} \mathcal{O}_{X,x}$ is in $D^{[-n,0]}$ for every x and $\mathcal{O}_{Y,f(x)}$ -module M . Thus the question is whether or not every $\mathcal{O}_{Y,f(x)}$ -module M is of the form $\mathcal{F}_{f(x)}$ for some sheaf $\mathcal{F}$. This is true because we can take the skyscraper sheaf of M at $f(x)$ and apply f^{-1} . $\square$

Example 2.2.57 Let $i: Z \rightarrow X$ be a closed embedding on the underlying spaces. By the projection formula, $i_*\mathcal{O}_Z \otimes_{\mathcal{O}_X}^{\mathbf{L}} \mathcal{G} \cong i_*Li^*\mathcal{G}$ for every $\mathcal{G}$ on X . Thus $i_*\mathcal{O}_Z$ has tor amplitude in $[-n,0]$ iff $i_*Li^*\mathcal{G}$ is in $D^{[-n,0]}(\mathcal{O}_X)$ for every $\mathcal{G}$, iff $Li^*\mathcal{G} \in D^{[n,0]}(\mathcal{O}_Z)$ for every $\mathcal{G}$. Thus the tor dimension of $i_*\mathcal{O}_Z$ coincides with the tor-dimension of i .

Lemma 2.2.58 *Let $A \rightarrow B \rightarrow C \rightarrow A[1]$ be a distinguished triangle in $D(\mathcal{O}_X)$*

- (a) *If A has tor-amplitude in $[a+1, b+1]$ and C has tor-amplitude in $[a, b]$, then B has tor-amplitude in $[a, b]$.*
- (b) *If A and C have tor-amplitude in $[a, b]$, then B has tor-amplitude in $[a, b]$.*
- (c) *B has tor amplitude in $[a, b]$ and C has tor-amplitude in $[a-1, b-1]$, then A has tor-amplitude in $[a, b]$.*

Proof Note that $(-)[1]$ takes $D^{[a,b]}$ to $D^{[a-1,b-1]}$. Thus it suffices by rotation to prove only one of the statements, and we'll do (b). For each sheaf $\mathcal{F}$,

$$A \otimes^{\mathbf{L}} \mathcal{F} \rightarrow B \otimes^{\mathbf{L}} \mathcal{F} \rightarrow C \otimes^{\mathbf{L}} \mathcal{F} \rightarrow A[1] \otimes^{\mathbf{L}} \mathcal{F}$$

is exact, and the first and third objects lie in $D^{[a,b]}$, so the middle one does too completing the proof. $\square$

Next we prove the main structure result relating perfect complexes to finite tor dimension. The main element is the following trivial observation: if $A \rightarrow B$ is a map in $D(\mathcal{A})$ and $B \in D^{\geq n}(\mathcal{A})$, then the map factors as $A \rightarrow \tau_{\geq n}A \rightarrow B$. *Proof:* Consider the canonical triangle $\tau_{\leq n-1}A \rightarrow A \rightarrow \tau_{\geq n}A \rightarrow$ and apply the cohomological functor $\text{Hom}(-, B)$ to get an exact sequence

$$\text{Hom}(\tau_{\geq n}A, B) \rightarrow \text{Hom}(A, B) \rightarrow \text{Hom}(\tau_{\leq n-1}A, B) = 0$$

where the 0 is because $\tau_{\leq n-1}A \in D^{\leq n-1}$ and $B \in D^{\geq n}$.

Lemma 2.2.59 *Let $E \in D(\mathcal{O}_X)$ have tor amplitude in $[a, b]$. Then the following are equivalent:*

- i. *E is $(a-1)$ -pseudocoherent.*
- ii. *E is perfect.*

iii. E is pseudocoherent.

iv. E is locally isomorphic to a strictly perfect complex concentrated in $[a, b]$.

Proof The obvious claims are $(iv) \Rightarrow (ii) \Rightarrow (iii) \Rightarrow (i)$, so assume (i) and we prove (iv) . The claim is local so assume there is a map $\mathcal{E}^\bullet \rightarrow E$ inducing isomorphisms on H^p for $p \geq a$ with $\mathcal{E}^\bullet$ strictly perfect and $\mathcal{E}^p = 0$ for $p < a$. Since $E \in D^{>a}$, this map factors as $\mathcal{E}^\bullet \rightarrow \tau_{\geq a} \mathcal{E}^\bullet \rightarrow E$ so the second map is an isomorphism. Moreover, by [Prop. 2.2.52](#) $\tau_{\geq a} \mathcal{E}^\bullet$ is a complex of flats. By definition we have an exact sequence

$$\mathcal{E}^{a-1} \rightarrow \mathcal{E}^a \rightarrow (\tau_{\geq a} \mathcal{E}^\bullet)^a \rightarrow 0$$

which shows that $(\tau_{\geq a} \mathcal{E}^\bullet)^a$ is (-1) -pseudocoherent, i.e. finitely presented. Thus $(\tau_{\geq a} \mathcal{E}^\bullet)^a$ is locally strictly perfect ([Prop. 2.2.46](#)), so by restricting further $\tau_{\geq a} \mathcal{E}^\bullet$ is strictly perfect.

Now we know that E can be locally represented by a strictly perfect complex $\mathcal{E}^\bullet$ concentrated in $[a, \infty)$. In particular it is exact in degrees $> b$, so we have an exact sequence $0 \rightarrow (\tau_{\leq b} \mathcal{E}^\bullet)^b \rightarrow \mathcal{E}^b \rightarrow \mathcal{E}^{b+1} \rightarrow 0$, and hence $(\tau_{\leq b} \mathcal{E}^\bullet)^b$ is flat. Since $\mathcal{E}^b$ and $\mathcal{E}^{b+1}$ are pseudocoherent, it is also pseudocoherent, hence finitely presented, hence locally strictly perfect. So by restricting, $E \cong \tau_{\leq b} \mathcal{E}^\bullet$ becomes an isomorphism with a strictly perfect complex concentrated in $[a, b]$. $\square$

Corollary 2.2.60 *An object $E \in D(\mathcal{O}_X)$ is perfect if and only if E is pseudocoherent and locally has finite tor dimension.*

Proof $(\Rightarrow)$ Clear because E is locally isomorphic to a strictly perfect complex. $(\Leftarrow)$ The question is local on X , and thus the claim follows from [Lemma 2.2.59](#). $\square$

Definition 2.2.61 If X is a ringed space, we define $\text{Perf}^b(\mathcal{O}_X) := \text{Perf}(\mathcal{O}_X) \cap D_{\text{tor}}^b(\mathcal{O}_X)$ to be the full subcategory of $D(\mathcal{O}_X)$ of perfect objects with finite tor dimension.

In particular, this is *not* the category of bounded perfect objects. However, in practice this is not a big restriction because e.g. if X is quasicompact, then $\text{Perf}^b(\mathcal{O}_X) = \text{Perf}(\mathcal{O}_X)$ by the above corollary.

Example 2.2.62 Perfect objects have the 2/3 property for distinguished triangle. Indeed, we showed pseudocoherent objects do and it holds for finite tor dimension too. Another property is that a direct summand of a perfect object is perfect, because we showed this for pseudocoherence, and it's obvious for finite tor dimension.

Now let's use this formalism to prove stuff. We showed earlier that if $C \in D_{\text{tor}}^b(\mathcal{O}_X)$, then there is a natural map

$$\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A, B) \otimes_{\mathcal{O}_X}^{\mathbf{L}} C \rightarrow \mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A, B \otimes_{\mathcal{O}_X}^{\mathbf{L}} C),$$

and that if A is pseudocoherent then it is an isomorphism.

Now assume that $A \in \text{Perf}^b(\mathcal{O}_X)$, so it has tor amplitude in $[a, b]$, and hence is locally a strictly perfect complex in $[a, b]$. It follows that $\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A, \mathcal{O}_X)$ is also locally a strictly perfect complex in $[-b, -a]$, i.e. $\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A, \mathcal{O}_X) \in \text{Perf}^b(\mathcal{O}_X)$. If (and only if) $A \in \text{Perf}^b(\mathcal{O}_X)$, we define

$$A^\vee := \mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A, \mathcal{O}_X) \in \text{Perf}^b(\mathcal{O}_X)$$

to be the *dual object* of A . Also, perfect objects (or just objects of $D_{\text{tor}}^b(\mathcal{O}_X)$) have the property that tensoring with them sends $D^b(\mathcal{O}_X) \rightarrow D^b(\mathcal{O}_X)$, which we will also use. First, by the tensor-hom adjunction,

$$\text{Hom}_{D(\mathcal{O}_X)}(A^\vee \otimes_{\mathcal{O}_X}^{\mathbf{L}} A, \mathcal{O}_X) \cong \text{Hom}_{D(\mathcal{O}_X)}(A^\vee, A^\vee)$$

if $A \in \text{Perf}^b(\mathcal{O}_X)$, which makes sense because of what we have just said. Thus we obtain a natural map $\text{ev}: A^\vee \otimes_{\mathcal{O}_X}^{\mathbf{L}} A \rightarrow \mathcal{O}_X$ called the evaluation map. Now, for $B \in D^-(\mathcal{O}_X)$, this gives a natural map

$$A^\vee \otimes_{\mathcal{O}_X}^{\mathbf{L}} B \otimes_{\mathcal{O}_X}^{\mathbf{L}} A \rightarrow B$$

by associativity. If we assume $B \in D^b(\mathcal{O}_X)$, then the tensor-hom adjunction makes sense and gives a natural map

$$A^\vee \otimes_{\mathcal{O}_X}^{\mathbf{L}} B \rightarrow \mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A, B).$$

We claim this is an isomorphism; indeed, on an open set where $A^\vee$ is strictly perfect, this reduces to [Lemma 2.2.45\(d\)](#). Now, if $C \in D^-(\mathcal{O}_X)$, consider the associativity isomorphism

$$(A^\vee \otimes_{\mathcal{O}_X}^{\mathbf{L}} B) \otimes_{\mathcal{O}_X}^{\mathbf{L}} C \xrightarrow{\sim} A^\vee \otimes_{\mathcal{O}_X}^{\mathbf{L}} (B \otimes_{\mathcal{O}_X}^{\mathbf{L}} C).$$

Thus if we assume $B, C, B \otimes_{\mathcal{O}_X}^{\mathbf{L}} C \in D^b(\mathcal{O}_X)$ to be safe, we obtain a natural isomorphism

$$\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A, B) \otimes_{\mathcal{O}_X}^{\mathbf{L}} C \xrightarrow{\sim} \mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A, B \otimes_{\mathcal{O}_X}^{\mathbf{L}} C).$$

E.g. if we assume $C \in D_{\text{tor}}^b(\mathcal{O}_X)$ then this is a special case of what we showed before, but otherwise this is more general.

Now return to the evaluation map. Since $A^\vee$ is in $\text{Perf}^b(\mathcal{O}_X)$, $(A^\vee)^\vee$ is also in $\text{Perf}^b(\mathcal{O}_X)$, so the tensor-hom adjunction also gives the natural double-dual map

$$A \rightarrow (A^\vee)^\vee$$

corresponding to $A^\vee \otimes_{\mathcal{O}_X}^{\mathbf{L}} A \rightarrow \mathcal{O}_X$. This is also an isomorphism: on an open set where A is a strictly perfect complex, the question reduces to the corresponding statement in [Lemma 2.2.45\(d\)](#). Thus perfect complexes are *dualizable* in $D(\mathcal{O}_X)$. Therefore, we have a natural isomorphism

$$A \otimes_{\mathcal{O}_X}^{\mathbf{L}} B \xrightarrow{\sim} (A^\vee)^\vee \otimes_{\mathcal{O}_X}^{\mathbf{L}} B \xrightarrow{\sim} \mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A^\vee, B)$$

For any $B \in D^+(\mathcal{O}_X)$.

Let's play around with this some more. Under our hypotheses, as long as A and C are bounded, the natural map

$$\mathbf{R}Hom_{\mathcal{O}_X}(A, B) \otimes_{\mathcal{O}_X}^{\mathbf{L}} C \rightarrow \mathbf{R}Hom_{\mathcal{O}_X}(A, B \otimes_{\mathcal{O}_X}^{\mathbf{L}} C)$$

is an isomorphism if $C \in \text{Perf}^b(\mathcal{O}_X)$ (meaning we don't need to assume A is pseudocoherent). To see this, recall that we showed this in [Lemma 2.2.45](#)(e) for C strictly perfect and underived, so the only issue is that $B \otimes_{\mathcal{O}_X}^{\mathbf{L}} C$ may not be a bounded above complex of injectives even if C is perfect and B is a bounded above complex of injectives. The trick is that $B \otimes_{\mathcal{O}_X}^{\mathbf{L}} C \cong \mathbf{R}Hom(C^\vee, B)$, and $C^\vee \in \text{Perf}^b$, so locally on X it restricts to $Hom(\mathcal{E}^\bullet, B|_U)$ where $\mathcal{E}^\bullet$ is strictly perfect. But since $\mathcal{E}^\bullet$ is a complex of flats, this is a complex of injectives. Therefore in fact $B \otimes_{\mathcal{O}_X}^{\mathbf{L}} C$ does compute $\mathbf{R}Hom$ after restricting to a small enough open set.

Now let's add in the sheafy derived tensor-hom adjunction. Assume that $A, B \in D^b(\mathcal{O}_X)$ and $E \in \text{Perf}^b(\mathcal{O}_X)$. Then:

$$\mathbf{R}Hom_{\mathcal{O}_X}(A \otimes_{\mathcal{O}_X}^{\mathbf{L}} E^\vee, B) \cong \mathbf{R}Hom_{\mathcal{O}_X}(A, E \otimes_{\mathcal{O}_X}^{\mathbf{L}} B) \cong \mathbf{R}Hom_{\mathcal{O}_X}(A, B) \otimes_{\mathcal{O}_X}^{\mathbf{L}} E$$

We sometimes like to write the first isomorphism in more perverse ways, like an inverse tensor-hom adjunction:

$$\mathbf{R}Hom_{\mathcal{O}_X}(\mathbf{R}Hom_{\mathcal{O}_X}(E, A), B) \cong \mathbf{R}Hom_{\mathcal{O}_X}(A, E \otimes_{\mathcal{O}_X}^{\mathbf{L}} B)$$

Next we investigate the consequences of the derived pull-push adjunction under the assumption of perfectness. First of all, let $f: X \rightarrow Y$ be a morphism and suppose $A \in D^b(\mathcal{O}_Y)$ is such that $Lf^*(A) \in D^b(\mathcal{O}_X)$ (for whatever reason, e.g. A or f have finite tor-dimension). Then by adjunction there is a canonical unit map $A \rightarrow Rf_*Lf^*(A)$ corresponding to the identity $Lf^*A \rightarrow Lf^*A$. In the special case that $A = \mathcal{O}_Y$, this gives a map $\mathcal{O}_Y \rightarrow Rf_*\mathcal{O}_X$, so

$$\mathbf{R}Hom_{\mathcal{O}_Y}(E, \mathcal{O}_Y) \rightarrow \mathbf{R}Hom_{\mathcal{O}_Y}(E, Rf_*\mathcal{O}_X) \cong Rf_*\mathbf{R}Hom_{\mathcal{O}_X}(Lf^*(E), \mathcal{O}_Y)$$

for any $E \in D^-(\mathcal{O}_Y)$. Now suppose $E \in \text{Perf}^b(\mathcal{O}_Y)$, so that the $\mathbf{R}Hom$'s on both sides are finite, since Lf^*E is also in $\text{Perf}^b(\mathcal{O}_Y)$. Then this by adjunction induces a map

$$Lf^*(E^\vee) \rightarrow (Lf^*(E))^\vee$$

which we claim is an isomorphism. Indeed, look on an open nbhd where E is strictly perfect; then $Lf^*(E)$ and $\mathbf{R}Hom(E, \mathcal{O}_Y)$ are strictly perfect, so they compute $\mathbf{R}Hom$ and Lf^* respectively. Therefore this reduces to [Lemma 2.2.45](#)(d). Now it's easy to generalize this isomorphism: For any $B \in D^b(\mathcal{O}_Y)$ for which $Lf^*B \in D^b(\mathcal{O}_X)$, this means

$$\begin{aligned} Lf^*\mathbf{R}Hom_{\mathcal{O}_Y}(E, B) &\cong Lf^*(E^\vee \otimes_{\mathcal{O}_Y}^{\mathbf{L}} B) \\ &\cong Lf^*(E^\vee) \otimes_{\mathcal{O}_X}^{\mathbf{L}} Lf^*(B) \\ &\cong Lf^*(E)^\vee \otimes_{\mathcal{O}_X}^{\mathbf{L}} Lf^*(B) \\ &\cong \mathbf{R}Hom_{\mathcal{O}_X}(Lf^*(E), Lf^*(B)), \end{aligned}$$

extending the already-known case when f is flat.

Now, we finish this section with a discussion of the projection formula. Let $f: X \rightarrow Y$ be a morphism, and let $A \in D^b(\mathcal{O}_X)$ be an object such that $Rf_*(A) \in D^b(\mathcal{O}_Y)$. Then by adjunction there is a morphism $Lf^*Rf_*A \rightarrow A$ in $D(\mathcal{O}_X)$, so we get a morphism

$$Lf^*(Rf_*A \otimes_{\mathcal{O}_Y}^{\mathbf{L}} B) \xrightarrow{\sim} Lf^*Rf_*A \otimes_{\mathcal{O}_Y}^{\mathbf{L}} Lf^*B \rightarrow A \otimes_{\mathcal{O}_X}^{\mathbf{L}} Lf^*B$$

for any $B \in D^-(\mathcal{O}_Y)$. Now assume $B = E \in \text{Perf}^b(\mathcal{O}_Y)$, so that $Rf_*A \otimes_{\mathcal{O}_Y}^{\mathbf{L}} E$ and $A \otimes_{\mathcal{O}_X}^{\mathbf{L}} Lf^*E$ are bounded. Thus we get by adjunction a morphism

$$Rf_*(A) \otimes_{\mathcal{O}_Y}^{\mathbf{L}} E \rightarrow Rf_*(A \otimes_{\mathcal{O}_X}^{\mathbf{L}} Lf^*(E))$$

called the projection map. We claim it is an isomorphism. The proof is easy:

$$\begin{aligned} Rf_*A \otimes_{\mathcal{O}_Y}^{\mathbf{L}} E &\cong \mathbf{R}\mathcal{H}om_{\mathcal{O}_Y}(E^\vee, Rf_*A) \\ &\cong Rf_*\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(Lf^*E^\vee, A) \\ &\cong Rf_*(A \otimes_{\mathcal{O}_X}^{\mathbf{L}} Lf^*E) \end{aligned}$$

by the previous result. As a concrete application,

$$R\Gamma(X, A \otimes_{\mathcal{O}_X}^{\mathbf{L}} Lf^*E) = R\Gamma(Y, Rf_*A \otimes_{\mathcal{O}_Y}^{\mathbf{L}} E)$$

on the level of sheaf cohomology.

Remark 2.2.63 The remark about tor-dimension also applies to derived pull-back along morphisms $f: X \rightarrow Y$: f having finite tor dimension, and $Lf^*(B)$ being bounded for some B , is a delicate question relating to singularities. For example, if $p \in X$ is a point of a scheme, then $\text{Spec } \kappa(p) \rightarrow X$ has finite tor-dimension essentially exactly when p is singular. However, asking for Rf_*A to be bounded is not a real condition; e.g. its satisfied for any bounded A with X Noetherian, or any A with quasicohherent cohomology and f a qcqs morphism of schemes.

To conclude this section, we will summarize all of the results that we just proved for perfect objects.

Proposition 2.2.64 *Let $E \in \text{Perf}^b(\mathcal{O}_X)$ and $A, B \in D^b(\mathcal{O}_X)$. Then we have the following canonical isomorphisms as described above:*

- (a) $E \otimes_{\mathcal{O}_X}^{\mathbf{L}} B \xrightarrow{\sim} \mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(E^\vee, B)$.
- (b) $E^\vee \otimes_{\mathcal{O}_X}^{\mathbf{L}} B \xrightarrow{\sim} \mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(E, B)$.
- (c) $\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(E, A) \otimes_{\mathcal{O}_X}^{\mathbf{L}} C \xrightarrow{\sim} \mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(E, A \otimes_{\mathcal{O}_X}^{\mathbf{L}} B)$ assuming $A \otimes_{\mathcal{O}_X}^{\mathbf{L}} B \in D^b(\mathcal{O}_X)$.
- (d) $\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A, B) \otimes_{\mathcal{O}_X}^{\mathbf{L}} E \xrightarrow{\sim} \mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A, B \otimes_{\mathcal{O}_X}^{\mathbf{L}} E)$.
- (e) $\mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A \otimes_{\mathcal{O}_X}^{\mathbf{L}} E^\vee, B) \cong \mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(A, E \otimes_{\mathcal{O}_X}^{\mathbf{L}} B)$.

Assume that $f: X \rightarrow Y$ is a morphism, $E \in \text{Perf}^b(\mathcal{O}_Y)$, $B \in D^b(\mathcal{O}_Y)$, $A \in D^b(\mathcal{O}_X)$. Then:

- (a) $Lf^* \mathbf{R}\mathcal{H}om_{\mathcal{O}_Y}(E, B) \cong \mathbf{R}\mathcal{H}om_{\mathcal{O}_X}(Lf^*E, Lf^*B)$, assuming $Lf^*B \in D^b(\mathcal{O}_X)$.
- (b) $Rf_* A \otimes_{\mathcal{O}_Y}^{\mathbf{L}} E \cong Rf_*(A \otimes_{\mathcal{O}_X}^{\mathbf{L}} Lf^*E)$, assuming $Rf_* A \in D^b(\mathcal{O}_Y)$.

2.3 The classical theory

2.3.1 Cohomology of quasicoherent sheaves

Theorem 2.3.1 *Let X be an affine scheme and $\mathcal{F}$ a quasicoherent sheaf.*

- (a) *For an arbitrary cover $\mathcal{U}$ of X by quasicompact opens, $\check{H}^p(\mathcal{U}, \mathcal{F}) = 0$ for all $p > 0$.*
- (b) *$H^p(X, \mathcal{F}) = 0$ for all $p > 0$. More generally, if $\{\mathcal{F}_\alpha\}$ is a family of quasicoherent sheaves on X , then $H^p(X, \prod_\alpha \mathcal{F}_\alpha) = 0$ for all $p > 0$.*

Proof (a) First assume that the cover $\mathcal{U}$ by quasicompact opens is finite. Then in the Čech resolution of [Lemma 2.1.16](#)

$$0 \rightarrow \mathcal{F} \rightarrow \prod_{\ell(I)=0} j_{I,*}\mathcal{F}|_{U_I} \rightarrow \prod_{\ell(I)=1} j_{I,*}\mathcal{F}|_{U_I} \rightarrow \cdots,$$

we can write each product as a direct sum. All opens in an affine scheme are separated, so for each I the inclusion $j_I: U \rightarrow X$ is in particular a qcqs map, so $j_{I,*}\mathcal{F}|_{U_I}$ is quasicoherent. Thus each term in the exact sequence above is quasicoherent, so taking global sections shows that $\mathcal{F}(X)[0] \rightarrow C^\bullet(\mathcal{U}, \mathcal{F})$ is a quasi-isomorphism. We'll come back to the case that $\mathcal{U}$ is infinite momentarily.

(b) If $\{\mathcal{F}_\alpha\}$ is a family of quasicoherent sheaves on an affine scheme, then $\check{H}^p(\mathcal{U}, \prod_\alpha \mathcal{F}_\alpha) = \prod_\alpha \check{H}^p(\mathcal{U}, \mathcal{F}_\alpha) = 0$ for every (e.g.) finite affine open cover $\mathcal{U}$, by (a). Since X is separated, the affine opens form a basis of X which is closed under finite intersection. Therefore by [Lemma 2.1.15](#) we conclude $H^p(X, \prod_\alpha \mathcal{F}_\alpha) = 0$ for all $p > 0$.

(a) Now we do the infinite case. Consider the same Čech resolution as written above; even though the terms aren't all quasicoherent, it is a resolution of $\mathcal{F}$ by products of quasicoherent sheaves, namely a Γ -acyclic resolution. Thus $H^*(X, \mathcal{F})$ coincides with the (complex) cohomology of $\Gamma(X, \check{\mathcal{C}}(\mathcal{U}, \mathcal{F})) = \check{C}(\mathcal{U}, \mathcal{F})$, i.e. $\check{H}^*(\mathcal{U}, \mathcal{F})$, so we are done by (b). $\square$